Meeting of the Board
23 – 26 October 2023
Tbilisi, Georgia
Provisional agenda item 12

GCF/B.37/02/Add.09

2 October 2023

Consideration of funding proposals – Addendum IX

Funding proposal package for FP218

Summary

This addendum contains the following seven parts:

a) A funding proposal titled "Building climate resilience in the landscapes of Kigoma region, Tanzania";
b) No-objection letter issued by the national designated authority(ies) or focal point(s);
c) Environmental and social report(s) disclosure;
d) Secretariat's assessment;
e) Independent Technical Advisory Panel's assessment;
f) Response from the accredited entity to the independent Technical Advisory Panel's assessment; and
g) Gender documentation.

Table of Contents

Funding Proposal

Project/Programme title:	Building climate resilience in the landscapes of Kigoma region, Tanzania
Country(ies):	Tanzania
Accredited Entity:	United Nations Environment Programme (UNEP)
Date of first submission:	2020/01/23
Date of current submission	2023/09/27
Version number	*V.007*

GREEN CLIMATE FUND

[1] The project title has been changed from "Climate Resilient Development in Refugee Camps and Host Communities in Kigoma Region, Tanzania" to "Building Climate Resilience in the Landscapes of Kigoma Region, Tanzania. Both titles refer to the same project.

Contents

Note to Accredited Entities on the use of the funding proposal template

- Accredited Entities should provide summary information in the proposal with cross-reference to annexes such as feasibility studies, gender action plan, term sheet, etc.
- Accredited Entities should ensure that annexes provided are consistent with the details provided in the funding proposal. Updates to the funding proposal and/or annexes must be reflected in all relevant documents.
- The total number of pages for the funding proposal (excluding annexes) **should not exceed 60**. Proposals exceeding the prescribed length will not be assessed within the usual service standard time.
- The recommended font is Arial, size 11.
- Under the GCF Information Disclosure Policy, project and programme funding proposals will be disclosed on the GCF website, simultaneous with the submission to the Board, subject to the redaction of any information that may not be disclosed pursuant to the IDP. Accredited Entities are asked to fill out information on disclosure in section G.4.

Please submit the completed proposal to:

fundingproposal@gcfund.org

Please use the following name convention for the file name:

"FP-[Accredited Entity Short Name]-[Country/Region]-[YYYY/MM/DD]"

A

A. PROJECT/PROGRAMME SUMMARY

A.1. Project or programme	Project	**A.2. Public or private sector**	Public
A.3. Request for Proposals (RFP)	*If the funding proposal is being submitted in response to a specific GCF Request for Proposals, indicate which RFP it is targeted for. Please note that there is a separate template for the Simplified Approval Process and REDD+.* Not applicable		

A.4. Result area(s)

*Check the applicable GCF result area(s) that the overall proposed project/programme targets below. For each checked result area(s), indicate the estimated percentage of **GCF and Co-financers' contribution** devoted to it. The total of the percentages when summed should be 100% for GCF and Co-financers' contribution respectively.*

	GCF contribution	Co-financers' contribution[2]
Mitigation total	Enter number %	Enter number %
☐ Energy generation and access	Enter number %	Enter number %
☐ Low-emission transport	Enter number %	Enter number %
☐ Buildings, cities, industries and appliances	Enter number %	Enter number %
☐ Forestry and land use	Enter number %	Enter number %
Adaptation total	Enter number %	Enter number %
☒ Most vulnerable people and communities	10 %	Enter number %
☒ Health and well-being, and food and water security	30 %	32 %
☐ Infrastructure and built environment	Enter number %	Enter number %
☒ Ecosystems and ecosystem services	60 %	68 %

A.5. Expected mitigation outcome *(Core indicator 1: GHG emissions reduced, avoided or removed / sequestered)*	n/a	**A.6. Expected adaptation outcome** *(Core indicator 2: direct and indirect beneficiaries reached)*	Total beneficiaries: 1,281,857	
			Direct: 570,341	Indirect: 711,516
			1.3% of the population of Tanzania (Direct beneficiaries)	1.6% of the population of Tanzania (Indirect beneficiaries)
A.7. Total financing (GCF + co-finance[3])	23,618,646 USD	**A.9. Project size**	Small (Upto USD 50 million)	
A.8. Total GCF funding requested	19,007,353 USD *For multi-country proposals, please fill out annex 17.*			

[2] Co-financer's contribution means the financial resources required, whether Public Finance or Private Finance, in addition to the GCF contribution (i.e. GCF financial resources requested by the Accredited Entity) to implement the project or programme described in the funding proposal.
[3] Refer to the Policy of Co-financing of the GCF.

A.10. Financial instrument(s) requested for the GCF funding	*Mark all that apply and provide total amounts. The sum of all total amounts should be consistent with A.8.* ☒ Grant $ 19,007,353 ☐ Loan Enter number ☐ Guarantee Enter number ☐ Equity Enter number ☐ Results-based payment Enter number		
A.11. Implementation period	5 years	**A.12. Total lifespan**	Up to 20 years
A.13. Expected date of AE internal approval	*11/23/2021*	**A.14. ESS category**	B
A.15. Has this FP been submitted as a CN before?	Yes ☒ No ☐	**A.16. Has Readiness or PPF support been used to prepare this FP?**	Yes ☐ No ☒
A.17. Is this FP included in the entity work programme?	Yes ☒ No ☐	**A.18. Is this FP included in the country programme?**	Yes ☒ No ☐
A.19. Complementarity and coherence	Yes ☐ No ☒		
A.20. Executing Entity information	Government of the United Republic of Tanzania acting through its Vice President's Office (VPO) and United Nations High Commission for Refugees (UNHCR)		

A.21. Executive summary (max. 750 words, approximately 1.5 pages)

Kigoma is a region with a population of ~2.3 million people[4] that spans ~37,000 km^2 in the north-western corner of Tanzania. It is currently hosting ~250,000 refugees from neighbouring countries, the majority of whom are living in the refugee camps of Nduta and Nyarugusu. These settlements — which were rapidly established in response to critical humanitarian needs — have added to the population pressures on the surrounding degraded agro-ecological landscapes[5].

[4] Statistics received via personal communication with the Kigoma Regional Secretariat, March 2020.

[5] The presence of the refugee population has also contributed to reduced forest cover, largely as a result of re-establishing and expanding the Nduta and Mtendeli camps in October 2015 and January 2016, respectively. However, the rate of forest loss in these areas is less than the average rate of loss in the Kigoma region. To isolate the additional effect — or lack

Approximately 90% of the host community[6] population in the areas adjacent to the refugee camps and a substantial number of refugees are engaged in small-scale agriculture. As a result, they rely heavily on the goods and services provided by the region's ecosystems. Climate and non-climate drivers have degraded and are further degrading these ecosystems. For example, between 2011 and 2018, Kigoma lost ~5.3% of its tree cover (~108,000 ha of forest), a proportion of which is attributable to climate impacts such as increased fire incidence, rainfall intensity and flooding, as well as rising temperatures and hazards such as erosion, long term land degradation and the destruction of water resources. During community consultations host communities directly attributed some of these impacts to climate change and further highlighted climate change as a significant threat to natural resources and their livelihoods. Other development drivers of degradation include demand for fuelwood, increased by refugee presence, and agricultural expansion that is partly attributable to reduced yields from climate factors. Moreover, the enforcement of established guidelines and village bylaws intended to protect water resources and mitigate flood impacts remained a challenge. Community consultations highlighted that even for villages where Village Land Use Plans had been developed, village-level environmental committees did not have the capacity to enforce community guidelines and their ability to raise awareness among community members on land degradation was constrained. These development factors, together with climate change impacts such as increases in the intensity and frequency of unmanaged wildfires, flooding, erosion, and disease, are placing considerable pressure on the region's natural resources. Projected climate change impacts will further increase the intensity of this pressure.

Historical climate trends for the country show that temperatures are rising and rainfall is becoming more erratic and intense, with Tanzania's Nationally Determined Contribution (NDC) highlighting the severity of climate-related disasters and the impacts associated with rainfall variability, droughts and floods. In Kigoma, despite precipitation having increased over the long-term (an average of +7.9 mm per year since 1954), rainfall has become increasingly variable since 2000. There have also been changes to the rainfall profile, including more prolonged dry spells and more intense rainfall events. Both host communities and refugee camps reported an increased susceptibility of water resources to heightened rainfall variability. For example, since 2000, flood events have occurred at a higher frequency and intensity, with associated greater negative impacts on infrastructure and livelihoods than in the past. In particular, there has been an increase in the occurrence of large magnitude flood events by 7.6% for the 1:10-year flood event and an increase by 7% for the 1:100-year flood event. Host communities reported exacerbated flood impacts in areas along the Nyangwa River as well as in some zones of the former Mtendeli camp and Nyarugusu camp, as well as marshland areas developing in Nduta. In all three camps areas the main issue is storm water, not river flooding – causing gully erosion. Additionally, for the same period there was an increase in the average number of dry months[7] per year – a metric which correlates with drought events. Prior to 1999, the number of wet and dry months in a year were almost equal; in recent times, the dry months exceed the wet months by eight times. Mean annual temperatures in Kigoma have also increased over this period, rising by 0.18°C per decade since 2000. This increase in temperature has led to increased evaporation and evapotranspiration of surface water resources. These drying trends have threatened local livelihoods which rely on rainfall and ecosystem goods and services in the region.

The above-mentioned changes have exacerbated the existing environmental degradation and had profoundly negative impacts on agricultural outputs, incidence of pests and diseases, forest resources, surface- and groundwater availability, soil erosion and soil fertility. For example, as a result of the impacts of climate variability in the last 20 years, farmers in Kigoma region now harvest 40 to 53% less maize – a staple food critical for the region's food security[8]. Projections show that maize yields are expected to decrease by a further 17% because of climate change, with other crops being similarly affected[9]. For other crops like beans and cassava, years with abnormal rainfall patterns have also been associated with poor crop yields, and the increasing unpredictability and unreliability of rains has made planning for agricultural activities difficult. Reduced yields and harvest uncertainty drive further environmental degradation by farmers, through clearing of fertile forest land for agricultural expansion or as they are forced by necessity to engage in alternative livelihood activities, like charcoal production.

thereof — of the refugees on the ecosystems surrounding the camps relative to host communities and rural populations in Tanzania, a control site ~85km south of Nyarugusu was selected for comparative purposes. Despite having similar physical and climatic conditions, the forest loss around the counterfactual control site was found to be significantly greater than in the area surrounding the refugee camps.

[6] The Tanzania local communities living within 10-15km of the camps. For clarity in the project, the host community includes various groups such as farmers, agropastoralists, and other groups engaged in various livelihood activities. For shorthand, the term host communities refer to all these groups including agropastoralists that are not formally settled in villages.

[7] SPEI (drought index) values

[8] Mwakaje AEG Yanda PZ, Mung'ong'o CG, & Kangalawe RYM. 2013. The cost of climate change on smallholder farmers' adaptation and mitigation: The case of Kasulu district. International Journal of Applied Agricultural Research, 8(1): 83–101. Declines in harvest corresponds to times when variable rainfall was perceived.

[9] Agrawala S, Moehner A, Hemp A, Van AM, Smith J, Meena H,& Mwaipopo OU. 2003. Development and climate change in Tanzania: Focus on Mount Kilimanjaro. Paris: Organisation for Economic Co-operation and Development.

The proposed project responds to the climate change impacts affecting an already vulnerable and degraded landscape, whilst considering the complex needs of host communities and refugees in an environment where climate, humanitarian, and development approaches to problems and solutions intersect. Proposed interventions will be implemented in the districts of Kibondo, Kasulu and Kakonko, which host the refugee camps of Nduta, Nyarugusu, and the former camp of Mtendeli respectively. The outcomes of these interventions are designed to address the unique climate adaptation needs of host communities and refugees using an integrated landscape-level approach and promoting gender mainstreaming.

The project will contribute to shifting the paradigm from a baseline of 1) addressing refugee and host community needs separately and not recognizing that they are part of the same integrated socio-ecological landscape; 2) fragmented sectoral responses focused on small projects and initiatives on forestry and cookstoves, climate smart agriculture, livelihoods, and financial inclusion without integration; and 3) promoting initiatives that are not informed by projected climate impacts. Instead, the project will promote 1) inclusive solutions to adaptation needs of host communities and refugees through adaptation solutions at the landscape level; 2) integrated interventions across sectors that are designed to be mutually reinforcing; and 3) implementation of activities that are informed by projected climate impacts where the people and landscape would be better able to cope with, and adapt to, climate shocks.

Moreover, the project has significant strategic contributions to environmental and social issues in the region by enhancing ecosystem services provision that impact a broader biodiverse landscape, minimizing resource conflict and promoting peaceful coexistence, lessening exposure of women and girls to gender-based violence while collecting fuelwood, and increasing the protection space for refugees. In addition, the project will have mitigation co-benefits through sequestration of 762,073 million tCO_2e and emission reductions of 3.2 million tCO_2e over 20 years.

All project interventions will be informed by comprehensively assessing and planning for land use, forestry, cooking energy requirements (fuelwood) and livelihood interventions that are relevant to host communities and refugees. These actions will complement the estimated US$ 45 million of development investments in the region that are addressing non-climate baseline drivers of degradation and vulnerability.

To address the climate adaptation needs of the targeted beneficiaries, the project builds on bottom-up approaches, anchors to local governance processes, and promotes adaptive management in implementation. These project approaches will also help to manage the risks associated with the potentially variable movement of people and an evolving policy environment. Ultimately, the project will contribute to: i) enhancing the resilience of individuals and households to the impacts of climate change, particularly in terms of food security and livelihoods; and ii) disrupting the patterns of degradation of critical ecosystems and the services they provide. The project strategy is to ensure that the landscape and ecosystem of the Kigoma region can continue to support its residents, including refugees, and host communities, including agro-pastoralist groups that pass through the region as it faces challenges of climate change. The project directly contributes to GCF adaptation results areas on most vulnerable people, communities and regions and ecosystems and ecosystem services in the target host community and refugee areas. It will also have indirect impacts and benefits to the broader landscape by supporting increased ecosystem functions and services and policy interventions in Output 4.

The project will be executed by the Government of Tanzania, acting through its Vice President’s Office (VPO) and the UN High Commission for Refugees (UNHCR) in collaboration and coordination with key Tanzanian government agencies, including at the local level. This includes, the District Councils, the President’s Office Regional Administration and Local Government, Ministry of Home Affairs, National Environmental Management Council, Tanzania Forest Service, and other relevant agencies, local government authorities, humanitarian and development partners, and NGOs working in the region, promoting an approach that is participatory, gender-responsive and socially inclusive.

B.1. Climate context (max. 1000 words, approximately 2 pages)

Kigoma Region Context

Kigoma is one of the poorest regions in the country, with more than 32% of its population living below the poverty line, compared with ~26% nationally[10]. In some districts within Kigoma, the poverty rate exceeds 50% — including Kibondo district[11]. The region is characterised by generally low levels of economic development and its economy only contributes 3% to the national GDP[12]. Moreover, ~34% of the population of ~2.3 million people[13] is unemployed or underemployed[14]. This underdevelopment is compounded by the region's relatively high population growth rate of 4.8% — higher than in the rest of the country (at ~3%[15]). The average size of households in Kigoma (seven members) is also much higher than in the rest of Tanzania (4.8 members). Contributing to population pressure in the region is the influx of refugees into Tanzania from neighbouring countries as a result of civil unrest and conflict. Most of Tanzania's current refugees are hosted in the Kigoma region, which is home to a population of ~250,000 refugees (as of September 2021[16]) accounting for ~10% of the region's total population. In 2021, most of these refugees (~206,000) were accommodated in three UNHCR camps, namely: i) Mtendeli in Kakonko District, with a population of ~18,000; ii) Nduta in Kibondo District, with a population of ~60,000; and iii) Nyarugusu, in Kasulu District, with a population of ~128,000[17]. In January 2022, UNHCR and the Government of Tanzania completed a camp consolidation exercise to close Mtendeli refugee camp. The majority of Mtendeli refugees, with the exception of those opting for voluntary repatriation, were relocated in Nduta camp in Kibondo district. As part of the decommissioning process, UNHCR will embark on camp rehabilitation to facilitate environmental restoration.

The presence of these refugees, the rising population of the region, as well as the continual subdivision of land within families have resulted in over-intensive agricultural practises which are placing increasing pressure on limited natural resources. These camps, and their host communities are the target of this project (Figure 1).

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[10] The World Bank. 2019. Tanzania Mainland Poverty Assessment. Available at: https://www.nbs.go.tz/nbs/takwimu/hbs/Tanzania_Mainland_Poverty_Assessment_Report.pdf
[11] The World Bank. 2019. Tanzania Mainland Poverty Assessment. Available at: https://www.nbs.go.tz/nbs/takwimu/hbs/Tanzania_Mainland_Poverty_Assessment_Report.pdf
[12] Enabel. 2009. Support to Income Generating Activities for Coast and Kigoma Regions. Available at: http://www.diplomatie.be/oda/11768_PROGDESCR_Technisch_en_Financieel_Dossier.pdf
[13] Kigoma regional secretariat
[14] Enabel. 2009. Support to Income Generating Activities for Coast and Kigoma Regions. Available at: http://www.diplomatie.be/oda/11768_PROGDESCR_Technisch_en_Financieel_Dossier.pdf
[15] Worldometer. 2020. Tanzania Population. Available at: https://www.worldometers.info/world-population/tanzania-population/
[16] Government of Tanzania and UNHCR Tanzania Refugee Situation Statistical Report as of September 2021.
[17] Ibid

B

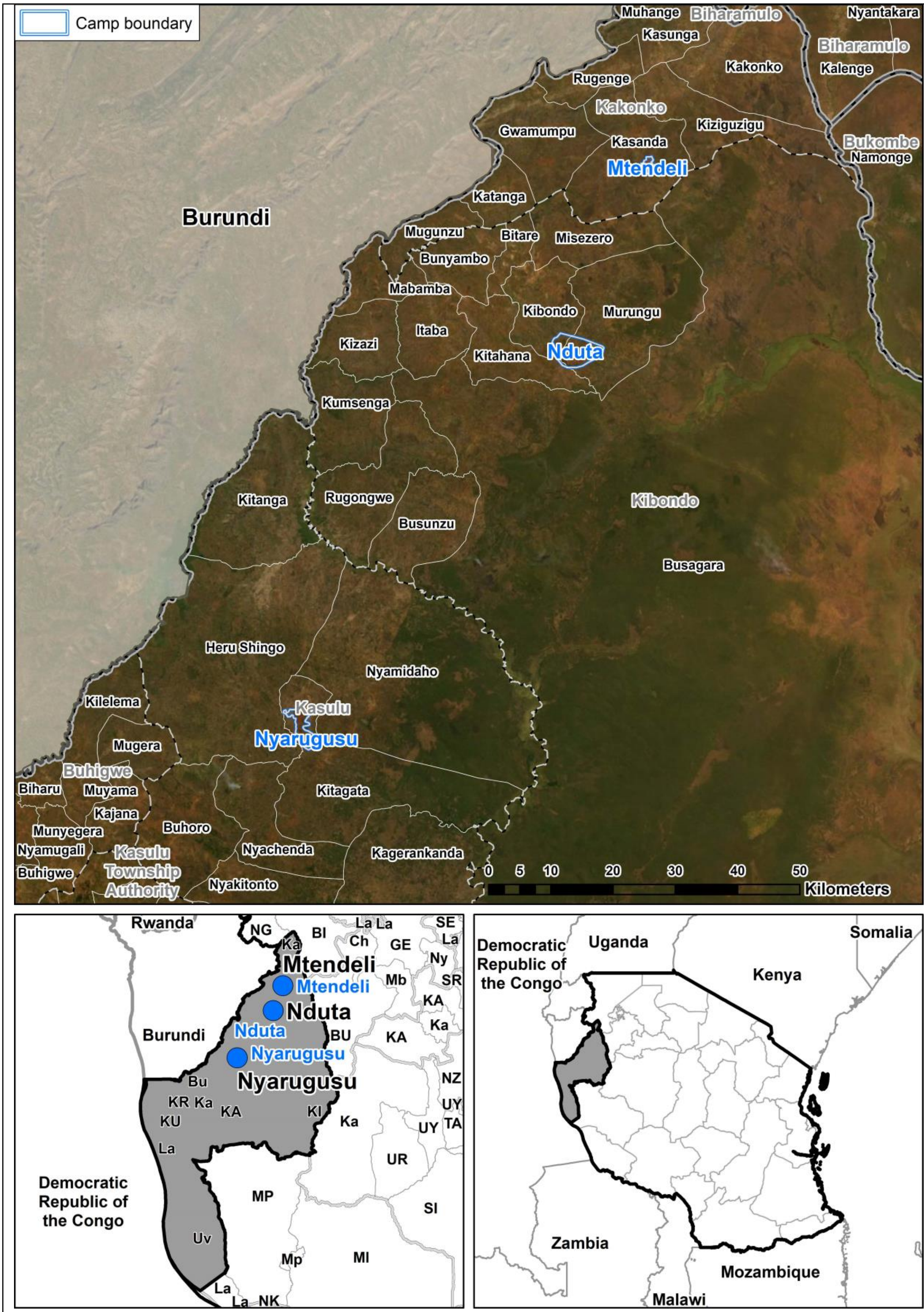


Figure 1. Map location of project sites

Ecosystem and ecosystem services

Kigoma Region is part of the Lake Tanganyika basin, a global biodiversity hotspot. To the east of the project sites are Moyowosi and Kigosi Game Reserves that have floodplain and wetland ecosystems that support large populations of buffalo, topi, hartebeest, kudu, waterbucks, and various bird species. A Ramsar site of international importance, the Malagarasi-Moyowosi wetland system, is located towards the south of Moyowosi game reserve. Further south in the region, Gombe National Park is home to the chimpanzee conservation program of the Jane Goodall Institute.

The project areas of Kakonko, Kibondo, and Kasulu Districts are characterized by a mosaic landscape or land area with patches of land cover types and land uses. In the project areas, these patches range from miombo forests, agricultural areas for maize, beans, and other crops, rangelands, woodlots for fuelwood, and built up areas for human settlements (Figure 2). At the landscape level, multiple ecosystem services are provided include provisioning services such as food from crops and fodder for livestock, wood fuel, non-timber forest products such as honey and mushrooms, fresh water; regulating services such as climate regulation, flood control, erosion control, water purification; cultural services such as wildlife viewing and recreation; and supporting services such as oxygen production, biogeochemical cycling, and others.

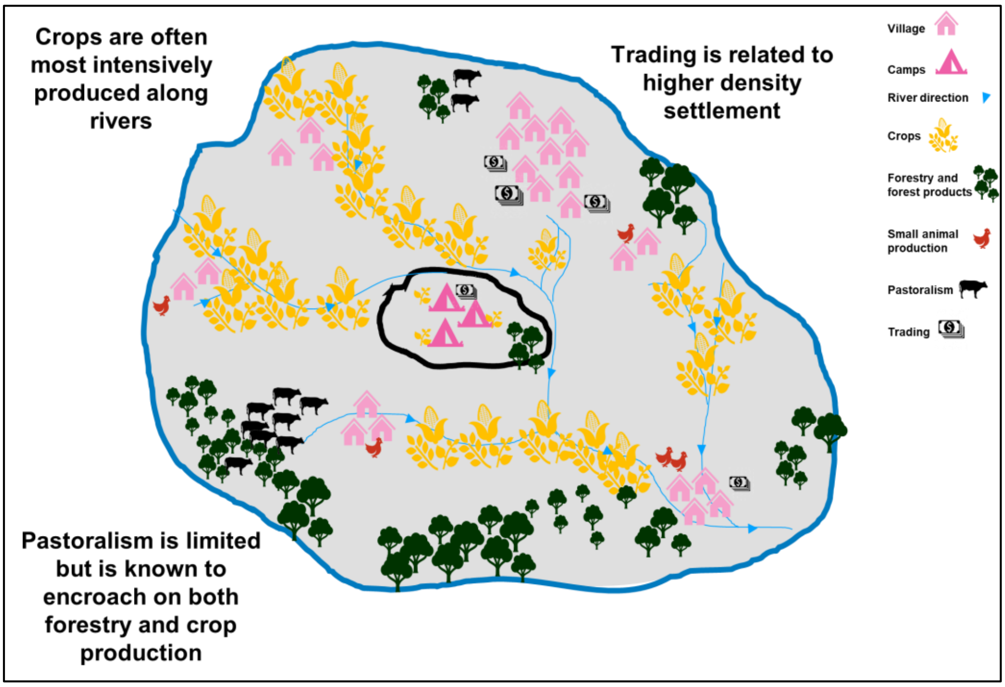
Patches of forest nested within the mosaic landscapes are mainly composed of miombo woodlands dominated by the genus *Brachystegia*. Characteristics of these forests include high endemism particularly for bird species and biodiversity as the woodlands provide food and cover for mammals such as elephants, wild dogs, and antelopes. These characteristics reflect high conservation value of miombo ecosystems in the face of climate change. One study found evidence of retraction of the range of miombo woodlands in Southern Africa in part due to temperature and precipitation changes and estimate that these ecosystems in Zimbabwe and Mozambique would retract by 30.6% and 47.5% respectively by 2050[18]. Potential increase of fire incidence under climate change can also negatively impact these woodlands, although miombo woodlands have high regenerative capacity if undisturbed and not encroached upon. The practice of setting fires to promote regeneration of grass is prevalent in each of the host areas. Fire events predominantly occur between June and September — the later part of the dry season to the early part of the wet season — when temperatures are higher, and vegetation is starting to lose moisture. The risk of uncontrolled fires is expected to be exacerbated under climate change conditions when longer dry spells and warmer temperatures are expected. These forests provide services that are critical to communities including source of fruits, honey, mushrooms, medicinal plants and other foraged forest products, energy and fodder, microclimate modification, carbon sequestration, pest control, biodiversity, regulate hydrological cycles, maintain soil fertility, provision of timber for tools, erosion control, and other important services[19]. Wild foods provide supplementary income of up to 42% in a household and support to nutrition. These forests also act as reserves in lean times, as consumption of wild foods can increase up to 30% of calorie intake during droughts[20]. As such, these ecosystems are critically important to livelihoods, household economies, and culture of communities in East and Southern Africa.

Protection of miombo woodlands through community-based forest management and planting of diverse native species support structure and functions of these forests and make them more resilient to disturbances. Without these interventions, these ecosystems are more vulnerable to unsustainable fuelwood harvesting, agricultural conversion, disturbance from grazing animals, unsustainable charcoal production and other factors contributing to their degradation that are expected to be intensified by climate change. Degraded landscapes are more vulnerable to forest succession or the replacement of one dominant plant community by another. Tanzania's National Communication to the UNFCCC also refer to observed transition of miombo woodlands into closed woodlands and evergreen forests.

Enhancing ecosystem services in mosaic landscapes are important in supporting resilience of both environment and people to climate shocks and other disturbances. The presence of native forests in close proximity to agricultural land, for example, has the potential to provide bee pollination services, particularly from Miombo woodlands, and support natural insect pest control and seed dispersal from birds in the forest, thereby increasing crop quality and yields. Heterogenous land cover helps enhance resilience by maintaining high diversity of different functional groups and patches of native vegetation may enhance dispersal of beneficial organisms such as pollinating birds and insects by

[18] Pienaar et al. 2015. Evidence for climate-induced range shift in Brachystegia (miombo) woodland. South African Journal of Science 111.
[19] FAO. 2018. Sustainable Management of Miombo Woodlands: Food Security, Nutrition, and Wood Energy. Available at: http://www.fao.org/3/i8852en/I8852EN.pdf.
[20] Ibid.

providing steppingstones across the landscape[21]. Native and planted forests in the landscape can also support landscape connectivity that is important for different species.



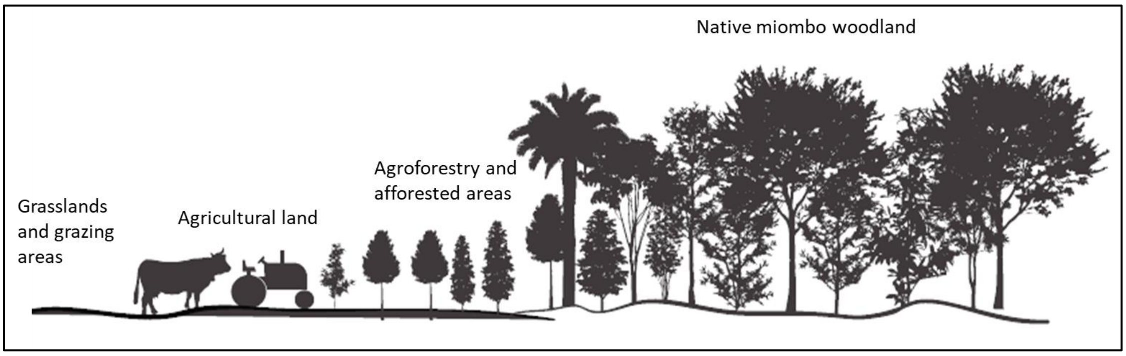


Figure 2. Schematic mapping of mosaic landscape with multiple land uses depicted in top (aerial) and side (linear) views.

Preserving and restoring native forest landscapes under community management regimes yield benefits in flood control and erosion reduction not only at the immediate intervention areas, but more broadly at the basin level. One of the threats to the biodiversity and health of Lake Tanganyika, home to hundreds of endemic aquatic life, is increased sediment and nutrient runoff. Increasing forest cover through planting of native species, afforestation, and agroforestry activities support water regulation and increased infiltration to groundwater supply in addition to providing carbon storage and cooling microclimate conditions amid projected temperature rise.

The project's approach to adaptation is adopting ecosystem-based approaches that support ecosystem functions and services. The interventions in degraded areas within and around refugee camps, as illustrated in Figure 1, are expected to support the broader resilience of both ecosystems and communities. Having locally led land management systems and climate smart agricultural improvements can reduce some of the drivers of deforestation while at the same time improving the condition of the forest and soils[22]. Managing forests for climate change impacts also safeguard forest-based climate resilient livelihoods, such as beekeeping, further encouraging sustainable forest use.

[21] Tscharntke, et al. 2005. Landscape perspectives on agricultural intensification and biodiversity—ecosystem service management. Ecological Letters 8: 857-874 as cited in Turner and Gardner. 2015. Landscape Ecology in Theory and Practice. Pattern and Process. 2nd Ed. Springer, USA.

[22] World Bank 2012. Using Forests to Enhance Resilience to Climate Change. What do we know about how forests can contribute to adaptation? Working Paper. Available at: http://documents1.worldbank.org/curated/en/753061468341334437/pdf/807830WP0PROFO0Box0379820B00PUBLIC0.pdf.

Agricultural practises and land-use change

In Kigoma region, agriculture is the primary economic activity and accounts for ~80% of the total economic output. Agricultural production in Kigoma is dominated by small-scale, subsistence farmers practising low-input farming, with average farm sizes ranging between 0.2 and 2 ha. Uptake of agricultural technology in Kigoma is low, with 62% of all cultivation done by hand[23]. Most farmers use traditional farming technologies such as hand hoes and bush knives, which limit agricultural efficiency and result in low yields. Seed quality is generally also poor and there is limited availability of improved seeds[24]. Agro-inputs, such as fertilisers and agrochemicals, are not widely used by farmers because they are generally not affordable and input distribution networks are not well developed. Crops grown in Kigoma are varied, with the commonest being maize, cassava, beans, rice, banana and groundnuts. Other crops produced in the region include peas, coffee, sugar cane, cotton and tobacco. Agriculture is the primary livelihood activity in Kasulu district — where Nyarugusu refugee camp is situated — which is a major food-producing area for the region. The main crops produced in the district are maize, cassava, sorghum, cabbage, beans and tomatoes[25]. In the project target districts (Kibondo, Kakonko, Kasulu), agroforestry is not widely practised, and few examples of on-farm tree planting were observed, mainly involving banana trees, some small-scale fuelwood plantations on farms, and natural elephant grass (*magugu*) planted in the areas around the camps.

Besides the small-scale agriculture practiced by host communities, agro-pastoral Sukuma people are also present in the project target area. They pass through the Kigoma region from the north in search of grazing land. Most of the Sukuma people are subsistence farmers and cattle herders. Some agro-pastoralist practices have been perceived to result in environmental degradation such as land clearing for agriculture including in village and national forest reserves and setting fires to trigger regeneration for pasture and getting rid of tse tse flies. One of the challenges related to community land use planning and enforcement undertaken by local authorities to date is the exclusion of (agro) pastoral communities such as the Sukuma, who are generally not part of the community planning and awareness raising processes due to their transient lifestyle and not being part of the local communities' social fabric.

Farming activities are also taking place within the camps. Refugees grow crops (mainly vegetables) individually around their homes in the form of kitchen gardens and in buffer zones; or as groups at the farm training centres (FTC) and community gardens. The harvested crops contribute to meeting household food and nutrition needs and complement the food rations provided by the World Food Programme. Crops are consumed by the household and/or (partially) sold to meet basic household needs. Sale takes places directly among households or in the small scattered markets in the camps. Refugees are also farming outside the camp through a range of informally negotiated land lease/land use or share cropping arrangements. Under these types of arrangements, refugees may pay some money or provide another form of payment (e.g. sharing of harvest) to host community landowners in exchange for land to cultivate. In some cases, refugees work on host community gardens and in return are given access to some land to cultivate. These agreements are generally on a seasonal/annual basis and are normally renegotiated after the harvesting period. In rare cases, host community landowners avail their idle land to refugees to cultivate for free. In addition, Burundian refugees are regularly engaged as casual agriculture laborers on agriculture plots owned and/or managed by both host communities and Congolese refugees. It is important to underline that these arrangements are informal and thus do not confer any tenure rights to the refugees.

Food security is a growing concern in Tanzania and while Kigoma currently has a surplus food supply, demand is increasing for food in other regions within the country as well as in neighbouring countries. Total grain equivalent for human consumption produced in Kigoma is ~1.5 million tonnes, with a required grain equivalent of only ~550,000 tonnes[26], resulting in a total food surplus of nearly 950,000 tonnes[27]. However, future climate change conditions — in combination with developmental challenges such as high population growth — will strongly affect the food situation in Kigoma in the future.

The resilience and sustainability of the agricultural sector in Kigoma is strongly affected by changes in food prices, which is primarily the result of fluctuations in regional demand for staple crops. Political instability in neighbouring countries decreases their agricultural productivity and results in food shortages, which are often augmented with supplies from Tanzania. As Kigoma is the agricultural heartland of western Tanzania, shortages in neighbouring

[23] FAO. 2016. Rapid Agriculture Needs Assessment in response to the "El - Niño" effects in the United Republic of Tanzania
[24] Seeds that produce higher yields than conventional seeds.
[25] Kibondo District Council, 2017, p. 11.
[26] Kigoma regional secretariat
[27] Ibid

countries are increasing demand for commodities from Kigoma[28]. For example, conflict in Burundi has resulted in an annual food deficit in the country of approximately 350,000–450,000 tonnes, most of which is being augmented by imports of maize, cassava and beans from Tanzania — particularly from Kasulu district in Kigoma[29]. Compounding these existing socio-political factors, climate change is resulting in reduced crop yields in Southern and East Africa, influencing the demand for crops grown in Tanzania, and exacerbating the challenge of price fluctuations in Kigoma and impacting food security in the region[30,31,32]. For example, the occurrence of two consecutive droughts in Southern Africa from 2014–15 and 2015-16 — exacerbated by a particularly severe ENSO cycle during the period — resulted in a significant reduction in the supply of white maize from the region[33]. Consequently, global stocks of maize decreased, demand for maize from Tanzania increased and prices in the country rose. By July 2016, Tanzania's wholesale white maize price rose to 32% above the five-year average — 10% higher than in June of the same year[34].

Increases in the price of crops result in reduced purchasing power of low-income households, inhibiting their ability to buy a greater variety of food, and contributing to malnutrition — particularly among children of 15 years and younger. Although there is a regional food surplus, these excess crops are not well distributed and are not nutritionally diverse. Moreover, farmers often rely on underdeveloped markets and weak institutions to support their livelihoods, and higher food prices — in combination with the reduced agricultural productivity of soils in the region — result in the increased need for farmers to expand their cultivated area or to over-rely on other resources such as wood from natural forests.

Expansion of agriculture has been cited as among the leading causes of degradation in the region[35]. This is observed as farm productivity declines, particularly for maize. Vegetable production, for example, has expanded to valleys that are more soil moisture rich and near streams and waterways for ease of irrigation[36]. Inadequate land-use planning results in communities competing for productive land and natural resources. The increasing reliance on these resources is driving land-use change in Kigoma region such as the conversion of forests to farmland. The trends are projected to accelerate under future climate change conditions if crop productivity within farms do not improve. Conversion of land to permanent agriculture reduces the area available for grazing livestock and results in increased competition between farmers. To supplement incomes, farmers and livestock keepers may also resort to selling charcoal and fuelwood in lean times, resulting the deterioration of miombo forests.

As regional prices rise, those within refugee camps are also directly affected. In refugee camps, food supply is influenced by market conditions outside of the camps and prices within the camps are already higher than in surrounding areas[37]. The limitations within the current markets in the camps and in the districts where they are hosted often result in market distortions within the camps. Most refugees do not produce their own food and rely on food aid and trade with outside producers. The fluctuation of food prices in the region increases prices for basic goods within camps even further and places pressure on aid agencies tasked with the provision of food aid in the camps. Refugee agencies rely largely on commodities produced within the region and the rise of prices has adverse consequences for the already limited budgets of these organisations[38]. As prices rise in the region, the purchasing power of refugees is reduced more significantly than in surrounding areas, leading to an even greater reliance of food aid[39]. Therefore, limited access of

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28 Enabel. 2009. Support to Income Generating Activities for Coast and Kigoma Regions. Available at: http://www.diplomatie.be/oda/11768_PROGDESCR_Technisch_en_Financieel_Dossier.pdf
29 WFP. 2016. Market Assessment: Nyarugusu refugee camp, United Republic of Tanzania. Available at: https://documents.wfp.org/stellent/groups/public/documents/ena/wfp286682.pdf?iframe
30 Arndt, C. et al., 2012. *Climate change, agriculture and food security in Tanzania*. The World Bank.
31 FAO.2017. Monthly report on food price trends. Available at: http://www.fao.org/3/a-i6829e.pdf
32 WFP. 2016. Market Assessment: Nyarugusu refugee camp, United Republic of Tanzania. Available at: https://documents.wfp.org/stellent/groups/public/documents/ena/wfp286682.pdf?iframe
33 WFP. 2016. Market Assessment: Nyarugusu refugee camp, United Republic of Tanzania. Available at: https://documents.wfp.org/stellent/groups/public/documents/ena/wfp286682.pdf?iframe
34 WFP. 2016. Market Assessment: Nyarugusu refugee camp, United Republic of Tanzania. Available at: https://documents.wfp.org/stellent/groups/public/documents/ena/wfp286682.pdf?iframe
35 FAO and UNHCR. 2018. Cost–benefit analysis of forestry interventions for supplying wood fuel in a refugee situation in the United Republic of Tanzania, by A. Gianvenuti & V.G. Vyamana. Rome, Food and Agriculture Organisation of the United Nations (FAO) and United Nations High Commissioner for Refugees (UNHCR). Available at: http://www.fao.org/3/ca0164en/CA0164EN.pdf
36 Kangalawe et al. 2017. Climate change and variability impacts on agricultural production and livelihood systems in Western Tanzania. Climate and Development, vol.9: 3.
37 Ibid
38 Ibid
39 de Bruin N & Becker P. 2019. Encampment and cash-based transfer: concord and controversy in the World Food Programme's pilot project in Nyarugusu refugee camp in Tanzania. *Journal of Immigrant & Refugee Studies*, *17*(4):492–508.

refugees to markets outside of the camps inhibits them from purchasing a wider selection of, and possibly less expensive, food[40].

For many of the refugees in Tanzania, several drivers of conflict in their countries of origin remain present, including continued regional instability, persistent ethnic tensions and an enduring culture of violence within communities[41]. Moreover, the insecure environment in these countries is being exacerbated by chronic poverty, malnutrition, low levels of development and young populations. In addition, environmental drivers of conflict are prevalent in the region, often related to scarcity of natural resources — particularly land. With agriculture being the primary livelihood for the majority of East and Central Africa, challenges related to land have significant potential to contribute to further instability.

Pests and diseases affecting agricultural yields and human health

Several agricultural pests and diseases are periodically recorded within Kigoma. Examples of the most common crop pests and diseases as well as their occurrence severity are presented below in Table 1.

Table 1. The most common pests and diseases and their occurrence severity for major crops in Kigoma[42].

Crop affected	Name of pest or disease	Occurrence severity
Maize	American bollworms	Average
	Army worm	Average
	Bacterial leaf spot	Average
Beans	American bollworms	Small
	Aphids	Average
	Anthracnose	Average
	Down mildew	Average
Cassava	Angular leaf spot	Small
	Cutworm	Small
	Mosaic disease	Average
	Streak disease	Average
	Wilt	Average
Banana	Banana Xanthomonas	Average

To date, evidence suggests that crop pests and diseases within Kigoma do not currently pose a considerable threat to the nutrition and food security of local communities[43]. Moreover, there have been no recent major outbreaks in the region that result in extensive crop damage. The most recent army worm outbreak, for example, occurred more than fifty years ago. Climate change, however, has the potential to increase the occurrence of agricultural pests and diseases beyond the capacity of local communities to manage them. The presence, population dynamics and persistence of these pests and diseases are influenced by changes in climatic conditions, which is more pronounced under circumstances with high rainfall amounts. Farmers in Tanzania, for example, have suggested that there has been a notable increase in stalk borers (*Calidea dregii*) — known locally as Mpipi — as a result of warmer temperatures.

An increase in the use of synthetic pesticides or insecticides in the control and suppression of pests and diseases poses a risk to both local communities and the environment. Recently, the use of pesticides for crops has increased substantially, resulting in the contamination of consumer products such as fruits and vegetables and presenting a major public health concern in Kigoma[44]. Inhabitants of the region are exposed to pesticides indirectly through the consumption of contaminated food and through aerosols released into the atmosphere, as well as through direct exposure via occupational, agricultural or domestic use. Consequently, farm workers are most exposed and are at a higher risk of acute intoxication and long-term adverse health conditions. Diseases and health conditions in farm workers that have been associated with exposure to pesticides include diabetes, neurological disorders such as Alzheimer's disease,

[40] Kigoma regional secretariat
[41] UNICEF. 2014. Conflict Analysis Summary: Burundi. Available at: https://reliefweb.int/sites/reliefweb.int/files/resources/Burundi-UNICEF_AID1.pdf
[42] From consultation with UNHCR Environment Officers, December 2018.
[43] Ibid.
[44] Massomo SMS. 2019. Vegetable pest management and pesticide use in Kigoma, Tanzania: Challenges and way forward. Huria Journal, 26: 195–227.

impaired reproductive fertility and numerous types of cancer[45]. Minimal education and training amongst rural farmers, limited alternatives to pesticides, insufficient information on hazards related to the use of pesticides and the unwillingness of farmers to risk crop losses contribute to the unsafe use of these synthetic chemicals to control pests.

Human disease epidemics are a major concern affecting communities within Kigoma. Vector-borne and water-borne diseases are the most significant climate-sensitive diseases reported in Tanzania. These include malaria, Rift Valley fever (RVF), dengue fever, cholera, Human African Trypanosomiasis (HAT), plague, schistosomiasis and diarrheal diseases. In Nyarugusu the community reported the prevalence of waterborne diseases including diarrhoea, malaria, Bilharzia, and urinary tract infections during the rainy season. Downstream, in the host community of Kalimongoma, there has been an increase in the incidence of cholera and urinary tract infections, the community believes are as a result of refugees upstream using the Makere River for bathing and washing of clothes. These diseases have been amplified during periods of severe weather, including floods and droughts. In recent years, the region has had several outbreaks that have had considerable impacts on public health, in refugee camps and host communities. In 2015, cholera affected refugee and host populations in four of Kigoma's districts, including Ujiji, Kigoma, Uvinza and Kasulu. In total, 5,581 cholera cases resulted from the 2015 outbreak, of which 1,051 occurred within host populations and 4,528 in refugee communities[46]. The Kigoma region is additionally impacted by zoonotic[47] and vector-borne diseases, many of which are influenced by changes in climate. For example, Human African Trypanosomiasis (HAT) or sleeping sickness is a potentially fatal disease transmitted by the tsetse fly, with most cases in Kigoma being contracted when individuals in the host communities travel into unmanaged forested areas for provisions or to conduct livelihood activities such as collecting firewood or producing charcoal[48]. Refugee populations are also particularly at risk, owing to the overlap in cases of different strains of HAT brought into the region from neighbouring countries. Within Tanzania, Kigoma is particularly at risk of HAT, with the region accounting for 81% of the country's cases[49]. All districts in Kigoma show evidence of this disease, with Kibondo and Kasulu containing the highest proportions of cases, as well as the refugee camps of Nduta and Nyarugusu, respectively.

Forest resources and deforestation

Western Tanzania, where Kigoma is located, is more densely forested than other parts of the country, with forest and woodland covering 62% of the land area. Across Tanzania, ~23% of land area is occupied by production forests — used for, *inter alia*, charcoal production — and ~7% is classified as shifting cultivation, which contains forests in various stages of regrowth that will not fully recover because of the practised farming system[50]. Literature estimates that deforestation in Tanzania has increased from ~130,000 ha[51] to 500,000 ha[52] of forest loss per year, between 1998 and 2010. In the Kigoma region specifically, observations demonstrate changes in land use, where closed woodland had diminished to only 5.6% in 2010 from 23.2% in 1990, while cultivation increased to 17% in 2014 from 2% in 1990.

Most of the refugee and host communities in Kigoma live within or near forests (Figure 3), and while lower than the national average, existing demand for natural resources by households within this region is remains high. The Kigoma Household Budget Survey 2017-2018 indicates that in Kigoma 57.2% of the population indicated to use fuelwood as the main source of cooking energy (against 60.9% national average), with 35.7% relying mainly on charcoal (against 28.8% for the national average). Use of solar for cooking was relatively developed in the region, with some 4.3% indicating they use solar as the main source of energy (against only 1.1% on average nationwide). Each of the host communities identified deforestation in the village and communal areas as a major environmental issue compromising ecosystem service provision. The trees are cleared predominantly for fuel wood and charcoal making, with community members highlighting that this occurs primarily from June to September each year. Within the refugee camps, the

[45] Gangemi S, Miozzi, E, Teodoro M, Briguglio G, de Luca, A, Alibrando C, Polito I & Libra M. 2016. Occupational exposure to pesticides as a possible rick factor for the development of chronic diseases in humans (Review). Molecular Medicine Reports, 14: 4475–4488.
[46] From consultation with UNHCR Environment Officers, December 2018.
[47] A zoonotic disease is an infectious disease that can be transferred from animals to humans.
[48] Kibona SN, Nkya GM & Matemba L. 2002. Sleeping sickness situation in Tanzania. Tanzania Health Research Bulletin, 4: 27–29.
[49] Nzalawahe J. 2010. Role of domestic animals in the epidemiology of human African trypanosomiasis (HAT) in Kigoma — Tanzania. Master of Veterinary Medicine Thesis, Sokoine University of Agriculture, Tanzania.
[50] Ministry of Natural Resources and Tourism and Tanzania Forest Services Agency. 2015. National Forest Resources Monitoring and Assessment of Tanzania Mainland (NAFORMA). Available at: http://www.tfs.go.tz/uploads/NAFORMA_REPORT.pdf
[51] United Republic of Tanzania, URT (1998) National Beekeeping Policy. Ministry of Natural Resources and Tourism, Dar es Salaam.
[52] Food and Agriculture Organisation, FAO, Global Forest Resources Assessment 2010

average daily consumption of wood fuel is ~1.8 kg per person[53]. Deforestation within the camps themselves is an acute problem that has left some areas "desert-like" in Nyarugusu and limits flood mitigation. Mtendeli former camp has less extensive deforestation, with reforestation being undertaken in certain areas in the camp. Numerous adverse impacts are caused by deforestation, including the reduced provisioning of ecosystem services such as water regulation, flood regulation and soil fertility and stabilisation, which further directly impact agricultural systems and threatens ecosystem-based livelihoods.

Forests within Tanzania are also increasingly becoming degraded by wildfires, with consequent changes in vegetation composition[54]. Kigoma ranks second as the most affected region by wildfires, with at least 1 million ha burned annually[55]. In the district ranking of area burnt, Kibondo ranks 6th and Kasulu 21st among the surveyed sites[56]. Most of the fires occur in forest and game reserves from the later part of the dry season to the early part of the wet season, when vegetation is starting to lose moisture and is more prone to combustion. However, as Tanzania's climate becomes hotter and drier, the frequency and intensity of uncontrolled forest fires are expected to increase. Fires can negatively impact ecosystem service provisioning by, *inter alia*, reducing soil organic matter which affects soil aggregation — one of the major factors influencing the chemical, physical and biological processes that contribute to soil productivity and agricultural sustainability. Moreover, fire-related impacts on Tanzanian ecosystems are expected to increase under climate change with higher temperatures and lower rainfall increasing vegetation flammability, which will result in a shift in species composition because of greater fire frequency[57].

As traditional agriculture becomes less sustainable and natural vegetation becomes less suitable for agricultural purposes – as a result of climate impacts – this places pressure on these forest ecosystems and results in decreased agricultural yield and fewer biomass stocks. To meet this growing productivity gap, host community and camp residents have relied more heavily on the natural tree stocks to an unsustainable degree and expanded farming areas to meet their needs, leading to significant deforestation throughout the Kigoma region.

The additional burden of the growing population, including through refugee influx, exacerbated by land clearing to reopen the two of the project's target camps — Nduta in October 2015 and Mtendeli in January 2016 — has had a discernible impact forest cover. During UNHCR and FAO field surveys, it was observed that fuel wood collection and clearing for agriculture are the main causes of forest degradation and deforestation in the areas surrounding the refugee camps[58]. Between 2011 and 2018, Kigoma lost ~108,000 ha of tree cover (~5.3% decrease). Several studies have pointed to the fact that the establishment of the Mtabila[59] and Nyarugusu refugee camps have contributed to accelerated deforestation, as a result of increased fuelwood harvesting and illegal cultivation. This has also created complex host community-refugee relations. Community consultations confirmed that deforestation is still prevalent in many of the project areas, with land being cleared for agricultural development, fuelwood or charcoal production and the illegal mining of limestone in the national forest reserve areas in Kasulu District.

The majority of the additional forest loss after 2015 around the three refugee camps, however, occurred within the camp boundaries. It is likely that this initial acceleration of forest loss is attributable to the reestablishment and expansion of the camps, to accommodate the additional population (Figure 4). The land granted (in the 1970s for Nyarugusu and in the 1990s for Mtendeli and Nduta, initially) by the Government of Tanzania for the emergency settlement of refugees in the camps was selected to be fit-for-purpose at the time, located near to the border with Burundi and well situated topographically and ecologically. As described below in Section B.3, the initial rapid settlement at these sites could not consider the long-term implications of prolonged development pressures or climate change on land-use planning and practices, which have led to maladaptation as a result. Since the camps' reopening in 2015/2016, the forested area within 4 km of the camps is still considerable, particularly around Nduta and Nyarugusu. In these areas, the rate of

[53] Quigley P. 2016. Providing energy for cooking: an analysis of fuel options in refugee camps in Tanzania. Geneva, Switzerland, UNHCR.

[54] Mussa KR, Mjemah IC & Malisa ET. 2012. The role of development projects in strengthening community-based adaptation strategies: the case of Uluguru mountains agricultural development project (UMADEP) – Morogoro – Tanzania. International Journal of Agricultural Sciences, 2: 157–165.

[55] FAO. 2013. A Fire Baseline for Tanzania. Available at: http://www.fao.org/forestry/39605-016494740dc4dd315b0b298b573b083b.pdf.

[56] Information on fire impacts is limited and a further area of study.

[57] Sangeda, A., and Malole, J. 2014. "Tanzanian rangelands in a changing climate: Impacts, adaptations and mitigation." *Journal of Agricultural Science* 2 (1).

[58] FAO and UNHCR. 2018. Cost–benefit analysis of forestry interventions for supplying wood fuel in a refugee situation in the United Republic of Tanzania, by A. Gianvenuti & V.G. Vyamana. Rome, Food and Agriculture Organisation of the United Nations (FAO) and United Nations High Commissioner for Refugees (UNHCR). Available at: http://www.fao.org/3/ca0164en/CA0164EN.pdf

[59] The Mtabila refugee camp was closed in 2012.

forest loss post-2015 is comparable to that of pre-2015 and is less than the average rate of acceleration in the Kigoma region (see Section 7.3 of the Feasibility Study). The implication of this is that the increase in the refugee population post-2015 exerted a negligible impact on the surrounding forests, as a result of clearing trees to expand settlements, beyond the camp boundary.

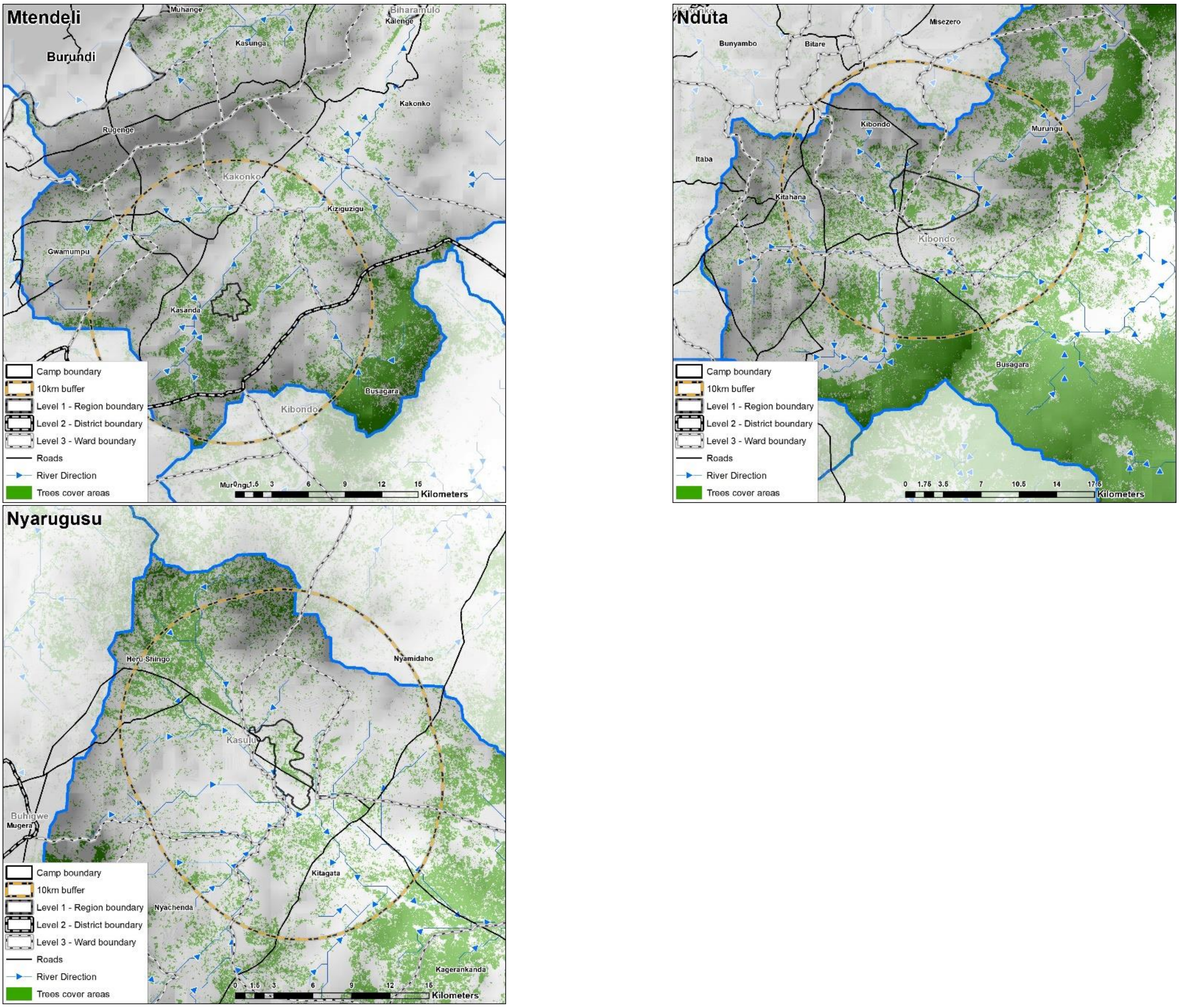

Figure 3. Forests around Mtendeli former Camp (top left), Nduta Camp (top right) and Nyarugusu Camp (bottom).

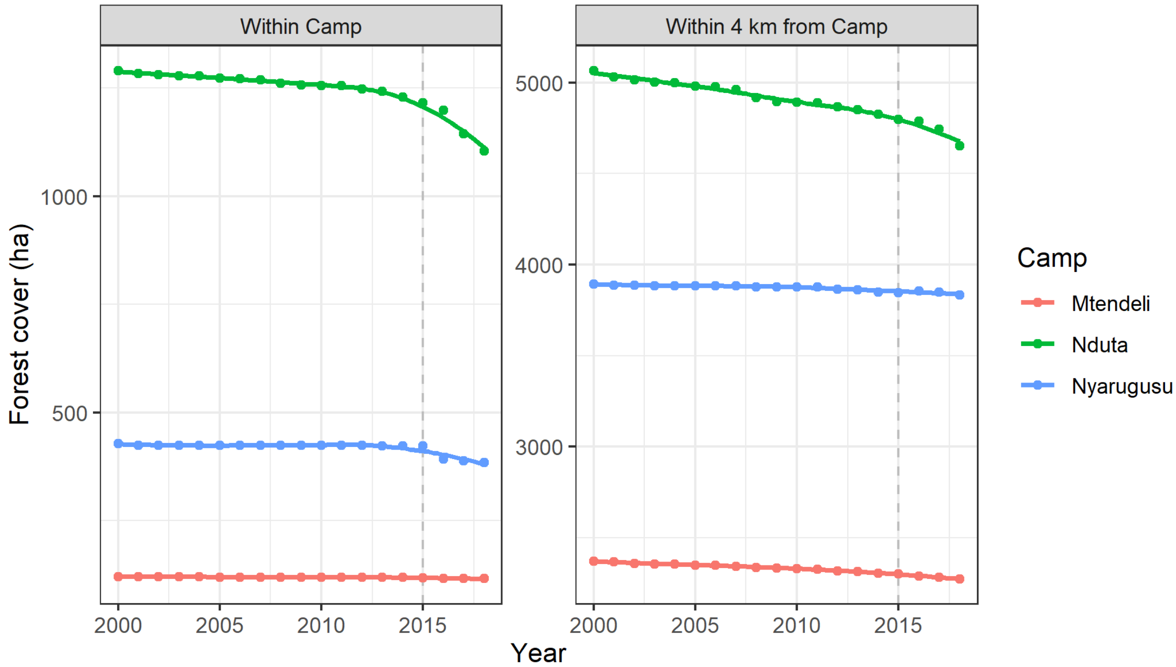

Figure 4. Forest cover (ha) between 2000 and 2018 for each camp within the camp boundary (left) and outside the camp boundary, within four km (right)[60]..

The result of these pressures on the forest cover is a degraded landscape with low soil stability, reduced soil nutrient availability, and reduced soil moisture. Additionally, traditional methods of protecting forests used by government forest departments — involving forest guards patrol forest areas — have had limited success owing to limited resources and supervision. These factors make the landscape more susceptible to the impacts of climate change and less able to support agricultural and other livelihood activities, less able to retain water resources required for these livelihoods and more susceptible to flooding. This has been highlighted by all the host communities and refugees who noted vulnerabilities in: i) zones F, G, H and I of Mtendeli; ii) zones 2, 3, 4, 6, 8, 9, 11 and 12 in Nyarugusu; and iii) zones 10, 15 and 16 in the Nduta camp (See Annex 2.1 Section 2.2.1 for maps of the camps' zonation). Several projects in the region (see Table 2) aim to address these demand-side developmental drivers of degradation, whilst others provide supply-side solutions and interventions that the proposed project aims to build upon and scale.

In addition, resource scarcity is also exacerbating the vulnerability of marginalised groups such as women and children. For example, community consultations revealed that female refugees were particularly vulnerable to incidences of GBV during fuelwood collection beyond the borders of their camps. Rural populations, and women in particular, generally have limited technical skills and technological support — such as access to information and improved agricultural inputs — which leads to reduced productivity, limited post-harvest storage for preservation of crops, and loss of harvests. Moreover, women are often excluded from decision-making on access to and the use of land and resources critical to their livelihoods. These challenges are expected to be exacerbated by the future impacts of climate change in the region[61].

Surface and groundwater

Kibondo and Kakonko districts are part of the Lake Tanganyika Basin which is experiencing impacts of climate change on water resources in various ways, including a decline in the flow of major rivers and drying of the Katubuka, Kilubu and Katosho wetlands located in Kigoma Region[62]. The national averages of Potential Evapotranspiration (PET) and Actual Evapotranspiration (AET) are 1,633 mm/yr and 771 mm/yr, respectively[63]. Kigoma region's PET averages 1,625 mm/yr. As a result, catchments are characterised by lower run off and higher evapotranspiration due to increased temperatures[64].

[60] Hansen/UMD/Google/USGS/NASA, "Global Forest Change 2000–2018. Version 1.6."
[61] Ibid
[62] United Republic of Tanzania, Ministry of Water. 2013. Water resources management strategic interventions and action plan for climate change adaptation.
[63]JICA. 2018. The Project on the Revision of National Irrigation Master Plan in the United Republic of Tanzania
[64] Ibid.

There is limited data available on actual flow rates of the rivers surrounding the camps and in the wider districts. However, considerable reductions of surface and groundwater resources have been observed according to anecdotal evidence. For example, the levels of the rivers surrounding Nduta, Nyarugusu and Mtendeli former camp have fallen markedly over the past 10+ years. Despite precipitation having increased in the long-term, trend fluctuations since 2000 show a reduction of ~7.5% from the historical climatological mean. Over this period, there has also been a statistically significant increase in daytime temperatures which have led to increased evaporation and evapotranspiration of surface water resources. These climatological factors may have been responsible for some of the streams in the districts having become seasonal and others have completely dried up, compounded by upstream agricultural activities that are putting unsustainable pressure on these rivers and streams. As a result, in Kakonko district as a whole, there is a heavy dependence on pumped groundwater. The district has a local, shallow aquifer, although groundwater resources are considered very low to medium, with only ~20–100 mm annual recharge. Groundwater is, consequently, not a reliable water source and previous efforts at extracting groundwater from boreholes had a low success rate, with only 6 out of 21 boreholes yielding water. Moreover, there are only two 'formal' irrigation schemes (commissioned by the District and limited to paddies) in Kibondo and Kakonko Districts.

Soil erosion and soil fertility

Land degradation has been ranked as the major environmental challenge in Tanzania for more than 60 years[65]. Of particular concern is soil erosion, which has occurred across more than 61% of Tanzania's total area over this timeframe. Historically, the amount of soil lost through water erosion in Tanzania was 105 tonnes/ha/yr in the 1960s, increasing to 224 tonnes/ha/yr in both the 1980s and 1990s[66]. Erosion mostly occurs at the beginning of the rainy season, with 70% of high erosion days taking place within the first month after the onset of the initial rains[67]. This is because both cultivated and natural vegetation are sparser after the dry season, resulting in reduced water infiltration and increased surface runoff. The degree of erosion in an area is therefore susceptible to changes in climate, specifically related to alterations in rainfall intensity as well as increases in the length of drought periods that decrease vegetation cover and reduce the binding of the soil.

Although the refugee camps are topographically relatively well situated, Tanzanian authorities and camp planners were likely unable to take long-term development pressures and climate change impacts into account when designating[68] the area for the settlements or during land use planning in their necessarily rapid establishment during acute situations. Within refugee camps in the Kigoma region, the high population densities in a small area and the associated human activities such as agriculture and the construction of shelters results in extensive soil degradation. Shelters, for example, are constructed with mud bricks that are excavated from pits within the camp. These pits promote erosion by creating bare areas lacking a topsoil which further contributes to their enlargement. If they become waterlogged, the pits present a physical hazard to the camp inhabitants and have the potential to increase the prevalence of water- and vector-borne diseases. If not addressed, these large pits — along with other forms of soil degradation including rill erosion and the development of gullies — have the potential to cause considerable injury to members of the refugee community. Further landscape degradation within and around the camps, along with increases in the intensity of rainfall events, is expected to aggravate these impacts for refugees.

Outside of refugee camps, farming practises that lead to vegetation degradation contribute to increased soil erosion. This is the case even for farms that are relatively recent (under five years old) and that were originally on stable soils previously covered by forests[69]. Soils on these farms become shallow with a larger proportion of stone, which reduces crop performance, particularly of cassava. The erosion of soil on smallholder farms is aggravated by the use of handheld tools. These tools do not allow farmers to plough deep enough, consequently leading to the formation of a hardpan over their farmlands that prevents water from infiltrating into the ground. Additionally, fire and over-grazing are both large contributors to soil erosion because they cause the exposure of the soil to abrasion from water and wind[70]. These two events also expose the soil to heat from direct sunlight, which reduces biological activity in the near-surface zone,

[65] Assey P. 2007. Environment at the heart of Tanzania's development: Lessons from Tanzania's National Strategy for Growth and Reduction of Poverty. MKUKUTA (No. 6), IIED.
[66] Kirui OK & Mirzabaev A. 2014. Economics of land degradation in Eastern Africa, ZEF Working Paper Series No. 128. University of Bonn, Centre for Development Research, Bonn.
[67] Moore TR. 1979. Rainfall erosivity in East Africa. Geografiska Annaler, Series A, Physical Geography, 61: 147–156.
[68] in the 1970s for Nyarugusu and in the 1990s for Mtendeli and Nduta, initially.
[69] Wickama J. 2017. Soil fertility options necessary to facilitate integrated soil fertility management (ISFM) in Uvinza District Kigoma. A consultancy assignment and report written for WEMA Consult-Dar-es-Salaam and the Nature Conservancy in Kigoma, Tanzania.
[70] Pramova E, Locatelli B, Djoudi H & Somorin OA. 2012. Forests and trees for social adaptation to climate variability and change. WIREs Climate Change, 3: 581–596.

altering the structure of the soil and reducing its stability. Increasing temperatures and more frequent higher intensity rainfall events are likely to further reduce the stability of soils on agricultural lands, resulting in soil and nutrient loss through erosion.

Through the removal of topsoils in agricultural areas or localities where vegetation has been removed, soil erosion affects soil fertility levels, a factor that is already limiting in certain areas of Kigoma. Soil fertility is a product of the balance between nutrients entering the system and nutrients being extracted, either by plants or as erosion[71]. Increased erosion related to changes in climate will therefore increase nutrient depletion and further contribute to soil infertility in agricultural areas of Kigoma.

Soil infertility resulting from nutrient depletion can be attributed to the breakdown of traditional farming practises and the low priority given by governments to agriculture in rural areas[72]. Over decades, small-scale farmers have removed large quantities of nutrients from their soils through unsustainable farming practises without replenishing them with sufficient amounts of manure or fertilisers. The community of Kalimongoma has noted the increase in land becoming infertile and are needing the effective use of fertiliser to replace the lost nutrients. Despite this, Tanzania shows very low fertiliser use compared with neighbouring countries, using 9 kg/ha/yr instead of the 32 kg/ha/yr or 29 kg/ha/yr used in Malawi and Kenya, respectively.

In addition to loss of nutrients from erosion, irrigated cultivated lands that are poorly drained are impacted by increased salinity as a result of naturally high evaporation rates. This results in 27% of irrigated lands in Tanzania becoming unusable as of 2011[73]. Climate change will exacerbate this problem, with increasing evaporation rates and greater demands for irrigation because of rising temperatures leading to higher salinity levels on cultivated lands and further reductions in the area of land that is usable for agriculture.

Poor land use planning and degraded ecosystems in the areas surrounding the region's refugee camps are also interacting to increase the negative impacts of increasingly frequent floods. Flooding causes, *inter alia,* roads to become impassable, damage to shelters and communal facilities, overflowing of latrine pits, and standing water which creates health and sanitation problems[74]. Floods also worsen soil erosion that reduces yields in kitchen gardens and community plots, as well as leading to deep gully[75] erosions that place community safety at risk of unstable ground. Notably, erosion also reduces the landscape's capacity to provide ecosystem services, such as flood attenuation. During future extreme rainfall events, this reduced attenuation will amplify flood impacts[76], thereby creating self-perpetuating cycles of degradation.

Interactions among host communities and refugees

Despite Tanzania's encampment policy, interactions between refugees and host communities take place. There are informal employment arrangements and agriculture land lease/land use agreements. Interactions take place through some trade activities despite the closure of common markets. Refugees exit the camps frequently to collect fuelwood from host communities' land. Furthermore, refugees and hosts interact during peaceful coexistence meetings.

The host community population in the project areas are composed of farmers and livestock keepers. Around 70% of the population in the region are engaged in small-scale farming, with others involved in livestock keeping. The current lack of land use planning and management of customary land can lead to conflict between farmers and pastoralists. Lack of land assigned for grazing can also lead to encroachment of village and national forest reserves outside camps and villages.

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[71] Bationo A, Kimetu J, Ikerra S, Kimani S, Mugendi D, Odendo M et al. 2004. The African network for soil biology and fertility: New challenges and opportunities. In: Bationo A (ed) Managing nutrient cycles to sustain soil fertility in sub-Saharan Africa. CIAT.
[72] Bationo A, Kimetu J, Ikerra S, Kimani S, Mugendi D, Odendo M et al. 2004. The African network for soil biology and fertility: New challenges and opportunities. In: Bationo A (ed) Managing nutrient cycles to sustain soil fertility in sub-Saharan Africa. CIAT.
[73] Liniger HP, Studer RM, Hauert C & Gurtner M. 2011. Sustainable land management in practise-guidelines and best practises for sub-Saharan Africa. TerrAfrica, World Overview of Conservation Approaches and Technologies and FAO.
[74] WFP. 2016. Market Assessment: Nyarugusu refugee camp, United Republic of Tanzania. Available at: https://documents.wfp.org/stellent/groups/public/documents/ena/wfp286682.pdf?iframe
[75] Once vegetation is lost either through drought or poor land management, rainwater washes away the topsoil that is rich in organic matter, exposing subsoils. Heavy rains funnel water into channels that form deep gullies. Refer to section 7.3.3 of Annex 2. Feasibility Study for discussion
[76] See Annex 2: Feasibility Study section 2.2.2 for flood impact analysis.

The recently conducted community consultations in September 2020 have given more insights on the interactions between host communities and refugees. Both communities generally remarked that relations between the communities are good. Economic activities between the local communities and refugees take place. Some refugees have informal arrangements with host communities as agriculture laborers or through land lease/land use arrangements. It is important to note that these arrangements are informal and thus do not confer any tenure rights to the refugees.

Despite good relations in general, there are some conflicts between host communities and refugees. The typology of local conflicts within host community groups and between refugees, host communities and other beneficiary groups are set out in Annex 6 Environmental and Social Management System Section 7.3. Some examples include tensions around:

- benefit sharing on agricultural production on land informally leased to refugees;
- concerns of responsibility for protection by host communities in upstream areas and benefit and pollution from bathing and washing by refugees in shared streams and water sources; and
- gender-based violence experience during fuelwood collection.[77]

Existing mechanisms to manage conflict include peaceful co-existence meetings that are known to work in resolving issues between refugees and host communities.

Market conditions and available financial services

The climate-vulnerability of smallholder and refugee farmers in Kigoma is determined by their strong reliance on rain-fed agricultural systems which, under anticipated precipitation changes, are expected to become further exposed. Farmers' vulnerability is exacerbated by low levels of knowledge of, and capacity to implement, resilient long-term climate-smart agricultural practises and the subsequent application of suboptimal means of production. These farmers also do not have the appropriate technology required to implement climate-resilient practises and have limited alternative income opportunities to counteract the potential loss of crop production.

The limited availability of financial services in the region has exacerbated the difficulties regarding the uptake of climate-resilient technologies within the Kigoma region. Most residents of the host communities in the Kibondo, Kakonko and Kasulu Districts are reliant on Savings Associations (SA) to manage their finances, in the absence of formal banking services. Community consultations noted that there were variations in both the level of participation in SAs as well as in the financial contributions and gains generated from SAs. Women tended to be better represented in SAs in host communities and reported received values of up to 800,000 Tanzanian shillings (TZS). The primary purpose of a SA is to provide simple savings and loan facilities. Levels of capital in single SAs are generally extremely low, and 25–30 SAs — with under 10 members each — may exist within a single village. UNCDF has been supporting VSLAs in the host communities as part of the Kigoma Joint Programme. There is significant potential for SAs to be used to scale up climate change adaptation interventions in Kigoma, for the distribution of funds to farmers for the improvement of inputs and the uptake of water-efficient technology. For more information on the role of SAs in the region — as well as an overview of their role in the proposed project delivery structure — please refer to Section 3.5.1. of the Feasibility Study (entitled 'The Savings Associations').

Historic and Projected Climate Change[78]

Tanzania's climate is highly variable, but climate trend data already indicates that temperatures are rising. Year on year rainfall volume is increasing in variability and peak precipitation intensity is increasing. The sections below discuss these historical changes and projections in the project sites in detail.

Observed temperature and rainfall change and impacts

Historical climate trends for the country indicate that temperatures have risen from 1960–1990[79]. The long-term temperature trend for the three camp areas has a statistically significant increase in both the average maximum temperatures as well as the highest maximum temperatures[80]. This increase is ~0.16°C per decade from 1983 to 2016 and this long-term general increase is also apparent in the more extreme temperatures. The change in the 90th percentile maximum temperature has increased from the 1980s to 2016 by more than 0.6°C. Analysis of both the day-

[77] Refer to Annex 8. Gender Assessment and Action Plan, section 1.3.3. for more information.

[78] See Annex 2. Feasibility study for detailed description of observed and modelled climate impacts at the level of the project sites.

[79] USAID. 2012. Tanzania Climate Vulnerability Profile.

[80] Long term maximum day time temperature.

and night-time temperature profiles show a shift away from the lower temperatures, towards more extreme day- and night-time temperatures. These extreme temperatures have occurred more frequently in recent decades.

Historical climate trends for the country indicate that rainfall is becoming more erratic. Over the past three decades, Tanzania has experienced several major national droughts as well as several instances of localised regions having anomalously lower precipitation for several years. Annual rainfall in Tanzania has decreased at an average rate of 3.3% per decade from 1960-2006[81]. These decreases are predominately noted in the southern parts of the country. While there are noted general decreases in precipitation elsewhere in the country, precipitation in the camps of Nduta and Nyarugusu, the former camp of Mtendeli and surrounds has experienced a statistically significant (p value ~0.13%) increase of ~7.5 mm/ decade since 1901. The most prominent observed climate change trend has been a tendency towards lower rainfall volume during the early growing seasons (October to December) and increases later in the season.

The increase in precipitation volumes is occurring in the presence of increasing temperatures. This has the effect of enhancing the moisture storing capacity of the lower atmosphere and therefore increases the potential evapotranspiration. The Standardised Precipitation-Evapotranspiration Index (SPEI) drought index classifies years from +3 (exceptionally wet) to -3 (exceptionally dry), with varying severity of wetness or drought between these values and +0.5 to -0.5 representing near normal wet and dry respectively. There is a general decrease in SPEI over time (Figure). While there are natural oscillations in the SPEI magnitude and there will certainly be extremely and exceptionally wet years in the future, the long-term trend indicates a general decrease and leaning toward the more dry categories as a result of the increasing temperature's effect on potential evapotranspiration. The long-term magnitude is decreasing for both the SPEI wet and dry years. This will result in wet years that are more moderate and dry years that are more severe.

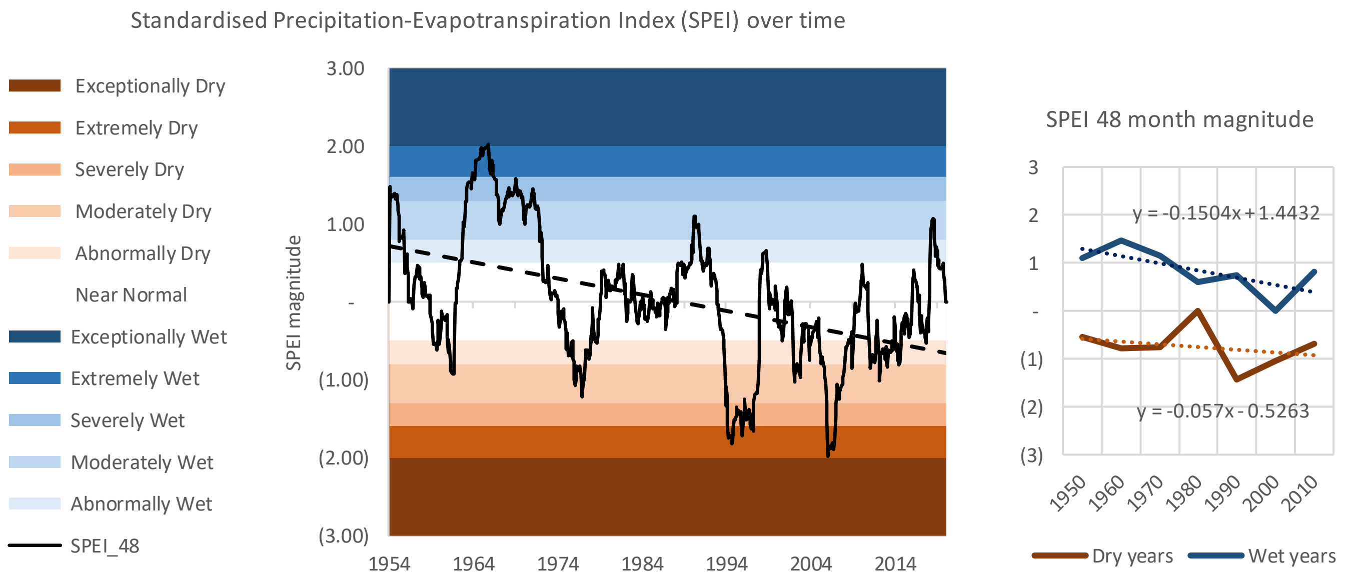
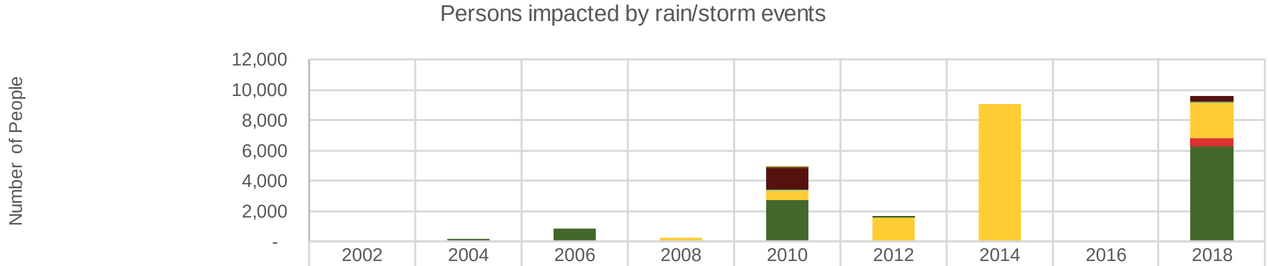


Figure 5. SPEI 48 month rolling variability magnitude over time (left), simplified SPEI wet and dry magnitude.

The historical changes in the climate (increased temperature and more intense precipitation) have already had noted impacts on the ground in Tanzania. Information from the United Nations Office for Disaster Risk Reduction (UNDRR) database[82] records events and their impacts on local populations. While these impacts are not spatially explicit, they do demonstrate an increasing trend of communities being affected by rainfall related hazards (Figure 6).

[81] USAID. 2012. Tanzania Climate Vulnerability Profile.
[82] UNDRR, DesInventar database.



	2002	2004	2006	2008	2010	2012	2014	2016	2018
Windstorm	-	-	-	-	-	1	-	-	-
Thunderstorm	-	-	-	-	2	-	-	-	-
Strong Winds	-	-	-	-	1,473	-	-	-	426
Rains	-	-	-	-	42	-	-	-	-
Lightning	-	-	-	-	-	-	-	-	70
Heavy Rainfall & Strong Winds	-	-	-	251	647	1,582	9,060	-	2,315
Hailstorm	-	-	-	-	-	-	-	-	558
Flood	-	192	853	-	2,695	-	-	-	6,243

2 year bins

Figure 6. UNDRR impact database for Tanzania. Rainfall-related impacts

Observational climate change control site
To quantify the additional effect — or lack thereof — of the refugees on the area surrounding the camps, a control site ~85km south of Nyarugusu was selected for comparative purposes. The town of Uvinza and its surrounds have similar reference land use, climate and geographical profiles to the project area, although no refugee communities are hosted in this area. The control area has experienced a similar change in observational climate, with notable increases in both day and night-time temperatures as well as increasing annual maximums from the 1980s to present. The area has also experienced an increase in total precipitation over time, as well as a strong decrease in SPEI magnitude as a result of increased potential evapotranspiration over the observational period.

Despite these similar physical and climatic conditions, the forest loss around the control site is significantly higher than in the area surrounding the refugee camps. This loss was, on average, ~850 ha per year between 2001 and 2018, although this increased over this period by ~102 ha per year. There are further losses through an ~22% increase in fire occurrence in the areas surrounding the control site. The cumulative NDVI surrounding the control villages shows losses similar to the areas surrounding Kasanda village close to Mtendeli former camp. The areas around Nduta and Nyarugusu camps have had even less forest loss.

Although the drivers of degradation are a challenge to attribute with certainty, given the similar climatic factors and basic land use profile, the presence of refugees in Kasulu, Kibondo and Kakonko do not appear to have resulted in greater degree of degradation compared with the control site of Uvinza. This reinforces the findings of the landcover changes for the target project areas, that suggests the presence of refugees are not a key determinant of accelerated forest loss, after initial clearing to establish/expand the camps. Further detail of this control study can be found in Section 2.1.4 of the Feasibility Study.

Projected temperature and rainfall change
Average maximum temperatures (currently ranging from 27–31°C in Kigoma region (Figure 7) are likely to increase under all projected climate scenarios. The projected changes under both RCP4.5 and 8.5 scenarios suggest increases of 1.2–2.2°C for the average maximum temperature and an increase of 2.5–4.5°C in the hottest month of the year from 2050 and 2070 respectively, with the area around Nyarugusu experiencing the greatest increases.

Average minimum temperatures of 14°C for the region will increase by ~1–1.1°C by 2050 and by 1.3-2.3°C by 2070 for RCP4.5 and 8.5, respectively. These increases will shift both day- and night-time temperature profiles away from cooler temperatures towards an increase in the occurrence of warm and hot days. Extremely hot days that currently occur rarely will become more frequent, and the future 'rarely occurring extremes' will be much warmer than current extremes. The mean number of days per year with temperatures over 35°C is projected to increase by an additional one to nine days (RCP4.5 and 8.5) by 2050 and ~five to 33 days (RCP4.5 and 8.5) by 2070. Projections also an indicate an increase from ~10 heatwaves[83] per year to an additional 19–24 per year by 2050 and 25–32 heatwaves more per year by 2070 for RCP4.5 and 8.5, respectively.

[83] three consecutive days above the highest average monthly daytime temperature

By the 2050s total annual rainfall is projected to increase in Kigoma by ~7–9%. Annual projected precipitation changes in this region show general increases for both 2040–2060 RCP4.5 (Figure 8, bottom left) and 2060–2080 RCP8.5 (Figure 8, bottom right). RCP4.5 projects a total increase in annual rainfall of 11 mm for Nyarugusu area and 16–27 mm for Mtendeli and Nduta areas by 2060. The latter have greater capacity for cloud formation and chance of rainfall in the future, having lower observed precipitation and currently being slightly cooler regions. This is further exacerbated under RCP8.5 with increases in the Kasulu area of 43 mm and 49–64 mm annually in the Kakonko and Kibondo areas. These increases account for between 5.3–9.7% of annual rainfall under future scenarios. There will be an increase in the dry spell duration because of the anticipated increase in total annual precipitation becoming more focused within a shorter timeframe. There will also be an increase in day- and night- time temperatures resulting in an increase in potential evapotranspiration. This will shift the SPEI measures more towards the dry and very dry categories overall, continuing the trend observed in the historical data.

While the projected change in total annual rainfall by the 2050s is small, seasonal fluctuations are larger, with increased variability in the early onset of rainfall events from October to December (~9% under RCP4.5 and 12% under RCP8.5 for 2050) and increased rainfall (5.8% under RCP4.5 and 3.9% under RCP8.5 for 2050) in March to May (the respective main growing seasons).

Higher average and extreme temperatures will enhance evaporation and evapotranspiration from the earth's surface, leading to an increase of water vapour in the atmosphere. This increased atmospheric moisture volume results in extreme rainfall events that can lead to flooding. Hourly peak precipitation intensity[84] will likely increase by up to 9% and 11% in 2050 and 11% and 17% by 2070 for RCP4.5 and 8.5, respectively. This will increase the precipitation profile, resulting in heavy and fast-occurring rainfall which is likely to increase the frequency and intensity of floods.

[84] The peak rainfall volume (mm) that occurs within one hour during a rain event. Measure of intensity

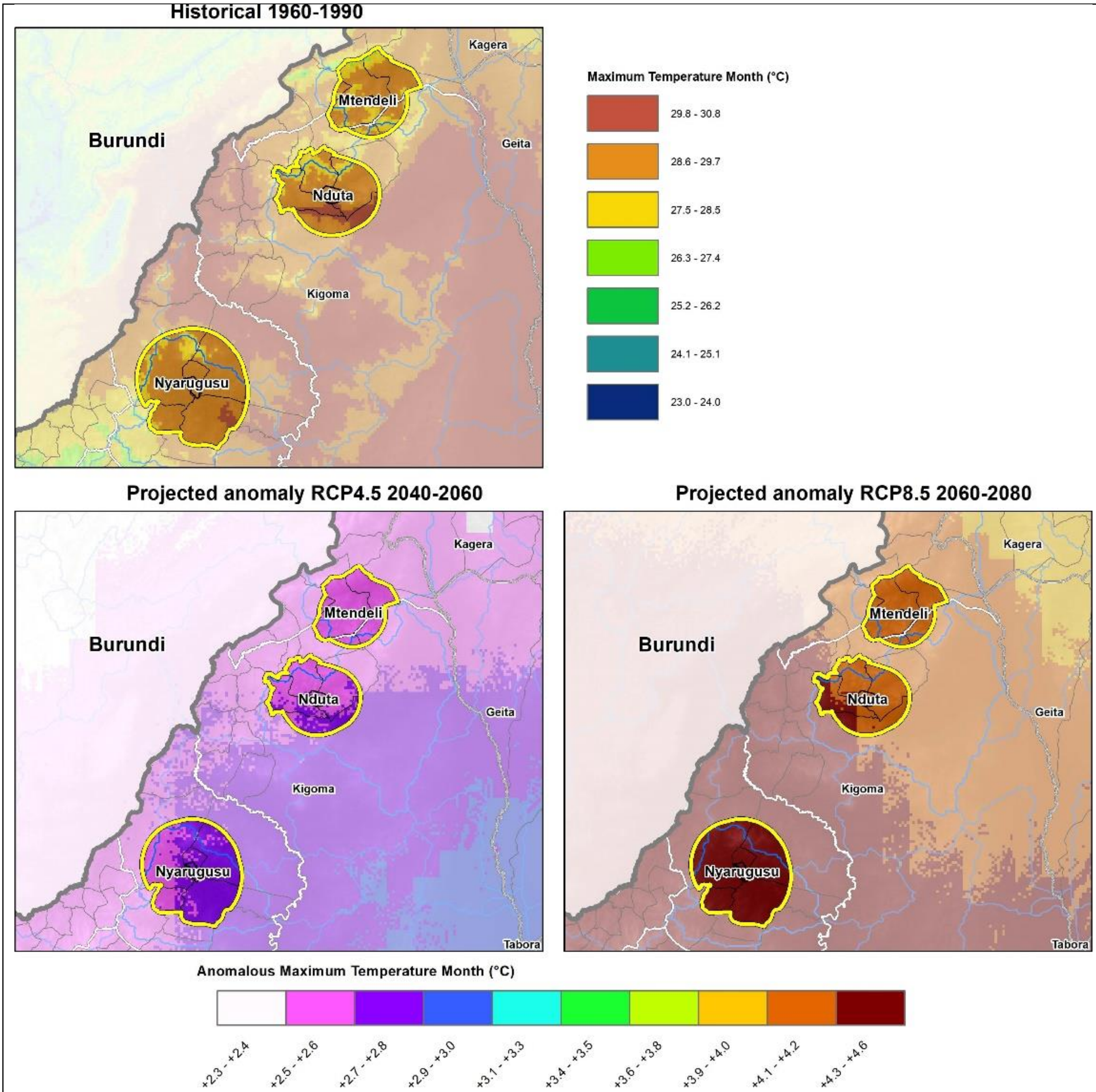


Figure 7. Localised average maximum temperature changes for the historical scenario (top-left) and projected for 2040–2060 RCP4.5 (bottom-left) and 2060–2080 RCP8.5 (bottom-right)[85].

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[85] Hijmans RJ, Cameron SE, Parra JL, Jones PG & Jarvis A. 2005. Very high-resolution interpolated climate surfaces for global land areas. International Journal of Climatology 25:1965–1978.

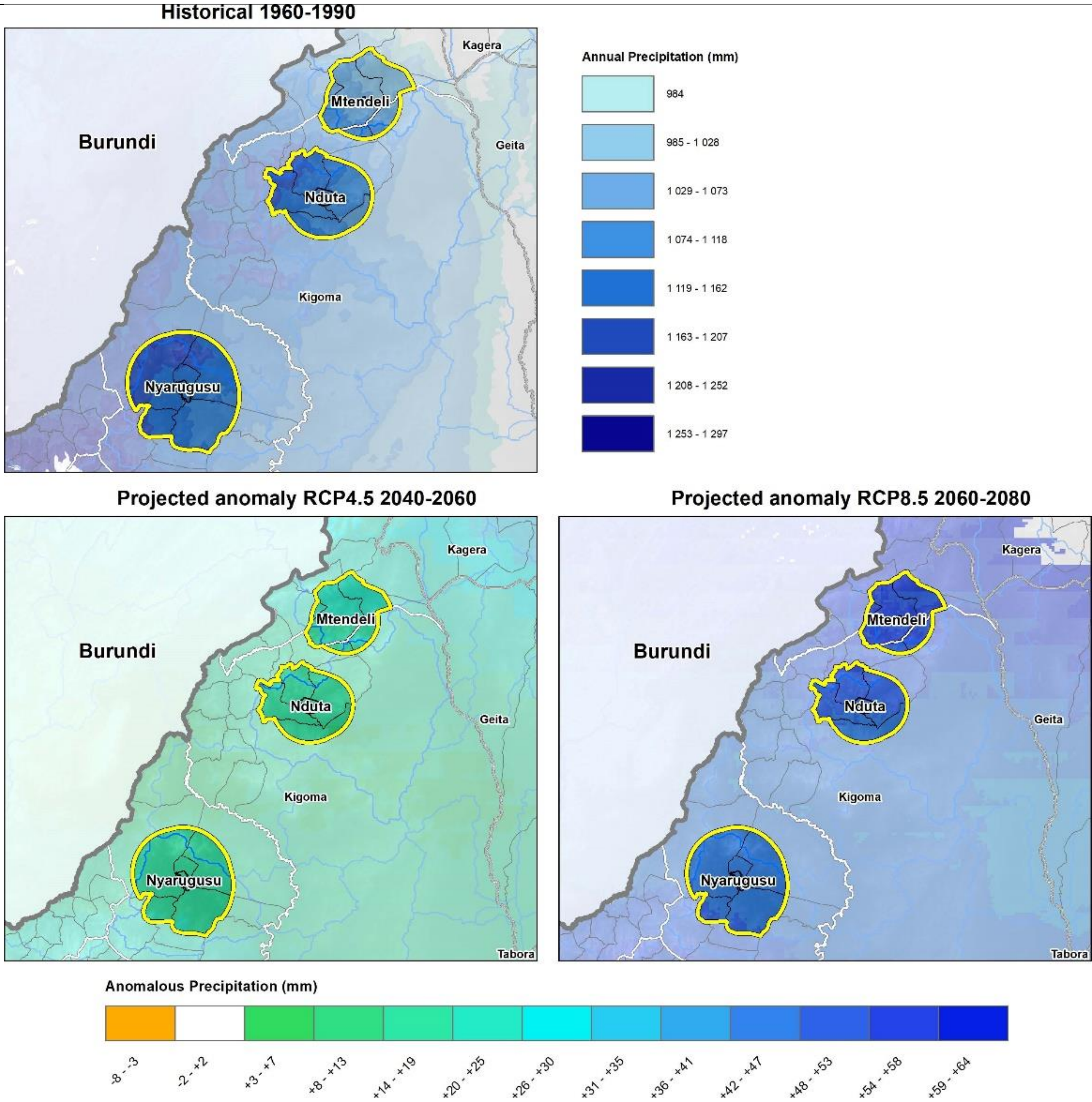

Figure 8. Localised precipitation changes for the historical scenario (top-left) and projected for RCP4.5 (2050) (bottom-left), and RCP8.5 (2070) (bottom-right)[86].

Climate Impacts

Climate change — in combination with non-climate drivers such as agricultural practises and land use changes — has already altered ecosystem functioning and affected livelihoods in the Kigoma region. The local-level impacts of climate change manifest through a cumulation of multiple, interrelated climate-forcing and meteorological-feedback mechanisms over the long term, interacting on socio- ecological systems, creating feedback loops that amplify climate hazards and their subsequent effects on the agro-ecological system.

Temperature and precipitation changes interact to amplify impacts. The potential failure of early season rainfall increases the likelihood of severe and prolonged drought. At the same time, floods are expected to increase in frequency and intensity. These effects will particularly impact upon the region's rainfall-dependent and flood-exposed livelihoods

[86] Hijmans RJ, Cameron SE, Parra JL, Jones PG & Jarvis A. 2005. Very high-resolution interpolated climate surfaces for global land areas. International Journal of Climatology 25:1965–1978.

and communities. Smallholder farmers in Kigoma region are particularly at risk of climate-driven impacts on their livelihoods, as their income and food supply depend on the rain-fed agricultural productive systems. In Kibondo district, 27% of the area is arable land, of which 8% is under predominantly rain-fed cultivation. Livelihoods are therefore reliant on the timely arrival of rainfall. These livelihood pressures and agricultural impacts also contribute to perpetuating ecosystem degradation such as soil erosion and deforestation. For example, fuelwood consumption is a historical driver of deforestation and ground cover loss but may also increase as a result of climate impacts on agricultural yields compelling farmers to seek alternative incomes. Agricultural land use conversion — as a response to meeting development demand or because of climate-related impacts — also contributes to soil erosion and localised water logging associated with high precipitation events. These impacts, in turn, exacerbate the secondary impacts of flood events, which are projected to increase. In addition, climate impacts create mutually reinforcing feedback loops that drive and accelerate degradation, placing pressure on ecosystems and communities.

The abovementioned challenges demonstrate the complex linkages between changes in climate variables and their direct and indirect impacts on agro-ecological systems, which support the livelihoods of host communities and refugees. The climate analysis and corresponding project design have been developed to address this complex interconnected system and to achieve transformative change at a landscape level. These impacts on the landscape, critical ecosystems and agro-ecological livelihoods are described below and demonstrated in Figure 9 to show the complex linkage between changes in climate variables and their direct and indirect impacts on the agro-ecological system, which supports the livelihoods of the communities.

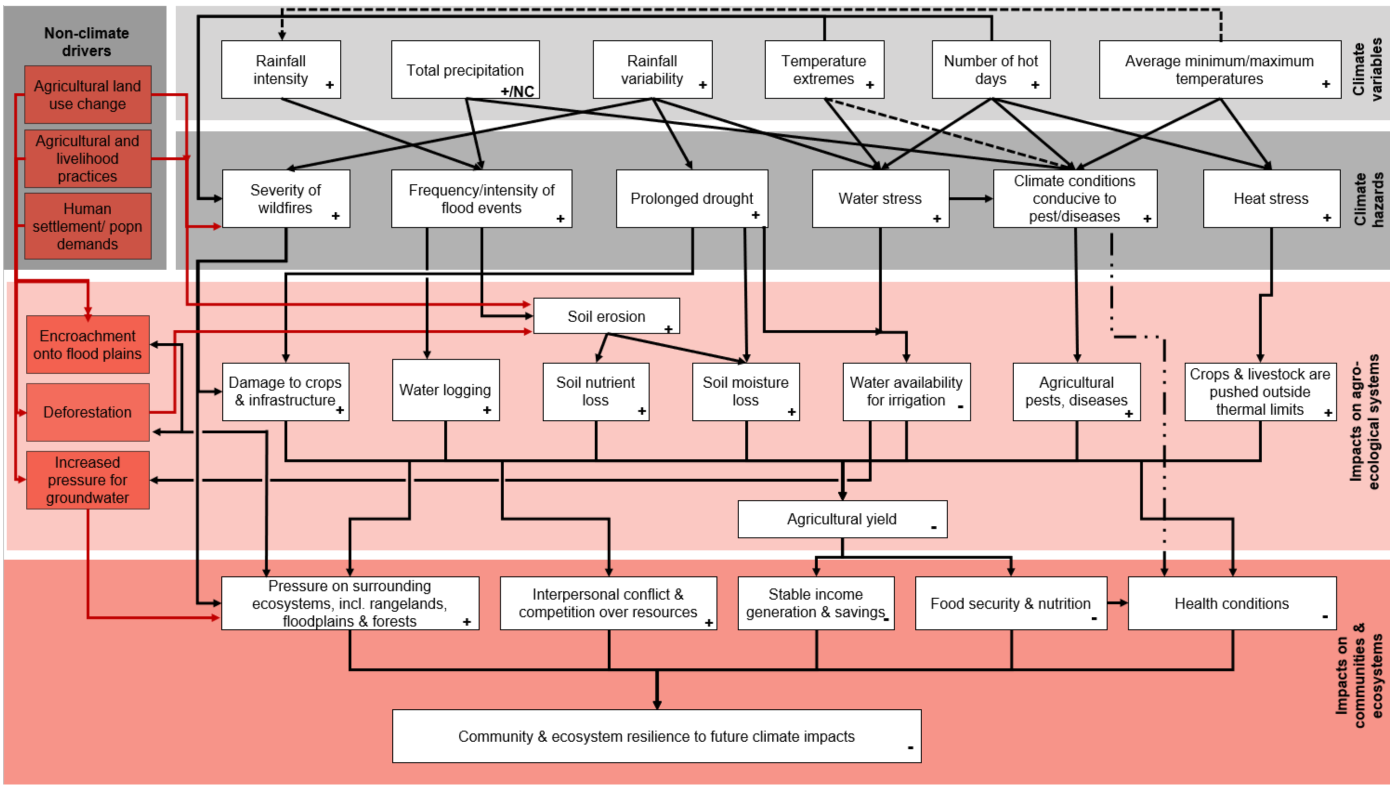

Figure 9. Climate change impact chain for agro-ecological systems and smallholder farmers in the host communities in Kigoma region. The +/- symbols in each cell represent the directionality of the projected changes. NC indicates 'no change'. Black lines indicate climate variable-driven impacts, red lines indicate the compounding impacts from non-climate drivers. Dashed lines indicate indirect relationships.

A large proportion of the country's GDP is associated with climate-sensitive activities, in particular subsistence agriculture. Climate-related production risks (such as drought, diseases and pest attacks[87]) are already costing Tanzania's agriculture sector at least USD 200 million per year, and without urgent adaptation these costs are likely to

[87] Some examples are stalk borers that are correlated with warmer temperatures and rift valley fever in livestock is mosquito borne and correlated with higher rainfall. Refer to Section 2.2.7 of Annex 2. Feasibility study for types of pests and diseases and influence of climate.

increase with rising climate variability[88]. If climate risks remain unaddressed, aggregate models estimate that the associated economic costs may be equivalent to as much as 2% of GDP per year, by 2030[89].
With agriculture being the predominant livelihood activity, the greatest risk from climate change is its impacts on agricultural yields and consequently food security[90]. In addition to diminishing the critical ecosystem services provided by forest systems — such as flood mitigation, soil moisture retention, soil erosion control and soil fertility — the impact of climate change on livelihoods is a general decline in agricultural productivity and agro-diversity[91]. Indeed, climate variability is already resulting in an average of 5–15% loss of maize crops per year because of the increased prevalence of pests and disease. Other crops have also been in decline. For example, a 50% decline in productivity of bananas and coffee has been reported between the 1970s and 1990s[92]. This occurred as a result of an increased prevalence of pests and disease, but also because of insufficient rotation of these crops resulting in inadequate return of nutrients to the soil. The decline in banana yields is accompanied by a concurrent increase in the demand for maize and root crops, putting greater pressure on land used to farm these crops[93]. The impacts of over-intensive agricultural practises have been exacerbated by an inheritance system which causes plots of land to continually be subdivided and fragmented — a factor directly related to population growth[94]. This is resulting in agricultural land being expanded into marginal areas. These factors are exacerbating the impacts of regional fluctuations in supply and demand for crops. As the population grows, this challenge is becoming more prevalent and will become increasingly severe under future climate change conditions[95], further compounding concerns of malnutrition. As a result, Kigoma will have to improve agricultural systems and diversify the variety of crops grown to meet the nutritional requirements of the growing population[96]. To achieve this, there is a need to improve productivity of existing agricultural production and supply systems and to reduce the vulnerability of these systems to the impacts of climate change by adopting climate-resilient agricultural practises[97].

Climate change is particularly likely to affect resource-poor households such as smallholder local farmers and refugee families, all of whom are unable to invest in or take advantage of alternative livelihoods or new agricultural strategies. They will likely transfer this pressure and cause greater reliance on the local ecosystem, further accelerating degradation. The subsequent effects of this are already observed by increases in severe erosion and enhanced flood impacts in the refugee camps, including zones F, G, H and I of the Mtendeli former camp and zones 3, 4, 8 in Nyarugusu camp (from December to April each year) where large gullies were eroded from significant overland flows. Projected increases in precipitation intensity will further exacerbate these impacts. Densely populated flood-prone areas are particularly exposed because of land clearance. Flat areas are prone to flooding, while high levels of erosion have occurred on slopes as observed in the project area. High intensity rainfall events following longer dry spells will further contribute to erosion on these slopes by displacing the topsoil layer. The overall result of the deforestation is a degraded landscape with low soil stability, reduced soil nutrient availability and reduced soil moisture. These impacts compound landscape susceptibility to further climate change impacts and means landscapes are less able to support agricultural and other livelihood activities, less able to retain water required for these livelihoods, and are more susceptible to flooding.

The provision of the landscape to continue to provide ecosystem services (such as flood attenuation) under climate change conditions will also be affected by other climate hazards, including fires. Although increased incidences of fires in forested areas are largely related to human activity, climate change is expected to further compound the effects. As the climate becomes hotter and drier, the frequency and intensity of forest fires are expected to increase. This will negatively affect the livelihoods of communities by burning of crops and important forest trees, as well as retreat of wild animals from certain areas[98].

[88] Arce CE & Caballero J. 2015. Tanzania: Agricultural Sector Risk Assessment. World Bank, Washington, DC. Available at: https://openknowledge. worldbank.org/handle/10986/22277 License: CC BY 3.0 IGO
[89] Global Climate Adaptation Partnership, 2011. The Economics of Climate Change in the United Republic of Tanzania.
[90] http://www.tzonline.org/pdf/kigoma.pdf p.25
[91] Kangalawe et al. 2017. Climate change and variability impacts on agricultural production and livelihood systems in Western Tanzania. Climate and Development, vol.9: 3
[92] Enabel. 2009. Sustainable improvement of the banana cropping system in Kagera region and Kibondo district in Kigoma region. Available at: http://www.diplomatie.be/oda/11765_PROGDESCR_Technisch_en_Financieel_Dossier.pdf
[93] Enabel. 2009. Sustainable improvement of the banana cropping system in Kagera region and Kibondo district in Kigoma region. Available at: http://www.diplomatie.be/oda/11765_PROGDESCR_Technisch_en_Financieel_Dossier.pdf
[94] Enabel. 2009. Sustainable improvement of the banana cropping system in Kagera region and Kibondo district in Kigoma region. Available at: http://www.diplomatie.be/oda/11765_PROGDESCR_Technisch_en_Financieel_Dossier.pdf
[95] Communication from UNHCR
[96] SNV. 2019. Climate Resilient Agribusiness for Tomorrow (CRAFT). Available at: https://snv.org/cms/sites/default/files/explore/download/final_amended_craft_factsheet_sept_2019_0.pdf
[97] SNV. 2019. Climate Resilient Agribusiness for Tomorrow (CRAFT). Available at: https://snv.org/cms/sites/default/files/explore/download/final_amended_craft_factsheet_sept_2019_0.pdf
[98] See Section 2.2.6 of Annex 2. Feasibility Study for further discussion.

This destruction of natural habitats and movement of animals will also increase the prevalence of pests and diseases, further compounding the direct impacts of climate change on these hazards. Future climate change conditions are expected to increase the risk and impact of crop pests and diseases as a result of higher temperatures as well as increased rainfall intensity and variability for the Kigoma region. Specifically, climate change is expected to have an influence by: i) changing the prevalence and distribution of crop pests; ii) increasing the frequency of new pest introductions; iii) increasing the occurrence of major pest outbreaks; and iv) encouraging the use of pesticides as well as increasing the resilience of pests to these chemicals, leading to food safety risks from pesticide residues. Human diseases in Kigoma are also expected to be influenced by climate change, particularly related to the unmanaged impacts of floods. Traditional experience and defences against these diseases will likely no longer be sufficient to respond to the unprecedented pest and disease burden. As the suitable conditions for the survival of pathogens increase because of changes in temperature or rainfall, this leads to an increased risk of emerging diseases, changes in migration pathways, carriers and vectors and changes in the natural ecosystems. Infectious agents are in a state of perpetual adaptation to their new hosts or vectors, which has the potential to lead to the emergence of ‘new’ diseases or the spread of known diseases to previously unaffected areas as climates shift.

The distribution ranges of water- and vector-borne diseases are expected to increase where warmer temperatures combine with heavy runoff from increased precipitation events. An increase in unmanaged standing water caused by greater rainfall amounts from climate change, for example, coupled with reduced infiltrability in areas with limited vegetation cover has the potential to increase the risk of cholera, a water-borne disease associated with limited sanitation that has resulted in a number of epidemics within Kigoma[99]. This can be mitigated however, by interventions to improve drainage, limit standing water and improve vegetation cover. Future long-term changes in rainfall variability and intensity, drought periods and temperature within an area are also expected to affect the suitability of localities within Kigoma to insect vectors such as tsetse flies and mosquitoes[100]. This will influence the epidemiology and transmission of diseases such as HAT or RVF[101]. Episodes of RVF, for example, are closely linked to variations in climate, particularly occurring after periods of abnormal drought, followed by abnormal heavy rains and the consequent emergence of large numbers of *Aedes* and *Culex* mosquitoes[102] [103].

In addition to experiencing the consequences of the immediate impacts of extreme climate events, the limited local adaptive capacity and opportunities results in an inability to recover following droughts, floods, diseases or other shocks. Refugees in Nduta and Nyarugusu practicing small-scale agriculture around homesteads are already reporting crop losses because of unexpected weather conditions[104], such as late rainy seasons, as they have little or no access to alternative water sources or water storage for supplementary irrigation. Increased early season onset variability will exacerbate this exposure. A further factor increasing the vulnerability of refugees is that they lack the traditional knowledge of local conditions which help to identify the warning signs for unusual weather. Negative impacts upon food security and nutrition of both refugee and host communities will result in additional pressures on the ecosystems which surround Kigoma’s refugee camps. In addition to socioeconomic conditions in the area, the lack of financial services available to host communities and refugees also means that they have a reduced ability to mitigate, cope with, and recover from climate shocks and to sustain their livelihoods.

[99] In 2015, cholera affected refugee and host populations in four of Kigoma’s districts, including Ujiji, Kigoma, Uvinza and Kasulu. In total, 5,581 cholera cases resulted from the 2015 outbreak, of which 1,051 occurred within the host populations and 4,528 in the refugee communities. Of the total cholera cases in 2015, 61 resulted in deaths. From consultation with UNHCR Environment Officers, December 2018.

[100] Cattle in Kigoma are susceptible to animal trypanosomiasis, particularly populations that are near the Malagarasi forest reserve which contains large numbers of tsetse flies. The prevalence of bovine trypanosomiasis in the Kasulu District was 11% in 2010. From Kgotlele T, Torsson E, Kasanga CJ, Wensman JJ & Misinzo G. 2016. Seroprevalence of peste des petits ruminants virus from samples collected in different regions of Tanzania in 2013 and 2015. Journal of Veterinary Science and Technology, 7: 394. Doi: 10.4172/2157-7579.1000394.

[101] Mboera LEG, Mayala BK, KWek EJ & Mazigo HD. 2011. Impact of climate change on human health and health systems in Tanzania: a review. Tanzania Journal of Health Research, 13: 1–23.

[102] Ibid

[103] Although historical epidemics of RVF in Tanzania did not affect Kigoma, there is evidence of the virus occurring in sheep and goats at small percentages of 12%, 2% and 1% within the Kigoma, Kibondo and Kasulu Districts respectively. This indicates that there is the potential for future epidemics of this disease in the region. From Kifaro EG, Nkangaga J, Joshua G, Sallu R, Yongolo M, Dautu G & Kasanga CJ. 2014. Epidemiological study of Rift Valley fever virus in Kigoma, Tanzania. Onderstepoort Journal of Veterinary Research, 81:1–5.

[104] Refer to Appendix A: Stakeholder interviews and focus group discussions in the Feasibility Study (Annex B) for more information.

Other climate change impacts include flood impacts, heat stress, reduced water availability and functional changes to agro-ecological zones and ecosystems[105]. These climate impacts interact with structural development challenges, such as those caused by the pressure of recurrent influxes of refugees (with fuelwood, food and water, and livelihood demands) into the region. Many of these structural development challenges are being addressed through complementary projects, as described below.

Investments in the Target Project Area

This project aims to build upon the ongoing development and humanitarian work that has, and is, being conducted in Kigoma to address the outstanding challenges through an integrated approach based on land use planning — specifically, to address the climate and non-climate drivers of degradation in a coordinated way. To avoid trade-offs and maximise co-benefits a "no-regrets" approach has been applied in the project design. Through this approach, measures have been developed using the precautionary principle. The aim is to allow refugees and host communities to respond immediately to the already existing impacts of climate change and other environmental pressures, while maintaining the maximum potential for on-going adaptation. This strategy is integrated into all of the project's investments in order to reduce the climate and other environmental risks associated with the communities' direct dependence on the surrounding ecosystems. Globally the no-regrets approach has been shown to increase resilience and enhance the potential to adjust to different types of hazards in a timely, cost-effective and equitable way because it works at the integrated landscape, or "whole system", -level[106]. Although the parallel financing and other projects have begun to address elements of the baseline, their implementation is fragmented as they address issues separately. The project will contribute to addressing this through the integrated land use planning in Output 1, which will further be scaled and replicated through Output 4 through developing policy recommendations and supporting mainstreaming to address more systemic coordinated issues across sectors and institutions.

In addition to committing co-financing to the project through its natural resources and shared environment programme, UNHCR will continue its activities over the project period to promote access to alternative energy for refugees and host communities. Furthermore, at a regional level, UNHCR is one of 16 UN agencies contributing to the One UN Kigoma Joint Programme (KJP). By linking the implementation of the project directly to the KJP activities, opportunities will be created to discuss and exchange with regional and district level authorities. Increasingly, the KJP is seeking to align with and contribute to local planning processes and development plans.

Several development and humanitarian projects are expected to be implemented in the next few years in the target area that will complement the GCF project activities, especially in addressing structural drivers of environmental degradation and building communities' resilience. Some of these initiatives are still under discussion and in negotiation stages and are therefore indicative of the interventions the project will coordinate with. These investments are in addition to public finance from national, regional and district allocations towards energy and environment interventions in the refugee hosting areas.

Section 2.2.6 of the FS provides detail on the current use of fuel wood in Kigoma that the project aims to address. Fuelwood will continue to be part of the energy mix for cooking, although other alternatives are being explored through other initiatives. Currently, fuel wood extraction accounts for 60% of forest resource usage. The proposed project will be addressing supply side factors, particularly in shifting extraction from slow-growing natural miombo ecosystems — that additionally provide important biodiversity and ecosystem services — toward faster growing species planted on marginal degraded land and/or toward sustainable, renewable offtake from agroforestry. The forestry activities under this project will be instrumental in scaling up supply side fuelwood interventions and complement UNHCR and others' activities addressing the demand side of fuelwood.

Several initiatives and projects are underway or planned to provide longer-term solutions and interventions to address demand-side problems of fuelwood extraction for heating and lighting in refugee camps and host communities. These include a briquette pilot and the continued provision of clean cookstoves and energy-efficiency training, as well as other UNHCR interventions that aim to reduce fuelwood demand. A new project funded by Denmark and to be implemented by the Danish Refugee Council will significantly scale up bio-briquette production as household level in the refugee camps and the host communities. The project will also focus on environmental and forest protection, addressing some of the supply-side elements highlighted above.

[105] Kangalawe et al. 2017. Climate change and variability impacts on agricultural production and livelihood systems in Western Tanzania. Climate and Development, vol.9: 3

[106] Heltberg IR, Bennett P, Steen S, Jorgensen L. 2009. Addressing human vulnerability to climate change: Toward a 'no-regrets' approach *Global Environmental Change* 19, 1, pp 89-99

These parallel projects and programmes will ensure, in the long run, that investments in the proposed project are not put at risk after project end. Several international and national NGOs are also expected to continue supporting complementary activities targeting either refugee or host communities, or both, that will be implemented during the project lifespan. The value of these projects, and other projects undertaking complementary activities, amount to almost USD 48 million. A summary of the major projects and their relevance and complementarity with the proposed GCF project is provided in Table 2.

Table 2. Current and planned projects in the region. 'Relevant Complementary Activities' refers to activities within the parallel projects, that are complementary to the proposed project's interventions.

Organisation and Project name	Funding	Duration	Relevant Complementary Activities	Related Output/ Activities	Complementarity and/or scaling up
UNHCR access to energy programme	USD 2,500,000 *(projected)*	2020–2024	UNHCR's other programming is facilitating access to alternative energy and promoting energy efficiency for refugee and host communities to help offset fuelwood consumption	Output 2 – Activities 2.1, 2.2	Investments would complement the proposed supply-side fuelwood and conservation project activities in CBFM, afforestation, and agroforestry by addressing demand side fuelwood drivers of deforestation.
Enabel – Sustainable Agriculture Kigoma Regional Project (SAKIRP)	EUR 8,800,000	2016–2021	SAKIRP is providing support to beans and cassava farmers and value chains through Farmer Field School (FFS) approach implementation modalities	Output 3, Activity 3.1	The proposed project will build upon the support delivery systems provided to small-scale bean and cassava farmers, by expanding savings activities and introducing climate smart agriculture inputs and training.
Enabel – Natural Resources Management for Local Economic Development in Kigoma Region (NRM-LED)	EUR 5,000,000	2014–2021	NRM-LED is supporting participatory village land use planning for sustainable NRM, CBFM in Kibondo, Kakonko and Kasulu; and providing institutional support to districts on natural resource governance and landscape coordination. The project is also supporting establishment and governance of CBFM	Output 1, Activity 1.1; Output 2, Activity 2.1; Output 4, Activities 4.2, 4.3	To scale and climate mainstream the land use planning approach, the proposed project would build on local capacities that have been strengthened in NRM planning and mutual benefit sharing. Scaling CBFM in target districts.
GIZ – Renewable Energy Service and Products as an Opportunity in National and Displaced Markets (RESPOND) through SNV	EUR 1,680,000	2018–2020	RESPOND is promoting results-driven and market driven approaches in woodlot afforestation (aligned with REDESO), fuel efficient cookstoves, alternative lighting solutions, and promoting energy efficiency in host communities and (to a lesser extent) refugee	Output 2, Activities 2.1, 2.2	The proposed project would learn from and scale up supply-side activities related to afforestation and woodlots. The RESPOND project also addresses demand side fuelwood needs for both cooking and lighting in host communities and camps.

			camps in Kasulu and Kibondo		
Danish Refugee Council (DRC) (funded by Denmark) – Improving access to alternative energy sources and promoting environmental conservation in refugee camps and host communities in Kigoma Region, Tanzania	DKK 25,000,000	March 2021 – December 2023	The project by DRC will focus on bio-briquette production at household level to increase access to alternative energy sources for cooking. Further, increased forest protection will be sought through establishment of tree nurseries and awareness and capacity building on environment and forest management will utilise existing structures like environmental/forest committees and student clubs in schools.	Output 2, Activity 2.2 Sub-activity 2.2.1	The proposed project complements the energy-related activities of the DRC project such as bio-briquette production at household level in the refugee camps and the host communities as well as environmental and forest protection activities.
Jane Goodall Institute (USAID funding) – Landscape Conservation in Western Tanzania	USD 19,900,000	2018–2023	JGI is implementing land use planning and supporting effective natural resource management, including in the Old Refugee Settlement of Mishamo	Output 1, Activities 1.1, 1.2; Output 4, Activities 4.2,4.3	Activities and partnerships under the proposed project would scale and climate mainstream the land use planning and CBFM implemented, whilst also learning from the LUP experience of JGI in the region and particularly in context of the old settlements.
FAO - Kigoma Joint Programme (KJP) Climate Smart Agriculture project	USD 500,000 *(projected)*	2020–2021	FAO is promoting climate smart agricultural practises in target districts, through FFS modality.	Output 3, Activities 3.1	Output 3 of the proposed project would build on the extension services provided to smallholder farmers, emphasising climate resiliency to future conditions.
UNCDF — Kigoma Joint Programme (KJP), Agriculture and Youth and Women's Economic Empowerment projects.	USD 700,000 per year *(projected)*	2020–2021 (but likely to be extended to 2022)	UNCDF is supporting savings groups and access to agricultural and livelihood inputs in Kibondo, Kasulu and Kakonko. Developing digital platform for disaggregation of data.	Output 3, Activities 3.1, 3.3	The proposed project would expand and institutionalize support provided to Village Savings and Loans Associations for both refugee and host communities and is considering the applicability of digital platform to support non-agricultural livelihoods.
EU — Livelihood beekeeping support project, (expected to be implemented by Enabel) – country wide	USD 2-3,000,000 *(estimated amount for Kigoma region)*	TBC	The EU project is supporting beekeeping activities	Output 3, Activity 3.3	Alternative livelihood enhancement would build upon synergies in implementing beekeeping activities and supporting beekeeping groups.
REDESO — through various donors (Oxfam, US, UNDP)	USD 1,500,000 *(projected)*	2020–2024	REDESO is implementing afforestation and woodlots in Kibondo and Kakonko (and soon, Kasulu),	Output 2, Activity 2.2; Output 3,	The proposed project would look for synergies for activities related to woodlots and sustainable livelihoods. Complementary activities addressing demand side of fuelwood

			including establishing nurseries, management and governance, and maintenance systems. Supporting mushroom cultivation and kitchen gardens in refugee camps.	Activities 3.1, 3.3	

As noted, some of the above investments and projects are still in discussion and negotiation stages and are only indicative of interventions the project will potentially coordinate with. However, the table clearly illustrates that baseline degradation issues — including those resulting from the recent influxes of refugees and already impacted by climate change — are already being addressed by development and humanitarian financing. As environmental degradation in refugee hosting areas continues to be a key priority of the government, it is expected that new initiatives will be initiated by the Government with other funding sources during the course of the GCF project implementation.

Table 2 does not include development financing allocated through district, regional, and national government investments and budgets in climate smart agriculture, alternative livelihoods, forestry, soil management, catchment and basin management, and land use planning activities in Kigoma. However, it is estimated that over the project duration public expenditure from local government authorities on related activities could be around USD 1,7 million.[107] The project will build on government policies and activities on environment and forest management and will closely coordinate with local government authorities throughout the project implementation to ensure project activities complement and reinforce government expenditures, plans and priorities.

B.2 (a). Theory of change narrative and diagram (max. 1500 words, approximately 3 pages plus diagram)

Problem Statement

Kigoma is experiencing rising temperatures as well as increasing rainfall intensity and variability, which will contribute to an increase in the frequency and severity of floods, droughts, heat stress and fires. Observed changes in weather patterns have produced impacts that have compounded environmental degradation, as well as negatively affecting agricultural outputs, incidence of pests and diseases, forest resources and deforestation, surface- and groundwater availability, and soil erosion and fertility. Projected climate changes will further exacerbate these impacts. These changes are threatening local rainfed and ecosystem-reliant livelihoods in the region.

Climate hazards interact to produce impacts on ecosystems and agriculture that include: i) soil erosion — and subsequent soil nutrient and moisture loss; ii) water logging; iii) damage to crops and infrastructure; iv) reduced water availability; v) proliferation of pests and diseases; and vi) geographical shifts of agro-ecological zones and ecosystems. In turn, these impacts create negative feedback loops that amplify the effects of future climate hazards on an already-pressurised landscape. These hazards and impacts are putting local livelihoods at risk and undermining the capacity of already-strained ecosystems to provide services for host communities and refugees. As the impact chain pathway (Figure 9 on page 23) above shows, the effects of projected climate changes on the agro-ecological system compound existing impacts from non-climate drivers, whilst simultaneously contributing to the factors that underlie the non-climate drivers themselves. Moreover, development (non-climate) impacts on ecosystems — which are likely to be exacerbated by climate change — are compounded by population growth, including the influx of refugees and host communities and refugees' limited capacity to access financing, technology and information for adaptation.

[107] This estimate is based on consultations with the Kibondo District Environmental Officer and Tanzania Forest Service that indicate that the Kibondo district annual environmental operating budget covering reforestation, soil conservation, solid waste management and similar activities normally stands at TSh 100,000,000 or around USD 43,500. For now, this budget has been taken as average budget across the three project districts in Kigoma. The Tanzania Forest Service (TFS) budget per district per year is estimated to be around TSh 100,000,000 for forest monitoring and beekeeping in Kibondo and Kakonko, whereas for Kasulu, the TFS budget is estimated to be slightly higher at TZh 300,000,000 per year. These figures were used to estimate government spending across five years in three districts. Discussions are ongoing with regional authorities to provide more accurate estimates of projected government contributions.

Insufficient access to natural resources and diminishing ecosystem services contribute to conflict both within and across refugee and host communities[108]. Accessing resources and inputs for adaptation is critical for maintaining livelihoods and ecosystem services in the project's target sites, particularly given the historical impact of 'business as usual' land-use planning and livelihood activities in refugee camps and host communities. In the absence of the intervention, the vital ecosystems of the Kigoma region may become irreparably degraded, which will be compounded by — and contribute to — the impacts of climate change on the agro-ecological system. Given the target districts' reliance on agriculture, high rates of poverty, and refugees' adaptive capacity being constrained by limitations on their livelihood activities, this will threaten the livelihoods and lives of most people in the project area.

While other activities in region are addressing discrete elements of these problems, there is limited landscape-level coordination of these interventions. Additionally, projects and activities addressing humanitarian or development concerns have limited consideration for the long-term impacts of climate changes or have insufficient scale to trigger a paradigm shift.

Preferred Solution

The theory of change for this project (Figure 10 below) has the following goal statement:

IF ecosystem-based adaptation is integrated into land use planning and development and humanitarian policies, key ecosystems restored/sustainably managed and resilient livelihoods practised, THEN climate-resilient sustainable development will be promoted thereby reducing community vulnerability BECAUSE of enabling policy and planning processes, healthy ecosystems and improvements to food security.

The project interventions aim to disrupt the abovementioned feedback loops by strengthening organisational capacity and providing knowledge, technologies and other resources to address worsening conditions under climate change and the barriers to adaptation for host communities and refugees. Concretely, by undertaking land use planning, restoration of degraded ecosystems, promoting climate-resilient livelihoods, and mainstreaming successful actions into policy; institutional systems will be strengthened and ecosystem and livelihoods resilience will improve among host communities and refugees in the refugee hosting areas of the Kigoma Region. This will be because communities will have the resources and knowledge to undertake adaptation activities, and local capacity will be created to accommodate needs of refugees and host communities in policy and planning.

In order to complete participatory climate resilient land use planning and related environmental management, the project will strengthen district and community planning and management capacities and create greater environmental awareness among communities.

The restoration of the degraded ecosystems will be largely informed by the completed land use planning processes. In addition, protection and monitoring support will be provided for selected impacted national forest reserve areas. Restoration of ecosystems will be achieved by facilitating and promoting Community Based Forest Management; undertaking agroforestry and village land afforestation. In the refugee camps, flood and erosion control measures will be introduced that will help avoid soil run-off and erosion.

To promote climate-resilient livelihoods and to provide incentives for sound environmental management and conservation, the project will promote climate resilient agriculture through the provision of inputs and transfer of skills and technologies to strengthen the capacities of farmers and district officials. Furthermore, through water harvesting and irrigation, water availability and water use efficiency will be improved. To promote livelihood diversification, including as a risk mitigation strategy, the project will strengthen skills and capacities to undertake beekeeping and mushroom cultivation.

To anchor the gains and positive outcomes of the project, knowledge and lessons will be mainstreamed into national and local government structures and wider humanitarian programmes. This will be achieved by generating and documenting evidence, disseminating information and providing inputs and knowledge to policy and planning processes.

In order to address the mutually reinforcing system of climate impacts with an equally networked response, the project proposes four interconnected outputs, described in detail in B.3 below, delivering direct benefits to 15,000 households (80,000 people) and restoring 42,000 hectares of land.

[108] Types of conflict include theft, skirmishes, and sexual and gender-based violence. Refer to Annex 6. Environmental and Social Management System for assessment of potential conflicts and their management.

Output 1 will involve the development of climate-resilient land-use plans (C-LUPs) in host community villages (Activity 1.1), applying a comprehensive landscape- and ecosystem-approach that includes for the needs of multiple stakeholders. Climate resilient land use planning within camps and villages will be coordinated through the project and maximize opportunities for mitigating climate impacts, such as through afforestation in key upstream areas and planning small-scale flood control infrastructure to reduce downstream flooding and erosion in host communities. Guided by the high-level perspective of these land-use plans, the land- and water-management interventions under the following outputs are therefore designed to cumulatively enhance the capacity and resilience of each catchment's mosaic of ecosystems to continue to provide services to the camps and host communities under future climate conditions. This will ensure coherence and complementarity between interventions and will enable the project to address the complex system of projected climate impacts and hazards anticipated at the landscape level. The C-LUPs developed under this output will inform the landscape-level monitoring of ecosystem benefits by defining the coverage of each project activity.

In Output 2, Community Based Forest Management (CBFM) interventions (Activity 2.1) will seek to improve the protection of natural forest systems and enhance ecosystem services. Simultaneously, the sustainability and ownership of this conservation is reinforced by the agroforestry and afforestation interventions (Activity 2.2), providing sustainable alternative sources for energy needs, thereby reducing pressure on the natural forests and allowing them to regenerate. Demand for fuelwood will be diverted from slow-growing indigenous Miombo ecosystems, towards fast-growing plantations that will be established on marginal land. Similarly, the agroforestry interventions, by providing multiple uses such as food, fodder, shade for crops, and sustainable fuelwood production, will further deter activities or behaviour that degrade the natural forest systems — including unsustainable fuelwood collection and charcoal burning[109]. The trees would be valued and protected due to their multiple uses including fodder, shade, and contribution to wood fuel supply. The shift from activities which degrade ecosystems towards more sustainable behaviour will also be facilitated by the improved income generation and savings associated with the enhanced agricultural and livelihood activities under Output 3. Consequently, pressure on the surrounding natural ecosystems will be further reduced. Reductions in the rate of deforestation and forest degradation will be measured by remotely sensed derived data relative to the baseline and projected business-as-usual deforestation rates (see Section 7.3 of the Feasibility Study). Soil and water conservation will benefit agriculture and livelihoods. Communities' direct involvement in the project and its participatory design is intended to support awareness and provide incentives for communities to contribute to their preservation, further enhancing the sustainability and interdependencies of the project's outputs.

The flood and erosion control measures (Activity 2.3) will also contribute to strengthening the climate resilience of the project's natural landscape. The interventions under this activity will substitute the services provided by ecosystems in areas that have been degraded to the extent of having insufficient natural flood-attenuation capacity. These measures will contribute to reducing the impact of floods on soil erosion, damage to crops and infrastructure, and waterlogging. Ultimately, this will help to support the resilience of the agriculture, forestry and agroforestry measures, contributing to restoring soil moisture and nutrients, as well as providing other ecosystem services and livelihood outcomes.

Under Output 3, agroforestry will also support the climate-resilient agriculture interventions (Activity 3.1). This will be achieved through strengthening of the capacity of agricultural landscapes to sustain or improve yields under various climate conditions, through intercropping and boundary tree planting (see Activity 2.2 project description for details). These agricultural interventions will also be supported by the rainwater-harvesting and irrigation interventions (Activity 3.2) which will help decrease risks to agricultural activities associated with increasing water stress and prolonged drought. The coverage of agroforestry adoption will be mapped in Output 1 and will be monitored by the changes in the remotely sensed seasonal vegetation index and productivity signals.

The alternative livelihood interventions in beekeeping (Activity 3.3) will operate symbiotically with the forest interventions under Output 2. Investments in forestry systems will also support the flood-mitigation impacts of the interventions under Activity 2.3, contributing to reducing waterlogging as well as soil erosion, nutrient loss and moisture loss.

The interactions described above will jointly contribute to strengthening the resilience of the landscape under future climate change conditions. This landscape approach will be institutionalised and upscaled to similar landscapes and scenarios under Output 4. Under this output, the results of the planning processes and physical interventions implemented under Outputs 1–3 will be used to demonstrate the economic, environmental and social value of this model to stakeholders. As a result, a shift in land-use planning and activities in target ecosystems will occur away from unsustainable practises leading to landscape degradation, towards climate-resilient landscapes and livelihoods. The activity will consist of capturing evidence of the ecosystem-wide benefits (Activity 4.1), developing communication products for policy- and decision-makers to appreciate and replicate the landscape interventions (Activity 4.2), and drafting revisions to key plans and policies and supporting their integration into national and district government planning processes, to promote up-scaling of the EbA model (Activity 4.3).

Proposed project approach

The project will adopt an integrated landscape EbA approach to help communities manage floods, droughts and higher temperatures. The project proposes an intervention strategy considers the mosaic nature of the landscape and the interconnectivity of various land uses. The complex human-ecological interface of the refugee context has also been considered by using a variety of tested and innovative tools at different levels across the development-humanitarian-climate nexus. This approach includes joint meetings between host communities and refugees, awareness-raising and participatory processes to manage benefit sharing arrangements between host communities and refugees and facilitate sustainable offtake, promote ecosystem protection, and minimise conflict over resource use. The solution would also be complementary to other projects operating in the area that focus on other dynamics of non-climate drivers (such as demand-side energy use).

Key barriers to reducing vulnerability and supporting resilience in the Kigoma region

The proposed project will overcome a number of identified barriers that impede building the climate resilience of host communities and refugees in Kigoma.

Institutional, planning, knowledge, and regulatory

<u>Barrier 1. Limited capacity in local institutions and processes to plan for and address long term climate vulnerabilities</u>.
The institutions providing support to the region are not well positioned to address long term climate vulnerabilities, as they have limited capacity and expertise to adequately promote adaptation. Whilst other districts in the region have a greater uptake of Village Land Use Plans (VLUPs)[110,111], very few VLUPs in Kibondo, Kakonko and Kasulu districts, have been finalised, although several are in the process of development. The low uptake of VLUPs is caused primarily because of budget constraints due to various competing priorities at the District level. An additional challenge is related to the limited capacity of District Environment Management Committees to support ward and village environmental management committees in developing and enforcing the VLUPs, a challenge noted by district government officials during consultations in September 2020. While VLUPs provide a participatory approach to decision-making and administration of natural resources, they do not typically support communities in planning for resilient ecosystems. VLUP processes also seldom benefit from formal climate risk and vulnerability assessments. This barrier is reinforced by insufficient awareness by planners of the value of ecosystems services, which constrains the adoption of ecosystem-based approaches to adaptation.

<u>Barrier 2. Refugees and the special needs of their host communities are not well articulated in adaptation policies, legislation, planning or programmes.</u>
Although policies and legislation in Tanzania provide an enabling environment for the adoption of adaptation interventions in some areas, refugees and the special needs of their host communities are not yet adequately considered in adaptation policies or activities such as National Adaptation Plans and Programmes, river basin plans, forest management plans, and health strategies. There are also opportunities for tailoring guidelines such as on climate smart agriculture, health and other plans and guidelines to the acute needs of host communities and refugees.

Technological

<u>Barrier 3. Limited access and uptake of agricultural, forestry or land management technologies or knowledge to improve climate resilience.</u>

Kigoma is predominantly rural, with very limited access to agricultural, forestry or land-management technologies that can enhance the resilience of the region to future climate impacts. Farmers' vulnerability is exacerbated by low levels of knowledge and capacity to implement resilient climate-smart agricultural practises such as those that promote soil

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110 Village Land Use Plans are tools for conserving priority areas, reserving land for investment, reducing land use conflicts and establishing market opportunities that benefit rural economies.
111 For example, Uvinza District has VLUP in 40% of its villages (25/60 villages) – Source: Regional Secretariat of Kigoma

conservation, minimum tillage, and other practices. In the long term, this results in the application of ineffectual means of production, unsustainable methods of cultivation. Access to inputs such as climate resilient varieties of crops and their seeds, materials for beekeeping and mushroom cultivation are limited. These farmers also lack resilient technology such as small-scale in-stream water infrastructure and pumping systems and have limited alternative income opportunities to substitute potential losses of crop production. For forestry-related activities, there is limited use of agroforestry in and around the project sites. Moreover, robust analysis of climate resilient species for afforestation and agroforestry is not present in Kigoma and requires specialist knowledge.

With regards to monitoring of natural resources within Kigoma, the TFS is constrained by limited technologies available for patrolling and incident recording/reporting. Owing to the vast size of national forest reserves within Tanzania and surrounding the project area, current patrolling methods (foot patrols) are unable to cover the full extent of forest reserves. This is of importance as ~64% of the forest area in Tanzania falls under this type of management system. In addition, forest management plans are often either non-existent or not properly implemented, owing to limited expertise among monitoring teams for monitoring and patrolling activities. Where such plans do exist, uncertainty among monitoring teams and communities around responsibilities, costs and benefits leads to a limited uptake of the plans. As a result, encroachment and illegal harvesting of natural resources in forest reserves is common. Where local governments also participate in the management of forest reserves, their planning, management and monitoring capacity is severely constrained by lack of technological resources — brought about by both limited access to financial resources and training/upskilling services as described below.

Financial

<u>Barrier 4. Low purchasing power of smallholder-farmers and refugees limits access to inputs for resilient agriculture and other long-term investments.</u>
The limited leverage of smallholder farmers and refugees reduces their access to equipment and improved planting materials for resilient agriculture and other investments in longer term solutions. Common markets allowing for exchange of goods and services between refugees and host communities were also closed in 2019, limiting flows and access to inputs. There are still markets within the camps although reduced in size and volume of transactions. More information is included in Annex 2.1. Feasibility Study Section 5.2.

<u>Barrier 5. Financial services are insufficient in the region, except for some savings associations.</u>
In the absence of formal financial services, mobile money services have been widely adopted by refugees and on a small scale by aid agencies to facilitate e-cash transfers. Digital cash is used to buy food in local markets both inside and outside the camps. The insufficient availability of local financial services and technical knowledge in the region limit the promotion and uptake of climate-resilient adaptation options. Although saving associations are present, they have limited capacity for high-level oversight and management to support savings and bulk procurement. This compounds the already limited capacity of smallholder famers and refugees to access finances or credit for investing in resilience-building inputs or livelihoods. Consultations undertaken in September 2020 noted that the scale of investment in savings associations in refugee camps was limited compared with host communities, and in some cases, savings associations required the monetisation of food rations.

<u>Barrier 6. Limited investment in Kigoma region and competing development priorities defer long-term adaptation planning or leave it underfunded.</u>
Although climate change adaptation is a priority for Tanzania, and whilst humanitarian activities are receiving international funding, the coordinated nexus of these clusters is underdeveloped at present, and as a result, generally underfunded. This discordance is reinforced by generally low investment in the region, as well as limited long-term planning, high levels of poverty, and more immediate social needs diverting priorities towards short-term interventions. District government budgets are very limited and nowhere near sufficient to address the climate adaptation needs in the region. Estimated amounts allocated to environment in district budgets are to the tune of $40,000. The investments in adaptation required are public goods and require grants. Revenue streams for some livelihood activities are available, but with relatively low returns per beneficiary making loan products not viable. More information per activity on the nature of investments, implementation and management arrangements, financial returns on investments and other information are available in Annex 2.1 Feasibility Study and Annex 3. Financial and Economic model.

Also, some development initiatives are struggling to mobilize resources to address environmental issues. To illustrate, the Kigoma Joint Programme that aims to provide development support to refugee hosting areas in the region has thus far not been able to mobilize any funding under its environment and energy theme. In addition to barriers to knowledge and access to technology, the actual funding is lacking.

Social

Barrier 7. Unique needs and challenges related to working with host populations and their refugee communities.
Refugees have limited adaptive capacity, as they have insufficient local adaptation knowledge as compared to the host community, cannot access the formal economy, have limitations in employment and livelihood activities due to the encampment policy, and have limited access to basic services (other than what is provided by humanitarian partners). Women smallholder farmers and refugees are particularly vulnerable to climate impacts because of limited access to capital, lack of land tenure, traditional gender roles, social and cultural norms, and limited access to training, decision-making process and financial resources. [112]

Overcoming the key barriers

Barrier 1 will be overcome by working with local government authorities to facilitate the development of climate resilient village land use plans (Output 1). These plans — and the participatory process by which they are developed — will provide the framework upon which subsequent interventions will be based, thereby ensuring their sustainability and effectiveness. This process, in combination with the upscaling interventions under Output 4, will also contribute to addressing Barrier 2 and strengthening the institutional and regulatory systems for climate-responsive planning and development. The participatory process and inclusion of both camps and host communities, will also help address Barrier 7, contributing towards ensuring the durability of the refugees in the medium- and long-term.

Outputs 2 and 3 will address Barrier 3, through activities in capacity building through the Farmer Field School approach, and hands-on engagement in the project interventions, livelihoods, and resource management, as well as physical interventions and inputs in forestry and agriculture. For the forestry sector, interventions will focus on implementing Community-based Forest Management (CBFM), including enhancing the technical and institutional capacity of the Tanzania Forest Service (TFS) to monitor and protect the North Makere, South Makere, Mwale and Buyungu National Forest Reserves. Inputs in agriculture will also help to address Barrier 4. Using the Savings Association model to distribute inputs for Output 3 in host communities, will contribute to the sustainability of the project whilst also addressing Barriers 4, 5 and 6.

Finally, Output 4 will contribute to scaling and institutionalising the mechanisms by which these barriers are being addressed, whilst also independently contributing to addressing Barriers 2, 5, 6 and 7. By specifically targeting institutions and key decision-makers, these activities will help subvert the underlying causes of these barriers to adaptation.

[112] Refer to Annex 8. Gender Assessment and Action Plan section 1.4 for more information.

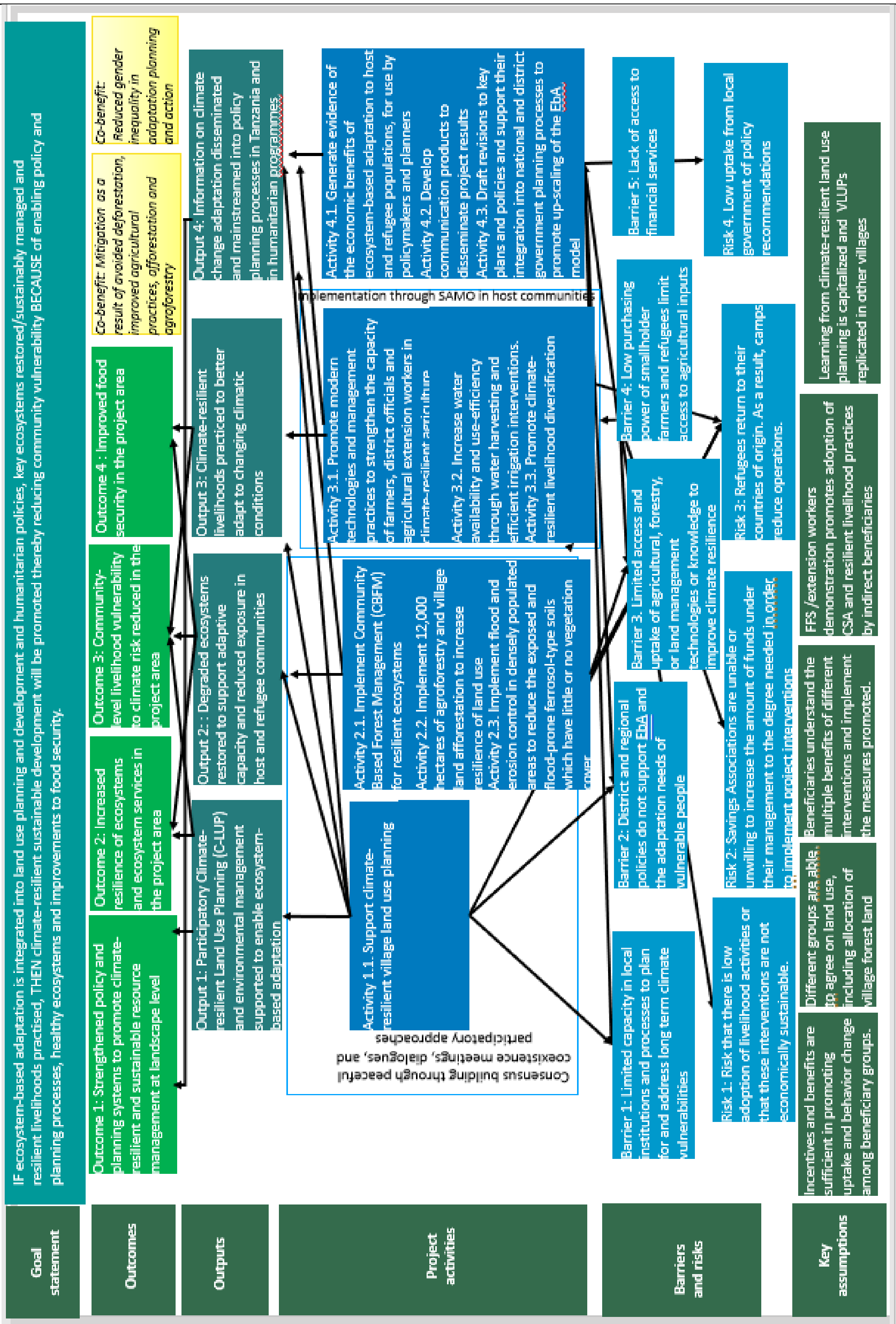

Figure 10. Theory of change diagram showing how the project design responds to barriers to achieve climate resilience. (To make the diagram readable, the linkages between the activities are illustrated in Figure 11 below).

Together, the abovementioned interventions will overcome the barriers to climate change adaptation in Kigoma, thereby contributing to a paradigm shift in host communities and refugees' climate-resilience. This shift will ultimately ensure that individuals, institutions and ecosystems move away from unsustainable practises and climate-vulnerable livelihoods, towards adaptation and climate-resilient sustainable development. In addition to the direct benefits to project beneficiaries the proposed project will also yield significant mitigation and gender co-benefits. The mitigation potential of the project has been estimated at a sequestration of 762,073 million tCO_2e and emission reductions of 3.2 million tCO_2e over 20 years — including a five-year implementation period — as a result of avoided deforestation, improved agricultural practises, afforestation and agroforestry (estimated using the FAO Ex-Ante Carbon-balance Tool). For further detail, please see Appendix C and the project descriptions, under Sections 7.3.1, 7.3.2, and 7.4.1 in the Feasibility Study.

Gender inequality in climate change adaptation planning and action is expected to be reduced through meaningful representation and participation of women in Climate-resilient Land Use Planning (C-LUP) to support ecosystem-based adaptation.

Additional environmental, economic, social and gender co-benefits are identified in section D3.

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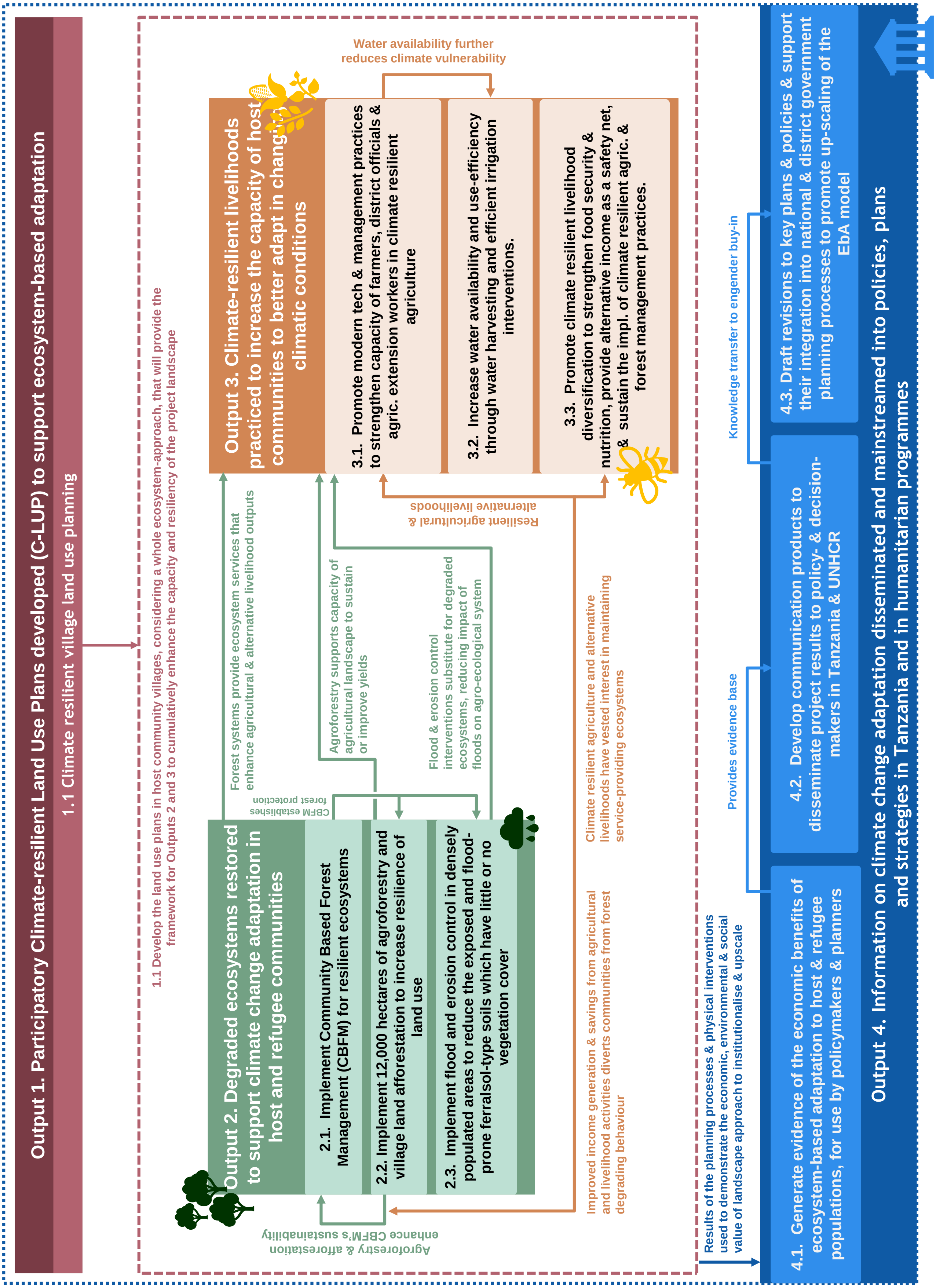


Figure 11. Proposed landscape approach to overcoming the barriers to adaptation showing how the proposed outputs and activities map onto the landscape's mosaic of land uses

The selection and distribution of interventions for each sub-catchment where the target host communities and refugees are located will be scaled relative to the attributes (including population) of each camp and host community, as well as by contextual land use practises and flood risk. This approach aims to provide equitable benefits from the agriculture, forestry and flood mitigation interventions under Outputs 2 and 3, that are appropriate to the on-the-ground context. The preferences of beneficiaries and the experience of local partners will be incorporated to ground-truth the intervention planning, but the initial distribution of interventions which this proposal describes is based on the relative population, number of host community settlements (villages), differential land use practises, and flooding risks in each sub-catchment.

- Population and village level analysis was undertaken using the NASA ~1 km Estimated Gridded Population dataset for 2020.
- Basic land use analysis was conducted using ESA CCI 30 m land cover product, highlighting areas of activity engaging agriculture, areas of grasslands and shrubs, as well as the allocation of woodlots or forests that may be providing EbA benefits.
- Flooding analysis (to inform distribution of interventions) was undertaken using the observed ERA5 hourly precipitation reanalysis to inform storm event characteristics in combination with SRTM 30 m topography data as the inputs for the HEC RAS hydrologic model[113] .
- These factors aid in intervention selection, design and prioritisation to build resilience of target beneficiaries that is appropriate, locally owned and has high prospects for sustainability[114].

B.2 (b). Outcome mapping to GCF results areas and co-benefit categorization

Fill in the GCF results area table below to map each project/programme outcome identified in section B.2(a) to the contributing GCF results area(s) by referring to the description of eight results areas provided in the guidance note.

Outcome number	GCF Mitigation Results Area (MRA 1-4)				GCF Adaptation Results Area (ARA 1-4)			
	MRA 1 Energy generation and access	**MRA 2** Low-emission transport	**MRA 3** Building, citiies, industries, appliances	**MRA 4** Forestry and land use	**ARA 1** Most vulnerable people and communities	**ARA 2** Health, well-being, food and water security	**ARA 3** Infrastructure and built environment	**ARA 4** Ecosystems and ecosystem services
Outcome 1:Strengthened policy and planning systems to promote climate-resilient and sustainable resource management at the landscape level	☐	☐	☐	☐	☒	☒	☐	☒
Outcome 2: Increased resilience of ecosystems and ecosystem services in the project area	☐	☐	☐	☐	☐	☐	☐	☒
Outcome 3: Community-level livelihood	☐	☐	☐	☐	☒	☐	☐	☐

[113] U.S. Army Corps of Engineers Hydrologic Engineering Center's River Analysis System (HEC-RAS), https://www.hec.usace.army.mil/software/hec-ras/
[114] Refer to Section 3.4 of Annex 2. Feasibility Study for an expanded discussion of selection and design.

vulnerability to climate risk reduced in the project area								
Outcome 4: Improved food security in the project area	☐	☐	☐	☐	☐	☒	☐	☐

If any co-benefits have been identified in section B.2(a), fill in the Co-benefit table below to map each co-benefit to the corresponding category as defined in the FP guidance note.

Please refer to section B2 a) for mitigation and gender co-benefits and section D3 for the description of other additional environmental, economic , social and gender co-benefits.

Co-benefit number	**Co-benefit**					
	Environmental	**Social**	**Economic**	**Gender**	**Adaptation**	**Mitigation**
Co-benefit 1: Mitigation as a result of avoided deforestation, improved agricultural practices, afforestation and agroforestry	☐	☐	☐	☐	☐	☒
Co-benefit 2: Reduced gender inequality in adaptation planning and action	☐	☐	☐	☒	☐	☐
Co-benefit 3	☐	☐	☐	☐	☐	☐
Co-benefit 4	☐	☐	☐	☐	☐	☐

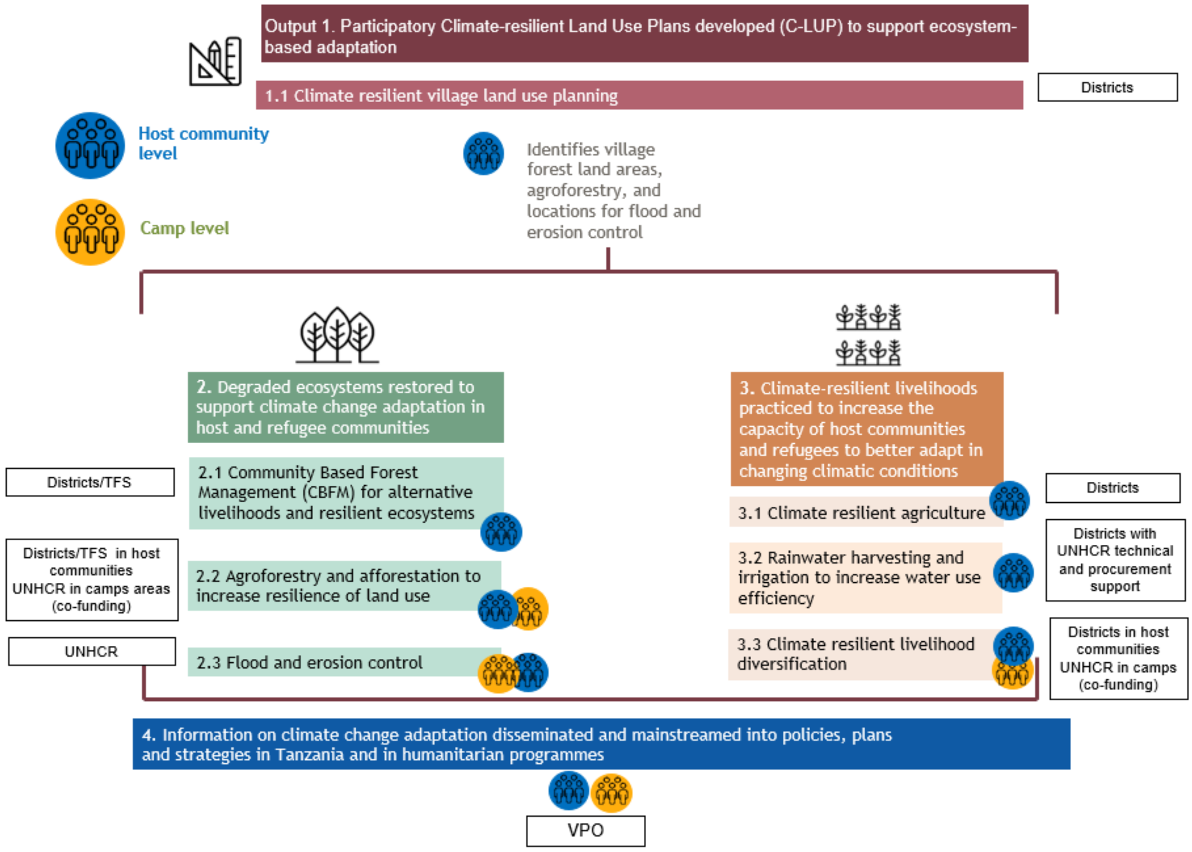
B.3. Project/programme description (max. 2500 words, approximately 5 pages)

This proposed project will address the urgent need for climate change adaptation for both host communities and refugees through four Outputs, namely: i) participatory, climate resilient land-use planning in villages; ii) improved land use and forestry management; iii) climate-resilient agriculture and livelihood diversification; and iv) information on climate change adaptation disseminated and mainstreamed into policies, plans and strategies in Tanzania and in humanitarian programmes. The project will focus on those areas that have been most degraded and are most vulnerable to climate change. These areas are the Nduta and Nyarugusu refugee camps, the former Mtendeli camp area and, within each camp's catchment, the host community settlements located within ~10–15 km of the camp boundary[115]. Refugees in the former Mtendeli camp, with the exception of those opting for voluntary repatriation, were relocated to Nduta camp and would therefore continue to benefit from the project activities planned in the camps. Therefore, the final beneficiaries under the project will be defined as the refugees in the Nduta and Nyarugusu camps. As part of the decommissioning process, UNHCR will embark on Mtendeli former camp rehabilitation to facilitate environmental restoration with the contribution of GCF project investments in flood and erosion control measures inside and around the camps, including Mtendeli former camp area. Figure 12a below shows the distribution of activities between host communities and the camps and Figure 12b shows how adaptation benefits accrue to different groups in response to climate change impacts and risks. The selection, scale and design of interventions have been informed by the needs,

[115] Although they are technically not allowed to travel beyond a 4 km buffer zone, 10km is the approximate maximum distance that refugees have been reported to travel on foot to collect firewood for cooking and so is the area most significantly impacted by their presence.

challenges, and available services in the respective sites and landscape connections. For example, concerns of erosion and flooding from increased runoff are more relevant in the high-density settlements of the camps where ground cover is limited and where people cannot always move out of flood prone areas — in contrast to the lower densities and greater mobility of communities around the camps. Conversely, the camps have more reliable water supplies and agricultural land is limited, so rainwater harvesting interventions are not as critical as in the host communities. The project maximises upon the mosaic of land uses, population densities and adaptation potential in the project area, targeting interventions geographically and by population to achieve the most effective, relevant and transformative impact.

Figure 12a. Project structure mapping activities at the host community and camp levels showing the project execution arrangements



Project interventions	Increased severity of wildfires	Increased frequency/intensity of floods	Increased occurrence of prolonged droughts	Increased water stress	Increased occurrence of pests and diseases	Increased heat stress
Output 1 - Participatory Climate-resilient Land Use Plans developed (C-LUP) to support ecosystem-based adaptation						
1.1 Village land use planning	Participatory planning to ensure success and sustainability of Outputs 2 and 3					
Output 2 - Degraded ecosystems restored to support climate change adaptation in host and refugee communities						
2.1 Community based forest management	X	X				

2.2 Agroforestry and village land afforestation		X				X
2.3 Flood and erosion control		X			X	
Output 3 - Climate-resilient livelihoods practised to increase the capacity of host communities and refugees to better adapt in changing climatic conditions						
3.1 Climate resilient agriculture practises		X	X	X	X	X
3.2 Rainwater harvesting and irrigation			X	X		
3.3 Livelihood diversification	X	X	X	X	X	X
Output 4 - Information on climate change adaptation disseminated and mainstreamed into policies, plans and strategies in Tanzania and in humanitarian programmes supported by UNHCR						
4.1 Economic evidence base for EbA	Establishes the institutional framework for the replicability and upscaling of the impacts under Outputs 1–3					
4.2 Communication products for scaling and mainstreaming project results						
4.3 Adaptation integrated with national and district planning processes						

Figure 12b. Climate change hazards addressed by each of the proposed interventions

The project area has been defined (see Section 3.3 Project boundary and site selection in the Feasibility Study for detail) taking into account land use, administrative boundaries, population distribution and geographic and hydrological features around the camps. The selection and distribution of interventions for each catchment will be scaled relative to the attributes (including population) of each camp and its host communities, as well as by contextual land-use practises and flooding risks. Within the sites, these will be distributed through the land-use planning processes in Output 1. This approach aims to provide equitable benefits from the agriculture, agroforestry and flood mitigation interventions under Outputs 2 and 3, that are appropriate to the on-the-ground context. The preferences of beneficiaries and the experience of local partners will be incorporated to ground-truth the intervention planning, but the initial distribution of interventions which this proposal describes is scaled based on the relative population, number of host community settlements (villages), differential land-use practises, and flooding potential in each catchment.

The project will be implemented by the Government of Tanzania, acting through its Vice President's Office (VPO) and UNHCR, particularly on interventions in the refugee camps, as Executing Entities (EEs). The Government of Tanzania, acting through its Vice President's Office (VPO), as lead EE will oversee and manage project implementation, with and through appropriate implementing partner organisations including local government authorities and, if and when required, engaging technical experts, and national and international NGOs. Selection of the technical experts and national and international NGOs will be undertaken by the executing entities VPO and UNHCR. UNEP will provide guidance to VPO to select partners in line with UN partnership policies and procedures. Selected partners will be subject to a due diligence process considering conflict of interest, human rights and gender equality, alleged involvement in cases of irresponsibility/litigation/green washing and reputational risk as well as specific technical capacities, strategic capacities, financial capacity to undertake the partnership and procurement procedures.

UNHCR will be the executing entity for the project activities 2.3 and sub-activities 3.2.1 and 3.2.2. The training/capacity development sub-activities 3.2.3 and 3.2.4 will be under the responsibility of the VPO to be implemented by the District Councils. Through committed co-financing, UNHCR will implement activities in the camps (contributing to activities 2.2and 3.3) and the infrastructures planned in Activity 3.2.

UNHCR will implement activities in consultation with MoHA. As the government authority responsible for overseeing all activities inside the camps and/or involving refugees, MoHA will have a supervisory role at government level. This role will be recognized in a cooperation agreement between VPO and MoHA. This legal instrument will govern the transfer of funds for supervision costs from VPO to MoHA and the reporting requirements. *Section B.4. Implementation arrangements* contains further detail on the nature of these partnerships, the selection process, existent operating partnerships, and indicative organisation's track records and suitability.

Component 1. Increased resilience of ecosystems and people to climate change threats

Output 1: Participatory Climate-resilient Land Use Plans developed (C-LUP) to support ecosystem-based adaptation[116].

The GCF funding request for the Output is US $ 1,062,805.

This is the starting point for all project interventions, developing participatory climate-resilient land-use plans that are consistent with an ecosystem-based adaptation approach (EbA) and which will inform the design and distribution of subsequent Outputs.

Activity 1.1. Support climate-resilient village land use planning

It is the intention that the Village Land Use Planning process, and the subsequent interventions that are planned through this process, are carried out in villages in the target area surrounding the camps, near Mtendeli former camp area, Nduta camp and Nyarugusu camp.

The Village Land Use Plan (VLUP) is an overall plan showing how land resources, located within the village boundaries, should be used to meet declared objectives. Lessons learned from these efforts suggest that while such participatory processes may require a considerable amount of time and resources, they successfully resolve conflicts around the use of natural resources, and they provide great opportunities for communities to take control and ownership of shared resources. The project will adapt this established approach, introducing the overall objective of increasing the climate resilience of the landscape into the planning process, resulting in a Climate-resilient Land Use Plan[117]. This will improve upon the VLUP process, by acknowledging and addressing the role of the changing climate in shaping local ecosystems and landscapes. This process will also help with identifying practises and behaviours that will become unsustainable under future climate conditions, in order to divert activities in anticipation. Participatory land-use planning is widely practised in the country and the project follows the effective approaches promoted by the UNDP-UNEP Poverty Environment Initiative in Kigoma[118] as well as numerous resources from FAO, International Land Coalition, Enabel, Jane Goodall Institute, and IUCN in the country. The overall goal of this activity is to strengthen the capacity of actors and institutions that deal with land tenure and to apply ecosystem-based planning, thereby providing a comprehensive approach for implementing the other activities envisioned under this project. This would foresee a central role for district authorities in the implementation of planning activities, as an Implementation Partner. Completing the land-use planning is pre-requisite for carrying out the climate-resilient practises implemented later under Outputs 2 and 3.

Technical assistance to facilitate the process and capacity building will be provided through experienced local partners with an understanding of the VLUP process. This process (and the interventions under Outputs 2 and 3) will further be supported by parallel awareness raising of the threats and opportunities of climate change in host and transient Sukuma communities.

Although village land use planning is typically a public good, in the case of Kigoma, limited financial resources and institutional capacity remain barriers to landscape level land-use planning. Consultations undertaken in September 2020 noted a lack of capacity amongst Village Environmental Management Committees to enforce village by-laws and regulations contained in existing land-use plans, as well as limited capacity amongst the District Environmental Management Committees to support the Ward and Village Environmental Management Committees. This has led to

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[116] Detailed activity descriptions covering descriptions of interventions and their climate resilience, lessons learned from other projects and best practices, analysis of alternatives and suitability of the proposed option, factors influencing design and scale of intervention, implementation arrangements, risks and exit strategy can be found in Section 7 of Annex 2. Feasibility Study.

[117] The project Ecosystem-based Adaptation for Rural Resilience financed by the Global Environment Facility and implemented by UNEP plans to mainstream climate resilience in land use planning in selected wards and districts in Tanzania. Lessons learned and experiences will be shared with the GCF project.

[118] http://www.unpei.org/sites/default/files/PDF/Uganda-report-study-tour-TZ-march09.pdf.

ad-hoc land-use patterns and an inefficient spatial distribution of economic activity, where much of the land is under-utilized or mismanaged. In addition to providing the public good of wide-scale land use planning, this activity will improve the definition and understanding of the property rights systems to ensure that markets can operate effectively.

The beneficiaries of this activity will be: i) population of villages participating in land use planning process and receiving climate change awareness-raising (~300,000 beneficiaries in approximately 20-25 villages, or equivalent and agro-pastoralist groups located in those communities); ii) local and regional government officials receiving capacity building; and iii) local and district management institutions being strengthened. Beneficiaries of other host community-based activities in Outputs 2 and 3), would be drawn from this population.

The VLUP processes supported by the project in host communities under output 1 are participatory in nature and the national guidelines provide suggestions to ensure that women are represented in the decision-making process. The VLUP seeks to include agreements that change traditional land management practices, thus improving the position of women in controlling land and natural resources.

<u>Sub-activity 1.1.1. Scale-up land use planning for villages, and support to issuance of Certificates of Customary Rights of Occupancy (CCRO)</u>

The objective of participatory decision-making is to ensure the most effective use of land resources through negotiation between different interests. The guiding principles of this approach are equity, efficiency, viability, conservation and sustainability. The development and creation of land-use plans therefore requires a framework and forum where stakeholders meet, communicate, formulate strategies and implement them collaboratively. Because such a plan requires consensus, it can also provide a means of resolving conflict, for example about the boundaries between adjacent villages or the demands of competing land uses (for example, forestry versus grazing; water-source protection versus agriculture).

Considering that the process of developing land use plans has already begun in some villages[119], the project will pursue a staged approach, helping to support the finalisation and enforcement of Land Use Plans already in development, in addition to developing new plans (see Annex 2.1 Feasibility Study section 7.2.1 for details). The 2011 National Land Use Planning Commission (NLUPC)'s guidelines describe six main steps to follow when developing VLUPs. As well as following these guidelines, the village and land use plans advanced or developed under this Activity will take into consideration the resource needs of camps which extend outside of camp boundaries. The project will also aim to develop benefit sharing schemes, to negotiate the complexities of serving camp needs from village forestry and afforestation. The first five steps of the NULPC's guidelines, as listed below.

1. Establishing a District-level Participatory Land-Use Management (PLUM) team
2. Carrying out a participatory rural appraisal (PRA)
3. A Village Land Use Management (VLUM) committee is set up
4. Formalising the Village Land Use Plan
5. Identification of appropriate land management measures

The sixth and last step of the VLUP process is the issuance of a Certificate of Customary Right of Occupancy (CCRO), which is the long-term, legal outcome of the VLUP — with the process typically taking between four and six months. The CCRO specifies rights conferred to a land occupier and user following tribal customs and traditions on land. Under the customary land tenure system, land belongs to the whole tribe, clan and family, while tribal leaders are the custodians on behalf of the members. The Village Land Act (1999) confers these custodial powers to Village Councils and Village Assemblies in registered villages. Once the VLUP has been prepared and adopted by the Village Assembly and approved by District Council, the issuance of the CCRO is a formal issuance, albeit with some cost implications to be covered by the project (for example formal mapping of boundaries and depositing of information with registrar of Lands). The project will support this relatively small incremental cost to promote project ownership, engender support and lead to sustainability of project outputs and outcomes. CCROs for communal lands will be allocated at village level to provide long-term security for the villagers and may be used as collateral if borrowing money for village infrastructure.

[119] Ongoing government processes to develop land use plans are outside the scope of this funding proposal (e.g. not supported by project financing). Additional support will be provided through the project to processes that have started but which have not been finalized. The information collected by VPO from District Officials in November 2021 indicates 5 villages out of 9 in the surroundings of Nduta camp have ongoing VLUPs, 1 village out of 6 in the surroundings of Nyarugusu camp have ongoing VLUPs and no villages in the surroundings of Mtendeli former camp have VLUPs.

Depending on the level and extent of the LUPs already in development, funding allocated should allow for the issuance of 20-25 CCROs by project end. For further information please see Section 7.2.1 of the Feasibility Study. Following the steps on land use planning, bylaws and governance arrangements regarding land use would be formulated by the communities, including guidance on land and water management, including fire risk management in the face of climate change.

Access of Sukuma people, agro-pastoralists, and other groups, to grazing land, will be given full consideration in the land use planning process of the project. Their participation in the land use planning process activities will be an important part of the process. Certificates of Customary Right of Occupancy will be used to secure communal land rights, including that for agro-pastoral use and grazing. Through the risk management framework, the project will routinely monitor any likely tension between refugees, host communities and other beneficiary groups.

<u>Sub-activity 1.1.2. Build capacity of local and regional government officials to support ecosystem-based adaptation and landscape restoration</u>

Capacity building will be provided to support district officials and Tanzania Forest Service officers in including climate impacts and risks for the region into the land-use planning processes, enabling PLUM teams (Participatory Land Use Management — see Implementation Arrangements) to support communities to embed these impacts in the allocation of land to different purposes. This will ensure that different land uses are mutually supportive. This capacity building will be provided by local consultants over a 2.5 years' period (Years 1, 2 and 3) and supported by the provision of hardware and materials to district officials, including *inter alia* GIS software and tools, crested paper for CCRO printing etc.

Since planning for climate resilience is a key project objective, ecosystem-based adaptation principles should be integrated into the land-use planning process. A risk-informed planning process considers future climate impacts and expected impacts on the agro-ecological zones and on the communities. District PLUM teams will require capacity building in advance, in order to facilitate the process with communities, which will include a community risk assessment process, the identification of hazards and their mapping on village land, the identification of specific siting of adaptation measures for each village (i.e. the activities suggested in this project) and intervention designs to account for climate change, for example for Activity 2.3 on flood and erosion control and Activity 3.2 on water harvesting and irrigation. The training material and the detailed nature of the technical assistance should be defined in the inception phase of the project.

<u>Sub-activity 1.1.3. Improve public awareness of climate change by sharing information on climate change risks in the land use planning meetings</u>
The project will also use the opportunity of the community organisation during the VLUP consultation process to conduct training and awareness raising on climate change and good environmental management practices. Led by the District Governments, topics covered in this training would include describing the projected climate changes, impact such as wildfire risks, adaptation strategies and the importance of local ecosystems and the services they provide. This will support the land-use planning process, by providing a greater understanding of the projected changes to the landscapes in the project area, which would also help with compliance and enforcement. By improving their awareness of climate change, this training will also help to support the capacity of the host communities to generally build their resilience, in addition to supporting the uptake and buy-in of the forestry, agricultural and livelihood activities under Outputs 2 and 3. Training and public awareness raising will also target Sukuma agro-pastoralist communities that contribute to deforestation and ecosystem degradation in the project area, but are not currently included in wider community conservation or land use planning processes. The refugee committees, who will be involved in the participatory process for Village Land Use Planning in the villages immediately surrounding the camps, will also benefit from the training and awareness raising on climate change and good environmental management practices under this sub-activity. Training would be conducted by project staff and district government, in coordination with the land-use planner and consultants.

<u>Sub-activity 1.1.4. Support District, Ward and Village Level Environmental Management Committees (EMCs)</u>
In order to overcome the challenges and sustainability issues of previous/ongoing VLUP processes, the project will strengthen the District-, Ward- and Village-level EMCs and improve coordination between the different levels. This will be achieved through the development of coordination protocols and structures, regular reporting and monitoring systems, and providing upskilling to EMC members at all levels. Training on conflict resolution and benefit sharing will also be provided to EMC members at all levels to ensure that the project contributes to reduced potential for conflict between natural resource users. This will capacitate EMC members to facilitate discussions when conflicts amongst beneficiaries have arisen, ensuring that suitable resolutions are identified and agreed on by all parties.
Furthermore, benefit sharing agreements regulating the use of, and access to, shared natural resources will be formalised in the protocols and structures developed under this sub-activity, which will be undertaken through

consultation with all stakeholders and validated during engagement workshops. These agreements may involve host community only, or host communities and refugees in villages bordering the camps, considering the resource needs of the camps extend outside the camp boundaries (ex. Fuelwood collection). The objective is to facilitate sustainable offtake, promote ecosystem protection, and minimise conflict over resource use. EMCs will have ownership of these agreements, and representatives of EMCs at all levels (including district, ward and village level) will facilitate their development in collaboration with appropriate stakeholders (further information is provided in Annex 6). The establishment of such agreements is supported by the Tanzanian Forest Service (TFS). In addition, the effectiveness of benefit-sharing agreements will be ensured by establishing strong linkages between EMCs and Water User Associations (WUA). Conflicts that cannot be resolved by EMCs will be addressed under the project's grievance redress mechanism (presented in Annex 6: Environmental and Social Management System). Combined with building capacity of local and regional government officials (Sub-activity 1.1.2) and awareness-raising (Sub-activity 1.1.3) these interventions will help to ensure the sustainability of VLUPs, whilst supporting buy-in for their guiding the implementation of Outputs 2 and 3.

Output 2: Degraded ecosystems restored to support climate change adaptation in host and refugee communities

The GCF funding request for this Output is $9,978,657. UNHCR co-financing of USD $3,060,086 will go toward technical staff support and activities in forestry and flood reduction infrastructure in the camps and with selected activities, such as tree planting, also being implemented in the host communities.

All of the host communities and refugees in the project area live within or near forests. Both host communities and refugees use timber and forest products construction of shelters and other uses as well as fuelwood. Efforts to manage deforestation and unsustainable offtake of fuelwood — such as patrol services from the Tanzania Forest Services (TFS) and community groups — are limited as a result of underfunding. The result of pressure on the forest cover by both host communities and refugees is a degraded landscape with low soil stability, increased soil erosion and soil loss, reduced soil-nutrient availability, and reduced soil moisture. This makes the landscape susceptible to climate change impacts, less able to retain water resources required for these livelihoods and making the landscape more susceptible to flooding. This in turn reduces its capacity to support agricultural and other livelihood activities.

The land-use and forestry interventions will restore critical ecosystems to meet climate adaptation needs and improve ecosystem services. The outcome of the C-LUP process in Output 1 will determine specific areas where afforestation, natural forest regeneration and agro-forestry interventions will be carried out. The starting point for all project interventions will be the development of climate-resilient land-use plans to support EbA in the previous output. This is a fundamental step in determining how the proposed interventions will be planned and delivered. Major steps of the C-LUP process will include the identification of specific siting of adaptation measures for each village and intervention designs to account for climate change — for example, for Activities 2.3 on flood and erosion control and 3.2 on water harvesting and irrigation. Additionally, as the process identifies village forest areas, it is also an entry point for the preparation of community-based forest management plans and for agroforestry interventions (see Output 2). The land-use plans that are generated will then become the basis for the delivery of project activities (as described in Outputs 2 and 3). For example, in consultation with TFS and District Councils that are key stakeholders in the planning process, villages that have identified a village forest will be supported to implement Community Based Forestry Management (see Activity 2.1).

In addition to the climate adaptation objectives, the activity has significant social co-benefits. Adequate management of resources, opportunities for joint planning and benefit sharing schemes particularly for fuelwood, increasing available fuelwood resources, and protection of forest reserves have the opportunity of reducing conflict over resource use between host communities and refugees. Significant gender co-benefits are also expected. Provision of fuelwood closer to the camps (for example, within a 4 km radius) reduces risks of gender-based violence as incidents are more common the further refugee women walk from the camps[120].

The land use planning in host communities and forestry related interventions are expected to lead to better managed natural resources which will substantially reduce women's work burden as less time will be spend collecting fuelwood. Moreover, and particularly for refugee women, the exposure to threat and GBV or other forms of violence will be reduced as they will no longer be compelled to move far from the camp boundaries in search of fuelwood. Finally, host community and refugee relations will improve as existing and expected future conflicts over fuelwood and deforestation will be mitigated.

[120] Refer to Annex 8. Gender Assessment and Action Plan for further discussion and Section G.2 below.

This Output includes the activities described below.

Activity 2.1. Implement Community Based Forest Management (CBFM) for resilient ecosystems

This activity will be conducted in village and national forests surrounding the three camps. District Councils, in collaboration with Tanzania Forest Service (TFS) will lead the implementation of this activity with support from the Technical Partner and local NGOs with experience in facilitating the establishment of VLFR (for example REDESO or JGI) and the Technical Partner.

The principal objective of this intervention is to implement CBFM in forest areas such as Miombo woodlands anchored to host communities. Miombo is acknowledged for its resilience in the face of disturbances such as tree cutting for wood fuel, a characteristic that boosts the ecosystem's potential to supply wood fuel over long periods, so long as the area is not repeatedly subject to fire or grazing pressure. Miombo woodland has the potential for a long-term supply of wood, when managed properly[121][122]. The extent of degradation in the area, as well as decreasing returns from woodland resources, demonstrate that common-pool resources (forested land) are not being efficiently managed. The underlying reasons for this are predominantly institutional and are outlined below. Market failures around the ineffective management of common-pool resources will be addressed, in conjunction with Activity 1.1, through the implementation of CBFM.

The concept of CBFM is already a central strategy of Tanzania's Forest Policy (1998), Forest Act (2002) and National Forest Programme (2001) and has been successfully rolled out across the country — although very limited in the Kibondo, Kakonko and Kasulu Districts of Kigoma Region. Under CBFM, villagers take full ownership and management responsibility for an area of forest as a Village Forest Reserve. Following this legal transfer of rights and responsibilities from central to village government, villagers may harvest timber and forest products, collect and retain forest royalties, and undertake patrols (including arresting and fining offenders). They are exempt from local government taxes on forest products and from regulations regarding 'reserved tree' species, and they are not obliged to remit any royalties to government.

Experience of CBFM has demonstrated effectiveness in improving ecosystem functions, as well as addressing poverty by enhancing livelihood conditions and providing alternative income to rural communities in Tanzania[123]. By integrating specific climate resilience objectives in CBFM — especially for threatened forest ecosystems, as in the case of the target regions — this approach can have a considerable climate adaptation potential by strengthening the adaptive capacity of forest-dependent communities. Community management is also preferential to central 'top-down' oversight for its advantages in reducing transaction costs, promoting local 'ownership' and responsibility for natural resources and promoting community cohesion and social capital. CBFM will replace conventional methods of protecting forest by government Forest Departments using Forest Guards to patrol and arrest local people found within forests. This conventional method alone has not been successful — Forest Departments are seldom sufficiently well-resourced to undertake the work and supervision is lacking. Figure 14 in page 45 shows the extent of forest loss in around Mtendeli former camp.

Village Land Forest Reserves (VLFR) are designated sites declared under sections 32 and 23 or gazetted under section 35 of the Tanzania Forest Act of 2002. According to this Act, the village council declares VLFRs based on the existing Village Land Use Plan. These forests are managed by the village forest management committees and are under general supervision of the village council. The village council may launch application to the National Director of the Forests and request gazettement of the forest. Upon clear commitment of the Village Council on the management of the forest, the Director in the Public note, declares the reserve to be a gazetted Village Land Forest Reserve. The management of the gazetted reserve is guided using the set of by-laws approved by the Director and implemented by the village forest management committees.

Most of the VLFRs in Kibondo, Kasulu and Kakonko District are still in the stage of declaration by the village council. Although public gazettement of the forest by the Director upgrades the forest to a high-level conservation status, less than 10% of the VLFRs have this status. The cost implication and difficulties in slowness of the approval process has previously discouraged the option. Therefore, sustainability of the VLFRs will be ensured by the project commitment to

121 Malimbwi RE. 2000. An Inventory Report of the Principal Timber Resources of the Miombo and Riverine Woodlands and Forests of Rufiji District.

122 Refer to Section 7.3 of Annex 2. Feasibility Study for further description of the role of Miombo woodlands in the landscape, and the project approach to provide alternative fuelwood sources and protection of natural forests.

123 Nzunda EF Luoga EJ & Mahuve TG. 2011. Community-based forest management in Tanzania: Strengths, weaknesses, opportunities and threats. Sokone University of Agriculture.

support the full chain VLFRs gazettement, strengthening community institutional monitoring systems, and promoting conservation benefits to the users.

This activity aims to bring 30,000 ha of land under CBFM, which is ~70% of the forested area within the project boundaries. This will provide benefits to most of the population of the project area covered by the land use planning in Output 1, through avoided degradation and associated ecosystem service provision. The major ecosystem services that could be promoted through Output 1 include *inter alia* soil stabilisation and reduced erosion, increased soil moisture and fertility, increased water infiltration leading to improved groundwater recharge, fuelwood collection and food provision. A co-benefit of the VLFRs is reduced exposure of local communities to forest areas that may contain high densities of tsetse flies, thereby minimising cases of sleeping sickness.

The specific villages and land area to be included in the CBFM system (as well as for the afforestation and agroforestry under Activity 2.1) will be chosen through the land use planning process under Output 1, although these will be chosen with the specific criteria in mind and in collaboration with the TFS, including:

- village forest land availability;
- level of degradation of village forest land
- whether the area falls within the sub-catchment of a camp;
- priority for those that are adjacent to, or within the 4 km ‘buffer zone’ around the camps (where refugees are permitted to travel within to collect firewood);
- the existence of provisional processes or interest for establishing CCRO or an existing forest reserve with management arrangements;
- whether there is strong potential for forest protection (forests are not severely and irrecoverably degraded); and
- if there is potential for collaborative benefit sharing schemes between refugees and host communities, where appropriate
- if bylaws or community management arrangements are in place
- consultations with stakeholders including agropastoralists, village leaders, and District Councils.

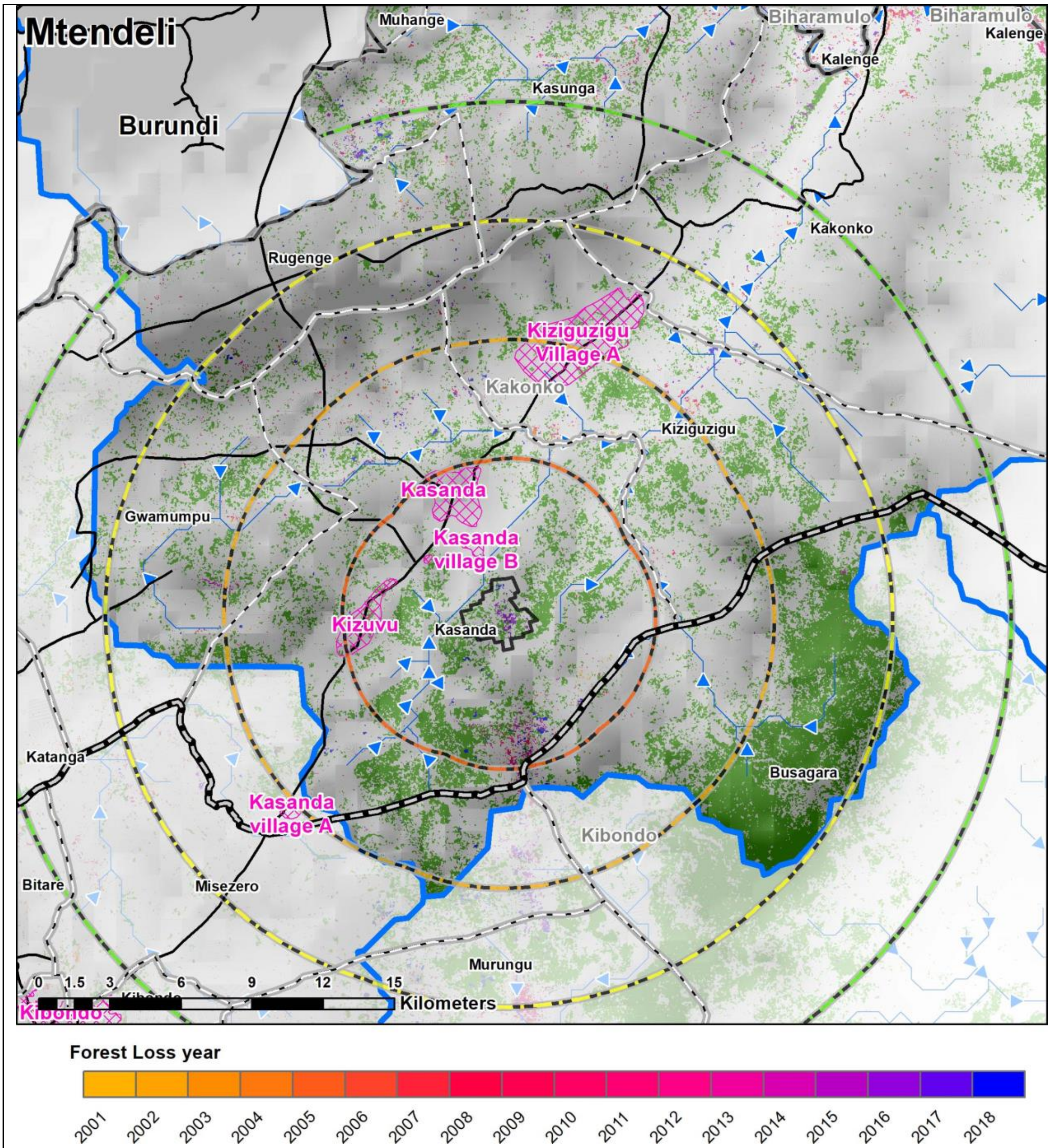


Figure 14. Forest loss in Mtendeli former camp[124]. The pink cross hatch represents the extents of local villages.

The underlying policy goal for CBFM is to progressively bring unprotected woodlands and forests under village management. Descriptions of the adaptation potential for forest regeneration and management through CBFM are given in Table 3.

Table 3. Adaptation potential for forest regeneration and management through CBFM.

Characteristics	Description
Geographic area for implementation	Village forests
Climate hazards and impacts addressed	• Increased damage to crops and infrastructure from more severe wildfires, more frequent and intense floods and prolonged droughts.

[124] Refer to Annex 2. Feasibility Study Section 2.2.6 for analysis of land cover change across the project sites.

	• Increased soil erosion from more frequent and intense floods and prolonged droughts, leading to the loss of soil nutrients and moisture. • The above impacts result in lower agricultural yields, leading to reduced income generation potential and diminishing food security • Potential impacts of shifts in temperature and precipitation to the range of Miombo woodlands[125],[126].	
Impacts addressed and adaptation potential[127]	Increased damage to crops and infrastructure from wildfires	Community forest management assists in preventing illegal ignitions, reducing the occurrence of severe wildfires.
	Increased damage to crops and infrastructure as well as soil erosion from floods	Healthy forest ecosystems increase infiltration and decrease surface water run-off, therefore reducing floods and soil erosion.
	Increased damage to crops and soil erosion from droughts	Through increased water infiltration, forests regulate hydrological processes, increasing groundwater recharge and reducing the impact of droughts.
	Soil nutrient loss	Healthy forests enhance soil retention and increase soil microflora and fauna, increasing soil fertility
	Soil moisture loss	Reduced runoff and the increase in soil fertility by healthy forests increases soil moisture
Additional benefits that increase resilience[128]	**The resilience of local communities —** by providing secure access to forest resources and to additional streams of income, CBFM strengthens livelihoods and increases food security, leaving them less vulnerable to climate change. The forests also perform a vital safety-net function by providing a range of non-timber forest products (NTFPs) that can be consumed directly or sold in local markets. NTFPs can increase the diversity of household income and nutrition. Forests, therefore, can help bridge the climate, humanitarian and development divide and build a resilient livelihood base, even in the face of climate change. **The social capital of local communities —** because community forestry is based on flexible and participatory management, it establishes decision-making structures that enable rapid and effective responses to change, climate or otherwise. These processes mobilise local knowledge and skills, allowing communities to connect to wider conservation efforts and to join in national planning and policies.	

Sub-activity 2.1.1. Establish and manage 30,000 ha of Village Land Forest Reserves

An established process towards CBFM exists and is described in the national CBFM Guidelines. As described in the national CBFM Guidelines, the process towards CBFM comprises six main stages, as described below.

1. Assessing the existing resource/deciding what is required to bring it under effective protection and management.
2. Preparing a provisional Management Plan — which is discussed and shared with the refugee communities and approved by the Village Council and Assembly.
3. Implementing the Plan: electing a Management Committee; appointing patrol teams; agreeing and demarcating perimeter boundary; zoning the forest for protection and use (e.g. seasonal grazing; beekeeping, grass cutting); keeping essential records.
4. Reviewing and revising the plan after at least one full year of implementation.
5. Legalising the Plan — agreeing and formalising rules, and enacting village by-laws to enforce them. Under this project, the legalisation of village forest land is planned under Output
6. 1, and this include formalising the by-laws under the CCRO
7. Declaring and registering the Village Land Forest Reserve and its by-laws with the District Council.

[125] Pienaar B, Thompson DI, Erasmus BFN, Hill TR & Witkowski ETF. 2015. Evidence for climate-induced range shift in Brachystegia (Miombo) woodland. South African Journal of Science. Volume 111 (Number 7/8):1–9.
[126] For more information, see Section 7.3.1 of Annex 2. Feasibility Study.
[127] RECOFTC, 2015. The role of Community Forestry in Climate Adaptation in the ASEAN region.
[128] Ibid.

District Councils of Kibondo and Kakonko Districts are expected to lead the implementation of this activity, in collaboration with Tanzania Forest Service and the technical advice from the Technical Partner [129]. In Kibondo and Kakonko Districts REDESO has been facilitating development of 16 community-based forest management plans for the village forests reserves. The project will support this established CBFM process by:

- training facilitators (District Council staff / TFS staff / extension staff); and
- identifying appropriate NGO support organisations as required[130];
- supporting field operations to build capacity of local Forest Management Committees to prepare implement and monitor land and resource management plans, including fire risk management plans. This support will be provided by the trained Facilitators. The project will support their training and field costs (travel and DSA).

Sub-activity 2.1.2. Provide equipment and training for forest monitoring in national forest reserves for 10 forestry and natural resources officers.

Two national reserves in close proximity to Nyarugusu in the Kasulu District — including North Makere (75,325 ha) and South Makere (59,500 ha) — have been greatly impacted by deforestation. Other forest reserves of concern include Buyungu Forest Reserve (74,430 ha) situated close to the Mtendeli former camp in Kakonko and the Mwale Forest Reserve (10,386 ha) in Kibondo. The Tanzania Forest Service Agency (TFS) has the mandate to protect the national forest reserves, as well as the forested areas on general land. The vastness of the reserve areas and the limited manpower and mobility constrain protection efforts.

Host communities and district officials confirmed there is substantial grazing, fuelwood collection, and agriculture activity ongoing in the national reserve areas, where soils are fertile. Fires are also used to clear and regenerate land, despite the increased risks especially under climate change. Land is cleared and cultivated for a few years, after which the soil fertility diminishes, and the land is abandoned. In Kalimungoma, respondents also mentioned deforestation liked to the lime industry, where lime stones are illegally excavated and fuelwood is required to produce lime powder, which is subsequently sold to the lime industry.

The sub-activity will empower TFS by increasing forest monitoring extent and improving the mobility of patrol teams to enhance the protection of the Makere, South Makere, Mwale and Buyungu National Forest Reserves. This will be achieved by developing forest management plans for the national forest reserves within the project area as well as setting clear boundaries for the conservation and monitoring of target areas. The existing monitoring systems of TFS will be strengthened with the development of a GIS-based forest monitoring data management system that integrates satellite imagery with high-resolution drone imagery and field data in a GIS platform. TFS is progressively incorporating the use of GIS for fire monitoring with support of a JICA project and is willing to expand the use of GIS to monitor forest cover and fire risk management in Kigoma region. The integration of drone surveillance is considered particularly relevant in the transboundary context of Kigoma to assist in the control of illegal deforestation activities in the National Forest Reserves (~219,642 ha) in the target districts. The monitoring system will build on the experience of Jane Goddal Institute (JDI) in the use of ArcGIS to monitor the impact of VLUPs on the Village Forest Reserves in the area of the Gombe National Park and the Community Forest Monitoring Dashboard launched in Kigoma in January 2016 by the JGI, Blue Raster, local and regional government officials and other partners.

To support these activities, TFS staff will be provided with equipment to maximise the coverage of monitoring in a cost-effective way, including monitoring of the National Forest Reserves (NFR) and Village Forest Reserves. Such equipment will include: i) drone hardware appropriate for the target NFR surface area and suitable for high altitude flight, as well as drone software and the required licenses; ii) smartphones/tablets that contain Open Data Kit (ODK) and GPS units for the capturing of geo-location data; iii) IT equipment and GIS software, iv) fire management gear and camping equipment to be used in the field by patrol teams and v) electronic calibres to monitor tree growth.

Training will also be provided to these staff on inter alia the efficient and effective use of drones for landscape monitoring, data collection techniques such as the use of transects to collect data and the use of the GIS -based monitoring system. In addition, training will be provided to TFS staff on conflict resolution and community engagement, to ensure that in situ conflict with and between land and resource users (both project beneficiaries and the broader population) is resolved effectively. This is in line with the TFS's mandate on enforcement of laws and regulations in areas under its jurisdiction, and its support for the design and implementation of appropriate benefit-sharing agreements under Sub-activity 1.1.4. Should TFS staff not be able to adequately resolve conflicts, the affected parties will be referred to the grievance redress mechanism established under the project (refer to Annex 6: Environmental and Social Management System for more

[129] The engagement of District Councils and other implementing partners will be subject to a due diligence process, also as described below in Section B.4.

[130] The identification of NGOs is subject to criteria and processes outlined in Section B.4.

detail). Together, interventions under this sub-activity will strengthen TFS staff's capacity for climate-responsive planning, monitoring and the protection of 219,642 ha of national forest reserves.This, in turn, will contribute to reducing the risk of fires in the project area, as well as preventing agricultural encroachment and land degradation in Tanzania's forest reserves. Through the Village Land Use Planning process under Activity 1.1.1, boundaries of national forest reserves will be demarcated, and land use around them will be agreed upon to be consistent with conservation objectives. The land use planning process and the support to Environmental Management Committees under Activity 1.1.4 will contribute to education, awareness raising, and effective communication of land use regulations and good environmental practices to local communities and agropastoralists.

Activity 2.2. Implement 12,000 hectares of agroforestry and village land afforestation to increase resilience of land use

This activity will be implemented in and around host community villages surrounding Mtendeli former camp area, Nduta camp and Nyarugusu camp as per the VLUPs. District Councils, in collaboration with Tanzania Forest Service (TFS), will lead the implementation of this activity with support from the Technical Partner and local NGOs/CBOs for community facilitation.

The principal objective of this intervention is to increase the number of trees across the agricultural (non-forest) landscape through agroforestry and village land afforestation. At landscape scale, agroforestry improves crop productivity and provides important ecosystem services including watershed protection and hydrological regulation, as well as biodiversity conservation and carbon storage. Most farms in the region are small-scale and rain-fed. Although some agroforestry systems — particularly agri-silviculture and silvopasture — have been adopted in the wider region, the benefits of agro-forestry do not appear to be well known to farmers in the target districts and uptake is presently low[131]. The training provided under this activity, and under the climate-resilient agriculture interventions in Activity 3.1, will help to address the presently low uptake and will promote ownership, sustainability and scalability of this intervention, by demonstrating the ecosystem and agricultural benefits of this approach. Complementary to this, village land afforestation can restore ecosystem services and establish a sustainable supply of fuelwood, taking the pressure off of native forest systems.

In providing interventions on the supply of fuelwood this project will complement others in the region focused on energy demand, including those that promote alternative energy sources (see Section B.1 Investments in the Target Project Area). By doing so the proposed project will address the degradation of slow-growing Miombo woodlands, which are being cleared for fuelwood. Through this intervention existing projects with a focus on CBFM and afforestation will be scaled up. For example, regeneration of natural forests through CBFM in Kibondo and Kakonko districts has been occurring under the auspices of REDESO and UNHCR since 2016 via the Village Natural Forest Reserves (VNFR) project, as well as under the SNV Netherlands Development Organisation project entitled Renewable Energy Service and Products as an Opportunity in National and Displaced (RESPOND) Markets of Kigoma Region. Before the introduction of these projects there was a large dependency on wood collected from forests as a primary fuel source for residents of both refugee camps and host communities, resulting in extensive deforestation of surrounding areas. Consequently, these projects intervened to enable villages and refugee camps to access alternative fuel sources as a means of conserving surrounding natural forests. Under the REDESO project, woodlots have been created in eight villages (four per District) with a total of 496,522 and 504,025 seedlings planted across 447 and 454 ha in Kibondo and Kakonko districts, respectively. The proposed GCF project will be able to build on the work already done in the project area to restore forest cover and reverse the drivers of a degraded landscape with low soil stability, reduced soil nutrient availability, and reduced soil moisture. This will therefore reduce the susceptibility of the landscape to the impacts of climate change and improve its ability to support agricultural and other livelihood activities, retain water resources required for these livelihoods and reduce susceptibility to flooding.

As with Activity 2.1, this activity will correct a market failure associated with the unsustainable management of common-pool resources. The direct beneficiaries of this activity will include those engaged in the growing, planting and maintenance of agroforestry across 12,000 ha (~35% of the total farmed area within project boundaries), some of whom may additionally benefit from other activities, including 2.3, 3.1, 3.2 and 3.3. Additionally, these agroforestry systems will provide ecosystem benefits to the broader population of beneficiaries, predominantly in the host communities. These ecosystem benefits include increased water infiltration and reduced runoff, consequently resulting in reduced risk from flooding and erosion as well as increased soil stabilisation and fertility. The final species chosen for planting will be determined by a Species Suitability Assessment, which will be informed by fire risk assessment, among other factors[132].

131 Findings from the Feasibility Study

132 Options for species selection are in Annex 2. Feasibility Study and informed by current SNV projects in Kigoma.

Sub-activity 2.2.1. Establish and maintain 10,000ha agroforestry systems to restore ecosystems that are critical to meet climate adaptation needs; and set up nurseries for production of tree seedlings

Tanzania is home to several traditional agroforestry systems that have been used for hundreds of years. Some have been documented: the Chagga home-gardens, the related Mara region home-gardens known as Obohochere and the traditional Wasukuma silvopastoral system called Ngitili. One outstanding aspect of these traditional methods is the use of multi-layered systems with a mixture of annual and perennial plants, which imitate natural ecosystems. In the target area, agroforestry does not seem to be widely practised and few examples of on-farm tree planting were observed, mainly involving banana trees, some small-scale fuelwood plantations on farms, and natural Elephant grass (magugu) planted in the areas around the camps. Interviews with farmers consulted for this study showed a lack of awareness of the benefits of planting fodder trees, and an underappreciation of their value in supporting animal husbandry.

Spatial land use analysis indicates ~28,000 ha of agricultural/crop land in the project area. Project ambition is to reach 35% of this area at a density of 500 trees/ha (see Appendix B for further detail on assumptions) for a total of 10,000 ha. This requires 2,500 ha/year in years 2–5 inclusive, although this planting may be most intensive in years four and five given the time taken to familiarise with methods and benefits of agro-forestry. Small-scale community nurseries, growing ~50,000 seedlings per year, are already a feature in the project area, and it is proposed that equivalent nurseries are established in 15–20 sites in project villages and in cooperation with the TFS in order to provide the number of seedlings required.

The deliberate integration of trees and shrubs in crop systems offer a means of increasing climate and livelihood resilience, as described in Table 4 below.

Table 4. The increased climate and livelihood resilience from integrating trees and shrubs in crop systems.

Characteristics	Description	
Geographic area for implementation	In villages, on-farm and off-farm	
Climate hazards and impacts addressed	• Increased damage to crops from more severe wildfires, more frequent and intense floods and prolonged droughts. • Increased soil erosion from more frequent and intense floods and prolonged droughts, leading to the loss of soil nutrients and moisture. • Heat stress leads to lowered crop productivity or crop death. • The above impacts result in lower agricultural yields, leading to reduced income generation potential and food security.	
Impacts addressed and adaptation potential[133]	Increased damage to crops and infrastructure from more intense rainfall and floods	Contour planting of trees will decrease exposure to intense rainfall and reduce run-off, thereby protecting crops.
	Increased damage to crops and soil erosion from droughts	Through increased water infiltration, forests regulate hydrological processes, increasing groundwater recharge and reducing the impact of droughts.
	Increased heat stress	Shading and increased soil moisture from trees reduces heat stress for crops.
	Soil moisture loss	Increase the retention of soil moisture and increase fertility of soil, reduces erosion.
	Soil nutrient loss	Keep the soil in place and stimulate soil microflora and fauna (i.e. encourage biodiversity) and increase fertility.
Additional benefits that increase resilience[134]	A co-benefit is the production of timber and fuel, thus providing an alternative means of renewable energy and reduces economic vulnerability by diversifying production as it acts as a security against total crop failure.	

Agroforestry activities will be facilitated by local government agriculture and forestry officials and sub-district extension and TFS staff, supported as required by contracted local NGOs/CBOs[135]. The Technical Partner will provide training

[133] Lasco et al. 2014. Climate risk adaptation by smallholder farmers: the roles of trees and agroforestry. Current Opinion on Environmental Sustainability, 6:83–88.
[134] Ibid.
[135] VPO as the executing entity will contract such organizations as implementing partners as described in Section B.4.

and technical advisory services to the local government and forestry officials in the technical design, implementation and monitoring of agroforestry systems. Small-scale community nurseries, growing ~50,000 seedlings per year, are already a feature in the project area, and it is proposed that equivalent nurseries are established in targeted villages to provide the number of seedlings required. These are manageable by a small number of part-time workers. Seed, basic equipment (watering cans, pots) and technical support is initially provided through local NGOs or CBOs. The nursery area should be enclosed with a stock-proof fence. Each nursery will be supported by in-stream irrigation interventions (under Activity 3.2) or other sustainable sources to provide sufficient water for the seed production.

<u>Sub-activity 2.2.2. Undertake afforestation on 2,000 ha of village land to a) restore critical ecosystem services to meet climate adaptation needs and improve ecosystem services such as flood control, b) establish a sustainable supply of fuelwood which will take some of the pressure off the native miombo forest and allow it to regenerate so that it can meet its ecological and livelihood support functions.</u>

Pressure upon natural forest systems is predominantly driven by the local demand for fuelwood, as the main source of energy for cooking in both the refugee and host communities. In the Kigoma region, 98% of households rely on fuelwood for cooking, with 81% using firewood and the balance using charcoal[136]. Within the refugee camps, the average daily consumption of fuelwood is 1.8 kg per person[137]. Only ~8% of the local fuelwood demand can be sustainably harvested from the Miombo woodland within 10 km of the refugee camps[138]. Over the past two decades a number of studies have been commissioned which examined the potential and feasibility of alternative energy sources to fuelwood for domestic cooking in the refugee camps and surrounding host communities. However, infrastructure for these technologies is still currently poorly developed, despite demand-side projects operating in the region[139]. Combined with local practises and a likely continued preference for fuelwood, these factors mean that pressure on the forests is anticipated to remain high for the near future.

Table 5 below describes characteristics of village land afforestation activity.

Table 5. Characteristics of village land afforestation activity.

Characteristics	Description
Geographic area for implementation	Village land
Climate hazards and impacts addressed	• Deforestation from activities such as fuelwood collection increase the vulnerability of surrounding ecosystems to climate hazards. • Increased damage to crops and infrastructure from more frequent and intense floods and prolonged droughts. Deforestation reduces the ability of forests to protect communities from climate hazards. • Anthropogenic deforestation exposes areas to increased soil erosion from more frequent and intense floods and prolonged droughts, leading to the loss of soil nutrients and moisture.

136 Kigoma Region Basic Demographic and Socio-Economic Profile 2016
137 Quigley P. 2016. Providing energy for cooking: an analysis of fuel options in refugee camps in Tanzania. Geneva, Switzerland, UNHCR.
138 See Appendix B of Annex 2. Feasibility Study.
139 The first energy strategy for the refugee hosting areas in Kigoma was published in 1997, at which time fuelwood supplies were presented as the main option for meeting energy needs. The solutions to energy provision have evolved and changed over the intervening years. Current interventions by local and international agencies in Kigoma include large scale distributions of improved cookstoves and provision of energy efficiency training, it is also moving towards a significant increase in the use of liquid petroleum gas. Further details on demand-side energy interventions being undertaken in the region can be found below and in Section 6 of the Feasibility Study. International studies have shown that to achieve successful rural development, biomass fuels like firewood must first be replaced by a liquid or gaseous transition fuel like liquid petroleum gas (LPG). Once these are available and energy-use behaviours related to modern fuels are established, communities may ultimately make a final transition to electricity which can come from non-fossil fuel sources. This stepwise transition through the energy ladder has also been identified as the most achievable approach to improving energy access in Kigoma A number of actors in the region are introducing transition fuel programs, primarily improving access to LPG, to help refugees and host communities in Kigoma begin the move up the energy ladder, these include the Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ), the Technical University of Denmark (DTU) and UNHCR. As these fuel transition initiatives do depend on the medium-term use of fossil fuels, they are not expected to be supported by climate finance. However, these projects will work synergistically with this GCF project by working to meet the medium-term energy demand and drastically reducing the Kigoma region's reliance on firewood, thereby creating an opportunity for the forest and ecosystem restoration components of this project to succeed. For further discussion, see Annex 2. Feasibility Study, Section 7.3.2.

	• The above impacts result in lower agricultural yields, leading to reduced income generation potential and food security.	
Impacts addressed and adaptation potential	Increased deforestation	Establish and maintain village forests that address non-climate drivers of vulnerability, by alleviating dependence on miombo forests for fuelwood harvesting.
	Increased damage to crops from more intense rainfall and floods	Afforested areas and managed forests reduce run-off, preventing floods and soil erosion.
	Increased damage to crops and soil erosion from droughts	Healthy forests under reduced pressure for fuelwood regulate hydrological processes, increasing groundwater recharge and reducing the impact of droughts.
	Soil moisture and nutrient loss	By improving soil structure, afforested areas and non-degraded forests reduce soil erosion and prevent water and nutrient losses from runoff.
Additional benefits that increase resilience	**The resilience of local communities:** By providing access to fuelwood and strengthening their livelihoods with a reliable and sustainable source of energy.	

Species selection for afforestation in degraded land will be undertaken at the project inception during a rapid species suitability assessment led by the Technical Partner in coordination with TFS and the local government. The assessment will form part of the land-use planning process. This will ensure that a robust site-specific analysis is done that takes into account the ecology of the areas that have been identified for afforestation and economic value. Criteria to be used for species selection include:

- Species suitability to environmental conditions (water availability, soil types, etc.),
- Ecosystem services provision (e.g. rooting systems that anchor soil, prevent erosion, and facilitate infiltration, nitrogen-fixing species, etc.)
- Productivity and growth,
- Cost-effectiveness,
- availability of local expertise for growing and management,
- availability of seeds and plant materials
- multiple uses / benefits of selected species (e.g. fuelwood, fodder, medicinal use, etc.)
- market demand for forest products
- community acceptance (e.g. traditional and cultural use)

The Tanzania Forest Service (TFS)[140], working together with Village Councils, will be provided with training and technical advisory from the Technical Partner to lead the technical design, implementation and monitoring of this activity. The land will remain under the control of the Village Assembly, and indeed, under Tanzania's Forest Policy, such plantations can be declared CBFM forests with 100% of income arising from them reverting to the community. In the host communities, the afforestation labour force for the village woodlots will come from the village community members and be mobilised by village leaders. Afforestation will be undertaken on marginal land identified during the Village Land Use Planning process. To mitigate the increasing risk of wildfire and human-induced fires in the afforested area, TFS will conduct an assessment to identify the required firebreak areas during the design of the afforestation intervention which is scheduled in the first quarter of the second year of project implementation. TFS will establish the necessary firebreak areas through casual labour and provide fire management training for patrol teams and communities.

The annual planting programme will require the production of 1,250,000 seedlings/yr in each of Years 2 to 5. An additional 20% should be added to these figures to allow for nursery losses and failure to establish — i.e. 1,500,000 seedlings/yr. To minimise costs of seedling transportation, nurseries should be sited as close as possible to planting

[140] TFS and a Technical Partner will be engaged by Tanzania VPO as implementing partners subject to the processes described in Section B.4.

sites. As with the previous sub-activity, each nursery will be supported by irrigation interventions (under Activity 3.2) to provide sufficient water for the seed production.

Activity 2.2 will complement the CBFM interventions implemented under Activity 2.1, which have the objective of protecting natural forest systems and enhancing the value that local communities place on the ecosystems they provide. The combined effect of these interventions will be that the demand for fuelwood from slow-growing, indigenous Miombo ecosystems will be diverted to fast-growing plantations and that local communities will become more involved in decision-making processes regulating the use of natural resources and enhancing the climate-resilience of agriculture. The shift from activities which degrade ecosystems towards more sustainable behaviour will also be complemented by agricultural and livelihood activities under Output 3 which will improve income generation and savings and reduce the need to produce charcoal as a livelihood source.

The direct beneficiaries of this activity will include those engaged in the growing, planting, maintenance and offtake of the afforested woodlots across 2,000 ha (~1.5% of the total marginal land area within project boundaries), some of whom may additionally benefit from other activities, including 2.2, 3.1, 3.2 and 3.3. Additionally, these afforestation systems may provide some ecosystem benefits to the broader population of beneficiaries, to both refugees and the host communities. These ecosystem benefits include decreased runoff and increased infiltration, resulting in reduced flooding and erosion as well as increased soil stability and soil fertility.

Activity 2.3. Implement flood and erosion control in densely populated areas to reduce the exposed and flood-prone ferrosol-type soils which have little or no vegetation cover

This activity will be carried out within and in the vicinity of the refugee camps of Nyarugusu, Nduta and Mtendeli former camp area.

As the camps were established rapidly, within a context of humanitarian pressures, limited land availability and no viable alternatives, this led to sub-optimal camp land-use planning. The establishment (and subsequent present layout) of the camps reflects a process which could not adequately consider the combined impacts of increased pressure from development and the increased intensity of rainfall events resulting from climate change on soils. Land (bestowed by MoHA) was cleared quickly, with vegetation being removed to provide space for the camps. This has amplified flood events in the area, leading to gully erosion, soil loss and degradation in and around the camps, as evidenced in Figure 15 below. These are further amplified by the camps being densely populated, without opportunity to move away from flood risk areas and without space for run-off to drain naturally. The increased severity of flood events predicted under climate change are likely to result in losses for subsistence farmers, damages to homes and community facilities including schools, and injury and potentially loss of life. Unmanaged clearing of land in particularly sensitive areas has exacerbated the problem. This activity will reduce the extent to which ineffective land-use planning is resulting to flood risk in the camps and in surrounding host communities.

In the areas outside the camps, the land-use and forestry interventions described above will increase soil stability and reduce the level of runoff and erosion, addressing flood impacts downstream, which are anticipated to worsen as rainfall intensity increases. Additionally, the interventions listed below will also contribute to flood and erosion control in host communities.

- Soil improvements in agricultural areas (Activity 3.1)
- Sensitising, awareness raising and training of local communities on how to manage local/micro (on household scale) storm water problems (rainwater harvesting) (Activity 3.2)
- Storing water for use in dry periods (Activity 3.2)

Figure 15. Erosion gullies in Mtendeli former camp

However, in the camps, as a result of the higher population density, these types of interventions are not feasible. The principal objective of this intervention is to provide solutions in the areas most susceptible to erosion and floods from high water flows (Figure 16). The solutions themselves will be designed and use materials that are able to withstand changing weather extremes such as temperature and rainfall intensities based on outline technical designs in the feasibility study. These designs have taken into consideration flood-risk modelling under climate change conditions but would need to be calibrated to specific sites based on slope, local soils, and other site conditions. The package of grey and green flood and erosion measures will therefore be tailored to the characteristics of each camp and designed based on field technical assessment and stakeholder consultations carried out in the first year of implementation.

Most of the refugee beneficiaries for this activity would be the inhabitants of the refugee camps of Nduta and Nyarugusu, with some flood-mitigation and erosion control benefits experienced by host communities located closest to the camp boundaries or immediately downstream. An additional co-benefit, particularly for inhabitants within the camps, is a reduction in the amount of standing water following high intensity rainfall events and flooding. This will reduce the prevalence of water- and vector-borne diseases such as cholera and Rift Valley fever.

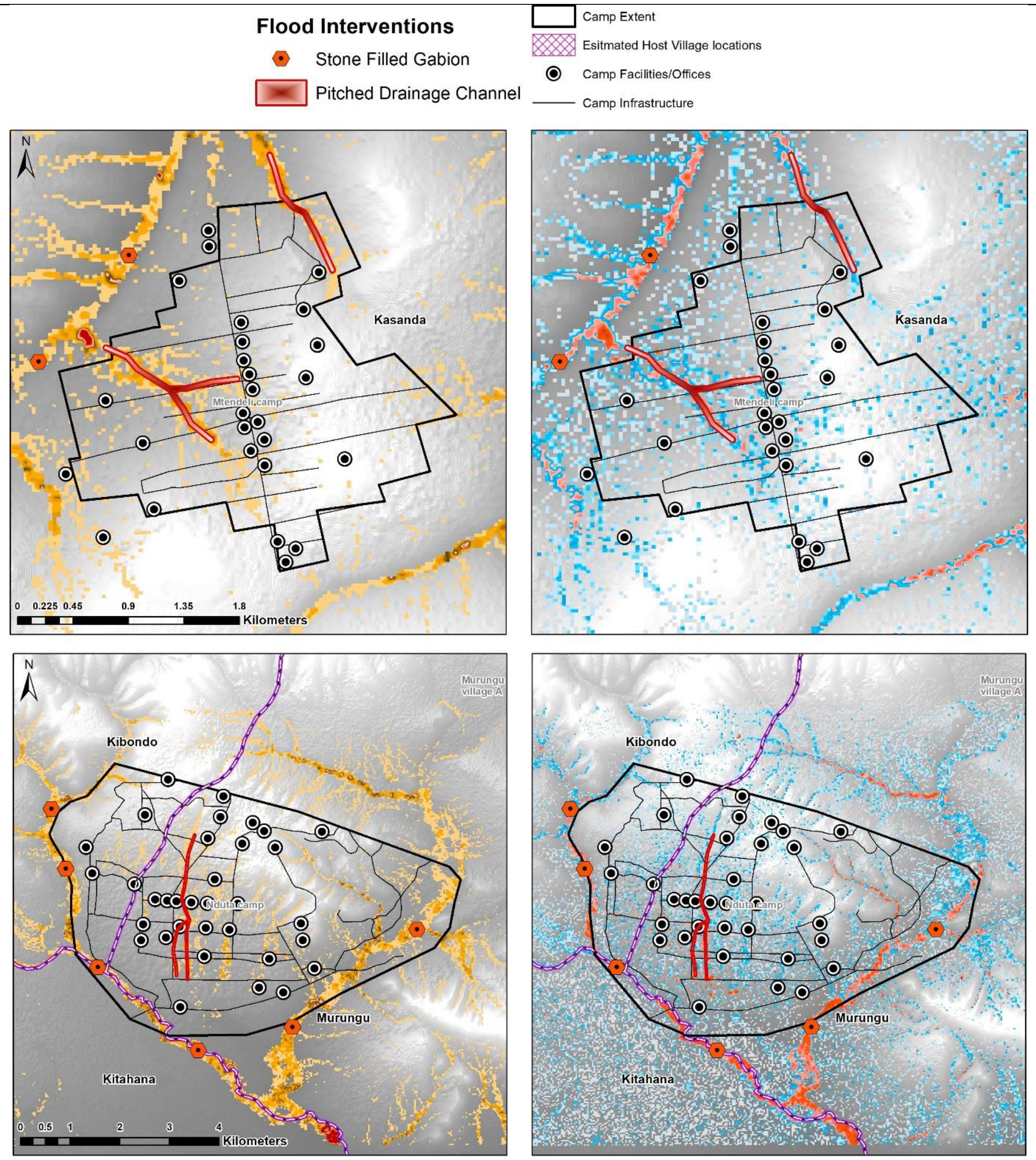
In Mtendeli former camp area, the flood and erosion control measures will be conducted after the camp closure in combination with other restoration interventions implemented by UNHCR. The decommissioning of the camp (closure process) will be undertaken by UNHCR and other partners operating in the camp in the scope of other programmes/other funding sources. GCF proceeds will not be used for Mtendeli camp decommissioning. The post-closure restoration will result in adaptation benefits for the host community in terms of i) reduced flood and erosion risk for downstream host communities, ii) potential for expansion of agricultural and agroforestry activities in the camp area and iii) enable District government plans to use the former camp infrastructure to improve access to education and health services. The beneficiaries of such services will be host community in the district, since all the refugees from Mtendeli camp were relocated to Nduta camp in Kobondo district where have access to basic services such as education and health.

As part of camp closure and consolidation, a taskforce, composed of MoHA at national and local level, District Commissioners of Kakonko, Kibondo and RC and RAS in Kigoma, UNHCR, DRC (camp management) and NRC (shelter &wash), has endorsed the following activities under camp clean up and environmental rehabilitation by relevant partners.

1. Demolition of refugee houses/infrastructure after the camp consolidation and recycle the materials
2. Decommission of latrines and bathing facilities
3. Tree plantation
4. Other activities related to environmental rehabilitations

A rapid technical assessment was conducted in early October 2021 by UNHCR Assistant Environment Officer, and other technical colleagues and several recommendations are proposed for environmental rehabilitation. A joint camp boundary demarcation was conducted with Kakonko district authority. Kakonko District authorities are discussing the use of land and facilities that will remain after decommissioning to be utilized by authorities as well as to provide services to the surrounding communities.

The feasibility study (Annex 2) included an analysis of land cover degradation through changes in NDVI (1981-2003), analysis of areas of erosion and hydraulic analysis to inform the flood profile of the camps, including Mtendeli former camp. The feasibility study also recommends designs and suitable areas for the location of gabions, stone-pit drainage channels and SUDs. A technical field assessment will be carried out in the first year of implementation to review and validate the final locations of the infrastructures and the technical designs required for each specific camp site to maximize results in flood and erosion mitigation in the camp areas and downstream villages. The results of this technical assessment and the stakeholder consultations will inform the Technical Dossier of the tender process for construction services.


Flood Interventions
Stone Filled Gabion
Pitched Drainage Channel
Camp Extent
Esitmated Host Village locations
Camp Facilities/Offices
Camp Infrastructure
Kasanda
Mtendeli camp
0 0.225 0.45 0.9 1.35 1.8
Kilometers
Kibondo
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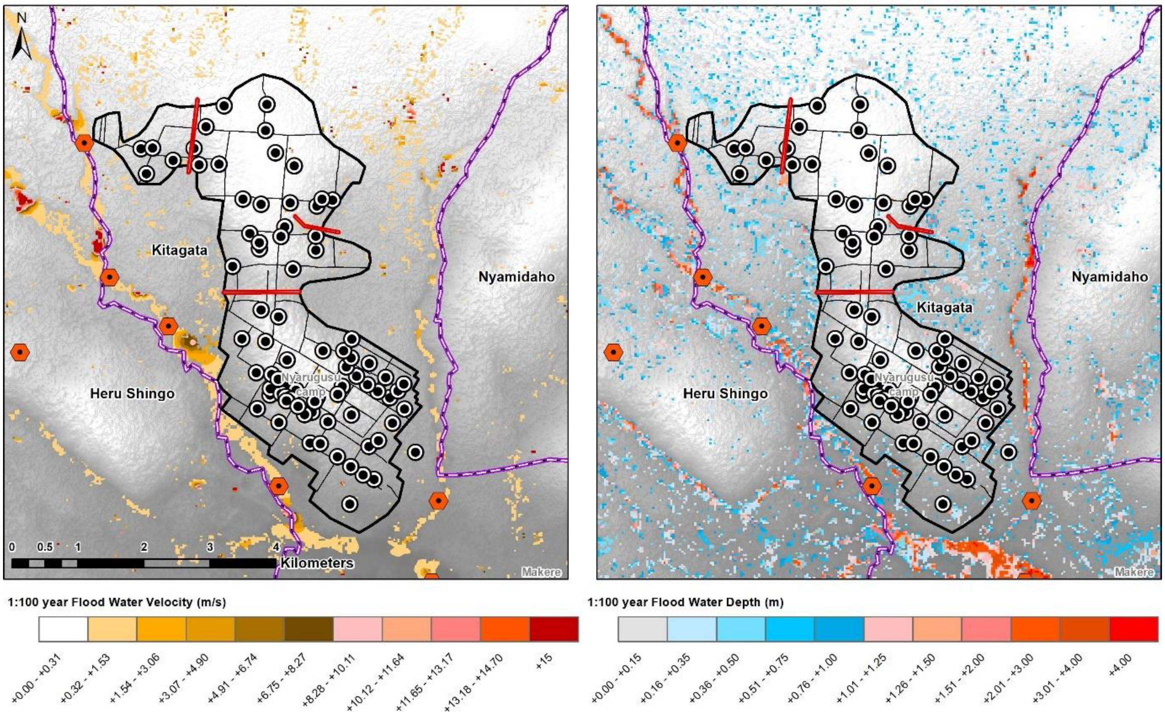

Figure 16. Potential locations for flood mitigation interventions in Mtendeli former camp (top), Nduta (middle), and Nyarugusu (bottom)

<u>Sub-activity 2.3.1. Construct 10 km of micro-scale stone-pitched drainage channels and Sustainable Drainage System (SuDS) interventions — including 16 stone-filled gabions and micro- infiltration trenches/ponds/soakaways — in three refugee camps.</u>

The beneficiaries for this intervention are primarily the refugees in Nduta and Nyarugusu camps and the host communities living in the vicinity of the Mtendeli former camp area, as well as the wider communities immediately downstream of the camp whose land and water sources may be impacted by flood and erosion events. Example technical designs accounting for the projected changes in precipitation, operations and maintenance requirements are included with the feasibility study.

Stone pitched drainage channels and stone filled gabions: ~10 km of stone pitched channels are estimated to be required to remove water efficiently from the priority facilities and infrastructure within the camps. The installation of 16 gabions will be needed in the main drainage channels just outside of the developed areas of the camps. These gabions will slow flood waters but will increase the horizontal propagation of water in doing so. These are therefore favoured outside of the camp area to limit this inundation. Hydrological modelling undertaken for the development of this funding proposal has provided indicative appropriate locations. The exact location of the drainage channels and gabions will be determined by the project's water officer in conjunction with the camp authorities (Ministry of Home Affairs) and the Lake Tanganyika Water Basin Board (LTWBB).

The traditional solutions will need to be designed and constructed by skilled technical contractors based on outline designs in the Feasibility Study, as they require specific technical knowledge and experience. The project's water officer will be able to identify suitable contractors, either using those that constructed the existing flood control structures in the camps, or other local qualified contractors[141]. Support can be provided by volunteer workers from with the camps.

Sustainable Drainage Systems (SuDS) are systems that are constructed or engineered using man-made materials and/or natural systems that tend to preserve existing open space, protect natural systems (groundwater, surface water) for improved drainage and filtration, and make use of existing land-use plans and maintenance to manage stormwater flows. These options include solutions including:

- infiltration trenches/ponds/soakaways;
- earthen bunds;
- rainwater gardens;
- filter strips; and

[141] Contractors will be procured under applicable UN policies, rules and regulations.

- vegetated swales.

Detailed hydrological modelling undertaken has provided the indicative appropriate locations for the Sustainable Drainage System interventions. The exact location of the infiltration trenches, swales and bunds will be determined by the project's water officer in conjunction with the camp authorities (Ministry of Home Affairs) (see Section 2.2.2 of the Feasibility Study).

The SuDs can be easily constructed by the local workers, using local materials and with limited technical 'know-how'. These will be constructed by the local communities after training and under supervision of the project's water officer, with support from local NGOs as required (REDESO, DRC, etc.). Periodic maintenance of both the traditional solutions and the sustainable drainage systems will be required to ensure their ongoing effectiveness. It is anticipated that this can be carried out at little-to-no cost using volunteer workers from the camps, overseen by the project's water officer.

A number of the traditional solutions (stone-pitched drainage channels, weirs and stone filled gabions in the drainage channel) have already been installed in Nduta, Nyarugusu and the former Mtendeli camp, although not at the scale that is estimated to be required. These structures have proven effective in controlling floodwater and reducing erosion, although, only to the extent that their scale allows. These structures are relatively inexpensive, easy to build and require limited maintenance. These traditional solutions are simple, inexpensive and relatively easy to construct and maintain. The main goal for these traditional solutions is to create a rapid exit route for surplus stormwater out of the camp. Some estimates suggest that the adoption of these types of measures could offer up to 40% flooding and erosion mitigation.

UNHCR will provide funds and oversight to its implementing partners to carry out operations and maintenance of water sanitation and hygiene and other infrastructure. The Implementing Partner, in turn, will utilise a combination of both national Tanzanian staff in managerial and technical roles, as well as incentive workers[142] in skilled and unskilled labour roles. MoHA will provide overall leadership in terms of land availability and selection of settlement areas and verifying the functionality of infrastructure and services across all sectors. Infrastructure will be built according to national standards, which will facilitate eventual hand over.

Output 3. Climate-resilient livelihoods practised to increase the capacity of host communities and refugees to better adapt in changing climatic conditions.

The GCF funding request for this Output is US $5,687,324. UNHCR co-financing of US $1,407,207 will go toward technical staff support and livelihood training and capacity building.

Activity 3.1. Promote modern technologies and management practises to strengthen the capacity of farmers, district officials and agricultural extension workers in climate-resilient agriculture.

This activity will be implemented by the District Councils in coordination with the Government of Tanzania Vice President's Office (VPO).
Sub-activity 3.1.1,3.1.2 and 3.1.3 will be conducted in villages surrounding the camps.

The two major crops are maize and beans which together represent 66% of total crop production. Farmers in Kigoma currently use locally available, unimproved varieties with low yield potential. Farmers with existing fields reserve viable seeds for replanting and sell seeds to vendors at local markets where farmers establishing new fields, including refugees, purchase them. As described in the Feasibility Study, the region has suffered extreme degradation caused by climate and non-climate drivers, which limits agricultural productivity. Kibondo uses only 23.3% of arable land, Kakonko uses 56.7% and Kasulu 30%, meaning that available land for agricultural production is not fully utilised and productivity is affected by the degraded ecosystems.

Tanzania's Resilience Strategy (2019–2022) identified Kigoma as one of the geographic regions in the country where improved agricultural techniques for enhanced climate resilience should be prioritised[143]. As part of the implementation of the Resilience Strategy, interventions include the establishment of integrated food-production systems and diversification of incomes in areas where small-scale agriculture is the predominant livelihood strategy. The Resilience Strategy also includes an acknowledgement of the need for sustainable strategies on land use, improved management

[142] Incentive workers are refugees engaged by UNHCR or partners to undertake jobs or tasks in connection with the provision of assistance and services to the refugee community. Such work is often characterised as volunteering rather than employment, with compensation (in cash or kind) being described as an 'incentive' rather than a wage
[143] FAO. 2019. The United Republic of Tanzania Resilience Strategy 2019–2022. Rome.

of natural resources and the facilitation of market mechanisms for improved livelihoods[144]. Services for improved crop production are currently limited. These services include crop forecasting, food security monitoring and early warning systems on transboundary challenges such as pests and disease, as well as extreme events such as droughts and floods.

A significant challenge to enhancing the climate resilience of local communities in Kigoma is that limited amounts of agricultural produce is sold at market[145]. Input costs for agriculture are high and compounded by inadequate physical market and other infrastructure — particularly road networks. The lack of sufficient transport and storage infrastructure reduces market access significantly, which leads to impacts on food production, as farmers do not have incentive to produce a greater variety of crops. This results in an increasing trend towards monocropping, which is more vulnerable to fluctuations in regional demand and at increasing risk of climate impacts. These factors are threatening the current food security in the region and exacerbating the existing challenge of underdeveloped and fragmented markets. In Kigoma, market mechanisms are inadequate and market information about prices insufficient — farmers are often unable to make an income or earn profits as a result of inadequate marketing and pricing policies. Market failures and high food prices, in combination with other systemic challenges, put pressure on livelihoods, food security and the nutrition of local communities.[146] Therefore, to support effective adaptation to climate change, the improvement of physical infrastructure should be accompanied by development of value chains through investments by commercial partners and multilateral development agencies[147]. Indeed, agriculture has been identified as one of the primary areas for investment to enable economic development in Kigoma[148].

Value-addition activities through the expansion of agro-processing are not well-developed in Kigoma. There are several underlying reasons for this, including insufficient access of farmers to funding for modern farming technology and methods, post-harvest losses, inadequate availability of farming capital and financing, as well as low productivity and low prices. Within the existing agro-processing industry, raw materials supplied by smallholder farmers are often of inadequate quality and the industry is characterised by insufficient availability of processing systems and equipment[149]. Access to credit should be provided to rural micro and informal enterprises, and mechanisms should be developed which are suitable for the access of farmers to funding for improved agricultural inputs[150]. These factors have been demonstrated in several existing projects. For example, the Climate Resilient Agribusiness for Tomorrow (CRAFT) project identified increased productivity and resilience to climate impacts as high priorities for enhanced resilience of local communities in the study area[151]. Moreover, the project identified the need for improved incomes, improved business performance for agribusiness, as well as the need to implement climate-resilient food production systems[152]. Improving institutional and technical capacity for climate-resilient agriculture is also necessary.

The climate-smart agriculture (CSA) guideline, a document produced by the FAO, notes that agriculture is one of the central pillars of the economy of Tanzania, with the sector employing ~78% of the population and contributing ~95% of the nations required food[153]. As a large proportion of the sector is rainfed crop production, and most households produce at subsistence level, agriculture is highly vulnerable to the impacts of climate change. The CSA guideline was developed in accordance with several national plans and programmes, including: i) National Adaptation Programme of Action (2007); ii) the National Climate Change Strategy (2012); iii) the Agriculture Climate Resilience Plan (2014–2019); iv) the National Climate-Smart Agriculture Programme (2015–2025); and v) the country's Nationally Determined Contributions (2016). These documents are closely aligned with the government's commitment to creating a resilient, climate-smart agriculture sector by 2030. The success of implementing and upscaling CSA in Tanzania is largely reliant

[144] FAO. 2019. The United Republic of Tanzania Resilience Strategy 2019–2022. Rome.
[145] Enabel. 2009. Support to Income Generating Activities for Coast and Kigoma Regions. Available at: http://www.diplomatie.be/oda/11768_PROGDESCR_Technisch_en_Financieel_Dossier.pdf
[146] FAO. 2019. The United Republic of Tanzania Resilience Strategy 2019–2022. Rome.
[147] SNV. 2019. Climate Resilient Agribusiness for Tomorrow (CRAFT). Available at: https://snv.org/cms/sites/default/files/explore/download/final_amended_craft_factsheet_sept_2019_0.pdf
[148] Enabel. 2009. Support to Income Generating Activities for Coast and Kigoma Regions. Available at: http://www.diplomatie.be/oda/11768_PROGDESCR_Technisch_en_Financieel_Dossier.pdf
[149] Communication from UNHCR
[150] Enabel. 2009. Support to Income Generating Activities for Coast and Kigoma Regions. Available at: http://www.diplomatie.be/oda/11768_PROGDESCR_Technisch_en_Financieel_Dossier.pdf
[151] SNV. 2019. Climate Resilient Agribusiness for Tomorrow (CRAFT). Available at: https://snv.org/cms/sites/default/files/explore/download/final_amended_craft_factsheet_sept_2019_0.pdf
[152] SNV. 2019. Climate Resilient Agribusiness for Tomorrow (CRAFT). Available at: https://snv.org/cms/sites/default/files/explore/download/final_amended_craft_factsheet_sept_2019_0.pdf
[153] Rioux, J et al. 2017. Climate-Smart Agriculture Guideline for the United Republic of Tanzania: A country–driven response to climate change, food and nutrition insecurity.

on institutional changes and a behavioural shift of the population[154]. The CSA guideline identifies several key areas where capacity-building interventions are required, which align with the interventions to be implemented under the proposed GCF project. These key areas include awareness raising among farmers of improved and climate-resilient farming techniques, training tailored to the needs of specific groups of stakeholders, easier access to climate information services, and improved agricultural risk management mechanisms[155]. Several projects in Kigoma have had a specific focus on capacity building for pro-poor economic development in the region as there is a high incidence of poverty in the population[156]. These projects also demonstrate the need to improve dialogue between stakeholders[157].

The Sustainable Agriculture Kigoma Regional Project (SAKIRP), implemented from 2015–2021is one example[158]. The project demonstrated the need for enhanced climate-resilience in agriculture and specifically the development of value chains for key commodities in the region — most notably cassava and beans. Moreover, the project involved technical capacity building for farmers through Farmers' Field Schools (FFS) in a coordinated manner, as well as the provision of extension services and other services with the aim of increasing productivity and improving livelihoods[159].

The interventions proposed under this activity will focus on promoting the adoption of climate-resilient agricultural technologies and management practises, as well as the procurement and distribution of agricultural inputs. The climate resilient agricultural practises that the project plans to promote include: i) use of pest, drought and heat stress resilient cultivars; ii) integrated soil management; iii) integrated soil fertility management; iv) intercropping; and v) water harvesting and management training — as are appropriate to the participatory needs and capacity evaluations undertaken with beneficiary farmers. These interventions will be supported by strengthening capacity of district officials and extension services, who can directly engage with local farmers to promote climate-smart agriculture (CSA) practises (training of trainers). Support provided will include training, recruitment of a local consultant to provide technical support, materials provision for demonstration plots, and per diem fees for extension officers. More details can be found in Annex 4. Project budget. Farmer field schools (FFS) in the villages will provide training to farmers and government extension workers. The FFS will function as centres of adaptive knowledge sharing methods can improve soil quality, water use efficiency and reduced erosion, as well as training in adaptive alternatives to crops, such as the production of honey and mushrooms. Project beneficiaries will also gain access to approximately 300,000 kg of climate resilient planting materials, among which at least 60% will be distributed to farmers through Savings Associations, prioritising resilient cassava and beans. The introduction of bean varieties such as Lyamungo 90 will out-yield more traditional varieties grown in the region and new varieties of cassava will bring disease resistance. The capital available in the Savings Associations will also stimulate markets for produce, incentivising local entrepreneurs to further establish small-scale nurseries for agroforestry seedlings. Through this, farmers will receive access to inputs such as drought-resistant seeds and a reliable market which will increase their incomes. For further information please Section 7.4.1 of the feasibility study.

In the host communities, improved climate resilient agriculture practices focusing on beans and cassava, greater access to inputs and enhanced water and soil management as supported by the project, will increase agriculture production and reduce farmers' vulnerability to climatic shocks and stresses. By improving agriculture production, food security will be improved both directly, by increasing the availability of food locally and indirectly by improving households' income from agriculture thereby improving their access to food by enabling them to purchase other, more diverse, food items.

A similar logic applies to the alternative livelihood activities related to beekeeping and mushroom production, which is planned to be implemented in both host communities and refugee camps. The production of honey, used as sweetener, and mushrooms, often used a meat substitute (as indicated during community consultations) brings more diversity to local diets, whereas increased incomes as a result of these activities improves households' purchasing power, enabling them to buy more and more diverse food items. In Kigoma Region, preferred mushroom species is *Pleurotus flabellatus*. Scaling up the growing of these species in the refugee camps and also strengthening market linkages in the context of the host community will serve to ensure sustainability in the near to medium-term whilst also ensuring livelihoods and

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154 Rioux, J et al. 2017. Climate-Smart Agriculture Guideline for the United Republic of Tanzania: A country–driven response to climate change, food and nutrition insecurity.
155 Rioux, J et al. 2017. Climate-Smart Agriculture Guideline for the United Republic of Tanzania: A country–driven response to climate change, food and nutrition insecurity.
156 Enabel. 2009. Support to Income Generating Activities for Coast and Kigoma Regions. Available at: http://www.diplomatie.be/oda/11768_PROGDESCR_Technisch_en_Financieel_Dossier.pdf
157 Enabel. 2009. Support to Income Generating Activities for Coast and Kigoma Regions. Available at: http://www.diplomatie.be/oda/11768_PROGDESCR_Technisch_en_Financieel_Dossier.pdf
158 The Sustainable Agriculture Kigoma Regional Project (SAKIRP). Available at: https://open.enabel.be/en/TZA/2157/p/sustainable-agriculture-kigoma-regional-project.html
159 The Sustainable Agriculture Kigoma Regional Project (SAKIRP). Available at: https://open.enabel.be/en/TZA/2157/p/sustainable-agriculture-kigoma-regional-project.html

nutritional diversity. A small mushroom growing pilot project has been undertaken in the refugee camps in collaboration with implementing partners and the response by persons of concern has been very positive. As such, the uptake is envisaged to be very good.

The barriers discussed in Section B.2 demonstrate the precarious position occupied by those practicing smallholder agriculture in the area. Limited access to technology and financial services, combined with limited knowledge and capacity, result in low returns to agriculture as well as limited resilience. While these issues would ordinarily be addressed by the market through agricultural extension services and access to finance, in the case of Kigoma these services are provided at a sub-optimal level because of inadequate transport links and poorly developed value chains. This activity will improve the level at which these services are provided by increasing the level of knowledge and technical competence among key role players in the agriculture and finance sectors.

Since this activity occurs on, or around agricultural land, some of the beneficiaries for this activity will likely overlap with those benefitting directly or indirectly from the agroforestry, afforestation and flood and erosion control measures in Output 2. The beneficiaries of Activity 3.1 will include: i) 1,200 farmers trained through 20 Farmer Field Schools; ii) ~75,000 beneficiaries in 15,000[160] households receiving agricultural inputs and on-training from those trained in the FFS; and iii) the Savings Association Management Offices (SAMOs) strengthened to support provision of agricultural and other climate-resilient livelihood inputs.

Pro-actively targeting women in the project’s alternative livelihoods activities and enabling their full participation, is expected to contribute to their economic empowerment. Women refugees will be specifically targeted for alternative livelihood interventions (at least 50%). The strengthening of Savings Association is expected to bring positive externalities as women are generally well represented in these groups. Money obtained by women in host communities through these savings associations is not only used to invest in productive assets and activities, but also to meet household needs and pay for school fees. Increased incomes may enable some women to purchase fuelwood or other energy sources to meet (part of) their cooking energy requirements. Women, like men, engage in agriculture activities and by ensuring they are able to benefit from accessing agriculture trainings and inputs and irrigation investments, women in the host communities will increase their resilience to current and projected future climate change impacts. Furthermore, for women in the host communities, improved environmental management including conservation of water sources will improve their year-round access to water for domestic and productive uses and thereby reducing the burden on women who have to fetch water.

<u>Sub-activity 3.1.1. Build capacity of 60 farmers per Farmer Field School (FFS) on climate-resilient agricultural practises, including use of traditional knowledge, through trainings co-designed in collaboration with the farmers, district officials and extension services.</u>

The specific package with climate-resilient agricultural practises will be co-designed in collaboration with the farmers in the villages where the project will be working. Agriculture district officials (representing Ministry of Agriculture at the district level) and extension workers will be involved in co-designing the training activities based on the climate-resilient agricultural practises and technologies, including integrated pest management, that best fit the selected context. The National Climate Smart Agriculture guideline developed by the Government of Tanzania will be used as an instructive tool to design the training activities and to provide instructions for best strategies suited for the selected agro-climatic zone. As part of the design of training activities, a needs assessment of district officials and extension services will be carried out to identify training requirements prior to the implementation of capacity building activities.

At the District level, a comprehensive training system will be provided to technical district staff and extension staff to build their capacity to plan for agricultural development, to facilitate community-level planning, and to sustainability support farmers in the long term. Standard procurement processes will be followed to identify the best-suited implementing agency.

Extension staff will be responsible for mobilising and training farmers to adopt climate resilient agricultural practises. This system will take advantage of synergies with farmer’s groups to tailor input packages for increased yield. The Farmer Field School (FFS) approach will allow extension officers to establish new farmers’ groups, for example those who focus on a selected crop (i.e. women who cultivate cassava) and to support existing ones with technical and organisational skills development.

[160] Which would include the households of the 1,200 farmers receiving training through the FFS, with selection of participants aiming to limit FFS attendance to one per household.

Extension workers will encourage farmers' learning through demonstration plots where they will have tools, start-up seeds and relevant crop husbandry training in pre-harvest, production and post-harvest practises. With the aim of addressing the lack of capacity of farmers to organise themselves and establish linkages to markets, the FFS will be used as a strategy to strengthen group cohesion, disseminate knowledge and ultimately create stronger farmers' groups. A total of 60 direct beneficiaries will be involved in each of the 20 FFS planned (with the distribution of FFS to be determined through the VLUP process), with two demonstration plots established per FFS. Starter kits for the FFS will be provided and knowledge exchange visits between FFS groups will be organized. Beneficiaries will be selected in a participatory way with a view to have equal representation of women and men as well as to select for members of the community that have the potential to influence and capacity to teach other farmers.

Climate resilient agricultural practises and technologies have been tested and adopted in several regions in Tanzania. For example, conservation agriculture practises are widely promoted as part of sustainable soil management and climate-resilient agriculture in Tanzania. The benefits of these practises on yield are the result of improved soil structure, as well as increased soil fertility, carbon content and water retention capacity[161] . As part of FAO Mitigation of Climate Change in Agriculture Programme (MICCA) in Tanzania there has been an agriculture extension component, various approaches have been used and FFSs have demonstrated to be very successful in Kolero and Morongo. Trainings have been organized in the villages — inspiring many people to participate.

Sub-activity 3.1.2. Provide 20kg of resilient seeds and 1 kit (at least 60% to be procured by SAMOs and distributed through Savings Associations) to each of 15,000 farmers for climate-resilient agriculture including seeds, tools and equipment necessary for implementing climate-resilient agricultural practises[162].

Once the training activities have been delivered and the capacities of extension officers have been enhanced, communities will be able to access inputs (seeds, tools, equipment[163]) to be procured by SAMOs (at least 60% of the total inputs) necessary for implementing climate-resilient agricultural practises. At the village level, procurement and distribution of agricultural inputs, such as climate and disease-resistant varieties of beans and cassava will be undertaken through the saving associations, who will receive requests from the farmers' groups on the types and quantity of inputs they need over the course of the project. In year 2, seeds and equipment will be procured from the local market by the District Councils and directly granted to smallholder farmer beneficiaries (6,000 households). In years 3 – 5, the agricultural inputs will be accessed through Savings Associations (9,000 households). Each household will have access to improved agricultural inputs worth 88USD, which may include agricultural tools, cassava cuttings, beans or improved maize seeds. This amount enables the cultivation of 0.5 hectares of cassava (10,000 cassava cuttings required per hectare at an average of 40TZS/cutting) or 1 hectare of beans in pure stand (50-70 kg beans/hectare depending on bean size at an average of 3,000 TZS/Kg) or 1 hectare of beans in intercropping with maize (40kg beans/hectare at an average of 3,000 TZS/Kg). Detail provided in Annex 4. Budget- detailed assumptions.

The project will prioritise two crops – cassava and beans – and will focus on increasing their productivity. The selection of these two crops is based on the following factors: i) the drought-tolerant and adaptation properties of these crops; ii) the availability of cuttings and seeds; iii) the preferences of potential beneficiaries expressed during focus groups discussions; and iv) the potential to create synergies with other projects in the area — such as agro-processing for cassava, supported by other agencies/NGO programmes. In the analysis of crops' supply chains, cassava and beans scored the highest in terms of market potential, potential for promoting gender equity and women's economic empowerment, contributing to food security, and income potential. This activity upscales the Sustainable Agriculture project by Enabel, incorporating the lessons learned from the implementation of that project[164] as well as lessons learned from the UNCDF assessments of savings groups in Kigoma[165].

Sub-activity 3.1.3. Build capacity of Savings Association Management Offices (SAMO) to manage savings associations.

Savings Associations (SAs) allow individuals to pool finances, to save and borrow money based on the consensus of members. This project seeks to strengthen and scale up these existing structures so that funds will be disbursed through a market-based approach[166]. Consultations undertaken in September 2020 indicated strong support for the proposed strengthening of Savings Associations, as focus groups recognised the role these associations played in facilitating access to finance and delivering agricultural inputs. SAs typically serve as cooperatives in the region to have improved

[161] Palm, Cheryl, Humberto Blanco-Canqui, Fabrice DeClerck, Lydiah Gaterea, and Peter Graced, 2014. Conservation agriculture and ecosystem services: An overview. Agriculture, Ecosystems & Environment 187: 87–105.
[162] For clarity, agroforestry and afforestation activities are not included in the operation of Savings Associations and in the financial model, as there are no viable revenue streams for these activities.
[163] Beneficiary farmers will own seeds, tools, and equipment.
[164] https://open.enabel.be/en/TZA/2157/p/sustainable-agriculture-kigoma-regional-project.html.
[165] UNCDF 2019. Strengthening resilience through savings groups www.uncdf.org.
[166] UNEP is accredited for grant awards. Details of the mechanism are in Appendix C: Operational Manual Outline for the Savings Association Management Offices in the Feasibility Study.

access to farm inputs, collectively negotiate prices, facilitate tax collection, improve marketing of products, and have access to microfinance. SAs in the project will have similar roles with the key difference from other SAs being capacity support for promoting climate-smart and adaptive practises and crop varieties. In the host community, there are numerous examples of self-organised SAs, and the NGO Good Neighbours is currently piloting provision of support to 4 host community SAs[167].

The project will support the SA and its members in managing funding decisions and building capacity to hold funds through 'Savings Association Management Office' (SAMO) and envisages a SAMO as playing a critical role in scaling up the funds available to participate in climate resilient activities. During early project implementation, barriers to SA membership as well as incentives and opportunities to increase enrolment will be investigated. In villages or areas where enrolment is low and not representative, direct provision of inputs will be considered. Options for supporting financial literacy can include digital platforms such as using tablets and SMS platforms together with in person trainings, similar to innovative approaches used by UN Community Development Fund (UNCDF) in Nyarugusu and Nduta. Options for building on experiences in the region on mobile-based financial services platforms for smallholder farmers will also be pursued. The SAMO can hold the accounts for multiple Savings Associations and can act as an intermediary between the Executing Entity and informal transactions with the host community.

Savings Association Management Offices (SAMOs) are a market-driven means to disburse funds for the equipment required for the climate-resilient activities set out in the Feasibility Study, rather than via direct procurement. The SAMOs are not new organisations but would be an extension of services provided by existing structures (likely NGOs). SAMOs will provide resilient seed types or tools available to members. Members reinvest income and savings from successful livelihood activities into the SA as well as the cash value of the seeds or equipment received. GCF grants will not be used to make cash payments to beneficiaries. The mode of delivery will be through purchased goods. Further details on how the SAMOs would operate can be found in Appendix B of the Feasibility Study: Draft Outline Operational Manual for SAMOs. The operational manual will outline procedures such as cash payments procedures, financial statements and internal controls, monitoring and evaluation procedures, and other processes. Agreements will be made between SAMOs and beneficiaries to govern the use of funds. The PMU, through the Districts Councils, will work closely with the SAMO to build its capacity to manage funds, procure the agreed set of equipment and materials and to ensure funds are disbursed to SAs in line with project objectives. The terms of reference for the Finance Officer in the PMU will support capacity building and monitoring of SAs.

Activity 3.2. Increase water availability and use-efficiency through water harvesting and efficient irrigation interventions.

Activity 3.2 will be done in villages surrounding the camps. UNHCR will lead the procurement of rainwater harvesting and irrigation infrastructure in coordination with the District Councils and the Government of Tanzania Vice President's Office (VPO). The District Councils will lead the implementation capacity building and awareness raising on water management and efficient water use.

Under intensifying climate change, water scarcity is expected to become an ever more limiting factor for activities such as agriculture, agroforestry and afforestation. To ensure that the contribution that these activities make to livelihoods is not compromised as a result of lack of water, it is imperative that mitigation against this scarcity is prioritised. Provision of water can be considered a mixed good, given that it does not fully align with either the definitions of a private or a public good. For this reason, mixed goods are particularly prone to the market failure of inefficient distribution of the resource. This activity will ensure increased and continued availability of water for livelihoods.

Interventions under this activity will ensure that water is available in closer proximity to agriculture areas, as well as for the nurseries for agro-forestry and village land afforestation described in Activity 2.2. The engineering solutions themselves will be designed and use materials that are able to withstand changing weather extremes such as temperature and rainfall intensities. The specific interventions that will be implemented will be determined through the C-LUP processes, based on sample technical designs (see Section 7.4.2 of the Feasibility Study).

As this activity intends to complement and support the agricultural and forestry outcomes in Activity 3.1 and Output 2 respectively, some of the beneficiaries for this activity will necessarily overlap with those benefitting directly or indirectly from the agricultural training and inputs provided in Activity 3.1 and the agroforestry, afforestation and flood and erosion control measures in Output 2. The beneficiaries of Activity 3.2 will include those involved in the construction, maintenance of the rainwater harvesting and irrigation systems, as well as those maintain the agroforestry and afforestation nurseries and those benefiting from complementary watering of rainfed crops and water collection during dry periods.

[167] Refer to Section 3.5.1 of Annex 2. Feasibility Study for further description of Savings Associations.

Sub-activity 3.2.1. Construct micro-rainwater harvesting systems: ~94 check dams ~20 run-off harvesting micro-dams (charco dams), ~120 unlined water-pans and ~120 lined micro-ponds in host communities in Kasulu, Kibondo, and Kakonko.

Water flows as sheet flow over the surface without stagnation or accumulation in some areas. In other areas, water is stagnating or accumulating to flow as a stream. These areas are suitable for installation of *inter alia* small check dams or charco dams.

Charco dams are off-channel permanent water retainers with anticipated sizes of ~5000m3. These would either use the natural landforms or require deliberate excavation. As these are off-channel, these would be topped up during rain events. These may have linings depending on the underlying soils, resulting in reduced water loss to the soil. During the wet season, these dams will act as reservoirs allowing for more effective water usage in the community and agricultural areas. It is unlikely that these dams will be full throughout the year as the water will be used by the communities. However, these dams are meant to augment the water supply during the dry season to bridge the rainfall gap (June to August) which is currently negatively affecting livestock and crop productivity.

Check dams are designed as temporary overtopping storage structures that slow water flow, decrease erosion, and increase infiltration. While these small dams will be full only for short durations (weeks) after rain events, unless there is a near consistent flow in the channel, the longer-term benefit will be through the infiltration these check dams promote. The situating of these check dams is therefore in channels and generally upstream of agricultural areas. This has the dual benefit of slowing water reducing erosion and increasing the height of the water table in the agricultural areas.

The dams and in stream structures require significant technical knowledge and experience to construct. This will require external contractors that specialise in these systems. Contractors will be procured from among those operating in Tanzania in accordance with applicable UN policies, rules, and regulations. Basic labour can be supplied by the technically trained members of the host communities.

The hydraulic modelling of current and projected storm analysis indicates that in order to reach the majority of agricultural areas and host communities, 20 charco dams and 94 check dams could be built for water harvesting purposes. Irrigation in local agricultural areas will be augmented by approximately 240 lined and unlined in-situ ponds fed by pumping systems from the charco dams. The length of size, number of gabions and locations will be determined by the project's water officer in conjunction with the camp authorities (Ministry of Home Affairs), (see Activity 1.2) and subject to change based on these surveys and community priorities.

These interventions are expected to have minimal impacts on water availability and flood profile downstream. For Nyarugusu and its surrounding host community, a hydraulic assessment was undertaken on the changes that these structures would have at the entrance of the Malagarasi-Moyowosi RAMSAR site and concluded to be very minimal. The same holds true for the other sites. The structures will result in a flood profile similar to the current day events but will act to slow down water flow and lessen erosion. The maximum volume of water stored by the structures is nearly insignificant compared to the total volume of water that falls over the area[168].

The charco dams will be situated in areas feasible for natural refilling and being close to agriculturally important areas. The check dams, in the flow channel will also be constrained in the potential intervention location. Where possible these interventions are either higher or on the same contour as areas needing irrigation. This allows for gravity fed irrigation using the solar and treadle pumps for water dissemination. The propagation of this will be reliant on in-situ topographic factors. Lined and unlined ponds with minimal gravity fed recharge outside of the main channels, reply on water being pumped to them. They will therefore have more flexibility in their location, allowing for them being placed in closer proximity to crops. These could also be daisy chained to propagate irrigation networks, further.

Figure 17 shows the suitable locations for rainwater harvesting (RWH) interventions around Mtendeli former camp area, Nduta and Nyarugusu camps and the surrounding host communities.

[168] Assuming all check dams are 500m^3 and all Charco dams are 5000m^3 and they all fill completely (ie Capture 100% of their maximum retention potential), this comes to 41,000m^3 53,500 m^3 52,500 m^3 stored for Mtendeli, Nduta, and Nyarugusu respectively. Assuming there is only 1mm of rainfall falling over each of these areas, the stored water accounts for 0.00569%, 0.00633%, and 0.00469% of the total areas upper catchment rainfall. This would then be the maximum influence of these dams as subsequent rainfall would serve to top up, rather than fill completely, the dams. Larger rainfall events would reduce this proportional influence.

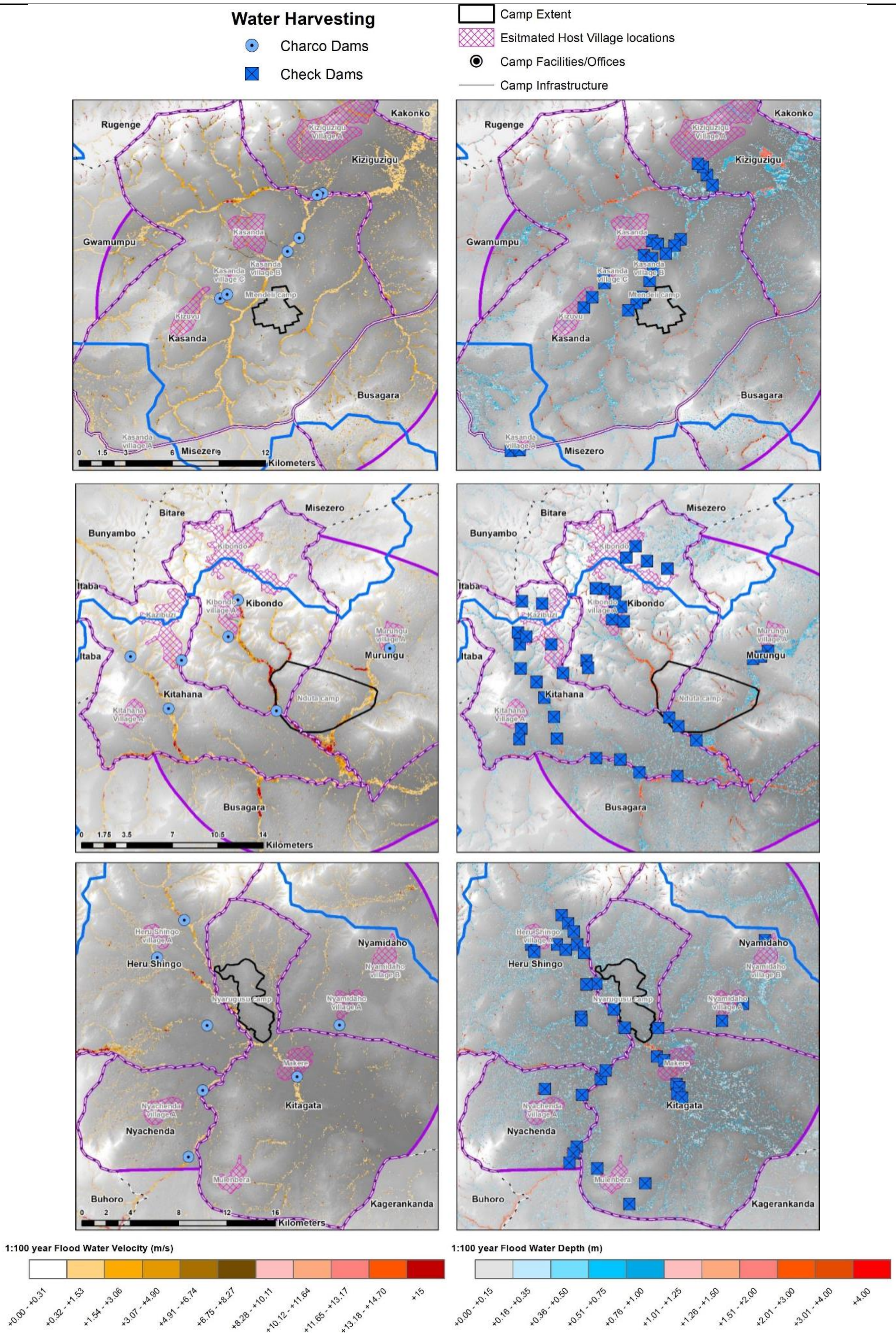

Figure 17. Indicative suitable locations for RWH interventions in and around Nduta camp (top), Mtendeli former camp (center), and Nyarugusu camp (bottom), based on hydrological modelling undertaken. The maps on the left show potential location of charco dams while those on the right show potential location of check dams.

Construction and installation of in-situ water ponds and sustainable infiltration systems technologies requires limited technical know-how and can be undertaken by the farmers themselves using local materials, with supervision from the technical teams of the local NGOs. The small lined and unlined ponds will be minor interventions sited as close as possible to the agriculture in the area. These will act as the local water storage for farm usage. The siting of these ponds will be done in the project implementation.

Fencing or other physical structures to prevent encroachment would likely not be viable due to the cost of installation and maintenance. Some of the areas may also be marshy so that fences would not work well. Community ownership of these structures would be essential to ensure water resources are not misused. To minimize evaporation, trees would be planted around charco dams. These would also act as wind breaks, lowering evaporation. Covers may be effective and options for using locally available materials to make and repair these would be explored during implementation. Where possible, charco dams will be constructed deeper and have a lower surface area. Check dams would not require protection for evaporation as their store water temporarily and facilitate infiltration. Charco dams would have silt traps to reduce sediments getting into the dams. Maintenance activities would also be done in the dry season when the dams are at the lowest levels. Check dams require less maintenance for sediment as these allow for overtopping during flood events. Additional clearing, if required, would be done in the dry season.

Training will be provided as part of Activity 3.1 (Resilient Agriculture). The specific run-off systems to use in each area will depend on the terrain, the population density, the size and type of the catchment system and how much water can be harvested, the type of soil and its properties, the groundwater level, geological information about the catchment, and the community priorities.

<u>Sub-activity 3.2.2. Establish water pumping and irrigation through ~50 micro-solar and ~172 treadle pumps that will draw water from the rainwater harvesting locations to where it is needed in host communities in Kasulu, Kibondo, and Kakonko.</u>

Water can be drawn from natural water courses and engineered rainwater harvesting locations (Activity 3.2.1) to where it is needed for irrigation. As well as the host community farmers, water pumps and irrigation systems are also proposed to be supplied to the nurseries supporting the agroforestry and afforestation activities.

For those farming activities located close (or near/below elevation) to water sources (max head 7.5 m), a collection of farmers will be provided with treadle pumps to deliver water to their gardens. This water will be stored in tanks or small in-situ storage which then flows by gravity to the drip irrigation kits. The treadle pumps can also provide sprinkler irrigation. A treadle pump system would include:

- the pump
- mains distribution hose
- irrigation kit (soil moisture testing kit, water tank, control head (valve), a filter and the drip lines)
- spray/sprinkler head.

Solar pumps will be provided for communities whose cultivated land is located further away from the water storage pond/dam/tank and where the network design is unable to propagate water to the needed locations (max head 150m). The solar pump system would include:

- the pump and secure pump house/control room
- the solar power system
- mains distribution hose
- irrigation kit (soil moisture testing kit, water tank, control head (valve), a filter and the drip lines)

The beneficiary profile and technology selection will be undertaken in a participatory manner that accounts for technical factors as well as appropriateness for community needs. It is proposed that farmers' drip irrigation kits are designed to irrigate up to 500m² are also provided. The technology is used for both high value crops (vegetables) and fruit trees/row/perennial crops like banana, coffee, cotton grown to Kibondo and Kakonko districts. The technology is available in the country and is part of the Climate Smart Agriculture guidelines, climate-smart agriculture activities in the GCF project in Simiyu, among others. In this method, water is applied frequently with volumes of water approaching the consumptive use/crop water requirements of the plant thereby minimizing losses.

<u>Sub-activity 3.2.3. Conduct training and awareness raising for 400 members of Water User Associations and farmers on water management and efficient water use through the FFS.</u>

Given the extent of the physical and administrative interventions planned in the project, an extensive programme of training and awareness is proposed. Project activities will only be sustainable in the long-term if they are supported by the community, which has the necessary skills to use and maintain the rainwater harvesting, storage and pumping as well as irrigation equipment.

As noted above, the C-LUP process will seek to gain consensus among the host communities around how water resources should be managed, and how effective water management supports a climate-resilient landscape. Specific training will then be provided to ensure that this management plan can be implemented. Details of activities included in the training are described below.

- Watershed and catchment protection
  - The advantages of protecting water sources and understand the watershed/catchment
  - Understanding how potential development and other land-use changes will impact source lands over time
  - The problems with and consequences of farming in the river floodplains
  - The available alternatives in terms of sources of water, better farming methods, water conservation practises, and boosting soil fertility.
  - The advantages of agroforestry and the tree line planted along the 60m mark

- RWH, storage and distribution
  - Advantages of RWH for domestic and agricultural uses.
  - Soil and water conservation methods (in-situ RWH system)
  - How to install and operate a rainwater harvesting system
  - Operation and maintenance of the technologies put in place like the ponds/dams, the solar pumping stations, and the basic irrigation systems (design and installation) and basic irrigation agronomy
  - Causes of erosion, how to prevent it and the hands-on training on construction of proper vegetative swales as drainage systems for their homes/villages and farms.

This training programme will enable community members to continue maintaining activities introduced by the project over the long term, even after the project's close and to ensure that water supplies are protected, practise soil and water-conservation methods in their farms, and harvest water for irrigation and domestic use. Training will be done by a dedicated water resources management trainer (during the first six months of the project) and then continued as needed by the project's water officer.

<u>Sub-activity 3.2.4. Support the establishment of and strengthen existing Water Users Associations (WUAs) through organizing and conducting 20 meetings/trainings.</u>

Although there have been considerable efforts by the host community to protect the major water sources within the project area, the enforcement of village environmental bylaws has been a challenge as there is still some agricultural activity occurring along the rivers. Pollution upstream of important river systems — including the Makere River near the Nyarugusu camp — has resulted in an increase in waterborne diseases such as urinary tract infections and cholera, limiting the ability of communities downstream to use the river for drinking water. The establishment of Water Users Associations (WUAs) has the potential to address challenges regarding water management, the enforcement of water-related bylaws and conflicts that may arise between water users. To reduce the potential for such water-related conflicts, WUAs — alongside staff from the Tanzania Forest Service (TFS) and EMC members — will be trained on conflict resolution and community engagement to ensure that in situ conflict with and between water users will be resolved, in line with the benefit-sharing agreements established under Sub-activity 1.1.4. Moreover, strong linkages between the WUAs and EMCs will ensure horizontal and vertical coordination on conflict resolution and benefit sharing. If the conflict cannot be easily resolved, it will be managed through the grievance redress mechanism as outlined in Annex 6: Environmental and Social Management System. This in turn e combined impact of these interventions will assist in maintaining sufficient water quantity and quality for host and refugee communities that are predicted to be negatively influenced by future climate change.

Currently, the Tanganyika Lake Basin Water Board has supported the establishment of a WUA in the area, however additional support and the establishment of more WUAs is required. The proposed project will strengthen the capacity of existing WUAs through mobilization and training in the area and support the establishment of new WUAs as needed working and coordinating closely with the Tanganyika Lake Basin Water Board. It will additionally enhance linkages between the WUAs and C-LUP processes as well as village environmental management committees. The WUAs have the potential to play a major role in supporting other sub-activities under Activity 3.2, including awareness raising on water management and efficient water use as well as the installation and operation of water pumping and irrigation.

Activity 3.3. Promote climate-resilient livelihood diversification to strengthen food security and nutrition, provide alternative income as a safety net, and to sustain the implementation of climate-resilient agricultural and forest management practises.

The District Councils, in coordination with the Government of Tanzania Vice President's Office (VPO), will implement the livelihood interventions in the host communities. UNHCR through the project co-funding will implement livelihood interventions in the camps, opening up to host communities if and where possible, with the final livelihoods activities to be coordinated and agreed with MoHA.
Both host and refugee communities in the Kigoma region have a limited suite of livelihood options available to them and are therefore vulnerable to climate-related hazards such as drought. Further to this, high levels of malnutrition among members of the communities demonstrate the need for diversified diets. While agriculture is the predominant source of livelihood in the region, other activities such as beekeeping and mushroom growing are also practised in the host and refugee communities. These activities are largely dependent on the health of natural resources (particularly forests, soils and water), but are more resilient than agriculture to projected climate variability. However, there are several barriers to the uptake of additional livelihood options. The barrier relating to insufficient financial services will be addressed through the strengthening of Savings Associations in host communities to distribute inputs and reinvest income or benefits. This will create an enabling environment for additional livelihood options. Stakeholder consultations found a strong level of support from focus groups for the strengthening of SAs. Additionally, this activity will provide communities the information and technical support needed to efficiently implement additional livelihood options. While this support would ideally be provided through public services including development initiatives, these services are currently being provided as a sub-optimal level or do not serve the populations that this project aims to target.

The primary objective of the livelihood diversification activities is to strengthen food security and nutrition in the event of hazards such as droughts or floods — which could reduce crop yields — provide alternative income as a safety net in case of yield loss and to sustain the implementation of climate-resilient agricultural practises. The selection and prioritisation of livelihood diversification activities has been discussed with potential beneficiaries in the target communities and with district officials, and interventions have been chosen to achieve the following objectives.

- Strengthening the resilience of the ecosystem services by linking livelihood diversification with other interventions proposed in this project — specifically linking forest conservation and the promotion of beekeeping and agroforestry.
- Increasing the resilience of households by targeting both assets and services that characterise their livelihoods. The livelihoods were selected because they provide both access to physical capital (e.g. beehives), knowledge (e.g. climate-resilient beekeeping practises) and improved services (e.g. training of extension officers).
- Targeting activities to vulnerable and marginalised groups, including women and youth, who are often excluded from other income-generating opportunities. These activities also aim to improve relations between host communities and refugees.
- Considering the synergies of livelihood diversification activities with existing programmes implemented in the region. These activities were selected because they build from pilots and lessons learned from other projects, and they integrate with and complement other programmes, such as Enabel's efforts to develop honey value chains.

As in Activity 3.2, some of the beneficiaries for these sub-activities are likely to be in the same communities benefiting from other interventions in Output 2 and in Activities 3.1 and 3.2.

Sub-activity 3.3.1. Promote beekeeping to incentivise forest conservation in the forests surrounding the camps and villages by providing equipment and training to 850 people.

Beekeeping and honey production are vulnerable to climate change and climate variability, which has negative impacts on the productivity of honeybees[169]. In addition to climate change impacts, two non-climate drivers of change have been linked to losses in honey production in Kibondo and Kakonko: the arrival of refugees and consequent demarcation of camps resulted in beekeepers moving into the nature reserves, while cattle grazing and tree cutting resulted in deforestation and hives loss. This project will tackle these challenges through participatory village land use planning which will mitigate conflicts between the two populations over shared natural resources, and consequently allow beekeepers to implement their activities on selected locations within the village forest reserves.

[169] These include altering plant flowering time, increasing water stress especially in situations of drought, thus reducing pollen and nectar availability, inhibiting movement, affecting bee communications, causing physical damage of hives, colony starvation and retarding bee forage activities. A study on the impacts of climate variability and change on beekeeping productivity in Tanzania found that higher rainfall (over 700 mm) and higher temperature (over 33º C) negatively influence honey production. Similar impacts have been observed in the project area. In response to climate change, beekeepers have adapted to reduce impacts by moving hives to other areas, providing food for bees, providing water, changing hive types, changing apiary location, putting hives in tree shadows, use of over-dimensioned wooden hive and changing harvesting methods and time.

The activity will be carried out by the District Councils and the Tanzania Forest Service in collaboration with the Kibondo Beekeepers Association who will provide technical support and capacity building for beekeeping and awareness on forest conservation value. The Kibondo Beekeepers Cooperative Society was founded in 1992. The cooperative provides processing, packaging and labelling for its members and supports them by linking them to markets and establishing contracts with local companies (for example, Jasmine Bee and Upendo Honey). There are four beekeepers' cooperatives in the selected districts which provide support to 43 beekeeping groups. At the district level, beekeepers are represented by a cooperative officer in Kibondo. A number of studies[170] have found that beekeeping in the country has not reached the commercial potential because of several factors, including lack of knowledge and skills to increase productivity, reliance on traditional hives and limited access to finance. When linked to community-based natural resources management, beekeeping has been proven as an effective strategy from an economic and environmental perspective [171].

Under this activity, beekeeping will be promoted as an alternative livelihood activity that contributes to forest conservation, food security and income diversification for both host communities and refugees. The activities will be implemented in the forests surrounding the camps and host community villages. Equipment and inputs will be distributed through the village saving associations.

The promotion of climate-resilient beekeeping activities will involve the distribution of modern hives, which protect colonies from heavy rains. Modern beehives and equipment to be owned by farmers will allow them to improve their productivity. Key adaptation strategies to increase climate resilience of beekeeping will include the provision of training on good beekeeping practises that include knowledge of climate change and their impacts, enabling farmers to select the right location, the appropriate maintenance and harvesting methods. The promotion of beekeeping as a climate-resilient diversification activity will therefore involve this type of training to both farmers and extension officers. The expected benefits from beekeeping are shown in Table 6 below.

Inputs include:

- identification of areas for beekeeping activities;
- provision of modern beehives and equipment; and
- capacity building and ongoing support to beekeepers and extension services.

Table 6. Economic benefits from beekeeping production for one group of about 10 people (20 beehives).

Benefits from beekeeping	In the first year	Over the course of the project (5 years)	Over 10 years	Assumptions
Honey produced (kg)	240 Kg	1,200 Kg over the course of the project	2,400 Kg over 10 years	1 beehive produces 12 kg per year. Each beekeepers group is provided with 20 beehives for 1 village
Honey production value	816 USD	4,080 USD	8,160 USD	1 Kg of honey is sold at 8,000 TZS (3.4 USD) to the local market

Demand for honey is established in the region and likely sufficient to support the business model. The target area, and particularly Kibondo district, leads as major producer of honey and beeswax in the region[172], with an annual production of honey of around 181 tonnes. Kibondo was the first district in Kigoma region to produce honey and beeswax. The Kibondo Beekeepers Cooperative Society was founded in 1992 and currently involves 43 beekeeping operations. The

[170] See for example: Tutuba and Vanhaverbeke (2018). Beekeeping in Tanzania: why is it beekeeping not commercially viable in Mvomero? Africa Focus, Vol. 31; Honey and beeswax value chain analysis in Tanzania (2007). A study conducted by Match Maker Associates Limited, commissioned by Traidcraft and SME Competitiveness Facility; Mwakatobe and Mlingwa (n.d.). Tanzania - The status of Tanzanian honey Trade- Domestic and International Markets.

[171] Hausser and Mpuya (2004). Beekeeping in Tanzania: when the bees get out of the woods. An innovative cross-sectoral approach to community-based natural resource management. Game and Wildlife Science, Vol. 21

[172] Kibondo Socio Economic Profile, 2016.

cooperative provides processing, packaging and labelling for its members and supports them by linking them to markets and establishing contracts with local companies (for example, Jasmine Bee and Upendo Honey). Enabel works to support honey value chains in the region. The project will support new beekeeping groups in improving product quality, packaging and marketing through training, equipment and advisory investments in order to facilitate access to the existing honey value chains in the Kigoma region.

More information on market constraints and opportunities are in Section 7.4.3 of Annex 2.1 Feasibility Study.

<u>Sub-activity 3.3.2. Promote mushroom cultivation for marginalised groups with training and mushroom growing facilities by providing equipment and training to 500 people</u>

This activity aims at supporting livelihoods diversification for both host communities and refugees, by establishing mushroom growing facilities and providing technical assistance to farmers and extension officers to effectively upscale this activity. This project will establish one mushroom growing facility for each village group (~20) and two facilities for each refugee group (~5 groups), for a total of 30 facilities established under the project.

Mushroom harvesting is a well-diffused practise in the region, particularly during rainy seasons. While mushroom have traditionally been eaten in Northern and Western Tanzania, most farmers do not grow them commercially. Mushroom harvesting in the miombo woodlands is a widely practised activity during rainy seasons, especially by women, who can earn considerable incomes (up to USD 650 per season)[173]. During consultations with women-only focus groups, several participants noted their preference for mushroom farming to beekeeping, as beekeeping is perceived to be more of a men's activity, but there is general interest in both activities. Small-scale mushroom farming has been used as a strategy for livelihood diversification in similar contexts where changes in rainfall patterns (more intense rainfall over shorter period of times) have affected communities[174]. In Tanzania, communities reliant on forests have been supported to develop sustainable enterprises around forest resources such as mushroom and beekeeping as a strategy to better manage forests[175]. Other programmes in different regions of Tanzania have promoted mushrooms growing as livelihood diversification for vulnerable groups and women, and as community-based income generating activity[176],[177].

Mushrooms are short-cycle, high-yield crops, and offer an alternative to currently cultivated crops both in terms of potential income opportunities and higher nutritional value. With continuous income from climate-friendly business activities different from crop cultivation, the communities will be better prepared for future climate change impacts. Existing studies and projects in the country have demonstrated that communities who harvest mushrooms have a higher seasonal income during rainy seasons, when crops are often damaged, leading to food shortages. Although climate and land conditions are not anticipated to present a risk to establishing growing facilities because of controlled humidity and temperature systems, the chosen species (*Pleurotus flabellatus*) was specifically selected because of its adaptability to warmer and wetter climates. In the selected districts, pilot projects that established mushrooms facilities have been implemented by a local NGO (REDESO) in the refugee camps and host communities, demonstrating the appropriateness of the species in the selected environment. Mushrooms are already a food crop in the region, and they are part of the local diets and used for medicinal purposes. Mushroom cultivation also provides opportunities for improving the sustainability of farming systems through the recycling of organic matter, which can be used as a growing substrate, and then returned to the land as fertilizer, or it can be used as fodder for poultry. Lastly, mushroom cultivation is a good livelihood option for farmers with limited or no available land. Existing pilots from REDESO served as stimulation of market demand. While host communities and refugees will be first targets, an additional large customer base is constituted by hotels and agencies in the region. While host communities and refugees will be first targets, an additional large customer base is constituted by hotels and agencies in the region. Fresh mushrooms need to be sold close to where they are produced, to reduce the amount spoiled during transportation. As there is a larger market for fresh mushrooms than for dried mushrooms in Tanzania, the limited availability of technology for preservation represents a significant constraint to expansion. The lack of reliable methods to transport mushrooms to markets also inhibits farmers from expanding their operations. However, dry mushrooms for food and medicine are still in demand in

[173] Chelela et al. (2014). Wild edible mushroom value chain for improved livelihoods in Southern Highlands of Tanzania
[174] https://www.ndf.fi/news/ncf-mushroom-farming-generates-new-income-streams-women.
[175] https://www.farmafrica.org/tanzania/forest-management-in-tanzania; https://www.reuters.com/article/us-tanzania-forest-mushrooms/taste-for-mushrooms-helps-tanzanian-farmers-protect-forest-idUSKCN0ZY0V8
[176] USAID's Innovations in Gender Equality programme, https://www.landolakes.org/resources/success-stories/more-than-a-mushroom
[177] Case study: A community mushroom business in the Kilimanjaro region, FAO (2009) FAO Diversification Booklet. Make Money by Growing Mushrooms. 2009

many areas and are sold in markets in multiple regions, including Kigoma[178 179 180]. Existing demand for mushrooms coupled with parallel activities by local organizations in stimulating market demand, should allow for the viability of the activity. More information on the market for mushrooms and references are in Section 7.4.3 of Annex 2.1 Feasibility Study. The expected benefits from mushroom growing are shown in Table 7 below.

Table 7. Economic benefits from mushroom growing for each group of about 20 people (1 facility per group).

Benefits from mushroom growing	In the first year	Over the course of the project (5 years)	Over 10 years	Assumptions
Harvested mushroom for each facility (kg)	900 Kg	1,210 Kg per year after the first year 5,740 Kg over the course of the project	11,790 Kg	10 kg of harvested mushroom per day, harvest each 3 days after the first 3 months
Mushroom production value	1,953 USD	12,455 USD	25,584 USD	5,000 TZS (2.2 USD) per Kg of mushrooms sold to the local market

Component 2. Strengthened policies for climate responsive planning and development

Output 4. Information on climate change adaptation disseminated and mainstreamed into policies, plans and strategies in Tanzania and in humanitarian programmes.

The GCF funding request for this Output is US$ 469,554. The Government of Tanzania Vice President's Office will be the executing entity responsible for the implementation of Output 4.

Responding adequately to increasing climate-related risks in the refugee camps and host communities requires an integrated approach to land-use planning, natural resource management and local livelihoods. Promoting an evidence-based policy-making approach to achieve this will require decision-makers and stakeholders, including those involved in National Adaptation Planning, NDC development, and subsequent iterations of the National Climate Change Strategy processes, to be well informed on the benefits of the ecosystem-based adaptation (EbA) approach. In addition, policymakers and institutions will need to be capacitated with tools and knowledge to mainstream adaptation into policies, plans and strategies in Tanzania and in humanitarian programmes.

Within the UNHCR, institutional changes to address environmental challenges more effectively — and particularly climate change — will be prioritised. Although environmental considerations are accounted for in most aspects of UNHCR's work with refugees and returnees, the Global Compact on Refugees (GCR) calls for strengthened partnership with development partners, private sector and other stakeholders to address challenges related to energy and environment beyond UNHCR's direct engagement in this domain. Within the UNHCR, the proposed GCF project is recognised as a key partnership effort for the operationalization of the GCR. Lessons from the project's approach and implementation will be documented and shared to guide UNHCR's global engagement with environmental and climate change actors, as well as to inform operations in other countries. Dedicated teams have been established to strengthen the operationalisation of the GCR as well as to support UNHCR's work on mitigation and adaption to climate change. Moreover, corporate commitments to climate change and environmental issues as well as organisational structures at UNHCR are in favour of the institutionalization of the approach set forth by the project.

Output 4 will use the results of the planning processes and physical interventions implemented under Outputs 1–3 to demonstrate the economic, environmental and social value of this model to stakeholders. This approach will contribute to the sustainability and scalability of the outputs and fund-level outcomes at the national level. At the global level, the model will contribute to the growing evidence base on ecosystem-based adaptation. UNHCR is well placed as EE to promote upscaling and replicability of this model in other contexts, and to other UN organisations. UNHCR supports approaches that focus on the integration of effective practises by States and (sub-) regional organizations into their own

[178] Tibuhwa, D.D., 2018. Edible and Medicinal Mushrooms Sold at Traditional Markets in Tanzania. Research Journal of Forestry, 12(1).
[179] Makoye, K. 2016. Taste for mushrooms helps Tanzanian farmers protect forest. Available at: https://www.reuters.com/article/us-tanzania-forest-mushrooms-idUSKCN0ZY0V8.
[180] Kivaisi, A.K. 2007. Mushroom cultivation in Tanzania. Available at: https://www.sustainableagtanzania.com/_webedit/uploaded-files/All%20Files/agronomy/Mushroom%20Cultivation%20in%20Tanzania.pdf

normative frameworks and practises, in accordance with their specific situations and challenges. As Tanzania is a frontrunner in applying the One UN working modus, UN Agencies, including UNHCR, are often invited to contribute to national policy and development dialogues, with UNHCR also participating in the Development Partners Group on environment. The lessons from this project will be used to inform UNHCR's contributions and inputs to these policy processes under Output 4. The broader environment forums would also be used to disseminate lessons and findings from the project, which would produce policy recommendations and briefs. Through this, the project will provide an adaptation model that can be replicated to ensure that the world's most vulnerable populations benefit from climate finance.

Direct beneficiaries will include those recipients of the reports, communication products, policy briefs, and guidelines produced under the activities, as well as those involved in the ongoing policy processes. These activities will also contribute to upscaling the activities of the other Outputs, to extend the range of beneficiaries to similar refugee-host community contexts elsewhere.

The project acknowledges the intersectional systems of privilege and access that operate within the target sites, that may exclude women — particularly refugees or poor farmers —from consultative and decision-making processes and spaces. Additionally, women's needs, interests and knowledge are often marginalised when planning for, and implementing adaptation interventions. To address this inequality, the project will work to ensure women are included in the activities of Output 4, their needs, knowledge and interests are captured, and communication products are gender-sensitive in their content, design and distribution.

Activity 4.1. Generate evidence of the economic benefits of ecosystem-based adaptation to host and refugee populations, for use by policymakers and planners

To prioritise an ecosystem-based adaptation (EbA) approach, decision-makers need to understand and appreciate the relevant interlinked ecological processes and how they directly produce economic benefits. The analysis would consider: i) avoided costs — impacts or losses that would otherwise occur in the absence of ecosystem services, such as reduced food security or decreased water quality; and ii) replacement costs — the implication of implementing 'grey' artificial systems (as opposed to 'green' nature-based solutions) that would be required to deliver the same services as ecosystems. The tangible services provided by local ecosystems include food provisioning, sustainable biomass used for fuel, flood reduction, and improving water quality and water harvesting. As described in Section D.3., additional long-term benefits of these services are carbon sequestration, soil nutrient cycling, waste decomposition and numerous cultural and social services. These benefits also include reduced risk of water- and vector-borne illnesses during flood incidents, as well as a reduced disease burden from exposure to tsetse flies and risk of contracting sleeping sickness.

By recognising the positive environmental, economic and social co-benefits to communities of these ecosystems, this acknowledgment will help promote cross-sectoral cooperation and integrated planning to conserve and augment these natural resources. The sub-activities under this activity will achieve this by determining the economic value of the ecosystem services provided in the project site, conducting assessments and modelling the added value of EbA benefits, based on local livelihoods and primary resource dependence.

<u>Sub-activity 4.1.1. Identify benefit and cost streams for valuing ecosystem services under different climate scenarios.</u>

Project-delivered benefits and costs to be evaluated will be identified, to measure and value the ecosystem services provided by healthy vegetated ecosystems, water ways and areas of ponding. These benefits include flood mitigation (increasing lag time and decreasing flood peak), changes to water quality and quantity, enhanced soil health and reduced erosion, primary production, and social and cultural services, such as health. Costs would include direct implementation expenses, core institutional and enabling costs, opportunity costs, and unintended social and environmental losses (if any).

<u>Sub-activity 4.1.2. Identify data streams and modelling methodology for valuing ecosystem services.</u>

This sub-activity will determine the data required to conduct these valuations, as well as the modelling methodology for determining: i) avoided costs under hydrologic impact mitigation; and ii) replacement costs to replicate benefits of effectively functioning ecosystems for water and soil quality, and primary agricultural productivity.

The identification and valuation of ecosystem services provided by the ecosystems referenced above will be conducted under different climate change projections and a sensitivity analysis will be carried out determining critical degradation thresholds for effective flood mitigation, and production enhancements. The valuation will be undertaken through a variety of market and non-market methods, such as direct damage assessment, institutional and stakeholder analyses, spatial and temporal analyses of changes in the landscape and studies on people's willingness to accept compensation for losses.

Sub-activity 4.1.3. Gather primary and secondary data from project interventions.

Following the identification of the methodology in Sub-Activity 4.1.2, primary and secondary data will be gathered to inform the valuation modelling in Sub-Activity 4.1.3. This will include quantitative and qualitative data, with elements such as spatial information, landcover and land-use data, socio-economic population data, livelihood analyses, survey responses, health statistics, market assessments, institutional analysis, stakeholder mapping and soil and vegetation surveys.

Sub-activity 4.1.4. Carry out valuation modelling exercises of benefit and cost streams identified under 4.1.1.

Based on the methodology determined under Sub-Activities 4.1.1–4.1.2 and the data collected under Sub-activity 4.1.3, a valuation process will be conducted, including data analysis, model calibration and computation.

Sub-activity 4.1.5. Consider temporal, spatial and distributional impacts and opportunities resulting from the costs and benefits evaluated in Sub-activity 4.1.4

This sub-activity would consider the rate at which habitat recovery (afforestation, agroforestry, flood interventions) restores ecosystem services as well as the timing of other benefit and cost streams, as identified in Sub-activity 4.1.1. This will consider gains and losses for up- and down-stream refugee and host communities. Additionally, this would evaluate changes in resource access or ecosystem services-linked income opportunities, amongst key population groups (gender, urban-rural, refugee-host community, youth etc.). In addition to considering the impacts, this sub-activity would also appraise opportunities in the landscape and affected communities, including potential for conflict and synergy, finance solutions to optimise benefit streams (such as payment-for-ecosystem-services (PES)[181] or redirecting revenues towards sustainable production modalities/conservation activities.

Sub-activity 4.1.6. Prepare full and summary reports.

The analysis and findings of the economic assessment will be compiled into full and summary reports — highlighting the interdependencies and sensitivities based on the redundancy hypothesis of the ecosystem — to be distributed to stakeholders. Briefing notes will be developed and working sessions held with key decision-makers, with the objective of communicating the evidence of benefits of EbA, providing specific policy recommendations and looking at opportunities for further engagement and investment. This knowledge base will help to mainstream these kinds of activities into local and district decision making, as well as being scaled globally through UNHCR and partners.

Activity 4.2. Develop communication products to disseminate project results

To scale the impacts of the proposed interventions, it is necessary to disseminate the results and lessons of the project to inform similar interventions in the future. This information, and the tools they inform, will help policy- and decision-makers in Tanzania and UNHCR's humanitarian programmes to mainstream climate resilience into village land-use planning, as well as to make use of Savings Associations to scale up climate-resilient agricultural practises. Products will also be shared more broadly to stakeholders involved in the National Adaptation Planning, NDC development, and subsequent iterations of the National Climate Change Strategy processes.

Sub-activity 4.2.1. Develop policy briefs on mainstreaming climate resilience in village land-use planning.

Capturing the lessons learned from the climate-resilient village land-use planning implemented under Activity 1.1, policy briefs will be compiled and distributed to key stakeholders, including the National Land Use Planning Commission, to inform recommendations for policy review and revision. Stakeholders targeted during distribution will include representatives from district and regional government, host villages, as well as refugees and camp authorities. These lessons are then scalable to other, similar, contexts within Kigoma and Tanzania at large, as well as in other refugee contexts globally.

[181] For example, this could consider diverting funding and resources for WASH in Kigoma (which is relatively readily available from humanitarian funders) towards PES interventions to support ecosystem conservation that improves water quality and quantity.

These policy briefs, and the guidelines developed under Sub-Activity 4.2.2 will improve upon existing land-use planning by addressing the gaps in mainstreaming climate change adaptation, planning integrated landscape approaches and incorporating the different needs and challenges of host communities and refugees. They will also allow for the approach piloted here to be scaled to other settings, to make them more climate resilient.

Topics for these briefs may include:
- capacity building of local institutions and anchoring of refugee-host community land use planning and adaptation activities to local governance in the hosting country
- effectiveness of peaceful coexistence meetings, dialogues, and design of benefit-sharing arrangements in building consensus and mitigating conflict to achieve common climate change adaptation objectives
- planning for, and implementing ecosystem-based adaptation (EbA) interventions;
- finance solutions to optimise benefit streams (such as payment-for-ecosystem-services (PES) or redirecting revenues towards sustainable production modalities/conservation activities;
- including agroforestry or similar interventions as a tool for climate resiliency in host community land use planning;
- including vulnerable groups (such as refugees and women) in participatory land-use planning processes;
- effectiveness of sustainable production and benefit sharing arrangements for fuelwood in reducing gender-based violence and conflict; and
- navigating the dynamics of refugee and host community relationships, to ensure durable and sustainable social-ecological ecosystems.

<u>Sub-activity 4.2.2. Develop a guidelines paper on mainstreaming climate resilience in village land-use planning.</u>

To support the implementation of the policy briefs developed under Sub-Activity 4.2.1., guidelines will be developed and distributed to inform decision-makers, planners and authorities on how to mainstream climate resilience into the design, implementation and monitoring of village land-use plans. The guidelines will include options for interventions in different contexts and institutional responsibilities for enforcement, monitoring and implementation. In addition, the guidelines will offer detailed guidance on the processes of:
- defining the climate problem threatening the camps/villages, including its impacts on women, men and vulnerable social groups;
- best practices and effective strategies in consensus building and negotiation in fair and equitable allocation of land use across different resource user groups, including pastoralists;
- selecting intervention sites and assessing impact scenarios with and without climate-resilient land-use planning;
- identifying how impact mitigation targets can be met using land-use planning;
- incorporating agroforestry, afforestation, and other EbA-type interventions into sustainable village planning and management;
- estimating costs and benefits of climate-resilient land-use planning;
- implementing and updating of climate-resilient land-use planning, including identifying appropriate sustainable financing strategies; and
- monitoring and evaluating climate-resilient land-use planning.

<u>Sub-activity 4.2.3. Produce lessons-learned and guidelines paper on the use of Savings Associations to scale up climate resilient agricultural practises.</u>

Capturing the lessons from the use of Savings Associations (SA) in host communities to scale up climate-resilient agricultural practises demonstrated under Activity 3.1, findings will be compiled and distributed to key stakeholders to inform future scaling up or replication of these interventions. Stakeholders targeted during distribution will include groups at various levels, from SAs themselves to locally-operating NGOs, high-level district planners and multi-lateral development partners.

In order to support the implementation of these findings, guidelines will be developed and distributed to inform users, decision-makers and development partners on planning, implementing and monitoring the use of SA's to scale up climate-resilient agricultural practises. The process to develop these guidelines would include a review of how SA's in different countries/contexts are being used for financing climate adaptation. The relative advantages and drawbacks of the Tanzania model will be benchmarked against other contexts, before and after implementation of project activities.

The guidelines will include options for adapting this model in different contexts (in Tanzania, and globally) as well as defining the institutional responsibilities for enforcement, monitoring and implementation. The guidelines will build upon or adapt the Operational Manual Outline for the Savings Association Management Offices included in Appendix C. In addition, the guidelines will offer detailed guidance on the processes of:

- defining the climate problems facing agriculture, including its impacts on women, men and vulnerable social groups;
- selecting intervention sites and assessing impact scenarios with and without the use of SA's to scale up climate-resilient agricultural practises;
- identifying how impact mitigation targets can be met using SA's to scale up climate-resilient agricultural practises;
- estimating costs and benefits of using SA's to scale up climate-resilient agricultural practises;
- implementing and maintaining the use of SA's to scale up climate-resilient agricultural practises, including identifying appropriate sustainable financing strategies; and
- monitoring and evaluating the use of SA's to scale up climate-resilient agricultural practises.

Activity 4.3 Draft revisions to key plans and policies and support their integration into national and district government planning processes to promote up-scaling of the EbA model

This activity will focus on the development-humanitarian-climate nexus (Figure 18), to bridge the gaps and discordance between clusters within this space. Various institutions — from government, international organisations and civil society — oversee the various elements of these intersecting clusters, with divergent priorities, timelines and target beneficiaries. This creates silos, overlaps and inefficiencies that produce barriers to achieving integrated, sustainable and inclusive development in Kigoma. The coordination of the entities involved in the design and implementation of this project — namely <u>the Government of Tanzania Vice President's Office (VPO)</u>, UNHCR and UNEP and — are well placed to facilitate this integration, in a manner that is scalable to other refugee settings. Linkages will be sought during project implementation with initiatives that are ongoing to close these gaps and to strengthen planning process with Local Government Authorities, such as the One UN Kigoma Joint Programme and the UNHCR co-chaired host community working group.

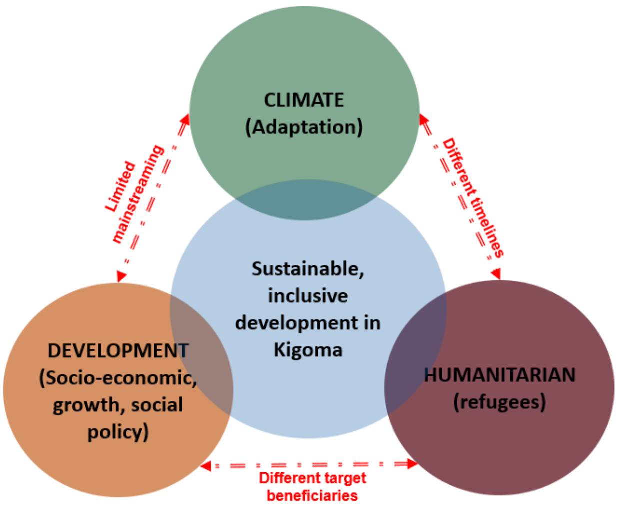

Figure 18. Development-humanitarian-climate nexus.

<u>Sub-activity 4.3.1. Review laws and policies that hinder adaptation in Kigoma region including political risks that need to be closely monitored, and identify entry points for where laws and policies require harmonisation to promote adaptation.</u>

As climate change has not yet been mainstreamed into the legal and policy frameworks governing host communities and refugees in Kigoma, related processes are not well oriented to advancing adaptation may create additional barriers to building climate resilience. This sub-activity will review the relevant laws and policies that govern or are relevant to adaptation in Kigoma region, to assess their role in hindering climate adaptation and identify entry points where harmonisation is required to promote adaptation, including analyses of how current policy targets should be revised to reflect adaptation the country's priorities.

The purpose of the review would be to determine how each sector's entities and plans, policies and practises are aligned (or divergent) from the other sectors. These include district development plans, forest management plans, livelihood development plans, river basin plans, health strategies and water and sanitation plans. The review would examine, for example, the extent to which UNHCR, the Ministry of Home Affairs and other entities operating within the camps/in humanitarian concerns, are embedded in district- and region-level development and spatial planning. Conversely, the review would include analysing the reciprocal role of institutions like the National Land Use Planning Commission (NLUPC), Tanzania Forestry Service (TFS), District government and others in mainstreaming development, adaptation and integrated land use planning and management, WASH and refugee livelihoods as well as the management of relations with the host community.

The process of this review will include a desktop review of relevant policies (see below) as well as engaging with public and private sector stakeholders involved in planning for, and implementing adaptation, in Kigoma region. Various focus groups, workshops and meetings with planners, decision-makers and policy practitioners will be conducted to inform this process. Using a participatory approach will contribute to gaining buy-in from the relevant stakeholders and ensure that the regulations drafted are aligned with national priorities and climate change objectives stipulated in Tanzania's NAPA, (I)NDC and national communications. The laws and policies impinging on adaptation will be determined and mapped, with recommendations developed for addressing these barriers. This review will be undertaken by an international expert working with a national policy expert who is well acquainted with the policies and processes in question.

The project context involves political risks that may impact the implementation of ecosystem-based adaptation activities in the region. These will be monitored by the executing entities following risk indicators (See Risk Factor 10) and in conjunction with regular risk monitoring by UNHCR Senior Risk and Compliance Officer based in Dar Es Salaam and the use of the UNHCR risk registry for monitoring and updating emergent risks.

<u>Sub-activity 4.3.2. Identify legal, policy and planning processes that can be capitalised on to advance the mainstreaming of adaptation</u>.

The Kigoma Regional Administration statutory function, among others, is to provide administrative and technical advice as well as creating a conducive environment for LGAs to deliver quality services. As such the Regional Administration plays a key role in coordinating and supporting environmental interventions, land use planning, as well as agriculture activities in the Kigoma Region. The Regional Administration is therefore an important entry point for planning and coordination of the proposed project interventions.

District planning to prepare the annual workplan and budget under the Medium-Term Expenditure Framework (MTEF) generally takes place in the second quarter of the government fiscal year, i.e. October to December, coordinated by the District Planning Officer. Budget requirements, including investment and recurrent expenses, are revisited and normally submitted to the central government in early in the calendar year. Project activities to be implemented by the local government authorities are to be incorporated and planned for the MTEFs.

To strengthen country ownership and promote uptake, the project will capitalise on the baseline legal, policy and planning processes in Tanzania, leveraging on existing processes at the district and regional levels and involving key stakeholders such as the Tanzania Forest Service and the Lake Tanganyika Basin Water Board. Through stakeholder consultation with policymakers, this sub-activity will identify processes to capitalise upon for implementing the recommendations produced under Sub-activity 4.3.1.

<u>Sub-activity 4.3.3. Prepare guidelines and other inputs for policy- and decision-makers to use to inform in the mainstreaming of adaptation in national and district government planning.</u>

Following identification in Sub-Activity 4.3.2, guidelines will be developed for utilising the processes to mainstream adaptation into village land-use planning. This would include activities under regular medium- and long-term strategic development planning, as well as the National Adaptation Planning (NAP) process, implementation of Tanzania's Country Programme (CP), production of National Communications to the UNFCC, and delivery of the priorities under the Nationally Determined Contribution (NDC). Action plans will then be developed, including timelines, roles and responsibilities for integrating these recommendations into the target policies, plans and strategies.

<u>Sub-activity 4.3.4. Co-host workshops, forums and consultations within ongoing policy and planning processes to showcase project results and implications for the legal, policy and planning frameworks of the country.</u>

Finally, this sub-activity will — in conjunction with the recommendations of the action plan developed under Sub-Activity 4.3.3 — co-host workshops, forums, and consultations to engage stakeholders. These events will aim to demonstrate the outcomes of the proposed project to decision-making stakeholders and distribute the materials and guidance developed under Activities 4.1–4.2. Additionally, in line with the action plan, implications for the legal, policy and planning frameworks of the country will be discussed and validated, to initiate the process of revising or initiating policies and legislation.

B.4. Implementation arrangements (max. 1500 words, approximately 3 pages plus diagrams)

The proposed project will be implemented over a five-year period. The United Nations Environment Programme (UNEP) will be the Accredited Entity (AE), the Government of Tanzania, acting through the Vice President's Office (VPO) and the United Nations High Commission for Refugees (UNHCR) will function as the Executing Entities (EE). The collaboration between these entities will combine UNEP's expertise as the leading global environmental authority, VPO leverage capacity at the national, regional and district-level and climate change adaptation expertise and UNHCR's unique capacity and experience in refugee protection, assistance and solutions.

Accredited Entity

As the Accredited Entity, UNEP will carry out fiduciary and safeguards oversight and provide the necessary scientific expertise and technical support to the project formulation, start up, implementation, evaluations and closure.

Experience and track record of the AE

UNEP has a broad global portfolio on climate change projects funded through, *inter alia*, the Least Developed Countries Fund (LDCF), Special Climate Change Fund (SCCF), Adaptation Fund (AF) and bilateral funds such as the European Commission. Through these projects, UNEP has supported selected national governments and local communities to adapt to climate change. UNEP's knowledge base is derived from completed and ongoing projects supporting: i) methods and tools for decision-making; ii) prioritising, designing and implementing adaptation interventions; iii) enhancing climate resilience by restoring vulnerable ecosystems that underpin community livelihoods; and iv) monitoring the socio-economic and environmental benefits of adaptation interventions. It will draw upon previous experiences and lessons learned during the implementation of the proposed project.

UNEP's comparative advantage over other accredited entities lies in its ability to provide robust scientific/technical advice regarding sustainable national planning and development processes. These skills are supplemented with UNEP's strong capacity in the field of climate change through implementation of climate change adaptation projects at global, regional and national levels. As the GCF AE for the proposed project, UNEP will therefore oversee the efficient and effective delivery of the project's objectives. Lastly, UNEP's in-house policies and expertise on gender mainstreaming as well as social and environmental safeguards will contribute towards gender- and environmentally-sensitive project implementation.

Executing Entities

The Government of Tanzania Vice President Office (VPO) will act as the lead Executing Entity (EE) of the project, ensuring alignment with the national policies and plans as well as complementarity with other existing adaptation projects and Tanzania's NAP process. As the lead EE, VPO will oversee overall project implementation and will be accountable to the AE for the effective use of resources during project implementation. All operating policies and procedures will follow applicable UN rules, regulations, and policies, which include provisions for financial management and procurement. A Partnership Cooperation Agreement (PCA) signed between VPO and UNEP will serve as the Subsidiary Agreement to regulate the obligations of the parties. Compliance with applicable UN rules, regulations, and policies will be reflected in the Subsidiary Agreement. VPO will be responsible for the efficient disbursement of financial resources for the implementation of the designated project activities for the compilation of the required information from all implementing partners, including UNHCR as second EE, to provide consolidated project progress reports and expenditure reports to the AE.

VPO will execute project activities through different mechanisms including direct execution by PMU, cooperation agreements with District Government and TFS as well as contracts with service providers and consultants. The District governments, under the supervision of VPO, will execute delegated activities as the district-level directly and may be authorised to contract NGOs, SAMOs and service providers.

The selection of such partners will be guided by UNEP to ensure that UN requirements are met and with the aim of speeding project implementation. UNEP will provide guidance and support to VPO in the selection of Personnel (personnel, agents, staff, contractors) as well as in the selection of implementing NGOs and the Technical Partner following UNEP partnership policy and procedures, including due diligence process. UNEP will review terms of reference and selection processes to ensure the eligibility criteria are met as well as contracts and agreements issued by VPO to ensure compliance with the procedures and regulations of the PCA, the ESMS (Annex 6) and the Gender Action Plan (Annex 8). The final selection of Personnel (personnel, agents, staff, contractors) and non -governmental implementing partners (NGOs and Technical Partner) will be taken by the VPO after confirmation of no objection by UNEP. This will be reflected in the Subsidiary Agreement (PCA).

Execution by UNHCR is necessary to leverage its mandate on refugee and host community relations and its convening power among the different actors on the ground. As second EE, UNHCR will execute GCF funded project activities (2.3 and 3.2) and contribute to Output 4 activities in collaboration with MoHA and VPO. UNHCR will be responsible for overseeing and managing the implementing partners that will be involved in the implementation of the designated project interventions in the camp areas, prioritizing implementation by local government authorities where possible.

UNHCR will be accountable to the AE for the effective use of resources during project implementation. All operating policies and procedures will follow applicable UN rules, regulations, and policies, which include provisions for financial management and procurement. UNHCR will be responsible for the efficient disbursement of financial resources for the implementation of the designated project activities. The transfer of funds between the AE and UNHCR will be governed by an UN-to-UN Agreement that will serve as Subsidiary Agreement

A Memorandum of Understanding (MoU) between VPO and UNHCR will define the principles of collaboration, team coordination and information sharing mechanisms between the two executing entities. This is particularly important for those activities that will require close coordination between the two executing entities such as 3.2, 4.1, 4.2 and 4.3. Moreover, the MoU will define the project reporting mechanism between the two executing entities. UNHCR will provide progress reports and financial reports to VPO concerning project activities and budget executed by UNHCR. As the lead executing entity, VPO will be responsible for providing consolidated progress and financial reports to the AE as per the PCA.

Experience and track record of the EE Government of Tanzania Vice President's Office

VPO has experience with various projects of similar size and scope with several bilateral and multilateral donors, including ADB, IFAD, UNEP. The current portfolio includes among others:

- IFAD/ GEF 9132 'Reversing Land Degradation trends and increasing Food Security in degraded Ecosystems of Semi-Arid Areas of Tanzania (LDFS' (July 2017- September 2022, US$ 7 M GEF grant plus US$ 42,94 M co-finance).
- UNEP/GEF 5691 'Sustainable Land Management of Lake Nyasa Catchment in Tanzania´ (February 2017 – December 2021 extended to December 2023, US$ 1,3 M GEF grant plus 4,4M co-finance),
- UNEP/GEF 9524 'Supporting the implementation of integrated ecosystem management approach for landscape restoration and biodiversity conservation in Tanzania (December 2020 – December 2026, US$ 11,2 M GEF grant plus US$ 64M co-financing)).
- UNEP/GEF5659 'Ecosystem-Based Adaptation for Rural Resilience in Tanzania approved September 2014 with UNEP as Accredited Entity. GEF financing US$ 7.5 M grant and co-financing of US$ 20.1M).

Other closed projects include:

- UNEP/ GEF LDCF 'Developing core capacity to address adaptation to climate change in productive coastal zones'. (November 2012-November 2019, GEF financing US$ 3.3 M grant from GEF LDCF and co-financing of US$ 41,943)
- UNEP/Adaptation Fund 'Implementation of concrete adaptation measures to reduce vulnerability of livelihoods and economy of coastal communities of Tanzania'. (November 2012-March 2017, US$ 4.6 M grant from the Adaptation Fund)

Experience and track record of the EE (UNHCR)

UNHCR's comparative advantage is in its expertise and mandate in the refugee context. As the vast majority of ~68.5 million 'persons of concern' are concentrated in 'climate change hotspots' around the world, UNHCR has a long history of working to address climate change issues in refugee-hosting contexts — especially through promoting the inclusion of refugees into local climate change adaptation programming. UNHCR has three field offices in the project area (in Kigoma town, Kibondo, and Kasulu) which works with the Ministry of Home Affairs (MoHA) and local government authorities. In addition to its humanitarian activities, UNHCR also supports local communities with a large portfolio of relevant projects including in forestry, water conservation and sustainable energy. Environmental considerations are taken into account in almost all aspects of UNHCR's work with refugees and returnees. Beyond UNHCR's direct engagement in this domain, the Global Compact on Refugees calls for strengthened partnership, including with development partners, private sector and other stakeholders to address issues related to energy and environment. The project with the GCF is recognised as a major partnership effort to operationalise the GCR. In line with the principles on the Global Compact on Refugees, UNHCR Tanzania has forged partnerships with a range of development partners, UN agencies and the private sector to address energy and environment challenges in the camps and facing the host communities. UNHCR will build on these experiences and partnership while implementing the project. Additionally, dedicated teams have been set up to strengthen the operationalisation of the GCR as well as to support UNHCR's work on mitigation and adaption to climate change. Quality assurance and technical support will be provided by UNHCR's Global Technical Environment Teams located in UNHCR HQ in Geneva.

UNHCR has a long-standing commitment to reducing the environmental impact of refugee crises, for example by launching climate change mitigation programmes which includes a shift to renewable energy options for refugees, reforestation activities, and enhancing refugees' access to clean modern fuels and technology for cooking to reduce dependence on charcoal and firewood. UNHCR seeks to increase the sustainable use of renewable energy sources to minimise environmental impact, in a way that includes host communities and other stakeholders, while improving refugees' protection and well-being. To further mainstream refugees' access to sustainable energy and minimize climate impacts impact across its operations worldwide, UNHCR launched a five-year *Global Strategy for Sustainable Energy* in 2019. In the strategy UNHCR commits to improving its environmental performance, climate compatibility and resource efficiency by reducing its use of fossil fuels, purchasing green energy from reliable suppliers and other sustainability initiatives. In line with the GCR, the strategy strongly promotes partnerships and innovation, expanding UNHCR's catalytic role to promote sustainability in displacement settings. This includes developing joint programs between humanitarian, development and environmental actors, as in the project proposed here. It further includes engagement with private sector and advocating for innovative financing mechanisms. For example, in 2019 UNHCR formed a partnership with Swedish International Development Cooperation Agency (Sida) to phase out fossil fuels from UNHCR operations. The partnership established UNHCR's first ever rotational fund to help operations transition to renewable energy. Specifically, the fund will provide de-risking measures and other incentives to encourage private sector renewable energy providers to move into underserviced refugee hosting areas.

UNHCR also engages to support climate change adaptation for refugees and host communities, working with partners to design and deliver climate change adaptation programmes in vulnerable countries. UNHCR through its environmental partner - REDESO, supported establishment of community and individual woodlots within host communities in Kakonko and Kibondo district. In Kasulu, in addition to UNHCR's interventions implemented by CEMDO, SNV supported an individual farmer-based woodlot management program focusing on future energy sustainability for both host and refugee communities. Through this intervention, 245,000 fast growing tree seedlings were planted[182]. Some key examples of UNHCR's global initiatives for climate compatibility in refugee hosting areas include: the establishment of solar farms in Jordan's Azraq and Za'atari refugee camps; biogas fuel and solid waste treatment plants for refugees in Bangladesh; renewable biomass cooking energy access for refugees in Rwanda, Tanzania and Ethiopia and land restoration projects such as the "Green Refugee Camp" programs in Cameroon and Sudan. UNHCR has also developed partnerships with agencies like GIZ to include refugees into their wider, climate and energy programs in countries including Uganda, Kenya, Ethiopia and Tanzania. This is reflected in this project's parallel finance through the GIZ RESPOND project in Kigoma, which seeks to provide renewable energy access and reforestation services in refugee hosting areas. Staff in UNHCR Tanzania relevant to adaptation and landscape management include a Senior Development Officer, Environment Officer and Energy Officer, among others. The partnership with UNEP will leverage their expertise in EbA.

Project compliance with the UNHCR policy on Age, Gender and Diversity (AGD)[183] is mandatory. UNHCR applies an age, gender, and diversity (AGD) approach to all aspects of its work, applying participatory methodologies to promote the role of women, men, girls, and boys of all ages and backgrounds as agents of change in their families and

[182] These interventions were done by partners outside of this project.
[183] https://www.unhcr.org/protection/women/5aa13c0c7/policy-age-gender-diversity-accountability-2018.html

communities. UNHCR issued key policies and tools such as the 2006 Tool for Participatory Assessment in Operations, the 2008 Manual on a Community-Based Approach in UNHCR Operations, and the previous 2011 Age, Gender and Diversity Policy and organized global consultations with women and youth, to better incorporate their views into the development of policies and tools. As part of its AGD approach, UNHCR has made progress in promoting gender equality in its operations: mainstreamed the inclusion of women and girls in decision-making processes, ensured individual registration for females and worked to prevent and respond to SGBV.

Implementing Partners

The Government of Tanzania, acting through its Vice-President's Office (VPO), as lead EE will oversee and manage project implementation, with and through appropriate implementing partner organisations including local government authorities and, if and when required, engaging technical experts, and national and international NGOs. Selection of the technical experts and national and international NGOs will be undertaken with UNEP guidance.

Government agencies and UN agencies

For the implementation of project activities, the Government of Tanzania Vice President's Office (VPO) will establish cooperation agreements with local government agencies and institutions selected on the basis of their mandate and their relevant role in facilitating project execution and sustainability of project outcomes.

The District Councils are the mandated authorities in land use planning and development at the district level and will therefore play a key role in convening all technical officers including agriculture to inform the project annual workplan and to discuss project implementation progress and adaptive management approaches, ensuring the project is anchored in local governance processes and that results are sustainable beyond the project duration. VPO will sign cooperation agreements with the three respective District Councils for the implementation of activities 1.1, 2.1, 2.2, 3.1, 3.2.3, 3.2.4 and 3.3. under the supervision of VPO. This type of project implementation arrangement between VPO and District Councils is already in place in other ongoing projects such as the GEF/UNEP project "Ecosystem-Based Adaptation for Rural Resilience in Tanzania".

VPO will also sign one cooperation agreement with the Tanzania Forest Service (TFS), authority responsible for forestry interventions and forest monitoring Tanzania, for the implementation of activities 2.1 and 2.2 in coordination with the respective District Councils.
Ministry of Home Affairs (MoHA) will play a key role in overseeing the implementation of activities in the refugee camps, coordinating with UNHCR the planning and monitoring of camp activities. This will include consultation on the location and design of infrastructure in the camps. MoHA will also provide technical guidance and oversight on the performance of NGOs and service providers executing activities in the camps.

National Environmental Management Council (NEMC), particularly NEMC office in Kigoma, will be responsible for ensuring compliance with relevant environmental laws and regulations as well as providing technical advisory and compliance guidance on all environmental aspects of the project to the two executing entities and the implementing partners.

The Kigoma Regional Authority will host the PMU, provide support to the delivery of activities in the region and oversee the implementation of activities delegated to the District Councils. The Kigoma Regional Administration, where the PMU will be located, will convene all technical officers including land use planning officers, agriculture officers (representatives of Ministry of Agriculture at district level), engineering officers, etc. at regular intervals, to inform the project annual workplan and to discuss project implementation progress and adaptive management approaches.

Research and technical expertise organizations

A Technical Partner with expertise in landscape approach, agroforestry, afforestation and GIS will be identified as a key project implementing partner to provide technical advisory and capacity building services to District officials, TFS officials and participant communities in the implementation of Output 1 and Output 2. Specific tasks of the Technical Partner include:

- Technical advisory on the establishment of the Village Land Forest Reserves and Community Based Forest Management.
- Technical advisory on the design and monitoring of agroforestry interventions.

- Technical advisory on the design and monitoring of afforestation interventions.
- Development of a GIS-based forest monitoring system integrating satellite, drone and field spatial data and capacity building to users (building on the existing TFS tools)
- Social and ecological baseline assessment.
- Assessment of tree species for afforestation.

The selected Technical Partner will be a non-for-profit competent and reputable organization with expertise in forestry, agroforestry and landscape restoration approach, proven experience and capacity to deliver on the above tasks with high quality standards. The Technical Partner will be selected by VPO through a comparative analysis process in which at least three short-listed candidates will be considered. UNEP will provide support for the selection of the Technical Partner in line with UNEP partnership policy and procedures.
A Pre-requisite for the Technical Partner is that the organization must align to UN values and must not be complicit in human rights abuses; tolerate forced or compulsory labour or the use of child labour or violate sanctions established by the UN Security Council. A partner due diligence assessment following UNEP Partnership policy and procedures will be carried out at two levels:

- Level one due diligence process assessing conflict of interest, human rights and gender equality, alleged involvement in cases of irresponsibility/litigation/green washing and reputational risk
- Level two due diligence process assessing specific technical capacities, strategic capacities, financial capacity to undertake the partnership and procurement procedures.

The engagement of a Technical Partner is expected to have an added value in terms of quality assurance and knowledge transfer to national stakeholders, thereby enhancing the sustainability of the project outcomes. The partnership will be governed by a cooperation agreement with VPO.

Non-government organizations

The contracting of qualified national and international non-governmental organization partners on the ground will be determined using a competitive selection process[184] and is not pre-determined. The general criteria for selecting NGO implementing partners include:

  - Non-profit organizations
  - Legally registered internationally or nationally, with registration and permit to operate in Tanzania.
  - Sector expertise and experience: the required specific skills, sector specialists, knowledge and human resources.
  - Project management: ability to deliver project objectives, accountability mechanisms and sound financial management,
  - Local experience and presence: ongoing programme in the area of operation; local knowledge; engaging refugees and other persons of concern.
  - Contribution of resources: evidenced and documented contribution of resources to the Project in cash or in-kind
  - Cost effectiveness: level of direct costs and administrative costs imposed on the Project in relation to project deliverables.
  - Experience working with UNHCR (only for UNHCR partners): Partners that have three consecutive qualified audit opinions for UNHCR-funded projects may not be considered.
  - The Partner's capacity to procure. Clarification is obtained on the capacity of the potential partners and their processes for conducting procurement in compatibility with UN procurement principles
  - Align to UN values, and must not be complicit in human rights abuses, tolerate forced or compulsory labour or the use of child labour or violate sanctions established by the UN Security Council.

VPO will be responsible for the competitive selection of NGOs in alignment with UNEP partnership policy and procedures. Due diligence assessment will be carried out at two levels:
-Level one due diligence process assessing conflict of interest, human rights and gender equality, alleged involvement in cases of irresponsibility/litigation/green washing and reputational risk
-Level two due diligence process assessing specific technical capacities, strategic capacities, financial capacity to undertake the partnership and procurement procedures.

Implementing partners engaged by UNHCR for the implementation activities 2.3 and 3.3 in the camps may be drawn from UNHCR's existing partnerships and other organisations operating in the region. This process ensures cost

[184] The selection processes are compliant with applicable UN rules, regulations, and policies.

effectiveness of implementation and taps into the subject matter expertise and depth of partners' experience in the Kigoma host community and refugee context, as the project's approach is to leverage lessons learned from previous and existing initiatives and anchor on local institutions. The project design has already benefitted from interactions for partners integrating lessons learned. Coordination mechanisms will be put in place to ensure lessons learned are leveraged in the project. Following the engagement of implementing partners, UNHCR implements risk-based approaches to implementation, including a strong monitoring framework for financial management and procurement by implementing partners, options for UNHCR to procure on behalf of implementing partners, and adequate controls on ensuring resources are used for intended purposes and right of use agreements. In Tanzania, UNHCR already engages, through national and international implementing partners, in activities such as nursery development, afforestation and other similar activities benefitting both refugees and host communities. Table 8 includes the list of UNHCR partnerships for 2020/21 activities in Tanzania. The flow of funds between UNHCR and implementing partners such as NGOs and other service providers is governed by a Project Partnership Agreement or in the case of firms or individuals other applicable contracting arrangements. The flow of funds between VPO and NGOs will be governed by cooperation agreements.

Coordination

Linkages and coordination with the Ministry of Home Affairs for activities in the camps, the National Environmental Management Council, National Land Use Planning Commission, Tanzania Forest Services, Regional Administration, District Councils and other relevant agencies will be pursued for all project activities, including the various policy activities under Output 4.

UNHCR coordinates and shares information with a range of other partners working on environment and energy issues. Partners such as Danish Refugee Council, Jane Goodall Institute (JGI) and Enabel are already working on the ground with interventions that complement proposed project activities. For example, Enabel has a track record in Kasulu, Kibondo and Kakonko in implementing CBFM, participatory VLUP, conducting FFS and value chain development in target crops of beans and cassava, and beekeeping capacity building and market development. SNV has experience in establishing and managing woodlots and afforestation in Kibondo and Kasulu, albeit with private modalities rather than on communal property. Jane Goodall Institute has 15+ years' experience in land use planning, focussing on natural resources management whilst making consideration for livelihood enhancement. JGI also has experience in refugee settings, having worked in the old refugee settlement of Mishamo, which hosts Burundians who left their country of origin in the 1970s. The proposed project will draw upon JGI's experience in Mishamo to pre-emptively address the maladaptation and maldevelopment that has built up in those contexts as a result of poor land use planning during early settlement.

The proposed project will also closely coordinate with the Kigoma Joint Programme (KJP) partners including the United Nations Capital Development Fund (UNCDF), Food and Agriculture Organisation of the United Nations (FAO), World Food Project (WFP) and International Trade Centre (ITC) through coordination. Drawing on the complementary activities in the KJP (as described in B.1), the proposed project will apply the experience and lessons-learned from these partners to inform the implementation of key activities.

Agency	Category	Sector/Description
African Initiative for Relief and Development (AIRD)	International NGO	Logistics and Fleet Management
DRC/DDG East Africa and Great Lakes region (DRC)	International NGO	Camp Management, Coordination & Community Empowerment to Refugees in Nyarugusu, Nduta Camps
HelpAge International UK (HAI)	International NGO	Strengthen services for Persons with Specific Needs through an integrated and community-based approach
Norwegian Refugee Council (NRC)	International NGO	Access to WASH – Shelter and Protection services in Nyarugusu, Nduta and Mtendeli former camp
International Rescue Committee, USA (IRC)	International NGO	Provision of Assistance and Protection to Refugees in Nyarugusu, Nduta and former Mtendeli camp
Plan International, United Kingdom	International NGO	Strengthening Child Protection & Youth Empowerment services in Nyarugusu, Nduta and former Mtendeli camp
Relief to Development Society, Tanzania (REDESO)	Local NGO	Environment and Urban Refugees' Assistance
Women's Legal Aid Centre (WLAC)	Local NGO	Access to Legal Assistance and Justice for Refugees and Asylum seekers in Tanzania
Medical Team International (MTI)	International NGO	Comprehensive Health Care and Nutrition Assistance for Refugees in Nyarugusu, Nduta and former Mtendeli camp.
Ministry of Home Affairs (MoHA)	RSD - MHA	Government agency

Table 8. UNHCR Tanzania Implementing Partners for Existing 2020 Activities[185]

Project arrangements

A series of agreements between project actors will allow GCF funds and their associated benefits to flow to beneficiary communities in the region. Figure 19 below summarises the headline agreements and engagements that are required. 'Service providers' include suppliers of equipment and materials, professional/contractual services and independent consultants.

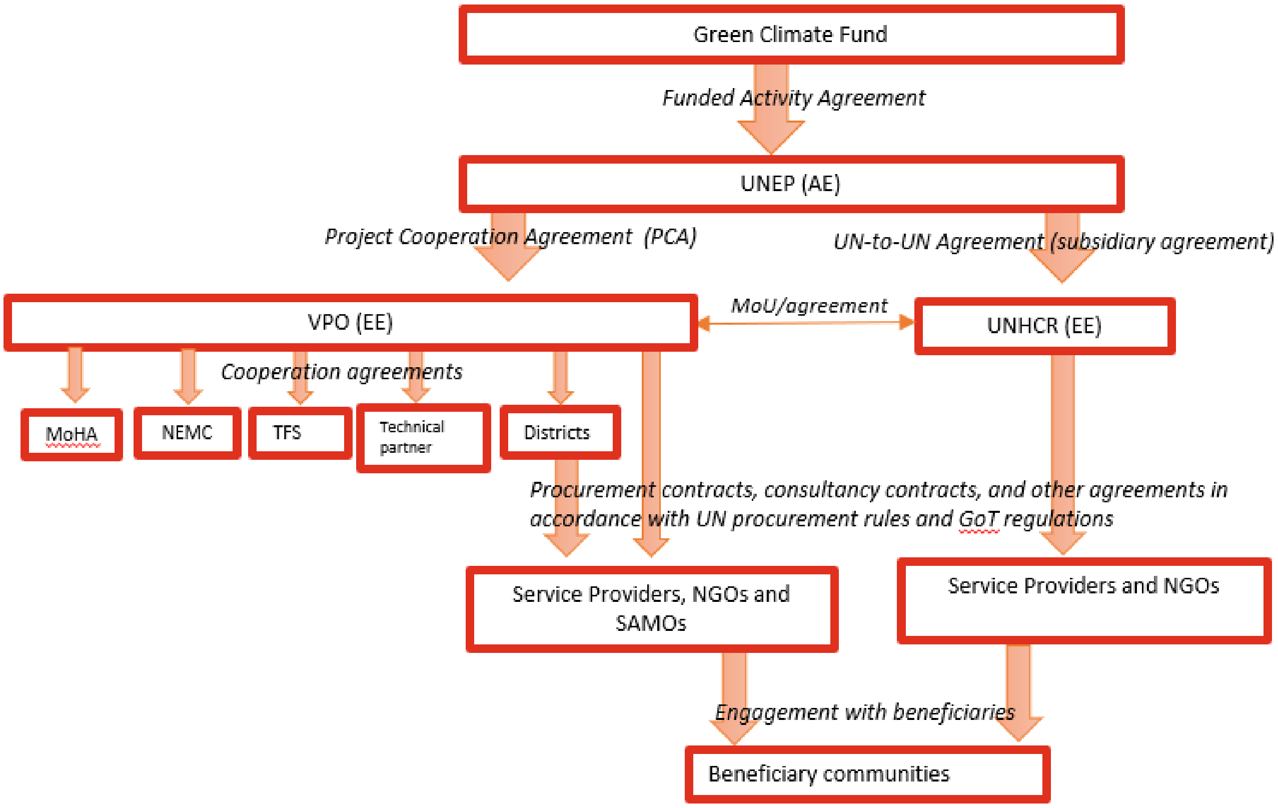

Figure 19. Flow of funds from the Green Climate Fund through to beneficiaries.

The diagram in Figure 20 below provides an overview of the implementation structure for the project.

The Subsidiary Agreements between the AE and the EEs are:
- Project Cooperation Agreement (PCA) between UNEP and the Government of Tanzania Vice President's Office (VPO)
- UN-to-UN Agreement between UNEP and UNHCR.

The coordination and reporting mechanisms between VPO and UNHCR will be specified in a Memorandum of Understanding.

UNEP, the Government of Tanzania Vice President's Office (VPO) and UNHCR will coordinate closely with government agencies including the Ministry of Home Affairs (MoHA), the National Environmental Management Council (NEMC) the Regional Government Authority of Kigoma and Local Government Authorities, the local district agencies and affected communities to deliver the project activities.

Ministry of Home Affairs (MoHA) will play a key role in overseeing the implementation of activities in the refugee camps, coordinating with UNHCR the planning and monitoring of camp activities. This will include consultation on the location and design of infrastructure in the camps.

The National Environmental Management Council (NEMC) will ensure compliance with relevant environmental laws and regulations as well as providing technical advisory and compliance guidance on all environmental aspects of the project.

The Kigoma Regional Authority will host the operation team of the PMU, provide support to the delivery of activities in the region and oversee the implementation of activities delegated to the District Councils. The Kigoma Regional Administration, where the PMU will be located, will convene through the District Councils all technical district officers including land use planning officers, agriculture officers, engineering officers, etc. at regular intervals, to inform the project annual workplan and to discuss project implementation progress and adaptive management approaches.

The execution of activities will be mostly decentralized to implementing partners through cooperation agreements with the District Councils, TFS, NGOs and SAMOs, whilst certain activities, particularly 3.2, 4.1, 4.2 and 4.3 will be delivered directly by services providers through procurement contracts issued by VPO and UNHCR.

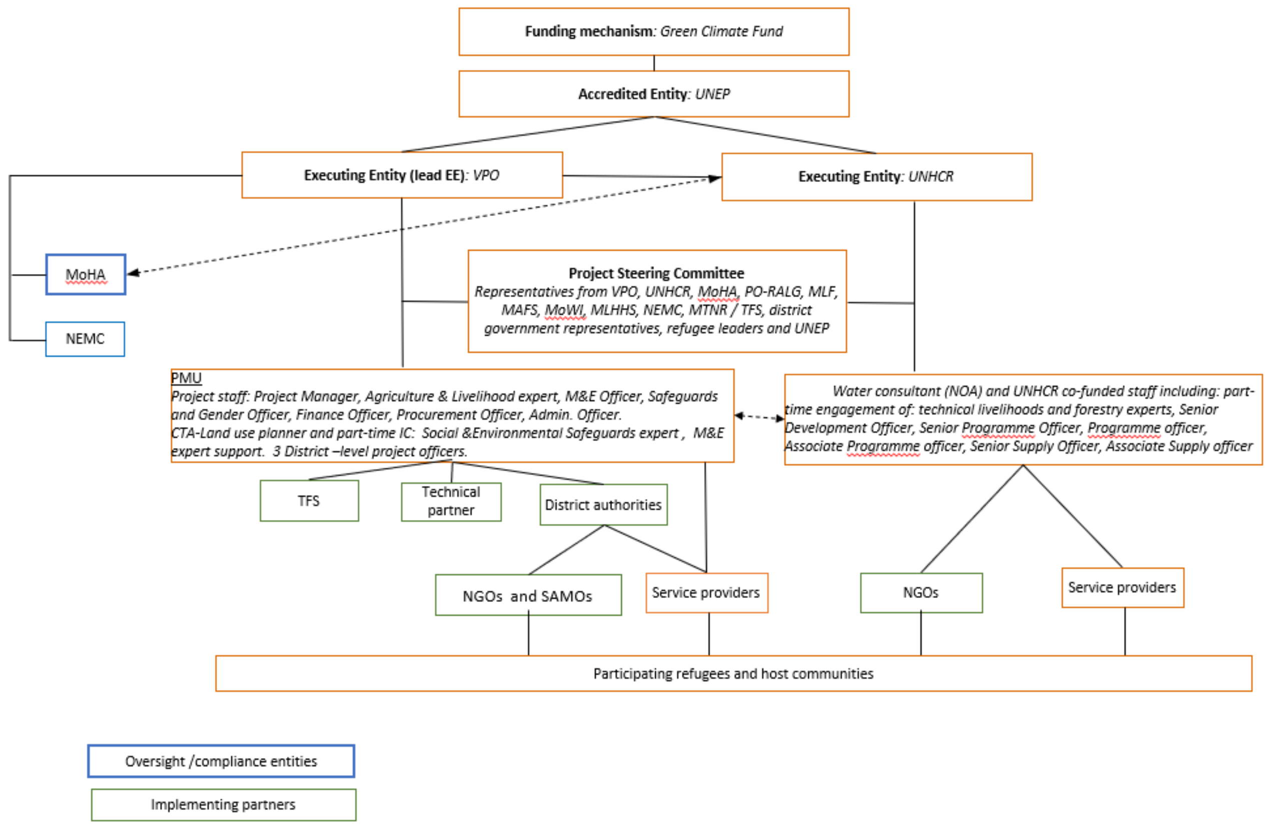

Figure 20. Implementation structure for the project

Project Steering Committee

The Project Steering Committee (PSC) will be formed during the inception of the project and is responsible for high-level project direction. It will also address concerns raised by any of its members (for example, transparency in procurement processes and changes in the humanitarian programming landscape) and will establish consensus amongst stakeholder groups on the best way to manage these concerns. The key responsibilities of the PSC will be to:

- Provide project oversight and advisory support by overseeing implementation;
- Review annual work plans, and project reports;
- Approve any changes to the project's targets, activities or timelines
- Support the Project Management Unit in managing emergent risks; and
- Meet bi-annually to discuss progress on the project and resolve any issues identified.

It will be composed of nominated representatives from the Vice President's Office as the NDA and lead Executing Entity, including the Ministry of Home Affairs of Tanzania (MoHA), President's Office — Regional Administration and Local Government, Ministry of Water and Irrigation, Ministry of Agriculture and Food Security, Ministry of Livestock and Fisheries, Ministry of Lands, Housing and Human Settlement, Ministry of Community Development, Women Affairs and Children, Ministry of Tourism and Natural Resources, National Environmental Management Council, Tanzanian Forest Service, District Government representatives and refugee leaders, UNHCR and UNEP. The PSC will be co-chaired by the Vice President's Office and UNEP. The Project Management Unit will serve as the secretariat and coordinate the meetings, with each represented group nominating a suitably qualified individual.

Project Management Unit (PMU)

The lead EE, the Government of Tanzania Vice President's Office (VPO), will establish a Project Management Unit with dedicated staff, which will be recruited competitively and responsible for the day-to-day implementation and management of the project as follows:

- *Project Manager* — day-to-day implementation and management, including project oversight, technical inputs and guidance, adherence to workplan and management of results, reporting and key point of liaison with UNHCR and between the project and PSC The PM will also be responsible for ensuring that all tasks are carried out according to UNEP and GCF policies and best practises;
- *Climate-resilience agriculture & livelihood expert* – technical oversight, implementation support to District Councils and monitoring of Output 3 activities. Responsible for ensuring the provision of quality inputs and adherence to Climate-Smart Agriculture guidance, linkages with value chains and integration of agriculture and livelihood interventions in the project landscape approach.
- *M&E officer* — responsible for the implementation of the project M&E Plan, monitoring and reporting on the project's activities in the field and production of project learning and communication outputs in coordination with relevant government agencies and implementing partners;
- *Safeguards and gender specialist/officer*— responsible for the implementation of the Social & Environmental Safeguards Plan, including Stakeholder Response Mechanism and the Gender Action Plan, ensuring equitable benefit distribution/minimal adverse impacts;
- *Finance officer* — managing the budget, expenditures, procurement plans and decisions throughout project and training and supervision of the Savings Association Management Offices;
- *Procurement officer* — lead procurement planning and procurement activities, oversee procurement of implementing partners and provide support as necessary;
- *Admin x 2* — support to PMU team in delivering project activities, including support in financial reporting;
- *Chief Technical Adviser (land use specialisation)* — lead the climate-resilient village land-use planning process with the District Councils and provide technical backstopping to sector specialist consultants throughout the project.
- *Social & Environment Safeguards expert*- A part time international expert consultant will be engaged to support PMU in the development and implementation of the Social & Environment Safeguards plan.
- *M&E expert* –A part time international expert consultant will be engaged to support PMU in the development and implementation of the project M&E plan and the document of project learning.

In addition to coordinating the land use planning process, the CTA will be employed on a full-time basis throughout project implementation to provide technical guidance on EbA and ensure that project activities result in building the climate-resilience of host communities and refugees in Kigoma. In the first two years of the project, during which Village Land Use Plans, technical designs and partnerships will be developed, the CTA will be a full-time international expert. In the last three years of the project the CTA position will shift to a national consultant contract. The responsibilities of the CTA include: i) providing overall technical support for the project; ii) supporting the annual planning process and budgets; iii) providing monitoring and operational support to the project; iv) coordinating and supervising the work of specialist technical consultants (including the water, forestry, livelihoods, agriculture and policy consultants) who will contribute to specific deliverables within each Output; and v) providing biannual reports to the PSC on project progress, performance towards objectives and recommendations.

For the execution of the activities 2.3 and 3.2 UNHCR will engage a full-time water expert/consultant for one year to assist with the consultative planning process, the final technical designs and the drawing up of the tendering documents for the procurement and installation of the flood and erosion (2.3) and water harvesting and irrigation (3.2) infrastructure. UNHCR funded staff positions will manage the designated project and co-funded activities in forestry and livelihoods in close coordination with the PMU and relevant government authorities.
The delivery of the project activities will incur a range of costs, from procurement of equipment and materials (for example, stone gabions) to costs associated with capacity building from NGOs on community-based forest management. The expected procurement costs for each year will be compiled in a procurement plan by the PMU's Finance and Procurement Officer, for sign-off by the PSC. A unified procurement plan will cover the budget executed by the two executing entities. Indicative costs per year, along with a detailed description of individual cost items associated with each activity can be found in the detailed budget (Annex 4). The PMU is proposed to be located in the Regional Government Office of Kigoma to maximize interface with local policies and programs, anchor project activities to local processes, and facilitate uptake and mainstreaming of produced economic valuation, policy revisions, and communication products through Output 4.

Key institutional actors

In Tanzania institutional arrangements and decision-making processes are concentrated at the national level. The project will benefit from the leadership of the Government of Tanzania Vice President's Office (VPO) as main executing entity. VPO's Division of Environment (DoE) has the mandate to implement climate change actions in the country and is the designated lead agency for environment and climate change-related activities in the country. The DoE is the National Climate Change Focal Point (NCCFP). A National Climate Change Technical Committee (NCCTC) is chaired by the Director of Environment (VPO), to provide technical advice to the NCCFP. The function of the Committee is to oversee all technical issues related to the implementation of climate change actions and ensure coordinated initiatives among the Ministries.

Climate change issues are currently addressed using this same institutional framework, following the mandate provided by the Environmental Management Act (2004). At the sub-national level, the Regional Administrative Secretariats that operate under the President's Office, Regional Administration and Local Government (PO-RALG), oversee the Local Government Authorities' (LGAs) activities providing supervision and administrative instructions. LGAs are responsible for service delivery concerning agriculture and natural resource sectors. Funding is provided directly from the Central Government and LGAs by the Ministry of Finance and Planning (MFP). District councils are mandated to provide extension services in the form of technical advice and support for local communities (wards and villages). Other key institutional actors relate to land-use planning and forestry.

Land administration in Tanzania is done jointly by the Ministry of Lands, Housing and Human Settlements Development, and District Land Offices that are under the PO-RALG.
The Tanzania Forest Service (TFS), a semi-autonomous organisation under the Ministry of Natural Resources and Tourism, is responsible for all forestry activities at District-level. Formed out of the Forest and Beekeeping Division (FBD), its remit includes improving the protection capacity of local (village) forest managers under community-based forest management.

Engagement with local institutions will be through coordination at the local level and signing of MOUs for the direct implementation of project activities.

B.5. Justification for GCF funding request (max. 1000 words, approximately 2 pages)

GCF involvement is critical for the project as projected and existing climate impacts are evident the Kigoma region. The region is among the poorest in the country and with the most vulnerable livelihoods, in its dependence on rainfed agriculture. The districts of Kibondo, Kakonko, and Kasulu are particularly vulnerable as they share resources with a large refugee population. Tanzania, as a least developed country, has limited financial and technical capacity to support adaptation activities. The majority of the activities and funding will benefit host communities in adapting their livelihoods and agroecosystems to the potential impacts of climate change.

Refugees are also vulnerable to climate change as, by circumstance, they are often fleeing to already climate vulnerable nations, such as Tanzania[186]. Their circumstances of limited movement, livelihoods, and access to financial services hamper their capacity to adapt to climate impacts. They are also a vulnerable group that is underrepresented in climate finance, as to the proponent's knowledge, this is the first climate project at this scale to include refugees as beneficiaries.

The refugee situation is protracted for approximately 115,000 refugees from neighbouring countries. Despite the Government's preference for voluntary repatriation as the durable solution, the protracted refugee situation is not likely to stop, considering that fewer refugees are opting for return to their country of origin. Given the current situation in neighbouring countries, the refugee population in Kigoma, notably the enduring insecurity in (Eastern) DRC means that repatriation for many refugees from this country is not a feasibly solution. While resettlement, mostly to the US, provides a solution to some of the DRC refugees, many are expected to remain in Tanzania in the foreseeable future. Given Tanzania's location in a region surrounded by countries that are experiencing insecurity and extremist insurgency, such as in the north of Mozambique, or the volatile security situation in the eastern Democratic Republic of Congo, it is expected that the country will remain an important refugee hosting country in the coming years. Hence, the project's approach is to promote long-term solutions in the Kigoma landscape that consider refugee presence that are anchored in the host communities and local institutions.

[186] New crises in 2018 caused refugee populations in Bangladesh to triple and in Sudan to double, two of the most climate-vulnerable countries.

The project provides an opportunity to address the climate change adaptation needs of both host communities and refugees in an integrated way, and lead in demonstrating complementarities in climate, development, and humanitarian finance. The project leverages the roles of UNEP, the Government of Tanzania Vice President's Office and UNHCR in mainstreaming climate change adaptation within humanitarian operations in a way that harnesses the organisations' strengths, while promoting collaboration with other non-humanitarian partners. Unless supported by activities such as those proposed in this project – such as strengthening the provision of ecosystem services and increasing the resilience of agro-ecosystems – forced displacement can further reduce the resilience of host communities and refugees to climate change. Table 9 below summarises alternatives to GCF funding for mainstreaming climate change resilience into refugee response and explains why they do not currently represent viable sources of funding.

Table 9. Alternatives to GCF funding for mainstreaming climate change resilience.

Alternative funding options	**Barriers and limitations**
Funding by the Government of Tanzania	The Government of Tanzania is unable to address fully the adaptation needs of the Tanzanian population, meaning that the complex question of how to increase the resilience of refugee camps and host communities is hardly discussed. International commitments to burden and responsibility sharing should also be considered as the project targets areas where refugees are hosted.
Funding from UNHCR core budget	UNHCR funding is largely comprised of annual non-earmarked core funding. Humanitarian funding priorities focus on basic human needs for example, security, food, water and shelter. Although UNHCR does have selected donor funding lines to support efforts of the Energy and Environment team, these would not be sufficient for an ambitious project such as the one proposed here.
Funding by the Government of Tanzania	The Government of Tanzania is unable to address fully the adaptation needs of the Tanzanian population, meaning that the complex question of how to increase the resilience of refugee camps and host communities is hardly discussed. International commitments to burden and responsibility sharing should also be considered as the project targets areas where refugees are hosted.
Funding from other multilateral and bilateral donor sources	Multilateral and bilateral donors have historically contributed to UNHCR through the lens of supporting humanitarian relief efforts. Over the longer term, we would expect to see a more holistic approach to funding the needs of refugees and their host countries, in line with the principles of the Global Compact on Refugees, through the mechanism of the UN Sustainable Development Goals. Whilst many of the proposed activities may align with multilateral/bilateral funding in the future, many of the modalities are still to be defined. Development institutions face constraints in addressing climate issues, which is within the mandate of the GCF and other financial mechanisms of the UNFCCC.
Funding from the private sector	There is very little private sector activity in the target region, mostly limited to small scale livelihoods and agriculture. Private sector activity in the camps is even more limited than that in the surrounding host communities since: i) refugees are not encouraged to engage in income-generating activities; ii) closure of common markets has reduced opportunities for some market-based solutions; and iii) UNHCR's country office have gathered from discussions with local private sector actors that they still view refugees as transient customers — with little assurance that they will form a long term, secure market, despite long-term residence of many refugees in the area. SAMOs would be engaged in market expansion efforts for crop, honey and mushroom production. The role of private sector in energy provision is being pursued by UNHCR through LPG and solar options, noting constraints such as affordability and very low incomes in the region and the policy environment limiting the operations of common markets.

While a significant amount of development activity has taken place in the Kigoma region, projects that have taken place to date have:

- not considered the long-term climate change impacts on water, agriculture and livelihoods and so have not been framed to adequately support adaptation;
- been dispersed throughout the Kigoma region and so have not concentrated on, or been designed for, the unique problems facing refugees and their host communities; and
- tended to look at water, livelihoods or agricultural systems in isolation, rather than considering the ecological interlinkages between these. Consequently, projects to date have not benefitted from mutual reinforcement of interventions with a landscape level approach.

Hence, there is clear value added for GCF funding for a project that will address climate change impacts in a holistic manner, with specific attention given to those most vulnerable, including the refugees in the camps and their neighbours.

Justification for level of concessionality

Tanzania is in need of concessional financing. Tanzania is currently classified by the UN as a Least Developed Country with an estimated GNI per capita of ~USD 900 in 2016. Consequently, government budgets do not provide any allocation for addressing the impacts of climate change at Tanzanian refugee camps and the lands surrounding them. Furthermore, the Kigoma region has suffered from chronic underinvestment over the years, having the among the highest poverty rates in the country. The project requires a high degree of concessionality as it focusses on delivering non-financial benefits.

Majority of the project interventions do not have significant revenue streams. While some project activities such as resilient livelihoods are expected to supplement food and contribute to incomes, these are not expected to provide sufficient revenue streams for formal debt repayments during the lifetime of the project. Access to financial services in the region is very low. The project aims to promote sustainability through the use of Savings Associations but loan repayments are not viable. GDP per capita in the host community is around USD 250, amongst the lowest in Tanzania. Livelihood activities are already threatened by climate change and since these are mostly for subsistence/basic needs rather than larger scale businesses, the resulting impact on household food security and resilience can be immediate. Furthermore, there are only five banks across the districts of Kibondo and Kakonko serving over 450,000 people. Most of these residents lack the assets which the banks request as collateral for transactions. This makes them unable to access finance to scale up their businesses and try new, climate-resilient practises.

Refugees do not form part of the formal economy. Refugees are required to remain in the camps and are not able to partake in formal employment either within or outside the camps. This means that their economic activities are mostly informal (for example, barter or in-kind payments) or are fully reliant on humanitarian assistance. Despite these limitations, they do have some engagement with the markets of the nearby host communities, creating a unique and complex economic context. Since almost half of the project's direct beneficiaries are refugees who cannot take part in the formal economy, this makes it challenging to stimulate economic activities that are robust enough to enable reliable loan repayments.

Reimbursable grants are not appropriate to use as a financial instrument for the project. A limited amount of the GCF funding requested is spent on equipment with the potential to directly generate revenue and even then, the benefits are primarily for subsistence or to supplement incomes of very poor, vulnerable populations, particularly in districts that host and share resources with refugees. Any increases in income and revenue will be re-channelled back to the operation of Savings Associations. In the face of climate shocks and its impacts such as low yields and damage to property, savings offer an important safety net for very poor households.

B.6. Exit strategy (max. 500 words, approximately 1 page)

The project has been designed so that interventions will translate into long term solutions which outlast the GCF's funding.

At the intervention level, long-term sustainability is ensured because of the inclusion of mutually supportive interventions in land use, agriculture, water and livelihoods, which are themselves supported by a revised institutional framework and sustainable financing mechanism. The basis for project sustainability and a successful exit strategy for GCF is underpinned by several factors, outlined below.

Anchoring interventions at the landscape level: While refugees are temporary residents in the Kigoma region, the region will most likely continue to host refugees in the long term. The project strategy is to ensure that the landscape and ecosystem of the Kigoma region can continue to support its residents, including refugees, and host communities, including agro-pastoralist groups that pass through the region as it faces challenges of climate change. The interventions are planned within the sub-catchments and contiguous landscape of host communities and refugees with the aim of strengthening the ecosystem's capacity to respond to climate shocks and continue to provide services such as provision of soil nutrients, prevention of flooding and erosion, creating shade for crops and others. Majority of the project interventions will be implemented in the host community, that will be less vulnerable to population movements.

Capacity building and empowerment of local government authorities: The project aims to involve local government authorities in the implementation of the project including the Regional Commission, District Councils, Tanzania Forest Service, Lake Tanganyika River Basin Authority, and other relevant actors. Training and equipment will be provided by the project to support their mandate in land use planning, forest monitoring and conservation, and expanding extension work climate resilient agriculture practices. The expert consultants and technical advisory organizations engaged in the project will work hand in hand with the local government authorities, NGOs and project staff to transfer technical knowledge and build capacity on-the-job and through training sessions. These include the CTA expert in land use planning, the Technical Partner, the Climate-resilient Agriculture & Livelihoods expert, the M&E expert consultant and the Safeguards expert consultant. UNHCR's field presence and its partnerships with other agencies and NGOs in the Kigoma region can continue to provide support and backstopping on project interventions beyond the project timeframe.

Introduction of a strategic tool for climate-resilient land-use planning*:* The Village Land Use Plan process has been the main tool for the integrated planning and management of natural resources in Tanzania since the 1990s. Its practise is enshrined in the legal framework for the management and administration of land within village boundaries. The project will introduce climate resilience considerations into the traditional planning process but work within this established framework to ensure its legitimacy.

Self-sustaining interventions*:* Many of the interventions introduced can be maintained or expanded by the host communities without further support after the initial five-year project period since they require minimal funding and repay the investment in inputs and labour through increases in productivity and income. The project aims to show the viability and impact of its activities, setting the context for the continuation of these interventions beyond the project lifespan.

Financial mechanisms*:* The majority of the physical inputs for project interventions will be funded through Savings Associations (SAs). This will be done by supporting the expansion of activities that SAs provide financial services. The host communities will have control over and be personally invested in these SAs. Profits from activities made possible by GCF funding will be channelled back to the SAs, thus making them self-sustaining.

Capacity building of the communities and inclusion*:* Project interventions include extensive training and capacity building, which will allow knowledge of the benefits of activities such as community-based forest management, resilient agriculture and alternative livelihoods to be sustained and dissipated through wider communities.

Gender sensitive approach: The project's gender sensitive approach seeks to achieve practical and tangible benefits to women as well as bring about strategic changes in gender relations. The gender action plan (Annex 8) outlines concrete steps and activities that will ensure inclusion of women in decision making processes and project activities. Through enhanced capacities, skills, access to assets transferred to refugee and host community women by the project, they will be capacitated to improve their decision-making power both at the household and the community level. This effect is expected to extend beyond the duration of the project.

Revised institutional and regulatory systems: The institutionalisation of climate change adaptation into regional, local and UNHCR plans, policies and programmes through Outputs 1 and 4 will contribute to the sustainability and replicability of these interventions, scaling to a broader project area as well as other refugee sites globally.
Beyond the direct engagement of UNHCR with refugees regarding environmental considerations, the Global Compact on Refugees (GCR) calls for strengthened national and international partnership with development partners, the private sector and other stakeholders to address challenges related to energy and the environment. The proposed project is recognised as a major partnership effort to operationalise the GCR. Lessons from the project's approach and implementation will be documented and shared to guide UNHCR's global engagement with environment and climate change actors as well as inform operations in other countries. Dedicated teams have been set up to strengthen the operationalisation of the GCR as well as to support UNHCR's work on mitigation and adaptation to climate change. Corporate commitments to climate change and environmental challenges as well as organisational structures at UNHCR encourage the institutionalisation of the approach proposed by the project.

Operations and maintenance (O&M): For assets outside the camps, ownership of equipment and infrastructure will be turned over to district or regional authorities for operations and maintenance after the project ends. Prior to this turnover, a budget will be identified to finance these O&M activities. Assets within the camps will be managed by MHA, with maintenance agreements being formulated with MHA and district councils prior to the implementation of relevant activities. Additionally, further insight will be given by UNHCR's and other partner NGOs, who have well established operations and presence in the region and districts.

The participatory land-use planning intervention in Output 1 is designed to be both sustainable in itself and to ensure the sustainability of other interventions. The main part of this intervention will conclude with the villages having: i) reached consensus on how the village land should be used to improve climate resilience; and ii) obtained legal status in the form of a Customary Certificate of Rights of Occupation. Achieving these goals will allow the host communities to assert their own authority around land-use planning (including by resolving any potential competing claims to land use) to adapt to be more climate resilient and in doing so, will ensure sustainability in the long term. In addition, through the capacity-building activities, the District Council and the host communities themselves will be better placed to adapt to future changes in circumstances. The land use plan forms the basis of the other interventions of this project and indeed of any other development activity, whether funded by government or other agencies. By selecting and designing these interventions in a participatory and consensus-based process, their long-term success is more likely.

With respect to the community-based forest management (Activity 2.1), experience from elsewhere in Tanzania and further afield shows that once the governance structures are in place villagers are able to ensure ongoing management with minimal external support. The project will promote and support local leadership by establishing strong community-level institutions in the form of Forest Management Committees to ensure this continuity. Agroforestry (Activity 2.2) will also become self-sustaining — as farmers realise the benefits of including trees in their farming systems, it is expected that the practise will both intensify and spread wider afield. For the tracts of afforested land, the Tanzania Forest Service will be able to provide continuing management support after the project has ended. Furthermore, recent increases in the local price of fuelwood (as reported by residents of the refugee camps) mean that the communities will have an interest in further extending the area under plantation to meet a larger part of fuel demand in the future.

The water harvesting, storage, distribution and irrigation interventions in Output 3 are designed to provide additional water to agricultural areas in the host communities, and so increase their resilience in the face of climate-related stresses on the land. This intervention is anticipated to help the host communities maintain, and even increase, agricultural production. Consequently, they are expected to be highly valued interventions. The water management systems will need to be maintained in order to ensure their ongoing effectiveness. Much of this will require only a basic level of skills that will be developed through the intervention's training activities. For the more advanced technologies, basic maintenance training will be delivered at village level, with more advanced maintenance training delivered to one engineer in each District. Funding will be made available through the Savings Associations for more advanced maintenance/repairs that may need to be carried out by the equipment suppliers. It is expected that the cost of such repairs will have reduced over the period of the project as the technologies become more prevalent in the region.

Output 4 will contribute to building strengthened institutional and regulatory systems for climate-responsive planning and development. These strengthened systems, and the evidence-base that underpins them, will further contribute to the sustainability of Outputs 1–3. This will be achieved by broadening the EbA and climate-resilient planning knowledge base — with economic evidence, policy briefs, guidelines — and by institutionalising adaptation in into policies, plans and strategies in Tanzania, as well as in humanitarian programmes. This will have an enduring legacy and once implemented (within the project cycle), will not require further updating in the medium-term, as the communication materials developed will help guide further developments, as needed.

Savings Associations as a Key Part of Sustainability

Savings Associations (SAs) are proposed as a key delivery mechanism for the project. SAs allow individuals to pool finances to save and borrow money based on consensus between members. SAs — which typically have 20–25 members — are a recognised mechanism for providing basic financial services in areas with limited banking services, including the Kigoma region. Most residents of the host communities and the refugees in the Kibondo, Kakonko and Kasulu Districts are reliant on SAs to manage their finances. However, their levels of capitalisation are low, and they have a general focus on saving rather than promoting climate-resilient activities specifically. This project aims to function as a critical tool to gather information on the financial services needs of the host communities, building on past assessments by agencies like UNCDF. This could demonstrate early results and pave the way for less concessional finance and broader access to financial services in the future.

To promote sustainability of financing for climate-resilient activities, the proposed project features an innovative approach which involves the channelling of funds to local communities through the SAs[187]. The project will enable these existing structures to be strengthened and scaled up which will allow beneficiaries to access seeds and seedlings, equipment, and tools for greater engagement in climate resilient activities. This will promote savings and reinvestment and boost markets for these inputs. In addition, the approach will allow for the barriers of limited purchasing power to be overcome and will provide an alternative to accessing formal banking services. For clarity, GCF funds will be used to procure tools, seeds, and basic goods to promote climate-resilient livelihood but will not be used to provide cash grants.

[187] UNEP is expected to be accredited for grant awards by the time the proposal is brought forward for GCF Board consideration.

The project is intended to develop the capacities, tools, and systems for the local and regional stakeholders, thereby sustaining a model that can continue beyond the project cycle. By developing diversified livelihoods and strengthening the capacities of farmers to benefit from these, the activities fostered by the project will become self-sustaining with time. This also applies for the village savings associations, which will continue to function, with the funds revolving. The capital available in the SAs will also stimulate markets for produce, incentivizing local entrepreneurs to further establish small scale nurseries for agroforestry seedlings. Moreover, the host communities will have control over and be personally invested in these SAs. Profits will be channelled back to the SAs, therefore making them more self-sustaining over time. Sustainability of the mechanism will also be ensured through the production of a lessons-learned and guideline papers on the use of SAs for upscaling of climate-resilient agricultural practises (Under Activity 4.2). The guideline papers will be provided to key stakeholders to inform users, decision-makers and development partners on planning, implementing and monitoring the use of SAs. The process to develop these guidelines will include a review of how SAs in different countries/contexts are being used for financing climate adaptation. Further details on the SAs and SAMOs can be found in Section 3.5.1 of the Feasibility Study (Annex 2.1) as well as in the operational manual outline for SAMOs that is appended to the Feasibility Study.

C. FINANCING INFORMATION

C.1. Total financing

(a) Requested GCF funding (i + ii + iii + iv + v + vi + vii)		Total amount		Currency	
		$19.01		million USD ($)	
GCF financial instrument		**Amount**	**Tenor**	**Grace period**	**Pricing**
(i)	Senior loans	Enter amount	Enter years	Enter years	Enter %
(ii)	Subordinated loans	Enter amount	Enter years	Enter years	Enter %
(iii)	Equity	Enter amount			Enter % equity return
(iv)	Guarantees	Enter amount	Enter years		
(v)	Reimbursable grants	Enter amount			
(vi)	Grants	$19,007,353			
(vii)	Results-based payments	Enter amount			

(b) Co-financing information	Total amount			Currency		
	$ 4.61			million USD ($)		
Name of institution	**Financial instrument**	**Amount**	**Currency**	**Tenor & grace**	**Pricing**	**Seniority**
UNHCR	In kind	4.61	million USD ($)	Enter years Enter years	Enter%	Options
Click here to enter text.	Options	Enter amount	Options	Enter years Enter years	Enter%	Options
Click here to enter text.	Options	Enter amount	Options	Enter years Enter years	Enter%	Options
Click here to enter text.	Options	Enter amount	Options	Enter years Enter years	Enter%	Options

(c) Total financing (c) = (a)+(b)	Amount	Currency
	$ 23.62	million USD ($)

(d) Other financing arrangements and contributions (max. 250 words, approximately 0.5 page)	
	In addition to the projects and financing described in this funding proposal under Table 2, which lists over US$ 48 million in parallel financing, the 'Overview of other relevant projects in the region' in the Feasibility Study (Annex 2.1) includes other climate change and development projects operating in the project area, whose activities complement those of the proposed project. The funding for these projects stems from GIZ (Deutsche Gesellschaft für Internationale Zusammenarbeit – German Development Agency), DFID (UK Department for International Development), BPRM (US Bureau of Population, Refugees, and Migration), GAC (Global Affairs Canada), KOICA (Korea International Cooperation Agency) and various UN agencies.

C.2. Financing by component

Component	Output	Indicative cost Options	GCF financing		Co-financing		
			Amount Options	Financial Instrument	Amount Options	Financial Instrument	Name of Institution
Component 1 Increased resilience of ecosystems and people	*Output 1: Participatory Climate-resilient Land Use Planning (C-LUP) to support ecosystem-based adaptation*	$1,062,805	$ $1,062,805	Grants	0		

to climate change threats	*Output 2: Degraded ecosystems restored to support climate change adaptation in host and refugee communities*	$ 13,038,743	$ 9,978,657	Grants	$3,060,086	In-kind	UNHCR
	Output 3. Climate-resilient livelihoods practised to increase the capacity of host communities and refugees to better adapt in changing climatic conditions	$ 7,094,532 478,767	$ 5,687,324	Grants	$1,407,207	In-kind	UNHCR
Component 2 Strengthened policies for climate responsive planning and development	Output 4. Information on climate change adaptation disseminated and mainstreamed into policies, plans and strategies in Tanzania and in humanitarian programmes	$ 469,554	$469,554	Grants	0		
Monitoring & Evaluation		$ 1,421,283	$1,421,283	Grants	0		
Project Management Cost		$531,730	$ 387,730	Grants	$144,000	In-kind	UNHCR
Indicative total cost (USD)		$23,618,646	$19,007,353		$4,611,293		

D.EXPECTED PERFORMANCE AGAINST INVESTMENT CRITERIA

This section refers to the performance of the project/programme against the investment criteria as set out in the GCF's Initial Investment Framework.

D.1. Impact potential (max. 500 words, approximately 1 page)

Impact potential
The primary impact potential of the proposed project can be summarized as follows:

- Approximately1,281,800 total beneficiaries, consisting of ~ 570,300 direct (292,800 women and 277,500 men) and ~7 11,500 indirect beneficiaries (366,635 women and 344,865 men)[188];
- 15,000 farmer households benefiting from the adoption of diversified, climate-resilient livelihood options. At least 50% of participants practicing climate smart agriculture, mushroom growing activities and beekeeping are women.
- 42,000 ha of ecosystems protected and strengthened in response to climate variability and change; and
- Approximately 363,745 people covered by village land use plans which improve incentives for climate resilience and their effective implementation.
- Approximately 30,000 ha of forest ecosystems protected under community-based forest management (CBFM).
- An additional 10,000 ha of agroforestry implemented and 2,000 ha of afforestation converted from monocrop or degraded land, equating to ~80% of forest area in the project area, and 2% of the total forest area in Kigoma region (noting that much of the forest in Kigoma is outside of the project area, in already-protected sites such as the Moyowosi Game Reserve Ramsar site and Mahale National Park).

Women will be specifically targeted for inclusion, seeking to reach at least 50% of women as part of the interventions. Further information on the gender targets and targeting strategy can be found in the Gender Action Plan (Annex 8).

Number of beneficiaries
Direct beneficiaries — drawn from both the host community and refugee camps — are those who directly engage with the project's activities, such as recipients of training. Although some interventions are targeted at specific groups of beneficiaries, all of the individuals listed in Table 10a below are expected to directly benefit from one or more of the interventions delivered under the project. Further details of the assumptions underpinning the number of beneficiaries can be found in the 'assumptions' tab of the project budget (Annex 4).

The project will contribute to GCF Result areas ARA1, ARA 2 and ARA4 and the following core indicators of the GCF Integrated Results Management Framework (IRMF):

Core indicator 2: Direct and indirect beneficiaries reached

Core indicator 4: Hectares of natural resource areas brought under improved low-emission and/or climate-resilient management practices

Core indicator 5: Degree to which GCF investments contribute to strengthening institutional and regulatory frameworks for low-emission climate-resilient development pathways in a country-driven manner

Core indicator 8: Degree to which GCF investments contribute to effective knowledge generation and learning processes, and use of good practices, methodologies and standards.

[188] Considering 51% women population in the camps (UNHCR Tanzania Refugee Statistical Report, September 2021) and 51.5% women population in the three target Districts (Tanzania population and housing census, 2012). Calculations can be consulted on Annex 4- detailed assumptions or sources.

Table 10 a. Project beneficiaries.

Community[189]	Direct beneficiaries	Accompanying remarks
Project population – Kibondo host community	159,311	This includes Tanzanians living in the settlements which fall within ~10–15 km radius of Nduta camp in the Kibondo District. The projected population at the end of the programme in 2028 is considered. (See Annex 3 and Annex 4 Detailed assumptions) These inhabitants (as well as those in the Kakonko and Kasulu districts) are the focus of the project as they are considered to be part of a host community which engages with the refugee population and are thus subject to a related set of climate challenges which the proposed project seeks to address. Beneficiaries are primarily individuals that benefit from resilient infrastructure installed in the host community (e.g. charco dams).
Project population – Nduta camp refugees	59,534	This is the population of Nduta camp as of September 2021.
Project population – Kakonko host community	38,519	As above for Kibondo – this is the population living within a 10-15 km radius of Mtendeli former camp area in the Kakonko District. The projected population at end of the programme in 2028 is considered. (See Annex 3 and Annex 4 Detailed assumptions)
Project population – Mtendeli camp refugees	17,777	This is the population of Mtendeli camp as of September 2021, prior to the closure and relocation to Nduta camp closure and are therefore considered as project beneficiaries in Nduta camp.
Project population – Kasulu host community	166,915	As above for Kibondo – this is the population living within a 10-15 km radius of Nyarugusu camp in Kasulu District. The projected population at end of the programme in 2028 is considered. (See Annex 3 and Annex 4 Detailed assumptions)
Project population – Nyarugusu camp refugees	128,285	This is the population of Nyarugusu camp as of September 2021.
Agro-pastoralist groups	Unknown numbers	Numbers of agropastoralist groups who are not residents in host communities will be estimated at the project inception phase.
Refugees as % of total direct beneficiaries	36 %	There is an approximately equal balance of beneficiaries between refugees and the host community, reflecting the project's objective of improving the climate resilience of refugee-hosting districts as a whole.
Total direct beneficiaries	**570,341**	Total population expected to directly benefit from project activities.
Indirect beneficiaries – broader Kibondo, Kakonko and Kasulu host community	**711,516**	This is the remaining population of the Kibondo, Kakonko and Kasulu Districts that does not include the host and refugee communities described above. The projected population at end of the programme in 2028 is considered (See Annex 3 and Annex 4 Detailed assumptions) They are expected to observe and learn from the climate-resilient activities being undertaken within the project's area of focus and will replicate these activities and approaches to increasing climate resilience in adjacent areas. The structures of Saving Associations may also be replicated in these neighbouring regions.
Total beneficiaries	**1,281,857**	**Total population expected to benefit directly and indirectly from the project activities and areas of Kibondo, Kakonko and Kasulu Districts which lie outside the project area.**

Table 10.b. Project benefits across beneficiary groups

Project Beneficiary/ Stakeholder Group	Main climate change impacts and risks	Socio-economic impacts	Type of adaptation intervention / project activities	Project activities	Benefits / incentives for participation	Assumptions
Refugees in Nduta and Nyarugusu camps	Increased precipitation and floods	Gully formation and enlargement leading to loss of land in already cramped areas	Flood and erosion control infrastructure	2.3. Flood and erosion control	**Climate adaptation:** Reduced flood damage to settlements, improved public	Refugees participate in infrastructure planning processes

		and risk of injuries; Cholera, Dengue fever and vector and water borne diseases.			health, safety, and well-being **Co-benefit:** Improved environmental health due to the reduction of stagnant waters and breeding sites for disease vectors such as mosquitos and flies. Reduced risk of latrine pit overflow and uncontrolled dispersion of solid waste	
	Prolonged drought and high temperatures;	Hunger and malnutrition: Refugees engage in small-scale agriculture such as planting beans and kitchen gardens but with poor harvests compared to previous years. Food security within camps has been reduced by cuts to food rations from lack of resources and closure of common markets in 2019	Climate resilient household level agriculture (kitchen gardens) and alternative livelihoods	3.1 Climate resilient agriculture practices 3.2 Mushroom-growing and beekeeping	**Climate adaptation:** Improved food security from nutritional supplement of crops, mushrooms (meat substitute) and honey (sweetener); improved income and resilience to shocks; improved skills and knowledge that can be used to rebuild livelihoods in case of repatriation **Co-benefit:** Contributes to empowerment of vulnerable groups such as women and elderly	Type and scope of livelihoods activities to benefit refugees as part of the UNHCR co-financed activities are to be firmly agreed with MoHA during project start-up
Host communities in Kasulu, Kakonko, and Kibondo districts, including agropastoralist groups	Forest fires; flooding; drought conditions	Soil loss and reduced agricultural production.	Land use planning and promotion of sustainable land management practices	1.1.1 Scale up land use planning in villages 1.1.2 Capacity building of local officials to support EbA 1.1.3. Improve public awareness of climate change	**Climate adaptation:** Improved agricultural production and productivity through reduced fires and improved soil health Optimised land use and management **Co-benefits:**	Planning meetings are successfully led by local government authorities that have sufficient capacity for planning, enforcement and monitoring

[189] Refugee numbers are based on the most recent available information and are subject to change (UNHCR Tanzania Refugee Statistical Report, September 2021). Host community population considers projected population at the end of the project in 2028 (Tanzania population and housing census, 2012). Population calculations and sources are detailed in Annex 4.

				Improved monitoring of forests 1.1.4 Support district, ward, and village level environmental management committees	Reduced conflict over communal land, particularly between farmers and pastoralists	
	Heat stress due to increased temperatures; drought conditions; flooding; increased fire incidence	Deforestation for wood fuel building materials and for agriculture	Forestry and agro-forestry	2.1 Community Based Forest Management 2.2 Agroforestry and village land afforestation	**Climate adaptation:** Improved ecosystem services (ES) that can buffer against climate risk. ES include improved soil health; erosion control; availability of non-timber forest products, tree cover providing cooler microclimate conditions and, longer term improvement of food security from improved land fertility Sustainable production of fodder for livestock. **Co-benefits:** Increased availability of agroforestry products, timber, and fuel	Identified co-benefits will motivate both groups to reach agreements and manage forest resources sustainably. Energy needs of host communities and refugee communities can be met through means that do not undermine forestry investments.
	Flood; drought; high temperatures and heat stress	Crop losses and poor productivity from maize, cassava, beans and nuts; poor productivity and quality; crops subjected to heat stress Degraded land and soil loss result in low production	Climate resilient agriculture and enhancing water security	3.1 Climate resilient agriculture 3.2 Water harvesting and efficient irrigation interventions 3.3 Livelihood diversification (beekeeping and mushroom growing)	**Climate adaptation:** Improved yields and income for cassava and beans Beekeeping and mushroom growing as alternative /diversified livelihoods to support income and nutritional needs Provision of water during dry periods for irrigation reduced reliance on rainfed agriculture and related climatic risks **Co-benefits:**	Successful interventions allow for income and savings that can be reinvested

					Savings can support household expenses such as food, education, and other needs Improved food security and nutrition outcomes Improved financial literacy Contributions to empowerment of vulnerable groups such as women and elderly	

D.2. Paradigm shift potential (max. 500 words, approximately 1 page)

Under the proposed project, a new innovative model will be implemented which brings together host communities and refugee camps through a mutually beneficial, ecosystem-based adaptation landscape planning approach that will improve the resilience of both groups to climate change. The project's innovation is that it promotes a paradigm of inclusivity in addressing the climate vulnerability of underserved populations in climate finance — refugees and their host communities facing unique challenges and layered vulnerabilities.

The sustainability of adaptation actions in the region requires addressing the needs of both groups as critical populations impacting the landscape. Addressing only host community needs and impacts on the landscape would not account for the impacts of refugees in the degradation of the environment, and vice versa. There are no examples or paradigm to date of adaptation policies in the refugee-host community setting yet, that involve the two communities, local and national governments as well as the humanitarian and development community in the implementation of adaptation activities and future planning. Current investments are limited to sectoral and fragmented solutions such as focus on energy or livelihood solutions only and operating only in limited land areas. The scale of investment in the project could be catalytic and bring together different actors and implementing partners in the region for activities to make a lasting impact at the landscape level. As described in further detail below, Outputs 1-3 will test this model of implementing adaptation activities at the landscape level with refugees and host communities and Output 4 will support policies and institutionalization of adaptation policy and planning across these levels, with lessons learned to be scaled up and applied by UNHCR in other operations. In Kigoma, after the exit of the GCF investment, it is expected that local government authorities, UNHCR, and implementing partners that have continued field presence in Kigoma over the long term will have the capacity to plan and implement adaptation investments at the landscape level based on the project's model and program their future investments from local, bilateral and multilateral sources accordingly.

This project contributes to broader strategic benefits bringing equitable benefits for host communities and refugees as well as increases the protection space for refugees. Kigoma has been home to refugees for at least five decades in variable numbers that depend on the security situation of neighbouring countries. As such, refugee presence is part of the landscape, social relations, and economy in Kigoma. The project recognizes their presence in the region and eligibility to receive adaptation finance as a vulnerable population. Addressing adaptation needs of both groups in a participatory way results in minimizing conflict over resources in the region, supporting self-reliance among refugees, limiting exposure of women and girls to gender-based violence while collecting fuelwood, and builds trust among different stakeholders. This is the first project of this kind and scale that is well-positioned to make a lasting change in the socio-environmental system in the region and with lessons informing broader humanitarian operations where climate change adaptation needs are present.

At the project level, the inclusive, landscape-level adaptation will be achieved through a comprehensive assessment of and planning for land-use, forestry, and livelihood interventions that consider refugee presence and provision of sustainable energy for cooking, complementing development projects in the region that are addressing non-climate drivers of degradation and vulnerability. Complementary activities are being implemented by UNHCR and others to address biomass energy demand by promoting distribution of improved stoves, promoting e-cooking, good cooking practises and examining the viability of market-based solutions to provide alternative cooking energy sources (although challenging in the context of poverty in the host community and lack of incomes among refugees). This planning-based landscape approach offers scalable solutions to address climate adaptation that considers resource use, management practises, landscape ecology and complex development-humanitarian needs and concerns.

Through this, a paradigm shift will be achieved whereby land-use planning and activities move away from unsustainable practises and climate-vulnerable livelihoods towards approaches that promote climate-resilient sustainable development. This will be achieved by:

i) climate informed, integrated land-use planning (Activities 1.1 and 1.2), technologies and practices (Activities 2.1-2.3 and 3.1-3.3) to build lasting local resilience to climate change by reducing the rates of land degradation, erosion and sedimentation, as well as by improving waterways, forests and agricultural resilience;

ii) improving access to sustainable funding mechanisms (i.e. Savings Associations) (Activities 3.1-3.3) and their capacity to provide micro-infrastructure, skills, knowledge and access to productive assets, use funds in a catalytic way; and

iii) supporting the mainstreaming of climate change adaptation into policies, plans and strategies in Tanzania and in broader humanitarian programmes, with capacity-building for policy- and decision-makers (Activities 4.1.-4.3). The project aims to integrate climate resilience within the planning functions of multiple sectors, including agriculture, water use, flood prevention and forestry and will systematically collect lessons learned for scaling up and replication.

To our knowledge, this project is the first of its kind that addresses adaptation needs of host communities and refugees in an integrated way by institutionalising climate-resilience and EbA practises. This approach has the potential to lead globally in mainstreaming climate change adaptation in refugee and host community settings as many countries that host refugees are also among the most vulnerable to climate change (Bangladesh, Jordan). The project will shift the paradigm in land use planning and practices in refugee and host community settings from fragmented, sector-based approaches that do not consider climate change impacts, to integrated solutions that build resilience to climate change impacts. This will move the paradigm of the refugee-host-ecosystem nexus to avoid circumstances where landscape degradation compounds climate change effects to produce vulnerability for communities in Kigoma. Locally, the demonstrated economic, social and environmental benefits of EbA interventions will help to change the mindsets and practises of village and camp planners and governing authorities — as well as refugees and local communities — towards embracing practises and behaviours that promote climate resilience.

Output 4 activities support replication and scaling up of the project results through sharing of lessons learned and mainstreaming more broadly in the in refugee and host community context within Tanzania and beyond. Activities in this final output that conduct ecosystem service valuations, develop guidelines and policy briefs, and undertake policy mainstreaming within district, regional, and national governance arrangements in Tanzania will leverage the project outcomes to achieve the broader adaptation objectives. They allow for the generation of evidence and scaling up of successful interventions. On a global level, the ecosystem management and adaptation model developed in the project will promote the dissemination of best practices in EbA and the contribute to the global adaptation knowledge base. The model can be scaled-up and replicated in other refugee hosting regions by UNHCR and other humanitarian partners, supported by the materials developed under Output 4, namely economic valuation products, policy briefs, guidelines, and other knowledge and communication products.

UNEP and UNHCR support approaches that focus on the integration of effective practises by States and (sub-) regional organizations into their own normative frameworks and practises in accordance with their specific situations and challenges. As Tanzania is a frontrunner in applying the One UN working modus, UN Agencies, including UNEP and UNHCR, are often invited to contribute to national policy and development dialogues, with UNHCR also participating in the Development Partners Group on environment. As a result, UNEP as AE and UNHCR as one of the EE of the project are well placed to promote upscaling and replicability of this model in other contexts, and to other UN organisations. The lessons from this project will be used to inform UNEP and UNHCR's contributions and inputs to these policy processes under Output 4. The broader environment forums would also be used to disseminate lessons and findings from the project, which would produce policy recommendations and briefs. Through this, the project will provide an adaptation model that can be replicated to ensure that the world's most vulnerable populations benefit from climate finance.

D.3. Sustainable development (max. 500 words, approximately 1 page)

The proposed project will contribute towards achieving 9 of the 17 Sustainable Development Goals (SDGs), namely: i) SDG 2 — Zero Hunger; ii) SDG 3 — Good Health and Well-being; iii) SDG 5 — Gender Equality; iv) SDG 6 — Clean Water and Sanitation; v) SDG 8 — Decent Work and Economic Growth; vi) SDG 11 — Sustainable Cities and Communities; vii) SDG 13 — Climate Action; viii) SDG 15 — Life on Land; and ix) SDG 16 — Peace, Justice and Strong Institutions. For monitoring, focus in given to SDG 2, to report under indicator 2.4.1 Proportion of agricultural area under productive and sustainable agriculture, and SDG 13 to report under indicator 13.b.1 Number of least developed countries and small island developing States that are receiving specialized support, and amount of support, including finance, technology and capacity-building, for mechanisms for raising capacities for effective climate change-related planning and management, including focusing on women, youth and local and marginalized communities.

In addition, as a result of the no-regrets approach, the project interventions will produce numerous environmental, social and economic co-benefits, through a gender-sensitive and -responsive development approach, as described below in Table 11 below. The monitoring and evaluation officer for the project will develop methodologies to establish baselines, track, and report on these co-benefits.

Table 1. Economic, environmental, social and gender benefits of the proposed project interventions.

Benefit type	Description of benefits
Economic	• An increase in provisioning services in the form of timber for wood fuel, associated with the establishment and maintenance of agroforestry systems (Activity 2.2.1) has been estimated to have an NPV over a 20-year period of US$9.8 million, with an associated IRR of 98%. • The increase in timber provision which will be generated by afforestation of village land (Activity 2.2.2) has been determined to have an NPV of US$2.9 million and an IRR of 50%. • Increased returns to agriculture under climate-resilient management (Activity 3.1.2) would result in a net present value (NPV) of US$5.8 million over a 20-year period using a discount rate of 9%. The internal rate of return (IRR) associated with this activity is 541 %. • Increase water availability and use-efficiency through water harvesting and efficient irrigation interventions (Activity 3.2) has an estimated IRR of 197% and an estimated NPV of US$3.5 million. • Beekeeping (Activity 3.3.1) would generate an NPV of US$1 million. The IRR for promotion of beekeeping under Sub-Activity 3.3.1 is undefined, given that there are no years in which net benefits are less than zero for this activity. This is of course a favourable indication, and the best possible value for the IRR. In addition, this activity would generate substantial co-benefits in the form of conservation incentives and improved ecosystem functionality, which have not been quantified. • Mushroom cultivation (Activity 3.3.2) would generate an NPV of US$554,000 and an IRR of 787%. • Comparing the with project scenario to the without project scenario, it has been estimated that the net present value of the overall project is US$10.5 million when greenhouse gas mitigation benefits are excluded from the analysis. When these benefits are included, the NPV for the whole project is estimated to be US$20.8 million. The project has been estimated to have an economic rate of return (ERR) of 13% when greenhouse gas mitigation benefits are excluded and 20% when these benefits are included. • In addition to the selected provisioning of services valued above, the project will result in ecosystem services which would consist of supporting (incl. nutrient cycling, soil formation), provisioning (incl. timber, non-timber forest products), regulating (incl. water regulation, flood control) and cultural (incl. recreation, education). The total economic value associated with these services is likely to be substantial. Zhang et al. (2015[190]) found that for a similar ecosystem type in Miyun County, China, raw materials accounted for 38% of overall value. It is therefore conservative to assume that the economic values quantified above are likely to make up less than half of the total economic value associated with these project activities. • The economic benefits associated with flood and erosion control in densely populated areas (Activity 2.3) will ensure that damages from flooding are attenuated. These damages would include a loss of crops to flooding, damage to houses and community facilities including schools, and injury and potentially loss of life. • Measures designed to increase water availability and use-efficiency will provide members of the host communities access to a more reliable source of water. This will extend the growing season as well as mitigate against drought by increasing storage capacity. The infrastructure put in place for these purposes will also have the added benefit of reducing the impacts of floods through attenuation. • Access to supplemental non-timber forest-foods. The 2019 studies by FAO in Ugandan refugee settings showed that a co-benefit of reforestation is the protection of refugees' and host communities' access to a range of non-timber forest products, primarily for supplemental nutrition[191]. The study found that 100% of surveyed households used non-timber forest products and it identified over 200 such products. Most contributed directly to the diets of refugees and host communities. The study noted, that while difficult to specifically value in economic terms, the huge diversity of non-timber forest products, their year-round availability, and the universal access to them, supports resilience at a landscape, community and

[190] Zhang, Ping, He, Liang, et al., "Ecosystem Service Value Assessment and Contribution Factor Analysis of Land Use Change in Miyun County, China," *Sustainability (Switzerland)* 7, no. 6 (2015): 7333–56, https://doi.org/10.3390/su7067333.
[191] FAO, 2019b Building Resilience With Non-Wood Forest Products: Assessing Opportunities For Nutrition And Livelihoods In The West Nile Sub-Region, Uganda FAO report for UNHCR

	household level. While not specifically focused on non-timber forest products, the 2018 FAO study in Kigoma similarly found that these products were used by refugees and host communities in the region and that these represented an important component of resilient local livelihoods[192].
Environmental	• Improved ecosystem services. Studies have shown that in Tanzania an increase in community-based land management results in a decrease in soil erosion, improvement in water quality and quantity and well as a reoccupation of beehives and an increase in ecological interaction diversity, which is critical for ensuring climate resilience in activities like honey production [193],[194]. • Sequestration of 762,073 million tCO_2e and emission reductions of 3.2 million tCO_2e over 20 years (including a five-year implementation period) as a result of avoided deforestation, improved agricultural practises, afforestation and agroforestry (estimated using the FAO Ex-Ante Carbon-balance Tool[195]). Establishing a robust monitoring system would not be cost-effective in the absence of revenue streams. In 2019, UNHCR's consultations with UNFCCC and FAO confirmed this is an important mitigation co-benefit though entry into the carbon market was not advised by UNFCCC due to the current status of the market.[196]
Social and health benefits	• The participatory process of village land use planning is an open, equitable system for involving the public in planning and decision-making and increases social cohesion and peaceful co-existence. • Improved food security and nutrition for vulnerable communities. • Increase in quality of life from reduced risk of water- and vector-borne illnesses during flood incidents. • Reduced exposure to tsetse flies and risk of contracting sleeping sickness, arising from livelihood diversification and woodlot provision reducing the need of local communities to enter forested areas for activities like harvesting wood or producing charcoal for income. • Reduction of risk conflicts and competition for resource use. • Improved social cohesion among host communities and refugees.

[192] FOA and UNHCR, 2018 Cost–benefit analysis of forestry interventions for supplying wood fuel in a refugee situation in the United Republic of Tanzania Public report

[193] Patenaude, G. & Lewis, K. 2014. The impacts of Tanzania's natural resource management programmes for ecosystem services and poverty alleviation. International Forestry Review, 16(4): 459−473.

[194] Kaiser-Bunbury CN, Mougal J, Whittington AE, Valentin T, Gabriel R, Olesen JN, Blüthgen N (2017) Ecosystem restoration strengthens pollination network resilience and function, Nature 9,542(7640):223-227

[195] FAO (2019). Ex-Ante Carbon balance Tool (Ex-ACT). Version 8.5.4. Accessed: 13 November 2019. URL: http://www.fao.org/tc/exact/ex-act-home/en/

[196] In 2019 UNHCR commissioned the FAO to develop "Guidelines for Managing Forest Resources in Bidibidi Refugee Hosting Area" in Uganda (FAO, 2019a). UNHCR further commissioned FAO to produce an assessment of the non-timber forest products in the same area (FAO, 2019b). The context in Bidibidi is comparable to that in the Kigoma region of Tanzania. These studies were commissioned to inform UNHCR on the best value-for-money, sustainable land-use and development options for land that has been earmarked for refugee use. The studies recommended that afforestation to increase the supply to fuelwood and reduce the harvesting of native trees for fuelwood was a viable and recommended activity, if undertaken to directly supply fuelwood for refugees. The studies by FAO did not recommend the production of fuelwood or timber for sale, there is no value chain for these products as they can be freely harvested from indigenous forests. Further, despite the large potential for reforestation, afforestation and forest restoration, the studies also did not specifically recommend engagement with carbon markets. Fuelwood sales and carbon markets were considered but also not specifically recommended by the FAO in their 2018 study, the "Cost–benefit analysis of forestry interventions for supplying wood fuel for refugees in the Kigoma region of Tanzania" (FAO, 2018). Nevertheless, as UNHCR invests approximately $10 million USD per year in reforestation/afforestation, UNHCR has investigated the potential to engage with carbon markets for the benefit of refugees and host communities. In particular, in 2019 UNHCR undertook extensive consultations with the UNFCCC on the potential to support afforestation and reforestation through the clean development mechanism (CDM). The consultations concluded that, given the UNHCR context, the costs involved in establishing and monitoring CDM projects and the current and expected mid-term global carbon prices, the UNFCCC did not advise UNHCR to pursue CDM forestry initiatives. Nevertheless, while it may not translate into a carbon-related revenue stream, the 2018 FAO study in Kigoma did shown that forestry in refugee areas will contribute measurably to carbon sequestration (FAO, 2018), we see this as and important mitigation co-benefit of this project that supports the overarching goal of climate compatibility.
References cited in this note include: FAO, 2019a Guidelines For Managing Forest Resources In Bidibidi Refugee Hosting Area Yumbe District, Uganda FAO report for UNHCR; FAO, 2019b Building Resilience With Non-Wood Forest Products: Assessing Opportunities For Nutrition And Livelihoods In The West Nile Sub-Region, Uganda FAO report for UNHCR; FAO and UNHCR, 2018 Cost–benefit analysis of forestry interventions for supplying wood fuel in a refugee situation in the United Republic of Tanzania Public report.

Gender[197]	• Reduction of risk for gender-based violence, particularly those that typically occur during firewood collection[198] [199]. • Opportunities for women to take leading roles in Savings Associations, livelihoods activities and participatory land-use planning processes. • Of the direct beneficiary farmers that engage with the project by becoming members of Savings Associations, the objective is that at least 50% will be female. The Savings Associations will be designed specifically to appeal to female farmers, and the Savings Associations Management Offices will be instructed to ensure a balance of member's genders (in line with existing Savings Association practises). An approximately equal distribution of male and female beneficiaries is assumed amongst the indirect beneficiaries in the neighbouring communities.

D.4. Needs of recipient (max. 500 words, approximately 1 page)

As described above, Kigoma is one of the poorest and most vulnerable regions in Tanzania. The adaptation requirements of the country's rural poor, including refugees, focus on increasing water availability and quality, supporting livelihoods that are resilient to climate change and strengthening the skills and knowledge of beneficiaries to implement climate-resilient agriculture. Furthermore, these requirements are strongly aligned to the self-identified needs of the recipients, as expressed and documented by UNHCR through ongoing consultation with host communities and refugees. Consultations are supported by numerous external studies undertaken in Kigoma.

Identified recipient needs are summarised below.

- **The need for improved economic opportunities:** On a macro level, Tanzania is modernizing rapidly, but this transformation is largely in urban areas and has limited linkages to the rural Kigoma region. The proposed project promotes an approach to climate proofing subsistence activities and small-scale livelihoods, ensuring local economic needs are met.
- **The need to address environmental degradation in Kigoma.** Improved planning and protection of natural resources, including forestry resources, is needed to halt the negative patterns of land and ecosystem degradation in the region. This environmental degradation negatively influences the availability and quality of natural food and water resources that host communities and refugees depend on. Complementary activities are being undertaken outside the scope of this GCF project to address the demand side drivers of degradation, as well as other development concerns for the region.
- **The need for financial services**: Within the Kigoma region, the availability of financial services is limited. Disbursing funds through Savings Associations to deliver climate-resilient interventions is a critical tool of the project and provides access to financial services, including savings and finance, which are currently limited in Kigoma. The proposed project will scale up financial inclusion activities to improve access to finance for the host community and climate-resilient equipment for refugees. This improved access will have a clear and direct impact on livelihoods, economic growth and inclusive social and economic development, and will help to build resilience to future climate impacts.
- **The need to address refugees' unique vulnerability**: Refugees are nearly universally excluded from national or local adaptation plans, education and interventions. This project will develop a model for meeting the adaptation needs of both host communities and refugees, which can be applied locally and globally.

Another aim of the proposed project is to address critical institutional capacity needs in Kigoma. Many of the project's interventions are designed to address future climate impacts that are included on the agenda of local government officials. However, local government and civil society capacity to deliver such complex, multi-stakeholder interventions is considerably limited. For example, district government officials noted that District Environmental Management

[197] Refer to Annex 8. Gender Assessment and Action Plan section 1.4.3 for more information

[198] A 2015 field report from Refugees International found that 25% of 224 GBV incidents reported by Burundian refugees in Tanzania over a five month period occurred during firewood collection trips, and 90% of these incidents were rape or sexual assault. Vigaud-Walsh, Women and Girls Failed: The Burundian Refugee Response in Tanzania, 2015. Available at: http://static1.squarespace.com/static/506c8ea1e4b01d9450dd53f5/t/5678aee07086d7cddecf1bab/1450749707001/20151222+Tanzania.pdf.

[199] More recent data report 2,900 incidents per year between 2016-2018. There are local mechanisms for reporting, providing support, and addressing incidents within the camps. Refer to Annex 6. Environmental and Social Management System and Annex 8. Gender Assessment and Action Plan for further discussion. UNHCR data from the Sexual and Gender-based Violence Information Management System as at December 2019. Participating actors include HelpAge International, International Rescue Committee (IRC), Médecins Sans Frontières (MSF), Ministry of Home Affairs, United Republic of Tanzania (MoHA), OXFAM, Plan International , Save the Children, Tanganika Christian Relief Services (TCRS), Tanzania Police Force, Tanzanian Red Cross Society (TRCS), The United Nations Children's Fund (UNICEF), United Nation High Commissioner for Refugees (UNHCR), United Nations Population Fund (UNFPA), United Nations World Food Programme (WFP), Women's Legal Aid Center (WLAC).

Committees did not have the capacity necessary to support the Ward and Village Environmental Management Committees. A critical intervention of the project to address the overarching institutional capacity deficit is climate-resilient land-use planning in villages. The District Agencies in Kigoma, along with refugee and host community representatives, will receive capacity building and training to develop climate informed land use plans that are owned at the local level. District officials will receive further capacity building to with regards to including climate impacts and risks in regional planning processes. Additionally, the policy review process will support the institutionalisation of the impacts and lessons learned of this project as well as its demonstrated benefits.

D.5. Country ownership (max. 500 words, approximately 1 page)

The Tanzania Development Vision 2025 aspires to have the country transformed to a middle income and semi industrialized nation by 2025. The Vision is implemented through three consecutive Five Year Development Plans. The second Tanzania National Five-Year Development Plan 2016—2021, currently under implementation, defines specific sector interventions in natural resource management and agricultural development for climate change adaptation.

Tanzania's NDC (2021) identifies adaptation as a national priority, which includes addressing the severity of climate related disasters and the impacts associated with spatial and temporal variability of rainfall, droughts and higher incidence of floods. The proposed project has been designed with the Environment Division of the Vice President's Office and aligns with their plans and strategies, including: i) the National Adaptation Programme of Action (NAPA)(2007); ii) National Climate Change Strategy (2012); and iii) Guidelines for Integrating Climate Change Adaptation into National Sectoral Policies, Plans and Programmes of Tanzania (2012).

Several of the project activities are listed in the proposed "topmost urgent and immediate projects" in the NAPA, including:

- increase irrigation by using appropriate water efficient technologies to boost crop production;
- develop water harvesting, storage and recycling programs for rural communities;
- community based catchments conservation and management programs;
- afforestation programmes in degraded lands using more adaptive and fast-growing tree species; and
- establish good land tenure systems and facilitate sustainable human settlements.

The project's activities are also well aligned with relevant national sector-level priorities in water, agriculture and forestry. In the Water Sector Development Programme (phase II), these priorities include: i) integrated water resources development and management practises; ii) enhanced protection and conservation of water catchments; iii) enhancing exploration and extraction of underground water; and iv) rainwater harvesting. The Agriculture Climate Resilience Plan (2014) and the Climate-Smart Agriculture Programme for Tanzania (2015-2025) have the objective of an "agricultural sector that sustainably increases productivity, enhances climate resilience and food security for the national economic development''. The Tanzania climate-smart agriculture (CSA) guideline — developed in accordance with numerous national plans and programmes — notes that the agriculture sector employs ~78% of the population and contributes ~95% of the nations required food[200]. As a large proportion of the sector is rainfed crop production, and most households produce at subsistence level, agriculture is highly vulnerable to the impacts of climate change. The documents used to develop the guidelines are closely aligned with the government's commitment to creating a resilient, climate-smart agriculture sector by 2030. The success of implementing and upscaling CSA in Tanzania is largely reliant on institutional changes and a behavioural shift of the population The CSA guidelines identify several key areas where capacity-building interventions are required, which align with the interventions to be implemented under the proposed GCF project. These key areas include awareness raising among farmers of improved and climate-resilient farming techniques, training tailored to the needs of specific groups of stakeholders, easier access to climate information services, and improved agricultural risk management mechanisms[201]. Several projects in Kigoma have had a specific focus on capacity building for pro-poor economic development in the region as there is a high incidence of poverty in the population[202]. These projects also demonstrate the need to improve dialogue between stakeholders[203].

[200] Rioux, J et al. 2017. Climate-Smart Agriculture Guideline for the United Republic of Tanzania: A country–driven response to climate change, food and nutrition insecurity.
[201] Rioux, J et al. 2017. Climate-Smart Agriculture Guideline for the United Republic of Tanzania: A country–driven response to climate change, food and nutrition insecurity.
[202] Enabel. 2009. Support to Income Generating Activities for Coast and Kigoma Regions. Available at: http://www.diplomatie.be/oda/11768_PROGDESCR_Technisch_en_Financieel_Dossier.pdf
[203] Enabel. 2009. Support to Income Generating Activities for Coast and Kigoma Regions. Available at: http://www.diplomatie.be/oda/11768_PROGDESCR_Technisch_en_Financieel_Dossier.pdf

The Tanzania Country Refugee Response Plan for refugees from Burundi and the Democratic Republic of the Congo January 2019 — December 2020 includes “wider efforts within the environment and energy sector and associated links with climate change adaptation and mitigation. This will be undertaken in the context of both the host community and the refugee community. Implementation priorities in this regard will include climate smart agriculture, forest landscape restoration, sustainable forest management, community based natural resource management, land use planning, water conservation through the promotion of water harvesting technologies, Integrated Water Resource Management, river catchment conservation and the adoption of renewable energy technologies”. In the project area, the Regional Commissioner and the Regional Administrative Secretary of Kigoma have identified the region’s highest priorities for development, which include improving extension services to farmers in order to increase productivity and community participation in forest management and reforestation, both of which are addressed through this project.

The regulatory framework in respect of refugees in Tanzania is set out in the Refugees Act, 1998 (the 1998 Act) and Tanzania’s 2003 National Refugee Policy (2003 Policy). Tanzania has ratified the 1951 United Nations Convention Relating to the Status of Refugees and the 1967 Protocol Relating to the Status of Refugees. The Tanzanian Government is committed to protecting refugees and asylum seekers by its international legal obligations. The project’s approach to addressing the needs of both the host communities and refugees follows principles in the Global Compact for Refugees[204], which has been endorsed by the Government of Tanzania, proposes coherent humanitarian and development responses to support both groups as effective management of refugee situations are based on the resilience of host communities. Moreover, in the context of this project and many refugee situations, host communities and refugees are embedded in the same landscape and benefit from the same ecosystem services and resources that are impacted negatively by climate change. Hence, their needs should be addressed in an integrated way.

In discussions on the project with the Tanzanian NDA (Vice President’s Office-Division of Environment) throughout 2017–2021, the NDA expressed continued support and prioritization of the project to UNEP, UNHCR and GCF and issued a no objection letter. Following consultations with the Government of Tanzania from June to September 2021, it was agreed that the Government pf Tanzania, acting through its Vice-President office, and UNHCR will co-execute this project in order to enhance country ownership, strengthen the climate-humanitarian nexus, facilitate policy and planning processes and promote the sustainability of project outcomes at the local, regional and national level. As described in Section B4 Implementation Arrangements, the Vice President’s Office-DoE and UNHCR will be executing entities of this project, with the Vice President’s Office- DoE acting as the lead executing entity. Additional consultations with the Vice President’s Office- DoE, the Tanzania Forest Service and UNHCR took place in the period September-October 2021 to inform the revision of the funding proposal and the new execution arrangements.

In addition to the executing entities, the Government of Tanzania (at ministerial, regional and district levels) will be represented in the Project Steering Committee of the project to ensure oversight and provide contributions.

A series of stakeholder consultations have been conducted throughout project preparation to gauge interest, solicit feedback and engender ownership, the outcomes of which have been incorporated into the project design. The stakeholder engagement during the second field visit (16–23 September 2018) included focus groups with representatives of host communities and refugees. These engagements were aimed at gathering information about existing programmes active in the region, discussing the feasibility of potential interventions and collecting information about the necessary management and arrangements structures for the implementation of the project. The scope of these engagements included validating baseline data, assess interesting and buy-in for potential interventions and providing opportunities to discuss needs and desires of the targeted population. Details of the stakeholders consulted, and the feedback provided can be found in the Feasibility Study.

D.6. Efficiency and effectiveness (max`. 500 words, approximately 1 page)

This project has been designed to use GCF funds as efficiently and effectively as possible, while recognizing the unique challenges faced by refugee-hosting regions. This is achieved through consideration of the points below.

- The selection and design of project activities is rooted in proven, effective methods to address the identified impacts of climate change on ecosystems, ecosystem services and agro-ecological systems. For example, Community-Based Forest Management is included as an activity as there is extensive evidence in Tanzania and neighbouring countries of its effectiveness in restoring miombo woodland and its associated ecosystems services. Each activity section in the Feasibility Study outlines where it has worked before, and what lessons can be learned.
- The exact locations for interventions and detailed intervention design will be agreed during the project itself, during the participatory climate-resilient land-use planning process. This approach will be driven by the current, localized needs of beneficiaries in the project area and reflects the rapidly changing nature of refugee-hosting regions. This

[204] UNHCR, 2018. Global Compact on Refugees, final draft as at 26 June 2018.

dual approach of being demand-led whilst also rooted in best practise will ensure that GCF funds are efficiently utilised where they are needed most and on proven interventions.

- The project prioritises traditional, low-cost solutions wherever possible. For example, the irrigation needs for agriculture of the project are met by a combination of rainwater-harvesting systems, solar water pumps and hand-operated treadle pumps. The pumps, both solar and treadle, would primarily draw water from the rainwater harvesting structures described above (although a limited number may draw directly from natural water courses). This approach promotes the rational and equitable use of water, which will be discussed and agreed upon in the village land use planning process. Treadle pumps represent a more efficient usage of GCF capital to meet irrigation requirements, pumping 4 litres per USD, compared with 8 litres per USD for solar pumps. Treadle pumps therefore play a key role in the project, with ~3 treadle pumps installed for every solar pump. However, solar pumps are still critical, for instance in supplementing water for the large-scale afforestation nurseries which require greater volumes of water of per day and would be unsuitable for a manually-operated system. Other examples of traditional, low-cost solutions include stone pitching and gabions to prevent flooding and erosion and prioritising resilient strains of native crops (cassava and beans) over foreign species.
- The project scales up successful best practises in the region practised and financed by different partners, including the following. These examples are discussed in the activity descriptions in Section B.3:
    - Resilient production of cassava and beans
    - Fuelwood production and forest protection
    - Savings Associations
    - Land use planning
    - Beekeeping and mushroom growing
    - Farmer field schools
- A critical aspect of the project's financial structure is the use of Savings Associations (SAs), accompanied by the Savings Associations Management Office (SAMO) which provide support and capacity building. These enable GCF grant funds to be used efficiently by providing them as farm inputs and tools for climate resilient practises. As members earn higher incomes from resilient activities, savings are reinvested into the SAs. The structure is intended to ensure that GCF funds are recycled after project close.
- Agroforestry and afforestation activities are included in the economic model, but a revenue stream would not be viable. Sub-activity 2.2.2 is intended to take the pressure of fuelwood harvesting in native miombo forests. Selling of fuelwood would provide a disincentive to protect natural forests.
- The project will be ~20% co-financed. Projected parallel financing is about double the project size.

In addition to the points above, the economic and financial analysis associated with the funding proposal (Annex 3) has revealed the following with regards to the project's efficiency and effectiveness.

Financial analysis of key project Activities and Sub-Activities was carried out, along with an economic analysis of the full project. Table 12 shows positive internal rate of return (IRR) figures for all activities analysed.

The IRR estimated for the establishment and maintenance of Agroforestry systems (Sub-activity 2.2.1) is 98% and the net present value is US$9.8 million over a 20-year period. It should be noted that Agroforestry provides a wide spectrum of ecosystem services including provisioning, regulating, habitat and cultural services[205]. However, the value measured in the financial analysis of agroforestry includes only two provisioning services, namely those of timber harvesting and fodder production. A broader array of the benefits from agroforestry will be assessed under Activity 4.1, which will provide a more complete picture of this activity's true IRR.

Afforestation is also shown to result in a relatively high IRR of 50%. However, the NPV is lower than that of forestry at US$2.9 million over the 20-year period. As with agroforestry, it is likely that his low NPV is the result of the partial nature of data considered in the analysis. Only one of the ecosystem services associated with woodland – timber production – has been measured under the analysis. For example, a systematic review across the tropics found that afforestation resulted in an average of a three-fold increase in the infiltration capacity of soils in the 14 comparative experiments

[205] Sileshi, G. Akinnifesi, FK. Ajayi, OC. Chakeredza, S. Kaonga, M. Matakala, PW. 2007. Contributions of agroforestry to ecosystem services in the miombo eco-region of eastern and southern Africa. African Journal of Environmental Science and Technology, 1(4). Jose, S. 2009. Agroforestry for ecosystem services and environmental benefits: an overview. Agroforest Syst 76, 1–10. https://doi.org/10.1007/s10457-009-9229-7; Buchelli, VJP, Bokelmann, W. 2017. Agroforestry systems for biodiversity and ecosystem services: the case of the Sibundoy Valley in the Colombian province of Putumayo. International Journal of Biodiversity Science, Ecosystem Services and Management, 13(1); Brown, S.E., Miller, D.C., Ordonez, P.J. et al. Evidence for the impacts of agroforestry on agricultural productivity, ecosystem services, and human well-being in high-income countries: a systematic map protocol. Environ Evid 7, 24 (2018). https://doi.org/10.1186/s13750-018-0136-0

Ilstedt, U. Malmer, A. Verbeeten, E. Murdiyarso, D. 2007. The effect of afforestation on water infiltration in the tropics: A systematic review and meta-analysis. Forestry Ecology and Management, 251. https://doi.org/10.1016/j.foreco.2007.06.014

considered. The implications for groundwater recharge and flood attenuation may be considerable. The economic analysis planned under Activity 4.1 is expected to reveal the value of a broader spectrum of ecosystem services associated with in-tact woodland ecosystems.

Climate-resilient agriculture is shown to provide an IRR of 541%, with an associated NPV of US$5.8 million over the 20-year period under the “with project scenario” relative to the “without project scenario”. The benefits measured under this analysis include improved yields from planting climate-resilient crop varieties. It is possible that other benefits, such as those associated with a higher consistency production relative to traditional agriculture, have not been captured as part of this analysis.

The water harvesting and irrigation interventions under Activity 3.2 have been found to generate an IRR of 197%. The NPV was estimated to be US$3.5 million over the 20-year period, which is equivalent to ~US$175,000 per year.

The IRR for promotion of beekeeping under Sub-Activity 3.3.1 is undefined, given that there are no years in which net benefits are less than zero for this activity. This is of course a favourable indication, and the best possible value for the IRR. Given the limited local market for honey, it is likely that only a portion of the value of this activity will generate monetary returns. The NPV is US$ 996,463 over the 20-year period, which equates to ~US$48,823 per year.

The promotion of mushroom cultivation was estimated to generate an IRR of 787% and an NPV of US$554,000 over the 20-year period. This equates to ~US$27,700 per year. Marginalised groups will be identified to receive this benefit.

All NPV figures presented in Table 12 have been estimated using a discount rate of 9%, in accordance with the rate used in FAO & UNHCR’s 2018 Cost–benefit analysis of forestry interventions for supplying wood fuel in a refugee situation in the United Republic of Tanzania by A. Gianvenuti & V.G. Vyamana.

Table 12. Internal rate of return and financial net present value for each of the listed activities and sub-activities. IRR for activity 3.3.1 is undefined, given that there are no years in which net benefits are less than zero for this activity.

Project sub-activity	IRR	Increase in NPV with project relative to without project (US$)	
		NPV	NPV per hectare
2.2.1. Establish and maintain (a) agroforestry systems; and (b) nurseries for production of tree seedlings	98%	9,828,099	343
2.2.2. Undertake afforestation on village land to a) restore critical ecosystem services, b) establish a sustainable supply of fuelwood	50%	2,933,973	1,467
3.1.2. Provide inputs for climate-resilient agriculture	541%	5,877,457	19,592
3.2. Increase water availability and use-efficiency through water harvesting and efficient irrigation interventions	197%	3,509,447	8,158
3.3.1. Promote beekeeping to incentivise forest conservation in the forests surrounding the camps and villages.	Undefined	996,463	N/A
3.3.2. Promote mushroom cultivation for marginalised groups with training and mushroom growing facilities.	787%	553,763	N/A

Table 13 displays the project’s ERR, which was estimated to be 13% when the value of greenhouse gas mitigation benefits are excluded and 20% when they are factored in. Also displayed is the project’s NPV over a 20-year period. The net benefit estimated to result from project implementation is US$ 10.5 million over the 20-year period, when mitigation benefits are excluded and US$20.8 million when they are included. These values equate to US$49 per hectare excluding mitigation benefits and US$ 98 per hectare including mitigation benefits. The economic analysis used the social rate of time preference as the discount rate. This was estimated to be 6.2%, following adjusted for Tanzania’s growth in per capita consumption[206].

Table 13. Economic rate of return and economic net present value of project

Economic CBA	ERR	NPV with project relative to without project (US$)	
		NPV	NPV per hectare
Excluding mitigation benefits	13%	10,516,821	49

[206] Further details are provided in Annex 3, which includes all financial and economic models used to generate the information presented here.

Including mitigation benefits	20 %	20,882,214	98

The results of financial and economic analyses were subjected to sensitivity analysis to assess their level of robustness to changes in key parameters, and to ascertain switching values. The results, outlined in Table 14, show that Sub-Activity 3.1.2 (inputs for climate-resilient agriculture) is most sensitive to changes in key cost and benefit streams. Under a 13% reduction in benefits, this activity results in a negative IRR. Similarly, a 30% reduction in benefits leads to a negative IRR for Sub-Activity 2.2.2 (afforestation). Also highlighted in the table, a 35% reduction in benefits results in a negative IRR for Activity 2.2.1 (agroforestry). The project as a whole is shown to be sensitive to changes in the magnitude of benefits. When mitigation benefits are excluded from the analysis, a 61 % reduction in project benefits results in a negative project ERR (71 % when mitigation benefits are included).

In contrast to changes in project benefits, the outcomes of the analysis appear to be relatively robust to changes in costs. Of note is that when costs incurred by beneficiaries in the 'with project' scenario are increased by 30%, the IRR for Activity 3.1.2 (inputs for climate-resilient agriculture) switches to negative. Also, when the costs associated with Activity 2.2.1 (agroforestry) are increased by 53%, the IRR switches to negative. The project as a whole, however, maintains a positive ERR up to a 156% increase in costs when mitigation benefits are excluded and a 239% increase in costs when mitigation benefits are included.

As mentioned previously, the full range of project benefits has not been included in the cost-benefit analysis. By comparison, it is likely that a relatively smaller proportion of the total costs has been omitted from the analysis. These findings should therefore be taken as representing a cautious outlook.

Table 14. Sensitivity analysis of financial and economic analyses.

Assumption and parameter	Values at which IRRs switch to negative						`Value at which project ERR switches to negative	
	2.2.1	2.2.2	3.1.2	3.2	3.3.1	3.3.2	Excluding mitigation benefits	Including mitigation benefits
Increase in activity implementation costs	53%	120%	30%	195%	901%	236%	156%	239 %
Decrease in magnitude of benefits	35%	30%	13%	61%	91%	71%	61%	71 %

E. LOGICAL FRAMEWORK

This section refers to the project/programme's logical framework in accordance with the ***GCF's Integrated Results Management Framework*** *to which the project/programme contributes as a whole, including in respect of any co-financing.*

E.1. Project/Programme Focus

Please indicate whether this proposal is for a mitigation or adaptation project/programme. For cross-cutting proposals, select both.

☐ Reduced emissions (mitigation)

☒ Increased resilience (adaptation)

E.2. GCF Impact level: Paradigm shift potential (max 600 words, approximately 1-2 pages)

This section of the logical framework is meant to help a project/programme monitor and assess how it contributes to the paradigm shift described in section D.2 above by applying three assessment dimensions - scale, replicability, and sustainability.

Accordingly, for each assessment dimension (see the definition per assessment in the accompanying guidance note), describe the current state (baseline) and the potential scenario (target) and rate the current state (baseline) by using the three-point-scale rating (low, medium, and high) provided in the guidance note. Also describe how the project/programme will contribute to that shift/ transformation under respective assessment dimensions (scale, replicability and sustainability). In doing so, please refer to section B.2(a) (theory of change).

Assessment Dimension	Current state (baseline)		Potential target scenario (Description)	How the project/programme will contribute (Description)
	Description	Rating		
Scale	Tanzania ranks 8th in the GDP ranking of Africa countries [207] and 26 in the HDI ranking of African countries (low human development). Kigoma region is one of 30 regions in Tanzania, with a population of ~2.3 million people, in a country of ~47 million people. Kigoma Region is part of the Lake Tanganyika basin, a global biodiversity hotspot. Forests in the Kigoma region are largely miombo woodland, representative of	<u>High</u>	The project landscape-based approach offers scalable solutions to address climate adaptation that consider resource use, management practises, landscape ecology and complex development-humanitarian needs and concerns. The project will promote a paradigm shift towards: 1) inclusive solutions to adaptation needs of host communities and refugees through adaptation solutions at the landscape level; 2) integrated interventions across sectors that are designed to be mutually reinforcing; and 3) implementation of activities that are	Outputs 1-3 will implement adaptation activities at the landscape level with host and refugee communities. Output 4 activities will support scaling up and replication of the project results through sharing of lessons learned and policy mainstreaming within Tanzania and beyond. Output 4 activities including ecosystem service valuations, development of guidelines and policy briefs, and policy mainstreaming within district, regional, and national governance arrangements in

[207] World Economic Outlook database SSA: April 2022 Nominal GDP". *imf.org*.

	Tanzania as whole. Approximately 10 million ha of forest was lost in Tanzania between 1990 and 2015, the majority of which were miombo woodlands. Miombo woodlands span across parts of: Angola, Botswana, Burundi, Democratic Republic of Congo, Malawi, Mozambique, Namibia, Tanzania, Zambia and Zimbabwe, Covering around 2.4 million km2, Today,~250,000 refugees are present today in Tanzania, with most accommodated in the two camps in Kigoma. There are now more than 100 million people globally displaced, around 75% of whom are hosted by low or middle-income countries and living in protracted circumstances. African countries host 5,7million refugees (24% of the refugees under UN mandate worldwide) and 14.7 million IDPs of concern to UNHC.[208] Global estimates are that climate change could contribute to the movement of 216 million people within their own countries by 2050, unless concrete climate and inclusive development actions are taken. According to the IPCC AR6-WG2 report, IDP and refugee settlements are disproportionately concentrated in regions that are exposed to higher-than-average warming levels and specific climate hazards, including temperature extremes and drought.		informed by projected climate impacts where the people and landscape would be better able to cope with, and adapt to, climate shocks. Experience from elsewhere in Tanzania shows that once the governance structures are in place villagers are able to ensure ongoing community-based forest management with minimal external support. The practise will both intensify and spread wider afield. with the villages having: i) reached consensus on how the village land should be used to improve climate resilience; and ii) obtained legal status in the form of a Customary Certificate of Rights of Occupation. Achieving these goals will allow the host communities to assert their own authority around land-use planning (including by resolving any potential competing claims to land use) to adapt to be more climate resilient and in doing so, will ensure sustainability in the long term. The Tanzania Forest Service will be able to provide continuing management support after the project has ended. We expect that communities will have an interest in further extending the area under plantation to meet a larger part of fuel demand in the future. The District Council and the host communities themselves will be better placed to adapt to future changes in circumstances. To our knowledge, this project is the first of its kind that addresses adaptation needs of host communities and refugees in an integrated way by institutionalising climate-resilience and EbA practises. This	Tanzania will leverage the project outputs to achieve the desired adaptation outcomes. The scale of investment in the project will be catalytic and bring together different actors and implementing partners in the region for activities to make a lasting impact at the landscape level. The project aims to involve local government authorities in the implementation of the project including the Regional Commission, District Councils, Tanzania Forest Service, Lake Tanganyika River Basin Authority, and other relevant actors. Training and equipment will be provided by the project to support their mandate in land use planning, forest monitoring and conservation, and expanding extension work climate resilient agriculture practices. Capacity building will be provided to support district officials and Tanzania Forest Service officers in including climate impacts and risks for the region into the land-use planning processes, enabling Participatory Land Use Management to support communities to embed these impacts in the allocation of land to different purposes. Knowledge and lessons will be mainstreamed into national and local government structures and wider humanitarian programmes. This will be achieved by generating and documenting evidence, disseminating information and providing inputs and knowledge to policy and planning processes. In Kigoma, after the ex the GCF investment, it is expected that local government authorities, UNHCR, and implementing partners that have continued field presence in Kigoma over the

[208] Data of 2022 available at UNHCR - Refugee Statistics

	There is evidence of a pathway towards better mainstreaming climate resilience in sectoral policies and plans and the consideration of host community and refugee needs as part of the same socio-ecological landscape. Example of this is the Kigoma Joint Programme II and the associated initiatives supporting the humanitarian-development-peace nexus in Kigoma. However, while other activities in the region are addressing specific development and humanitarian needs, there is limited landscape-level coordination of these interventions Additionally, projects and activities addressing humanitarian or development concerns have limited consideration for the long-term impacts of climate change or have insufficient scale to trigger a paradigm shift.		approach has the potential to lead globally in mainstreaming climate change adaptation in refugee and host community settings as many countries that host refugees are also among the most vulnerable to climate change (e.g. Bangladesh, Jordan). The project will shift the paradigm in land use planning and practices in refugee and host community settings from fragmented, sector-based approaches that do not consider climate change impacts, to integrated solutions that build resilience to climate change impacts. As Tanzania is a frontrunner in applying the One UN working modus, UN Agencies, including UNEP and UNHCR, are often invited to contribute to national policy and development dialogues, with UNHCR also participating in the Development Partners Group on environment. As a result, UNEP as AE and UNHCR as one of the EE of the project are well placed to promote upscaling and replicability of this model in other contexts, and to other UN organisations. UNHCR Global Compact on Refugees (GCR) calls for strengthened national and international partnership with development partners, the private sector and other stakeholders to address challenges related to energy and the environment. The proposed project is recognised as a major partnership effort to operationalise the GCR. Lessons from the project's approach and implementation will be documented and shared to guide UNHCR's global engagement with environment and climate change actors as well as inform operations	long term will have the capacity to plan and implement adaptation investments at the landscape level based on the project's model and program their future investments from local, bilateral and multilateral sources accordingly.

			in other countries. Dedicated teams have been set up to strengthen the operationalisation of the GCR as well as to support UNHCR's work on mitigation and adaptation to climate change. Corporate commitments to climate change and environmental challenges as well as organisational structures at UNHCR encourage the institutionalisation of the approach proposed by the project.	
Replicability	Though the districts of Kibondo, Kakonko and Kasulu in Kigoma region are among the poorest, and the region's per capita income — at ~USD 503 — is particularly low compared with other regions in the country, many of the development issues are representative of challenges in the country. Approximately 25% of Tanzania's population of ~47 million live below the poverty line, with subsistence agriculture being the primary livelihood of 75–80% of households in the country — including the majority of the rural low-income families. Agriculture is a dominant sector of the Tanzanian economy, accounting for 25% of the GDP. Despite having fertile land and water resources, the agricultural sector is hampered by low productivity and persistent poverty. This project will therefore be a flagship on the use of nature-based solutions for adaptation in Tanzania. Tanzania's Nationally Determined Contributions (NDC) identifies	<u>High</u>	The proposed project will overcome key barriers that impede building the climate resilience at the landscape level such as : • Limited capacity in local institutions and processes to plan for and address long term climate vulnerabilities • Refugees and the special needs of their host communities are not well articulated in adaptation policies, legislation, planning or programmes. • Limited access and uptake of agricultural, forestry or land management technologies or knowledge to improve climate resilience. • Low purchasing power of smallholder-farmers and refugees limits access to inputs for resilient agriculture and other long-term investments. • Limited investment in Kigoma region and competing development priorities defer long-term adaptation planning or leave it underfunded. During the implementation period the policy environment remains favourable to the mainstreaming of adaptation into policies, plans and programs as	The project will use the results of the planning processes and physical interventions implemented to demonstrate the economic, environmental and social value of this model to stakeholders. This approach will contribute to the sustainability and scalability of the outputs and fund-level outcomes at the national level. The project will contribute to building strengthened institutional and regulatory systems for climate-responsive planning and development. These strengthened systems, and the evidence-base that underpins them, will further contribute to the sustainability of field-based adaptation measures. This will be achieved by broadening the EbA and climate-resilient planning knowledge base — with economic evidence, policy briefs, guidelines — and by institutionalising adaptation in into policies, plans and strategies in Tanzania, as well as in humanitarian programmes.

	adaptation as a national priority. The interventions outlined in this Feasibility Study have been designed with the support of the Environment Division of the Vice President's Office and aligns with their plans and strategies including the National Adaptation Programme of Action (NAPA, 2007) and National Climate Change Strategy (NCSS, 2012), Guidelines for Integrating Climate Change Adaptation into National Sectoral Policies, Plans and Programmes of Tanzania (2012), Tanzanian Nationally Determined Contributions (NDC) of 2021, and the Tanzania Development Vision 2025. The is an increasing trend worldwide in the consideration of nature-based solutions to enhance climate resilience in refugee settings. However, the adaptation needs of host communities and refugees tend to be considered separately by development and humanitarian programmes. Through this project, UNHCR and the Government of Tanzania are promoting an innovative landscape approach that addresses adaptation needs of host communities and refugees in an integrated way		recommended by the Tanzania NAP stock take, the National Adaptation Compact and the National Climate Change Response Strategy (2021-2026). The replication of climate-resilient village land use planning is facilitated by the learning generated by this initiative and the new enabling policy framework, including the National Climate Change Response Strategy (2021-2026), the National Environment Policy of 2021 and the Environmental Master Plan (2022-2032), which promote the scale up of resilient land use plans for sustainable development (in at least 4,800 villages by 2032) and to mainstream climate change into land use planning by 2032. The draft 3rd edition of the Guidelines for Integrated and Participatory Village Land Use Planning, Management and Administration from the National Land Use Planning Commission, which include climate change adaptation and mitigation strategies and recognize ecosystem protection and conservation as an adaptation strategy is operationalized and capacity built through its implementation in target villages.	

Sustainability	More than 32% of Kigoma region's population lives below the poverty line. The rising population of the region, as well as the continual subdivision of land within families have resulted in over-intensive agricultural practises which are placing increasing pressure on limited natural resources. Farmers and livestock keepers sell charcoal and fuelwood in lean times, resulting the deterioration of miombo forests. Fuel wood extraction accounts for 60% of forest resource usage. In Kigoma, closed woodland had diminished to only 5.6% in 2010 from 23.2% in 1990, while cultivation increased to 17% in 2014 from 2% in 1990. Agricultural production in Kigoma is dominated by small-scale, subsistence farmers practising low-input farming, with average farm sizes ranging between 0.2 and 2 ha. Uptake of agricultural technology in Kigoma is low. Political instability in neighbouring countries results in food shortages, which are often augmented with supplies from Tanzania, producing price fluctuations in Kigoma and impacting food security in the region. Tensions between host communities and refugees result over shared natural resources — including forests, arable land and water resources — which are often the result of insufficient provision of	Low	The project strategy is to ensure that the landscape and ecosystem of the Kigoma region can continue to support its residents, including refugees, and host communities, including agro-pastoralist groups that pass through the region as it faces challenges of climate change. The interventions are planned within the sub-catchments and contiguous landscape of host communities and refugees with the aim of strengthening the ecosystem's capacity to respond to climate shocks and continue to provide services such as provision of soil nutrients, prevention of flooding and erosion, creating shade for crops and others. The interventions in degraded areas within and around refugee camps are expected to support the broader resilience of both ecosystems and communities. Having locally-led land management systems and climate resilient agricultural improvements can reduce the drivers of deforestation while at the same time improving the condition of the forest and soils. Managing forests for climate change impacts also safeguard forest-based climate resilient livelihoods, such as beekeeping, further encouraging sustainable forest use. Preserving and restoring native forest landscapes under community management regimes yield benefits in flood control and erosion reduction not only at the immediate intervention areas, but more broadly at the basin level. The water harvesting, storage, distribution and irrigation interventions will protect and	The project will promote 1) inclusive solutions to adaptation needs of host communities and refugees through adaptation solutions at the landscape level; 2) integrated interventions across sectors that are designed to be mutually reinforcing; and 3) implementation of activities that are informed by projected climate impacts where the people and landscape would be better able to cope with, and adapt to, climate shocks. Village land use planning culminating in the issuance of a Certificate of Customary Right of Occupancy (CCRO) is expected to empower communities to protect their land. Village Land Forest Reserves (VLFR) gazettement, strengthening community institutional monitoring systems will further ensure conservation benefits to the users. Community Based Forest Management (CBFM) interventions will protect natural forest systems and enhance ecosystem services. Conservation will be reinforced by the agroforestry and afforestation interventions providing sustainable alternative sources for energy needs, thereby reducing pressure on the natural forests and allowing them to regenerate. Demand for fuelwood will be diverted from slow-growing indigenous Miombo ecosystems, towards fast-growing plantations that will be established on marginal land. Agroforestry interventions, by providing multiple uses such as food, fodder, shade for crops, and sustainable fuelwood production, will further deter activities or behaviour that degrade the natural forest

	food in camps, and a strict encampment policy which has led to limited livelihood activities within the camps. The combination of the reliance on subsistence agriculture and the additional pressure on, and conflict over, natural resources as a result of hosting the refugees has resulted in the significant degradation of ecosystems in the region of the camps. Kigoma ranks second as the most affected region by wildfires, with at least 1 million ha burned annually. The risk of uncontrolled fires is expected to be exacerbated under climate change conditions. Increasing temperatures and more frequent higher intensity rainfall events are likely to further reduce the stability of soils on agricultural lands, resulting in soil and nutrient loss through erosion. A degraded landscape with low soil stability, reduced soil nutrient availability, and reduced soil moisture make the landscape more susceptible to the impacts of climate change and less able to support agricultural and other livelihood activities, less able to retain water resources required for these livelihoods and more susceptible to flooding. Kibondo and Kakonko districts are part of the Lake Tanganyika Basin which are experiencing a decline in the flow of major rivers and drying		increase agricultural production in the face of climate change. Local leadership is promoted by establishing strong community-level institutions in the form of Forest Management Committees to ensure this continuity. Agroforestry will also become self-sustaining — as farmers realise the benefits of including trees in their farming systems, it is expected that the practise will both intensify and spread wider afield. The capital available in the SAs will also stimulate markets for produce, incentivizing local entrepreneurs to further establish small scale nurseries for agroforestry seedlings. The project has targets on adoption of climate resilient farming practices, diversified livelihoods, protected forest area. After the exit of the GCF investment, it is expected that local government authorities, UNHCR, and implementing partners that have continued field presence in Kigoma over the long term will have the capacity to plan and implement adaptation investments at the landscape level based on the project's model and program their future investments from local, bilateral and multilateral sources accordingly.	systems — including unsustainable fuelwood collection and charcoal burning. The protection of the North Makere, South Makere, Mwale and Buyungu National Forest Reserves will be ensured through the technical and institutional capacity development of the Tanzania Forest Service (TFS). Climate resilient agriculture will be adopted through the provision of inputs and transfer of skills and technologies to strengthen the capacities of farmers and district officials. Furthermore, through water harvesting and irrigation, water availability and water use efficiency will be improved. Livelihoods will be diversified in respect of beekeeping and mushroom cultivation. In the refugee camps, flood and erosion control measures will help avoid soil run-off and erosion. The flood and erosion control measures will strengthen the climate resilience of the project's natural landscape, e.g. soil erosion, damage to crops and infrastructure, and waterlogging. The interventions under this activity will substitute the services provided by ecosystems in areas that have been degraded to the extent of having insufficient natural flood-attenuation capacity. Project interventions include extensive training and capacity building, which will allow knowledge of the benefits of activities such as community-based forest management, resilient agriculture and alternative livelihoods to be sustained and dissipated through wider communities. This approach includes joint meetings between

	wetlands in Kigoma, leading to a heavy dependence on pumped groundwater. Host communities reported exacerbated flood impacts. In the camps areas the main issue is storm water, not river flooding – causing gully erosion. Rural populations, including women, generally have limited technical skills and technological support. Women are often excluded from decision-making on access to and the use of land and resources critical to their livelihoods. The enforcement of established guidelines and village bylaws intended to protect water resources and mitigate flood impacts is a challenge. Human disease epidemics are a major concern affecting communities within Kigoma. and have been amplified during periods of severe weather, including floods and droughts.			host communities and refugees, awareness-raising and participatory processes to manage benefit sharing arrangements between host communities and refugees and facilitate sustainable offtake, promote ecosystem protection, and minimise conflict over resource use. Many of the interventions introduced can be maintained or expanded by host communities without further support after the initial five-year project period since they require minimal funding and repay the investment in inputs and labour through increases in productivity and income. The project aims to show the viability and impact of its activities, setting the context for the continuation of these interventions beyond the project lifespan. Scaled up Saving Associations which will allow beneficiaries to access seeds and seedlings, equipment, and tools for greater engagement in climate resilient activities. This will promote savings and reinvestment and boost markets for these inputs. The project's gender sensitive approach seeks to achieve practical and tangible benefits to women to improve their decision-making power both at the household and the community level as well as bring about strategic changes in gender relations.
E.3. GCF Outcome level: Reduced emissions and increased resilience (IRMF core indicators 1-4, quantitative indicators)				

Select appropriate IRMF core and supplementary indicators to monitor project/programme progress. More than one IRMF (core and or supplementary) indicators may be selected as applicable for each GCF results area and project/programme outcome (as defined in the table in section B.2(b)). If IRMF indicators are unable to measure any given project/programme outcomes, project/programme-specific indicators should be developed under section E.5 (project/programme specific indicators).						
GCF Result Area	**IRMF Indicator**	**Means of Verification (MoV)**	**Baseline**	**Target**		**Assumptions / Note**
				Mid-term	**Final[209]**	
All adaptation result areas	Core 2: Direct and indirect beneficiaries reached	Baseline reports, mid-term and final evaluation reports Population data records from the national census and UNHCR	0 direct beneficiaries 0 indirect beneficiaries	0 direct beneficiaries 0 indirect beneficiaries	570,341 direct beneficiaries ~50% of which are female These are the households in the host community with access to benefits, along with the refugees in the camps within the area of focus. 711,516 indirect beneficiaries ~50% of which are women	Benefits for direct and indirect beneficiaries are obtained upon completion of planned interventions in all target host villages and the refugees camps. Please refer to Table 10.a above for the detailed calculations and assumptions on direct and indirect beneficiaries. Estimated average population growth rate of 2.4% in host communities. No major variance in refugee population numbers

[209] The final target means the target at the end of project/programme implementation period. However, for core indicator 1 (GHG emission reduction), please also provide the target value at the end of the total lifespan period which is defined as the maximum number of years over which the impacts of the investment are expected to be effective.

ARA1 Most vulnerable people and communities	Core 2: Direct and indirect beneficiaries reached	Baseline reports, mid-term and final evaluation reports Population data records from the national census and UNHCR	0 direct beneficiaries 0 indirect beneficiaries	0 direct beneficiaries 0 indirect beneficiaries	570,341 direct beneficiaries ~50% of which are female These are the households in the host community with access to benefits, along with the refugees in the camps within the area of focus. 711,516 indirect beneficiaries ~50% of which are women	Benefits for direct and indirect beneficiaries are obtained upon completion of planned interventions in all target host villages and the refugees camps. Please refer to Table 10.a above for the detailed calculations and assumptions on direct and indirect beneficiaries. Estimated average population growth rate of 2.4% in host communities. No major variance in refugee population numbers
ARA1 Most vulnerable people and communities	Supplementary 2.1: Beneficiaries (female/male) adopting improved and/or new	Independent result verification reports to be produced at midterm and completion stages and compared to the initial situation (baseline) incorporating community	Number of people who adopted climate resilient farming 0 people 0 females 0 Males	Number of people who adopted climate resilient farming 1,500 people 750 females 750 males	Number of people who adopted climate resilient farming 15,000 people 7,500 females 7,500 males	Policies on livelihoods and labour regulations among refugees remain consistent

	<u>climate-resilient livelihood options</u>	surveys based on a random sampling plan. [210]	Number of people who adopted mushroom growing 0 people 0 females 0 males Number of people who adopted beekeeping 0 people 0 females 0 males Total number of people who adopted diversified, climate-resilient livelihood options 0 people 0 females 0 males	Number of people who adopted mushroom growing[211] 50 people 25 females 25 males Number of people who adopted beekeeping 80 people 40 females 40 males Total number of people who adopted diversified, climate-resilient livelihood options 1,630 people 815 females 815 males	Number of people who adopted mushroom growing 500 people 250 females 250 males Number of people who adopted beekeeping 850 people 425 females 425 males Total number of people who adopted diversified, climate-resilient livelihood options 16,350 people 8,175 females 8,175 males	The socio-cultural behaviours of beneficiaries will change as a result of project activities Incentives and benefits are sufficient in promoting uptake and behaviour change among beneficiary groups.

[210] Means of Verification for Fund-level Impact and Outcome indicators will be triangulated with the baseline surveys that will be undertaken by project consultants, and at project mid-term and project end with the latest available national census data. When data is available from the next national census, this will be used.
[211] . ~30 mushroom growing facilities have been established for ~25 mushroom growing groups (for marginalised people in refugee camps and host communities, sized 10-20 people per group). The estimates of males and females are indicated for planning purposes. The activity will likely have more female participants.

ARA2 Health, well-being, food and water security	Core 2: Direct and indirect beneficiaries reached	Independent result verification reports to be produced at midterm and completion stages and compared to the initial situation (baseline) incorporating community surveys based on a random sampling plan	Direct beneficiaries: 0 people 0 females 0 males	Direct beneficiaries: 1,630 (50% women) indirect beneficiaries: 34,839 (50% women)	Direct beneficiaries: 16,350 (50% women) indirect beneficiaries: 348,395 (50% women)	Direct beneficiaries of livelihood support including beneficiaries of climate smart agriculture and efficient irrigation. Indirect beneficiaries are considered the remaining population of the target host villages (population figures resented in Table 10.a) Incentives and benefits are sufficient in promoting uptake and behaviour change among beneficiary groups.
ARA4 Ecosystems and ecosystem services	Core 4: Hectares of natural resource areas brought under improved low-emission and/or climate-resilient management practices	Independent change analysis report based on GIS mapping of land cover change and project certifications	Total of 0 ha of forest ecosystems in Kakonko, Kasulu, and Kibondo Districts protected and strengthened in response to climate variability and change	Total of 620 ha of forest ecosystems in Kakonko, Kasulu, and Kibondo Districts protected and strengthened in response to climate	Total of 42,000[212] ha of forest ecosystems protected in Kakonko, Kasulu, and Kibondo Districts and strengthened in response to	Different groups are able to agree on land use, including allocation of village forest land

[212] Although not enumerated here (because of the difficulty of attribution) the flood management and rainwater harvesting interventions will also have indirect impacts on the scope of resilient ecosystems provided by the project, through improved infiltration, soil nutrient retention and freshwater ecosystem function, as well as reduced surface runoff, soil erosion, and siltation. Additionally, improved agricultural outputs may relieve pressure off natural ecosystems, by reducing land use change or the need for bridging income (from charcoal burning, wood fuel collection etc.) in the event of failing yields. Similarly, the sustainable provision of wood fuel from the agroforestry and afforestation interventions will also help relieve pressure off the natural forests, that are intended to be protected under CBFM. The economic valuation in Output 4 will be quantifying the benefits streams in relation to ecosystem services listed here.

				variability and change	climate variability and change	
ARA4 Ecosystems and ecosystem services	Supplementary 4.1: Hectares of terrestrial forest, terrestrial non-forest, freshwater and coastal marine areas brought under restoration and/or improved ecosystems	Independent change analysis report based on GIS mapping of land cover change and project certifications	0 ha under CBFM 0 ha under afforestation 0 ha under agroforestry Total of 0 ha of forest ecosystems in Kakonko, Kasulu, and Kibondo Districts protected and strengthened in response to climate variability and change	500 ha under CBFM 20 ha under afforestation 100 ha under agroforestry Total of 620 ha of forest ecosystems in Kakonko, Kasulu, and Kibondo Districts protected and strengthened in response to climate variability and change	30,000 under CBFM 2,000 under afforestation 10,000 ha under agroforestry Total of 42,000[213] ha of forest ecosystems protected in Kakonko, Kasulu, and Kibondo Districts and strengthened in response to climate variability and change	Different groups are able to agree on land use, including allocation of village forest land
E.4. GCF Outcome level: Enabling environment (IRMF core indicators 5-8 as applicable)						

[213] Although not enumerated here (because of the difficulty of attribution) the flood management and rainwater harvesting interventions will also have indirect impacts on the scope of resilient ecosystems provided by the project, through improved infiltration, soil nutrient retention and freshwater ecosystem function, as well as reduced surface runoff, soil erosion, and siltation. Additionally, improved agricultural outputs may relieve pressure off natural ecosystems, by reducing land use change or the need for bridging income (from charcoal burning, wood fuel collection etc.) in the event of failing yields. Similarly, the sustainable provision of wood fuel from the agroforestry and afforestation interventions will also help relieve pressure off the natural forests, that are intended to be protected under CBFM. The economic valuation in Output 4 will be quantifying the benefits streams in relation to ecosystem services listed here.

Select at least two relevant IRMF core (enabling environment) indicators to monitor and elaborate the baseline context and project/programme's targeted outcome against the respective indicators. Rate the current state (baseline) vis-à-vis the target scenario and select the geographical scope of the outcome to be assessed. Describe how the project/programme will contribute towards the target scenario. Refer to a case example in the accompanying guidance to complete this section.					
Core Indicator	**Baseline context (description)**	**Rating for current state (baseline)**	**Target scenario (description)**	**How the project will contribute**	**Coverage**
Core Indicator 5: Degree to which GCF investments contribute to strengthening institutional and regulatory frameworks for low emission climate-resilient development pathways in a country-driven manner	Current policies and legislation in Tanzania provide an enabling environment for climate change mainstreaming and adoption of adaptation interventions in some areas, however, refugees and the special needs of their host communities are not yet adequately considered in the relevant policy and planning frameworks such as the NDC, National Adaptation Plan, the National Climate Change Response Strategy (2021-2026); National Environment Policy of 2021 and the Environmental Master Plan (2022-2032) and National Land Use Planning Guidelines, as well as regional, district and village planning instruments in refugee-hosting areas like Kigoma. All these policy frameworks support the scale-up of resilient village land use plans that incorporate climate change adaptation	low	Legal and policy frameworks that govern or are relevant to adaptation in the Kigoma region are revised and harmonized to promote ecosystem-based adaptation considering the current and projected climate impacts and the adaptation needs of host and refugee populations.	Output 1 will involve the development of climate-resilient land-use plans (C-LUPs) in host community villages applying a comprehensive landscape- and ecosystem-approach that includes for the needs of multiple stakeholders Under output 4 the project will contribute to drafting revisions to key plans and policies and supporting their integration into national and district government planning processes, to promote up-scaling of the EbA model (Activity 4.3).	Single sub-national area within a country

	and mitigation strategies, recognizing ecosystem protection and conservation as an adaptation strategy. On the other hand, the regulatory framework with respect of refugees in Tanzania (Refugees Act, 1998 and Tanzania's 2003 National Refugee Policy) has limited consideration of climate impacts and adaptation needs of these displaced populations. The Global Compact for Refugees, been endorsed by the Government of Tanzania, proposes coherent humanitarian and development responses to support both groups as effective management of refugee situations are based on the resilience of host communities. Example of this is the Kigoma Joint Programme II under implementation. .				
Core indicator 8: Degree to which GCF investments contribute to effective knowledge generation and learning processes, and use of good practices, methodologies and standards	Evidence of the benefits of integrated ecosystem-based landscape approach to address the adaptation needs of both host and refugee communities in refugee-hosting landscapes is limited in Tanzania and worldwide due to the	low	Evidence generated on the economic, environmental and social value of this model supports mainstreaming of EbA into policies, plans and strategies in Tanzania and in humanitarian programmes, contributing to the scalability and replication of the project results at the national level	Under this output 4, the results of the planning processes and physical interventions implemented under Outputs 1–3 will be used to demonstrate the economic, environmental and social value of this model to stakeholders. The project will capture	Multi-countries

	limited number of experiences in the implementation of such approach. However, there is a global demand to expand the knowledge base on integrated approaches to build the resilience of host and displaced communities in a context of increasing displacement due to climate change impacts and conflicts.		and internationally in other refugee hosting landscapes through UNEP and UNHCR operations and policy contributions.	evidence of the ecosystem-wide benefits (Activity 4.1) and develop communication products for policy- and decision-makers to appreciate and replicate the landscape interventions (Activity 4.2)	

E.5. Project/programme specific indicators (project outcomes and outputs)

*This section should list out project/programme-specific performance indicators (outcomes and outputs) that are not covered in sections above (E.1-E.4). List down tailored indicators to monitor /track progress against relevant project/programme results (outcomes/outputs). AEs have the freedom to decide against which outcomes they would like to set project/programme specific indicators. If any co-benefits are identified in sections B.2(a)(b), and D.3, AEs are encouraged to add and monitor co-benefit indicators under the "**Project/programme co-benefit indicators**" section in table below. Add rows as needed.*

Please number each outcome and output as shown below to indicate association of outputs to the contributing outcome. The numbering for outputs under this section should correspond to the output numbering in annex 4 (detailed budget plan).

Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
		Sources of information and methods used to collect and report data/information to measure progress against targets	*The starting point or current value of the indicators before the implementation of the project*	*The estimated value of the indicator at the mid-point of the implementation*	*The estimated value of the indicator at the completion of the implementation*	*Externalities and factors outside project management's control that may impact on the Component* *Data sources and methodologies applied for estimating baseline and targets*

Outcome 1: Strengthened policy and planning systems to promote climate-resilient and sustainable resource management at the landscape level	Capacity of communities and local authorities to conduct ecosystem-based adaptation planning and implement strategies and activities to respond to climate change and variability	Capacity assessment scorecard measurements	Participants to the C-LUP process and supporting operational structures (Village Committees, WUAs, EMCs, District government) have Level 1[214] capacity to continue ecosystem-based adaptation planning and resource allocation.	Participants to the C-LUP process and supporting operational structures (Village Committees, WUAs, EMCs, District government) have Level 3 capacity to continue ecosystem-based adaptation planning and resource allocation. As a result, a shift in land-use planning and activities in target ecosystems will occur away from unsustainable practises leading to landscape degradation, towards climate-resilient	Participants to the C-LUP process and supporting operational structures (Village Committees, WUAs, EMCs, District government) have Level 4 capacity to continue ecosystem-based adaptation planning and resource allocation. As a result, a shift in land-use planning and activities in target ecosystems will occur away from unsustainable practises leading to landscape degradation, towards climate-resilient	Institutional support and engagement of stakeholders in the project is high thanks to the alignment with the national, regional and local priorities and policy frameworks. Political and institutional changes have limited impact on the C-LUPs process and its supporting operational structures at the district and village level. Different groups are able to agree on land use, including allocation of village forest land Beneficiaries understand the multiple benefits of different interventions and implement the measures promoted in the C-LUPs

[214] Level 1: communities and local authorities are aware of climate risks but have insufficient awareness on how landscape degradation and unsustainable practices contribute to increase climate risk and how ecosystem-based adaptation can enhance resilience; Level 2: communities and local authorities discourse reflects the arguments of ecosystem-based adaptation and show support to shift from unsustainable practises leading to landscape degradation, towards climate-resilient landscapes and livelihoods ; Level 3: resources are allocated and capacities built among communities and local authorities to implement ecosystem-based adaptation; Level 4: C-LUPs and EbA interventions are implemented by communities and local authorities resulting in a shift from unsustainable practises leading to landscape degradation, towards climate-resilient landscapes and livelihoods.

				landscapes and livelihoods.	landscapes and livelihoods.	
Outcome 2: Increased resilience of ecosystems and ecosystem services in the project area	Hectares of forest ecosystem protected and strengthened in response to climate variability and change	Independent change analysis report based on GIS mapping of land cover change and project certifications	0 ha under CBFM 0 ha under afforestation 0 ha under agroforestry Total of 0 ha of forest ecosystems in Kakonko, Kasulu, and Kibondo Districts protected and strengthened in response to climate variability and change	500 ha under CBFM 20 ha under afforestation 100 ha under agroforestry Total of 620 ha of forest ecosystems in Kakonko, Kasulu, and Kibondo Districts protected and strengthened in response to climate variability and change	30,000 under CBFM 2,000 under afforestation 10,000 ha under agroforestry Total of 42,000[215] ha of forest ecosystems protected in Kakonko, Kasulu, and Kibondo Districts and strengthened in response to climate variability and change	Different groups are able to agree on land use, including allocation of village forest land
Outcome 3: Community level livelihood vulnerability to climate	Average change in LVI-IPCC Index[216] or similar index	Secondary data on average standard deviation in monthly minimum and maximum temperatures and	To be determined during the baseline survey	N/A	Decrease in livelihood vulnerability expressed by an average increase	A similar methodology can be used at baseline and endline

[215] Although not enumerated here (because of the difficulty of attribution) the flood management and rainwater harvesting interventions will also have indirect impacts on the scope of resilient ecosystems provided by the project, through improved infiltration, soil nutrient retention and freshwater ecosystem function, as well as reduced surface runoff, soil erosion, and siltation. Additionally, improved agricultural outputs may relieve pressure off natural ecosystems, by reducing land use change or the need for bridging income (from charcoal burning, wood fuel collection etc.) in the event of failing yields. Similarly, the sustainable provision of wood fuel from the agroforestry and afforestation interventions will also help relieve pressure off the natural forests, that are intended to be protected under CBFM. The economic valuation in Output 4 will be quantifying the benefits streams in relation to ecosystem services listed here.

[216] The LVI-IPCC Index is scaled from −1 (least vulnerable) to 1 (most vulnerable) applying Hahn et al. (2009) LVI methodology and incorporating the IPCC vulnerability definition by grouping the seven major components under exposure, adaptive capacity, and sensitivity. It makes use of seven major components in three groups: Group 1 adaptive capacity (socio-demographic profile, livelihood strategies, social networks), Group 2 sensitivity (health, access to food, access to water) and Group 3 climate exposure (natural hazards and climate change).

change reduced in the project area		monthly rainfall over a 30-year period. Primary data will be collected with a tested questionnaire for LVI-IPCC Index administered by trained enumerators through a household survey to a statistically representative sample of households in host communities at baseline and endline.			of at least 0.3 points in the LVI range[217] in host communities by the end of the project	A tested questionnaire for LVI-IPCC Index be administered by trained enumerators through a household survey to a statistically representative sample of households
Outcome 4: Improved food security in the project area	% of beneficiaries under Output 3 eating 3 meals per day[218]	Primary data will be collected with a tested questionnaire by trained enumerators through a household survey to a statistically representative sample of households in host communities at baseline and endline.	0% of beneficiaries under Output 3 eating 3 meals per day[219]	7% of beneficiaries under Output 3 eat 3 meals per day[220]	70% of beneficiaries under Output 3 eat 3 meals per day[221]	A similar methodology can be used at baseline and endline No major economic crisis, political crisis or major climate hazard affects food accessibility and avaibility in the project implementation period.
Component 1: Increased resilience of ecosystems and people to climate change threats						
Output 1. Participatory Climate-resilient Land Use Plans developed (C-LUP) to support ecosystem-based adaptation	Number of villages in project area have adopted C-LUP plans and have implemented measures within them (interventions under Outputs 2 and 3)	Baseline report and project monitoring District and National Land Use Planning Commission registers of existing land use plans	0 villages in the project area have adopted C-LUP plans	~20-25 villages have adopted C-LUP plans	~20-25 villages have adopted C-LUP plans and have implemented measures within them (interventions	Consensus reached by Village Assemblies on land use allocations

[217] Target to be confirmed after the determination of the index at baseline
[218] Total 16,350 people.
[219] to be validated during project baseline survey.
[220] to be validated during project baseline survey.
[221] to be validated during project baseline survey.

		Independent assessment of the quality of the C-LUP plans in respect of climate change mainstreaming			under Outputs 2 and 3)	
Output 2: Degraded ecosystems restored to support climate change adaptation in host and refugee communities	Hectares of agroforestry, afforestation and flood in kms of erosion control measures implemented in priority locations identified, as identified in the C-LUP process Hectares of national forest reserve under increased protection	Independent change analysis report based on GIS mapping of land cover change Certification of hectares protected, afforested, under agroforestry or CBFM Certification of flood and erosion control infrastructure completed	~0 km of river channel under flood control[222] ~0 ha of forest area under CBFM 0 ha of agroforestry 0 ha of national forest reserve under increased protection	~2 km of river channel under flood control ~500 ha of forest area under CBFM ~100 ha of agroforestry ~50,000 ha of national forest reserve under increased protection	~25 km of river channel under flood control ~30,000 ha of forest area under CBFM 10,000 ha of agroforestry ~219,642 ha of national forest reserve under increased protection[223]	C-LUP process effectively identifies priority locations for flood and erosion control measures and these locations are available for construction
Output 3. Climate-resilient livelihoods practised to increase the capacity of host communities and refugees to better adapt in changing climatic conditions	Amount of water available for irrigation as a result of water harvesting and groundwater usage interventions	Baseline report and project monitoring Independent change analysis report encompassing community surveys based on random sampling plan.	0 L of water available to farmers for irrigation as a result of water harvesting and ground water usage interventions;	2,500 L of water /day/ household made available for ~50 plots as a result of water harvesting and ground water usage interventions;	2,500 L of water /day/ household made available for ~1,700 plots as a result of water harvesting and ground water usage interventions;	Farmers are convinced of the benefits of measures to improve climate resilience and decide to implement the measures on their farms. C-LUP process results in agreement on the selection of water use efficiency measures and their location.

222 Baselines to be confirmed through the social and environmental baseline survey to be implemented in year 1 of the project.
223 The 219,642 ha of national forest reserve land monitored is not included in the total hectares of ecosystem protected. The inputs towards the protection are minimal compared to other activities that support management, afforestation, agroforestry and more intensive inputs and activities.

	Amount of cassava, beans, mushrooms, and honey produced[224]	District and Region data on livelihoods Kigoma Joint Programme, WFP and UNHCR assessments on food security and nutrition	8,000 kg of cassava per ha[225] 400 kg of beans per ha[226] 15 kg of honey per beehive[227] 1,210 kg of mushrooms per facility [228]	11,000 kg of cassava per ha for 750 ha 1,048 kg of beans per ha for 750 ha 15 kg of honey per beehive for 170 beehives 1,210 kg of mushrooms per facility for 3 facilities	11,000 kg of cassava per ha for 7,500 ha 1,048 kg of beans per ha for 7,500 ha 15 kg of honey per beehive for 1700 beehives 1,210 kg of mushrooms per facility for 32 facilities	Labour is available for construction activities
Component 2 Strengthened policies for climate responsive planning and development						
Output 4. Information on climate change adaptation disseminated and mainstreamed into policies, plans and strategies in Tanzania and in humanitarian programmes	Planners in Ministry of Water and Irrigation, Ministry of Agriculture and Tanzania Forestry Service Agency at the National and District levels include economic valuation information in policy and planning documents.	Baseline report and project monitoring Availability of completed economic valuation of ecosystem services Workshop / consultation meeting reports	0 Planners at the National and District levels include economic valuation information in policy and planning documents Weak	None (Output 4 only begins implementation after mid-term)	Economic valuation information reflected in water, agriculture and forestry policy and planning documents	No major political upheaval/turnover in relevant authorities Ongoing policy and planning processes are taking place during project implementation and are accommodating of climate mainstreaming

[224] Amount of food produced combined with income indicators under 1.2 are used as proxies for measuring food security. An important component of food security related relates to food availability and income, as they influence the number of meals taken by households. Please see Gwalema 2010 reference, accessible from: http://repository.out.ac.tz/217/.
[225] Baseline production is based on unimproved varieties. Hectares under cassava production to be confirmed in inception phase.
[226] Baseline production is based on unimproved varieties. Hectares under bean production to be confirmed in inception phase.
[227] Baseline assumes no honey production beehives, exact baseline numbers to be confirmed in inception phase.
[228] Baseline assumes no mushroom production facilities available, exact baseline numbers to be confirmed in inception phase.

	Capacity developed at District level for EbA planning including cross-sectoral coordination.	Capacity/awareness scorecard[229] of representatives from key policy/planning authorities in Tanzania	Key representatives in Kigoma region capacitated to level 0	None (Output 4 only begins implementation after mid-term	Key representatives in Kigoma region capacitated to level 2	
	Political risk monitoring framework established and updated at the project level (See Risk Factor 10 including political risk indicator table)	Situation reports from VPO and UNHCR Tanzania (See other means of verification for political risk indicators under Risk Factor 10)	No political risk mitigation strategy established at project level	Political risk monitoring framework and mitigation strategy established and updated every 3 months	Political risk monitoring framework and mitigation strategy established and updated every 3 months	
Project/programme co-benefit indicators						
Co-benefit 1 Mitigation as a result of avoided deforestation, improved agricultural practises, afforestation and agroforestry	Estimated sequestration and reduced emissions in tCO2e over 20 years	Estimated using the FAO Ex-Ante Carbon-balance Tool as per the project Feasibility study	0	0	Sequestration of 762,073 million tCO2e from afforestation, agroforestry and Climate-Smart Agriculture and emission reduction from avoided deforestation of 3.2 million tCO2e over 20 years	The mid-term target for forest protection, afforestation and agroforestry is 1% at mid-term, therefore mitigation co-benefit will only be measured at the end of the project intervention. Estimated using the FAO Ex-Ante Carbon-balance Tool). For further detail, please see Appendix C of the Feasibility study

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Capacity/awareness scorecard[229] of representatives from key policy/planning authorities in Tanzania

. Level 0:Climate change and EbA not understood, or not considered meaningfully when developing policies, plans or strategies; Level 2: Training materials and/or draft policy briefs, guidelines or recommendations in climate change adaptation have been developed; Level 2: High-level representatives have attended training/workshops/forums and/or received policy briefs, guidelines or recommendations in climate change adaptation

Co-benefit 2 Reduced gender inequality in climate change adaptation planning and action	Women representation and meaningful participation in land use planning at the village level	Gender capacity assessment and gender analysis of completed Village Land Use Plans (VLUP) VLUPs and EMCs committees membership	Baseline values to be determined during the baseline survey	50% of VLUP committees and EMCs representatives are women. 100 % of VLUPs reflect women's needs and interests	50% of VLUP committees and EMCs representatives are women. 100 % of VLUPs reflect women's needs and interests	The project will promote these targets to the extent possible, but recognizing that the Village Councils are independently organized and the project may not be able to fully influence committees composition.

E.6. Project/programme activities and deliverables

All project activities should be listed here with a description and sub-activities. Significant deliverables should be reflected in annex 5 implementation timetable. Add rows as needed.

Please number the activities as shown below to indicate association of activities to the related outputs provided above in section E.5. Similarly, please number sub-activities as shown below to associate to the related activity.

Activities	Description	Sub-activities	Deliverables
1.1. Support climate-resilient village land use planning	This, and Activity 1.2. are the starting point for all project interventions, by the development of participatory climate-resilient land-use plans for villages that are consistent with an ecosystem-based adaptation approach (EbA). The Village Land Use Plan (VLUP) is an overall plan showing how land resources, located within the village boundaries, should be used to meet declared objectives. The project will adapt this established approach, by introducing climate resilience of the landscape into the planning process, resulting in a Climate-resilient Land Use Plan (C-LUP). The overall goal of this activity is to strengthen the capacity of actors and institutions that deal with land tenure and to apply ecosystem-based planning, thereby	1.1.1. Scale-up land use planning for villages, and support to issuance of Certificates of Customary Rights of Occupancy (CCRO) 1.1.2. Build capacity of local and regional government officials to support ecosystem-based adaptation and landscape restoration 1.1.3. Improve public awareness of climate change by sharing information on climate change risks in the land use planning meetings 1.1.4. Support District, Ward and Village level Environmental Management Committees (EMCs).	~20-25 village land use plans finalised or developed[230], with processes initiated for receiving CCRO ~20-25 revised or formulated bylaws 50 officers from Tanzanian Forest Service, District Agriculture, Livestock and Forest Officers, district water engineers, and land use management teams trained. Learning outcomes will be measured as part of the training package. 2,000 people from ~20 villages in Kasulu, Kibondo, and Kakonko receive training on climate change impacts 1 Workshop and 20 trainings provided to EMCs

[230] This is an indicative number, averaging the populations of the host communities in the three districts, since the exact number and sites of villages will only be determined during project implementation.

	providing a comprehensive approach for implementing the other activities envisioned under this project. Completing the land-use planning is pre-requisite for carrying out the climate-resilient practises implemented later under Outputs 2 and 3. This process will also consist of supporting the issuance of Certificates of Customary Rights of Occupancy (CCRO) and include concurrent climate change awareness raising for host communities as well as capacity building for officials including: representatives from the Tanzania Forestry Service; district agriculture, livestock and forest officers; district water engineers; and land use management teams. This capacity building will be supported by the provision of hardware and materials to district officials, including inter alia GIS software and tools, crested paper for CCRO printing etc.		
2.1. Implement Community Based Forest Management (CBFM) for resilient ecosystems	The principal objective of this intervention is to implement CBFM in forest areas anchored to host communities over the course of the GCF project activity. Under CBFM, villagers take full ownership and management responsibility for an area of forest as a Village Forest Reserve. Following this legal transfer of rights and responsibilities from central to village government, villagers may harvest timber and forest products, collect and retain forest royalties, and undertake patrols (including arresting and fining offenders), as well as receiving tax and regulatory benefits. The underlying policy goal for CBFM is to progressively bring	2.1.1. Establish and manage 30,000 ha of Village Land Forest Reserves 2.1.2 Provide equipment and training for forest monitoring in national forest reserves for 10 forestry and natural resources officers.	30,000 ha under CBFM management ~20 Approved management plan for each Village Forest Area, supported by by-laws approved by the District Council Active FMCs in ~20 villages[232] 10 Forestry and natural resources extension officers having received training in implementing CBFM 10 TFS staff equipped and trained in the use of drone and GIS-based monitoring system.

[232] This is an indicative number, averaging the populations of the host communities in the three districts, since the exact number and sites of villages will only be determined during project implementation.

	unprotected woodlands and forests under village management. National forest reserves will benefit from improved monitoring and management by the Tanzanian Forest Service through a GIS-based forest monitoring system and the provision of supplementary equipment in the form of drones, smartphone/ODK and GPS devices and camping and fire protection gear for TFS patrol teams. [231].		
2.2. Implement 12,000 hectares of agroforestry and village land afforestation to increase resilience of land use	The principal objective of this intervention is to increase the number of trees across the agricultural (non-forest) landscape through agroforestry and village land afforestation. At landscape scale, agroforestry improves crop productivity and provides important ecosystem services including watershed protection and hydrological regulation, as well as biodiversity conservation and carbon sequestration. Complementary to this, village land afforestation can restore ecosystem services and establish a sustainable supply of fuelwood, taking the pressure off of native forest systems.	2.2.1. To establish and maintain 10,000 ha of agroforestry systems to restore ecosystems that are critical to meet climate adaptation needs, and set up nurseries for production of tree seedlings 2.2.2. Undertake afforestation on 2,000 ha of village land to a) restore critical ecosystem services to meet climate adaptation needs and improve ecosystem services such as flood control, b) establish a sustainable supply of fuelwood by planting suitable tree species which will take some of the pressure off the native miombo forest and allow it to regenerate so that it can meet its ecological and livelihood support functions	10,000 ha of agroforestry established 2,000 ha afforested land established ~15 representatives from TFS and District Councils provided with technical advisory and capacity building services in the design, implementation and monitoring of agroforestry and afforestation
2.3. Implement flood and erosion control in densely populated areas to reduce the exposed and flood-prone ferrosol-type soils which have little or no vegetation cover	The refugee camps are exposed to a high risk of flooding and erosion, owing to the land clearance that has taken place. Whilst the land-use and forestry interventions described above will increase soil stability and reduce the level of runoff and erosion in the project area, these interventions not suitable within the camp boundaries because of their density.	2.3.1. Construct 10 km of micro-scale stone-pitched drainage channels, 16 stone-filled gabions and Sustainable Drainage System (SuDS) interventions — including micro- infiltration trenches/ponds/soakaways — in three refugee camps.	Drainage channels (~10.1km) Gabions (~16) Establish sustainable Drainage System (SuDS) interventions such as infiltration trenches and basins, the number and locations of which will be determined during project implementation.

[231] Equipment to be turned over to TFS at project completion.

	The principal objective of this intervention is to provide engineering solutions in the areas most susceptible to erosion from high water flows. The engineering solutions themselves will be designed and use materials that are able to withstand changing weather extremes such as temperature and rainfall intensities. The final design and distribution of the interventions would be guided by a technical assessment and stakeholder consultations.		1,500 Water user association members trained on how to construct and maintain flood and erosion control measures. Learning outcomes will be measured through assessments built into the training package.
3.1. Promote modern technologies and management practises to strengthen the capacity of farmers, district officials and agricultural extension workers in climate-resilient agriculture	The interventions proposed under this activity will focus on promoting the adoption of climate-resilient agricultural technologies and management practises, as well as the procurement and distribution of agricultural inputs. These interventions will be supported by strengthening capacity of district officials and extension services, who can directly engage with local farmers to promote climate-smart agriculture (CSA) practises (training of trainers). The specific package of climate-resilient agricultural practises and inputs to be distributed will be co-designed in collaboration with the farmers in the villages where the project will be working, District officials and extension services, based on the climate-resilient agricultural practises and technologies that best fit the selected context. This project seeks to strengthen and scale up Savings Associations (SAs) so that funds will be disbursed through a market-based approach. SAs typically serve as cooperatives in the region to have improved access to farm inputs, collectively negotiate prices, facilitate tax	3.1.1. Build capacity of 60 farmers per Farmer Field School (FFS) on climate-resilient agricultural practises, including use of traditional knowledge, through trainings co-designed in collaboration with the farmers, district officials and extension services. 3.1.2. Provide 20 kg of resilient seeds/cuttings and 1 agricultural tool kit (at least 60% to be procured and distributed through savings associations) to each of 15,000 farmers for climate-resilient agriculture including seeds, tools and equipment necessary for implementing climate-resilient agricultural practices. 3.1.3. Build capacity of Savings Association Management Offices (SAMO) to manage savings associations	~20 Farmer Field Schools established ~1,200 farmers receiving training through FFS 300,000 kg of seeds/cuttings distributed to farming households by Year 5 (at least 80% through SAs) 15,000 CSA kits distributed to farming households by Year 5 (at least 60% through SAs) ~ 3 SAMOs trained

	collection, improve marketing of products, and have access to microfinance as well as being capacity support for promoting climate-smart and adaptive practises and crop varieties. The project will support the SA and its members in managing funding decisions and building capacity to hold funds through 'Savings Association Management Office' (SAMO) and envisages the SAMOs as playing a critical role in scaling up the funds available for refugees to participate in climate resilient activities.		
3.2. Increase water availability and use-efficiency through water harvesting and efficient irrigation interventions.	Interventions under this activity will ensure that water is available in closer proximity to agriculture areas, as well as for the nurseries for agro-forestry and village land afforestation described in Activity 2.2. The engineering solutions themselves will be use materials that are able to withstand changing weather extremes such as temperature and rainfall intensities. The final design and distribution of the interventions would be guided by the C-LUP process. Rainwater harvesting will be supported by: • solar powered pumps (for societal/communal use to supply a number of farms from a single source) and manual pumps (e.g. treadle pumps) for individual household use • irrigation systems for nurseries and farms, including moisture testing kits • training on water management good practise, soil and water conservation techniques • technical training on RWH, pumping and irrigation system; and	3.2.1. Construct micro-rainwater harvesting systems: ~94 Check dams ~20 run-off harvesting micro-dams (charco dams), ~120 unlined water-pans and ~120 lined micro-ponds in host communities in Kasulu, Kibondo, and Kakonko. 3.2.2. Establish water pumping and irrigation through ~ 50 micro-solar and ~ 172 treadle pumps that will draw water from the rainwater harvesting locations to where it is needed in host communities in Kasulu, Kibondo, and Kakonko. 3.2.3. Conduct training and awareness raising for 400 members of Water User Associations and farmers on water management and efficient water use through the FFS. 3.2.4. Support the establishment of and strengthen existing Water Users Associations (WUAs) through organizing and conducting 20 meetings/trainings.	94 Check dams 20 Charco dams 120 Unlined and 120 lined water ponds 50 Solar pumps 172 treadle pumps Training of 1200 farmers in effective water usage and irrigation installation and management ~400 participants of WUAs trained over 20 meeting/training sessions

	• new and existing WUAs that will support water management and efficient water use practices		
3.3. Promote climate-resilient livelihood diversification to strengthen food security and nutrition, provide alternative income as a safety net, and to sustain the implementation of climate-resilient agricultural and forest management practises.	While agriculture is the predominant source of livelihood in the region, other activities such as beekeeping and mushroom growing are also practised in the host and refugee communities. These activities are largely dependent on the health of natural resources (particularly forests, soils and water), but are more resilient than agriculture to projected climate variability. The primary objective of the livelihood diversification activities is to strengthen food security and nutrition in the event of hazards such as droughts or floods — which could reduce crop yields — provide alternative income as a safety net in case of yield loss and to sustain the implementation of climate-resilient agricultural practises. This would entail establishing and capacitating beekeeping and mushroom growing groups with training, oversight and inputs (equipment and materials).	3.3.1. Promote beekeeping to incentivise forest conservation in the forests surrounding the camps and villages by providing equipment and training to 850 people. 3.3.2. Promote mushroom cultivation for marginalised groups with training and mushroom growing facilities by providing equipment and training to 500 people.	850 people trained and provided with materials for beekeeping, honey and by-products quality control, packaging and marketing. ~30 mushroom facilities constructed, supplied with the necessary inputs and associated with training 500 people
4.1. Generate evidence of the economic benefits of ecosystem-based adaptation to host and refugee populations, for use by policymakers and planners	To prioritise an ecosystem-based adaptation (EbA) approach, decision-makers need to understand and appreciate the relevant interlinked ecological processes and how they directly produce economic benefits and costs. The sub-activities under this activity will contribute to the evidence base to achieve this by determining the economic value of the ecosystem services provided in the project site, conducting assessments and modelling the added value of EbA benefits, based on local livelihoods and	4.1.1. Identify benefit and cost streams for valuing ecosystem services under different climate scenarios 4.1.2. Identify data streams and modelling methodology for valuing ecosystem services 4.1.3. Gather primary and secondary data from project interventions 4.1.4. Carry out valuation modelling exercises of benefit and cost streams identified under 4.1.1.	Full and summary reports of economic benefits and costs of ecosystem-based adaptation interventions Raw data used for modelling and valuing ecosystem services (e.g. economic values per ecosystem service)

	primary resource dependence. By recognising the positive environmental, economic and social co-benefits and costs to communities of these ecosystems, this acknowledgment will help promote cross-sectoral cooperation and integrated planning to conserve and augment these natural resources.	4.1.5. Consider temporal, spatial and distributional impacts and opportunities resulting from the costs and benefits evaluated in Sub-activity 4.1.4. 4.1.6. Prepare full and summary reports	
4.2. Develop communication products to disseminate project results	To scale the impacts of the proposed interventions, it is necessary to disseminate the results and lessons of the project to inform similar interventions in the future. This information, and the tools they inform, will help policy- and decision-makers in Tanzania and UNHCR's humanitarian programmes to mainstream climate resilience into village land-use planning, as well as to make use of Savings Associations in host communities to scale up climate-resilient agricultural practises. This activity comprises compiling and distributing these lessons to key stakeholders.	4.2.1. Develop policy briefs on mainstreaming climate resilience in village land-use planning 4.2.2. Develop a guidelines paper for mainstreaming climate resilience in village land-use planning 4.2.3. Produce lessons-learned and guidelines paper on the use of Savings Associations to scale up climate-resilient agricultural practises	Policy briefs on mainstreaming climate resilience in village land-use planning Guidelines paper on mainstreaming climate resilience in village land-use planning. Lessons-learned and guidelines paper on the use of Savings Associations to scale up climate resilient agricultural practises.
4.3. Draft revisions to key plans and policies and support their integration into national and district government planning processes to promote up-scaling of the EbA model	To achieve enduring and sustainable climate resilience amongst host communities and refugees, the local and high-level legal and policy frameworks for Tanzania will require updating and revision. In doing so, climate resilience and adaptation planning will become institutionalised within public operations, thereby addressing barriers to reducing climate vulnerability amongst host communities and refugees. This will be achieved by proposing policy revisions and updates of relevant reviewed policies and processes, in close consultation with government decision-makers and technical staff. Along with the mainstreaming work, policy briefs will be	4.3.1. Review laws and policies that hinder adaptation in Kigoma region, including political risks that need to be closely monitored (See Risk Factor 10 and risk indicators), and identify entry points for where laws and policies require harmonisation to promote adaptation 4.3.2. Identify legal, policy and planning processes that can be capitalised on to advance the mainstreaming of adaptation 4.3.3. Prepare guidelines and other inputs for policy- and decision-makers to use to inform in the mainstreaming of adaptation in national and district government planning	Consultancy report on laws, planning processes, and policies pertinent to adaptation in Kigoma region Guidelines paper, training materials and other inputs to inform mainstreaming of adaptation 1 workshop with policy- and decision-makers (20-40 pax) to identify and/or validate entry points for policy harmonisation 10 consultation meetings held with stakeholders (2-10 pax) to identify and/or validate entry points for policy harmonisation

	developed and working sessions will take place with key decision makers and stakeholders to bring forward specific policy recommendations and evidence to be considered in policy working groups. These revisions will be informed by the evidence base established under Activity 4.1 and supported by the demonstrated impacts of project interventions implemented under Outputs 1–3.	4.3.4 Co-host workshops, forums and consultations within ongoing policy and planning processes to showcase project results and implications for the legal, policy and planning frameworks of the country	1 workshop with policy- and decision-makers (20-40 pax) to identify legal, policy and planning processes 10 consultation/meetings held with stakeholders (2-10 pax) to identify legal, policy and planning processes 1 2-day workshop with policy- and decision-makers (20-40 pax) to develop guidelines for mainstreaming adaptation into national and district government planning 5 workshops (2 days, 20-40 pax) to showcase project results and validate implications for the legal, policy and planning frameworks of the country Political risk mitigation framework established

E.7. Monitoring, reporting and evaluation arrangements (max. 500 words, approximately 1 page)

An M&E officer will be recruited in the PMU for the duration of the project to conduct and supervise the monitoring and reporting on the project's activities. The M&E officer will be supported by an M&E expert international consultant (part-time) in the implementation of the project M&E plan which includes independent surveys and assessments to monitor the results framework indicators (see Annex 11 for further details). The M&E officer will coordinate the execution of the project M&E plan with UNHCR, implementing partners, relevant stakeholders at the district and regional level and with relevant ministries particularly the Regional Administration and Local Government, Ministry of Home Affairs, Ministry of Agriculture and Food Security and the Ministry of Lands, Housing and Human Settlement Development. Monitoring and evaluation reports, drafts of the Annual Performance Reports, the terms of references for the interim evaluation (mid-term review) and independent ex-post evaluation (terminal evaluation) will be circulated to the relevant ministries and the Project Steering Committee.

A UNEP Task Manager will be responsible for monitoring progress against output and outcome indicators during project implementation. At key points (i.e. baseline, mid-point and end of project) the project team will carry out evidence gathering exercises to verify this progress. During the ex-post evaluation (terminal evaluation) at the end of the project an evaluation consultant will validate a sample of the data collected through these monitoring tools.

UNEP will be responsible for managing an interim evaluation Mid-Term Review (MTR) at the mid-point of the project's duration. The Task Manager will oversee the process of hiring an external consultant to carry out the MTR, which will provide an assessment of project performance at the project's mid-point. This will be a formative exercise and will include analysing whether the project is on track, what problems and challenges the project is encountering, and which corrective actions are required so that the project can achieve its intended outcomes by project completion in the most efficient and sustainable way. The Project Steering Committee will participate in the MTR process and develop a management response to the review's recommendations along with an implementation plan. It is the responsibility of the UNEP Task Manager to monitor whether the agreed recommendations are being implemented during the remainder of the project's operational life. The Independent Evaluation Office of UNEP will provide an assessment of the quality of the Mid-Term Review report.

An independent ex-post evaluation (terminal evaluation) (TE) will take place once the project has reached operational completion. The Evaluation Office of UNEP will be responsible for the TE, which is a summative evaluation, and will liaise with the UNEP Task Manager and relevant stakeholders throughout the process. An independent assessment of project performance against standard evaluation criteria (e.g. strategic relevance, effectiveness, efficiency, likelihood of impact and sustainability) will be made based on documentary evidence, stakeholder interviews and, in most cases, a field mission. Each evaluation criterion will be rated using a six-point rating scheme and a weighted average will be determined to provide an overall performance rating for the project as a whole. Where there are any differences in ratings between the evaluation team and the Evaluation Office a final determination will be made by the Evaluation Office when the evaluation report is finalised.

The draft ex-post evaluation (terminal evaluation) report will be sent to project stakeholders during a commenting process managed by the Evaluation Office. Formal comments on the report will be shared by the Evaluation Office in an open and transparent manner. This evaluation report will be publicly disclosed and will be followed by a recommendation compliance process.

The total 5-year cost associated to project monitoring and evaluation is $1,421,283, including:

- Safeguards and gender specialist $228,000
- M&E Officer $240,000
- Social and environmental safeguards expert (international consultant part-time 30-35 days/year) $99,000
- M&E expert (international consultant part-time 30-35 days/year) $99,000
- Implementation of Safeguards management plan ESMS $75,000
- Implementation of Gender action Plan $ 50,000
- Implementation of M&E plan $ 222,783
- Baseline assessment $150,000
- Interim evaluation/ Mid-Term Review $114,500

- Ex-post evaluation/terminal evaluation $143,000

The Vice-Present's Office, as lead EE, will compile UNHCR reports and will submit consolidated semi-annual progress reports and quarterly financial statements to the UNEP as the AE. In turn, UNEP will submit annual performance reports and semi-annual financial reports to GCF. The detailed reporting timelines are as follows:

Under the PCAs, the EE will be required to report to UNEP as follows:

a. Progress reports: by 30 July for January to June;
b. Annual Performance reports on or before 15 January;
c. Quarterly financial reports by 15 January, 15 April, 15 July, and 15 October;
d. Annual audited statements by 30 April;
e. Final report: within 3 months of project completion.

UNEP (AE) reports to the GCF:

a. Annual Performance Reports by 28 February;
b. Semi-annual Financial Information by 1 March and 30 Sept;
c. Mid-Term Assessment report: halfway through Programme;
d. Final APR: within 6 months of project completion;
e. Ex-post evaluation (terminal evaluation) report: within 12 months of Programme completion.

Monitoring will also be undertaken by the AE through supervision visits and field missions to track implementation progress and challenges and strategically plan the way forward.

E

F. RISK ASSESSMENT AND MANAGEMENT

F.1. Risk factors and mitigations measures (max. 3 pages)

The main risk factors include political, technical and operational, social and environmental and financial risks. Section G.2 discusses the environmental and social risks identified in the Environmental and Social Assessment and how these risks are mitigated. The project design minimises project risks and includes mitigation measures for the identified risks, taking into account lessons learned from previous interventions in the area. Risks are continuously monitored at the levels of the Project Management Unit, Government of Tanzania Vice President's Office and UNHCR as the Executing Entities, and UNEP as the Accredited Entity.

VPO established a risk management policy and framework in 2018, this was driven by the fact that managing risk has been an increasingly important aspect of public sector governance and a significant factor to support the achievement of public objectives. Risk coordinators are charged with the overall responsibility of reporting risk management issues to the Permanent Secretary and audit committee as well as consolidating risks and periodically updating the risk register.

VPO will consolidate, analyse and triangulate risk information reported by different project partners including UNHCR, MoHA, NEMC, TFS and the District Council, as well as information collected by PMU during field monitoring visits. The role of the Safeguards and Gender officer at PMU will be to monitor risks as they arise, and facilitate avoidance, mitigation, and management of risks. The Safeguards and Gender officer will be supported by a Safeguards expert international consultant (part-time) to develop, implement and monitor the Environmental and Social Management System and the Gender and Social Inclusion Plan. The recruited person will work closely with the Project Management Unit, the Project Steering Committee, UNHCR, and UNEP in managing such risks.

In the event of significant risks to the project, VPO will engage in consultations with the relevant parties (District Councils, Kigoma Regional Authority, TFS, MoHA, UNHCR or others) to agree on adequate measures for conflict resolution and mitigation of impacts on the project. This will be done through the project steering committee or ad-hoc project stakeholder consultations.

The Government of Tanzania Vice President's Office Project Management Unit will coordinate risk management with UNHCR who will apply its approach to management of risks, including political and safeguard risks. As of 2018, UNHCR has embarked on a process to strengthen risk management across the agency, through the launch of the Risk Management 2.0 initiative. Risk Management 2.0 is an initiative which strengthens the accountability and integrity of UNHCR's programmes and addresses the root causes of corruption and fraud, and other types of critical risks to UNHCR. It aims at improving programme delivery, making operations more efficient, and in doing so, maximizing UNHCR's our ability to achieve its objective of providing protection and assistance to people forced to flee. Within the UN, UNHCR co-leads the UN High Level Committee on Management Taskforce on Risk Management and recently was assessed as having an "established" level of risk maturity. Based on a request by the UNHCR High Commissioner, work is currently underway to develop a new risk management strategy that will allow UNHCR to achieve an "advanced" level of risk maturity by the end of 2025. This initiative has led to the expansion of professional resources across all levels of the organization, providing targeted support to specific operations who needed it most.

UNHCR Tanzania has staff, specifically the Senior Project Control Officer to support facilitation of risk assessments, updating and maintenance of the country risk register, undertaking risk reviews, following up on risk treatments, sharing risk information, overseeing risk escalation, and providing risk management training and support to the entire operation. This capacity is in place to support the entire spectrum of work undertaken by UNHCR Tanzania, including any future GCF projects. In addition to the support of the Senior Project Control Officer's risk management tasks, UNHCR has also introduced a new position of Senior Protection Officer in Kigoma to support overall protection activities, including mitigation of protection risks.

At the Accredited Entity level, the Task Manager, with the support of the Portfolio Manager, the Safeguards Advisor, and the GCF Coordination Office, will take the leading role in monitoring risks in close cooperation with the Project Management Unit and UNHCR.

Selected Risk Factor 1

Category	Probability	Impact
Technical and operational	Low	Medium

Description		
Water-related infrastructure fall into disrepair, lack of spare parts or lack of skills in the host communities impedes maintenance of irrigation equipment. As a result, irrigated areas are abandoned and cultivation of riparian areas increases further, therefore degrading water courses. Operations and maintenance (O&M) costs are embedded in the contractual services costs during the project implementation period. Beyond the lifetime of the project, O&M costs will be assumed by asset owners (e.g. District Councils and water user associations, community members in the case of charco dams). Majority of the maintenance work required are inspection, cleaning, desilting, and weeding (through community labour). Materials for replacement in case of need for repairs are widely available on site or in the local markets. These include stones, brushwood, wooden posts, sandbags, wire mesh. Costs of maintenance depend on the structures, with stone gabions at around 10% of the cost, infiltration strips at between $0.3 – $1.3 per square meter of surface, and infiltration basins at $0.1 - $0.4/ square meter of detention basin area. The total costs will be estimated during the implementation stage once the interventions and their extent have been plotted on the project area. It is highly likely that the costs in Tanzania will be at the lower end of these estimates (based on references in US and UK. Please See Annex 2.2 for references), based on local market prices.		
Mitigation Measure(s)		
UNHCR, as he EE of this project activity will monitor effectiveness of training provided to water engineers and communities and adapt as necessary. The project will work with suppliers to ensure parts can be made available in Kibondo/Kakonko/Kasulu and develop operations and maintenance plans. Annex 21. Operations and Maintenance has already been prepared to address relevant issues. Within the camps, assets will be turned over to the Ministry of Home Affairs for management and maintenance oversight. The water and sanitation humanitarian cluster (WASH) development agencies can support MHA in this role. Water harvesting infrastructure in the host communities will be turned over to district and regional authorities, with regular maintenance activities to be performed by trained water user associations (WUAs) and in cooperation with the Lake Tanganyika River Basin Authority. WUAs can leverage on savings associations or devise schemes for small user fees to support O&M. This will be determined during project implementation for payment structures and viability.		
Selected Risk Factor 2		
Category	Probability	Impact
Technical and operational	Medium	Low
Description		
Savings Associations are unable or unwilling to increase the amount of funds under their management to the degree needed in order to implement project interventions. As a result, project funds are not able to be spent or are spent ineffectively.		
Mitigation Measure(s)		
The project will support the substantial expansion foreseen in the funds management by SAs through the establishment of a Savings Association Management Office (SAMO), which will be supported by a dedicated member of the Project Management Office. The SAMOs and the Savings Association Expert will work together to enable the SAs to gradually increase the amount of funds handled. To mitigate this risk, the proportion of agricultural inputs supplied to farmers through SAs is limited to 60% of the total cost of agricultural input. This percentage will be adjusted based on the SAs performance in the delivery of agricultural inputs to farmers in Years 3, 4 and 5.		
Selected Risk Factor 3		
Category	Probability	Impact
Technical and operational	Medium	Medium
Description		
Extreme weather events such as heavy rains and wildfire risk may result in implementation delays.		
Mitigation Measure(s)		
Timing of the project activities will consider rainy season and maximum and projected rainfall parameters when planning and designing interventions. The latest information from the met services and other information providers will be sought and used. Firebreak areas will be established in the afforested land area and wildfire risk monitoring will take place in		

coordination with the Tanzania Forest Service. Capacity building activities with the TFS include fire management response plans. Awareness raising in communities under Activity 1.1 on wildfire risk and prevention measures will take place.		
Selected Risk Factor 4		
Category	Probability	Impact
Governance	Low	High
Description		
Women are not adequately engaged in decision-making structures.		
Mitigation Measure(s)		
The Gender and Social Inclusion action plan for the project will set forth the guidelines for engaging women at various levels of decision-making including in village land use planning, Savings Associations, and other participatory approaches adopted by the project. Representation of the Ministry of Community Development, Women Affairs and Children (MCDWAC) included in the steering committee to advise project stakeholders on gender and women empowerment mainstreaming and implementation of the Gender Action Plan. Adequate representation of women representatives in the steering committee will be encouraged. The project will support villages that have not initiated the land use planning process, as well as those that have initiated but not completed it. For the latter, the project will ensure meaningful representation of women at various stages of the process and may review previous steps in the land use planning process if a gender-sensitive approach has not been applied and correct the gap based on an assessment of whether all planning processes have followed the guidelines set by the National Land Use Planning Commission of Tanzania, titled 'Guidelines for Integrated and Participatory Village Land Use Planning, Management, and Administration,' which apply to all stakeholders involved, including local institutions, NGOs, and the private sector. These guidelines emphasize a gender-sensitive approach, which is an integral part of the participatory approach, ensuring the consideration of different roles, interests, and expectations of both men and women. Specifically, the guidelines call for 'ensuring full participation of socio-economic groups in a village, including gender, which have different interests and expectations.'		
Selected Risk Factor 5		
Category	Probability	Impact
Other	Low	Medium
Description		
Refugees return to their countries of origin. As a result, camps reduce operations.		
Mitigation Measure(s)		
UNHCR will keep close attention to the situation, should major developments take place that would prompt large-scale voluntary repatriation. In the unlikely event of refugee numbers dramatically decreasing or other camp(s), (besides Mtendeli camp closure and land allocation to the host communities), project activities have been designed to be of 'no regret' and not sensitive to the fluctuation or transience of the refugees and anchored on the landscape and host community population. Additionally, the landscape interventions proposed in the camps (flood and erosion control) are sustainable measures that result in adaptation benefits for host communities in the long term. Adaptive management measures and flexibility in implementation will be undertaken to redistribute interventions to host communities, if necessary, with camp benefits and livelihood activities being absorbed by the host communities (who are receiving similar interventions), and the flood and erosion control measures will help to protect future occupants against climate-driven flood events and erosion damages. The Tanzania's Refugee Act of 1998 and the National Refugee Policy of 2003 serve as the legal and policy framework in the country. The Government of Tanzania also has regional commitments to refugee protection under the Organization of African Unit, the East African Community, and the Southern African Development Community and the government has expressed support Global Compact on Refugees. Government actions and support for the camps are governed by a complex interplay of law, policy, norms, practice, and politics. There is an existing policy of voluntary repatriations. Any forced evictions of refugees would violate the government's commitments.		
Selected Risk Factor 6		
Category	Probability	Impact
Other	Medium	Medium
Description		

Tension between those within the project area and those outside. The benefits received by host communities in the project areas create a source of tension with those outside of the project area.

Mitigation Measure(s)

As much of the project is based on capacity building, the awareness and benefits can be transferred outside of the project area. 'Priming' with initial investment, for example for irrigation, will spur markets in the project areas, increasing accessibility of them to a wider area in the community. The conflict sensitivity assessment planned under output 1 will contribute to further assess the risk and identify suitable measures to mitigate conflicts between the target area population and the host community outside the target area.

Selected Risk Factor 7

Category	Probability	Impact
Other	Low	Medium

Description

Risk of tension between refugees following relocation of refugees from Mtendeli camp to Nduta and/or surrounding host communities due to arrival of relocated refugees. As existing tensions and occasional conflicts (including GBV incidences) are often related to the collection of fuelwood, an increase in camp population in Nduta may exacerbate these conflicts.

The population relocated from Mtendeli to Nduta are of the same profile of those in Nduta. Both populations are predominantly Burundian refugees who were displaced following the 2015 upheavals in Burundi. Host communities around Nduta have been coexisting for years with these Burundian refugees in an overall peaceful manner and have built relationships through various formal and informal interactions. The main issue for tension and conflict has been related to fuelwood. Other concerns from refugees related to the consolidation adding pressure to repatriate, the new housing situation and access to services in Nduta camp. It can be anticipated that an increase in refugee population will result in increased competition for fuelwood. Also, longer distances may need to be tracked by refugees to meet cooking energy needs which may lead to increased SGBV incidences.

Mitigation Measure(s)

UNHCR risk assessment, using UNHCR risk register methodology, was undertaken for Mtendeli consolidation exercise and mitigation measures identified. A Task Force has been established to plan and oversee the relocation consisting of MoHA at national and local level, District Commissioners of Kakonko, Kibondo and RC and RAS in Kigoma, UNHCR, DRC (camp management) and NRC (shelter &wash). Furthermore, Camp Consolidation Coordination teams (CCCT) has been established consisting of MoHA, UNHCR, DRC CCCM to lead and supported by Registration, Protection WASH, Shelter, AIRD, Health, Education, Environment, Refugee leadership, CWT, Police and Host community leadership. The Protection working group (UNHCR, MoHA, protection partners) has also been involved and specific protection actions/activities have been identified planned in the context of the relocation. This includes specific measures to deal with protection risks of vulnerable individuals as they move. After relocation specific measures have been identified to be implemented to ensure inclusion and protection of vulnerable groups. UNHCR and partners are stepping up their activities in support of scaling up alternative energy sources for cooking through different interventions complementary to the GCF project, primarily through increasing production and use of biomass briquettes. The afforestation of 2,000 hectares of land in villages (sub-activity 2.2.2) aims to establish fast-growing fuelwood species on degraded areas of village lands adjacent to refugee camps. It is calculated that 12% of the current fuelwood demand can be met (see detailed assumptions in Annex 4: Detailed Budget Plan). Though this is a modest direct contribution, it is a critical part of overall fuel mix for the region.

UNHCR supports regular peaceful co-existence meetings which bring together community leaders from refugees and host communities. These forums are used to discuss and resolve any conflict that may arise from the arrival of relocated refugees from Mtendeli. With regards to conflicts among refugees in Nduta, community structures are in place to resolve tensions and conflicts among community members. Sensitization of refugee leadership on the consolidation process will be undertaken. Refugee leaders play a key role in communicating with and disseminating information to the wider refugee community and creating awareness on a range of issues. By maintaining a dialogue and sharing information during physical meetings with refugee leaders in practicalities as well as risks and mitigating actions (for example as identified by the protection working group) the wider refugee community will be informed and sensitized.

Mass communication campaigns were carried out in both Nduta and Mtendeli. Engagement and communication with the host communities is done through formal and informal peaceful coexistence activities whereby leaders from both communities (refugees and hosts) are represented. In the host communities, the Village executive officers and the Ward executive officers have a mandate in dispute resolution and reporting of incidences of local conflicts over land or resources to the District Councils. This includes reporting of local conflicts between host community and refugees. The District Councils will in turn report conflict incidences to VPO PMU for project risk monitoring and risk management. The conflict sensitivity assessment planned under output 1 will contribute to further assess the risks associated with the relocation of refugees from Mtendeli to Nduta and identify suitable measures to mitigate conflict within the camps and with host community in the scope of the project intervention.

Selected Risk Factor 8		
Category	Probability	Impact
Financial	Low	Medium
Description		
Risk that there is low adoption of livelihood activities or that these interventions are not economically sustainable. The livelihood activities do not achieve the intended uptake nor generate the anticipated economic impact.		
Mitigation Measure(s)		
The Kigoma Joint Programme establishes the framework for long-term economic development in the region, and Kibondo and Kakonko regions are becoming more connected to markets — a new paved road connecting Kigoma to Mwanza (through Kibondo, Kakonko and Kasulu) is due to open in 2020. The funding framework proposed for the project uses Savings Associations that have the potential to reinvest resources to continue to support resilient activities. Savings Associations from UN CDF activities have shown success.		

Selected Risk Factor 9		
Category	Probability	Impact
Other	Medium	High
Description		
Large-scale refugee in-migration. Further influx of large numbers of refugees from adjacent countries into project area due to political instability/climate-stress.		
Mitigation Measure(s)		
Liaison with government/UNHCR. Use of NGOs and bilateral agencies to support project implementation.		

Selected Risk Factor 10		
Category	Probability	Impact
Governance	Low	Medium
Description		
Low uptake from local government of policy recommendations under Result/Output 4.		
Mitigation Measure (s)		
At the project design stage, local government agencies have been involved in the development of interventions and community-level consultations as reflected in Annex 7. There is a very high level of engagement and support for all the project interventions. Output 4 activities are design to show evidence of benefits of ecosystem-based adaptation to facilitate inclusion into development and sectoral planning. The potential location of some project positions in the Regional Office, can also facilitate understanding of the project activities and policy recommendations with the Regional Administration. Additionally, with the revised execution arrangements proposed, the implementation of activities will be partially delegated by the Government of Tanzania Vice-President's Office to District Councils and TFS through cooperation agreements, further enhancing project ownership from local government.		

Selected Risk Factor 11

Category	Probability	Impact
Other	High	Medium
Description		
Political risks. Tanzania's policies towards refugees have been more restrictive in recent years. UNHCR Tanzania has identified a number of risks related to government policy. These risks relate to Tanzania's implementation of its asylum policy in line with international standards, access to asylum seekers fleeing conflict from neighbouring countries, risks related to voluntary repatriation, refugee data, and an overall restrictive protection environment.		
Mitigation Measure (s)		
UNHCR Tanzania develops a yearly risk register to support risk identification and treatment. The most recent Risk Register was updated on 1 December 2020, for risks foreseen in 2021. The Risk Register includes specific proactive and reactive treatment measures to ensure that UNHCR can proactively mitigate them and, in the event they materialise, that UNHCR has the ability to react and address them. To address the political risks noted above, UNHCR has and will continue to put in place measures to ensure that government capacity to apply international refugee standards are continuously strengthened. Initiatives are underway to train government officials and build their capacity in management of refugee data and application of international protection standards and to jointly develop Standard Operating Procedures with government, particularly around voluntary repatriation and data management. UNHCR Tanzania continually undertakes advocacy with government at all levels to ensure that refugees are able to live in an environment where their needs are met. Such advocacy is undertaken at the country level by UNHCR Tanzania as well as at the regional and Headquarters level and, where necessary, also at the UN Secretary-General level. Where appropriate, joint advocacy is also conducted with other UN agencies and with the international donor community in Tanzania. UNHCR has paid particular focus to bolstering relations with the Kigoma regional authorities (where the project will be based) as a direct measure to mitigate risks that government policies may lead to increased restrictions on refugees and the humanitarian community. In 2020, significant progress was made in this regard and as of 2021, UNHCR's presence in Kigoma will be strengthened with the opening of a Sub-Office and the deployment of some senior management positions to that office. Through its own risk analysis, while UNHCR notes that government policies towards refugees will continue to be restrictive, efforts implemented in 2020 and treatments to be implemented in 2021 will likely reduce the impact these policies have on UNHCR's ability to continue supporting refugees in 2021, especially in the Kigoma region. Furthermore, it should be noted that the GCF project will, upon realization, contribute to the recognition that refugees can contribute to host communities, therefore ensuring greater local acceptance of refugees. Evidence on the benefits of the project to climate adaptation of both groups in the region will be collected to inform these discussions. The GCF project also addresses a key government priority, namely environmental degradation in refugee hosting areas. In doing so, the project is expected to positively influence the discourse on refugee issues and management, also through the involvement of a wider range on government ministries. At the regional and district level, there remains overwhelming support for the project and its activities as it engages directly with District officials and communities to promote harmonious relationships between host communities and refugees and project benefits to host communities. Through these people-to-people dialogues and benefit sharing arrangements in the project, the project aims to minimize and mitigate conflict in the area and promote dialogue. Overall, the positive outcomes of this projected are expected to contribute to a more positive outlook on refugee hosting and peaceful coexistence, which in turn will further contribute to improvement of government policies related to refugees. It is expected that project leadership from VPO may contribute to an improved policy dialogue involving MoHA and sector ministries, which will mitigate any possible (refugee related) political risks and may even have a positive impact on the existing policy environment – as the project is believed to be able to highlight the shared benefits and advantages of a more inclusive approach.		

G. GCF POLICIES AND STANDARDS

G.1. Environmental and social risk assessment (max. 750 words, approximately 1.5 pages)

Project categorisation

The project is being proposed as Category B or moderate risk according to the GCF project classification. This is because the project has potential limited adverse environmental and/or social risks and impacts that are few, generally site-specific, largely reversible, and readily addressed through mitigation measures. Justification for this is provided in the Environmental and Social Management System in Annex 6. The main risks anticipated are summarised below in Table 15, along with mitigation measures to address them.

Table 15. Summary of risks and mitigation measures.

Summary of risks	**Mitigation factors or measures and management strategies**
Environmental	
Risk that project has potential impact on national game reserves (the proposed project area surrounding Nduta camp is located within 10 km of the Moyowosi Game Reserve, and within 30 km of the Kigosi Game Reserve) Low risk	An overriding objective of this project is ecosystem restoration, including improving the quantity and quality of water in the region. Consequently, net environmental impacts will likely be positive rather than negative. Restoration and conservation in forest reserves within a 10 km radius to the Moyowosi Game Reserve would be conducted. This is expected to have positive benefits. No afforestation with fast-growing species for fuelwood will take place in this area. Hydrological connectivity to the Malagarasi- Moyowosi wetlands is present, but the drainage obtains water from many sources external to the project. Water extraction from in-stream rainwater harvesting schemes would likely have a negligible impact on the wetland. There are expected net benefits in improvement to water quality in the wetland due to the project. Modelling and discussion are available in more detail in the Feasibility Study. The project team will monitor any potential impacts of the rainwater harvesting structures on streamflow into the wetland by monitoring extraction rates and streamflow over time relative to a baseline that will be determined at the start of the project.
Risk that project has potential impact on areas considered protected or important by local communities Low risk	The project will be initiated through the land use planning processes (Output 1), which will work with local communities to identify the specific areas of land where interventions can be sited. This process will be used to identify any areas of land that are locally important and ensure these are demarcated in the land use planning process to reflect the wishes of the local communities. This process is fully participatory, involving multiple stakeholder groups, and interventions will only be implemented on host community land with the full support of the host communities.
Risk associated with the introduction of non-native species, but which are non-invasive Moderate risk	For species related to Activity 2.2, the final species in different sites will be selected during the Species Suitability Assessment conducted within the first year of the project. In the preparation stages, a mix of species was explored, including reference to species used in recent SNV afforestation projects in the country. Species selection will be informed by properties as being fast growing and highly productive – which will help to meet the fuelwood demand and by doing so reduce the pressure on the native Miombo forest. Plantation areas will be chosen by experienced foresters, focussing on degraded land where planting is required to reduce erosion and run off etc. Distribution of non-native species will be sensitive to the ecosystems in the project area, making particular consideration for avoiding riparian land and or where the water table is <5m, to ensure there is no negative impact on groundwater levels Species chosen for the afforestation would constitute only a small element of the mosaic landscape provided in Activity 2.2 and is restricted to already degraded lands. This will be complemented by the use of indigenous species such as Acacia auriculiformis, A. crassicarpa, A. leptocarpa, A. mangium, Albizia lebbeck, Glyricidia sepium, Leucaena pallida and Senna siamea in agroforestry, which will reduce landscape degradation and encourage the regeneration of biodiversity.

	The final species chosen will be determined by the Species Suitability Assessment, which will be informed by fire risk assessment, among other factors. The Feasibility Study describes the types of characteristics to be considered (see Annex 2.1 Feasibility Study). A specialist forest consultant will be engaged to select species for each site. Species selection will be benchmarked against other practises in the area such as activities promoted by the District Officers, GIZ and SNV projects.
Risk of impact on passage of fish from in-stream structures Low risk	The in-stream structures have been designed to incorporate fish ramps to allow for the passage of fish (see Activity 3.2).
Risk of impact on water resources in the project area from water pumping and irrigation of farmland and nurseries Low risk	Project objectives include increasing overall water infiltration and reducing flooding restoring degraded land, improving soil stability and providing rainwater harvesting infrastructure. This will reduce the pressure on surface and groundwater resources. While the project will look to supply water pumping equipment, this is intended to draw upon water stored in rainwater harvesting systems, as well as surface water sources. The solar and treadle pumps proposed have relatively low capacity. Capacity building on water conservation as well as soil moisture testing practises will be put in place to use water resources efficiently. Hydrological modelling in the feasibility study indicates sufficient availability of water relative to extraction under climate change. In addition, the project team will monitor any potential impacts of the rainwater harvesting structures on streamflow into the wetland by monitoring extraction rates and streamflow over time relative to a baseline that will be determined at the start of the project.
Risk of impact on ecosystems from more intensive agriculture Low risk	The project is funding training and tools only –pesticide use will not be provided/promoted in the project. Pest resistant crop varieties will be promoted. There may be some risk of farmers having increased production, thereby increasing their incomes and resources available for farm inputs. The implementation of activities will be consistent with the Climate Smart Agriculture guidelines of Tanzania that promote sustainable practises and will be done in coordination with the Agricultural District Officer. Sustainable practises will be promoted in the Farmer Field School approach.
Social	
Risk to health, safety or labour rights of workers involved in the construction of structures in the project (for example, mushroom structures, rainwater harvesting structures, pump houses) Low risk	The project will implement health, safety and labour rights safeguards. The procurement process for the project will include contractors' processes to manage health, safety and labour rights at the construction phase.
Safety impacts of in-stream structures Low risk	The intended structures are small scale only, which reduces the likelihood of integrity issues, the safety impact in the event of any damage to structures, and risk of children or people falling into drainage infrastructure. The procurement process for the project will include consideration of contractors' experience in building similar structures. Local communities will be trained by the contractor on maintenance and safety near water sources.
Risk of disputes between villagers during village land use planning Moderate risk	Joint land use planning process will used to develop land use plans for villages that share resources such as grazing land. These are based on existing land use management approaches that are enshrined in Tanzanian national regulations and policies. Lessons learned from these efforts suggest that while such participatory processes may require a considerable amount of time and resources, they successfully resolve conflicts around the use of natural resources, and they provide great opportunities for communities to take control and ownership of shared resources. While not usually accounting for climate change risks, the land use planning process is well-established in Tanzania, and it provides a proven approach for establishing the basis to formalise land tenure under the CCROs.

	The conflict sensitivity assessment planned under output 1 will contribute to further assess the risk of potential conflicts associated with Land Use Planning and the identification of suitable mitigation measures in the specific context of the project target areas. The land use planning process will engage separately with a range of potentially under-represented groups (e.g. women, pastoralist representatives) to ensure full participation. The project will engage NGOs experienced in delivering the planning processes and will follow the provisions of the Voluntary Guidelines on the Responsible Governance of Tenure (as applicable) to ensure best practise in this process.
Risk of water and vector borne diseases from standing water in rainwater harvesting structures Low to moderate risk	Water in rainwater harvesting structures is for irrigation only, not intended for human consumption and hence not expected to result in increased prevalence of cholera, diarrhoea, and other illnesses from drinking contaminated water. Risks will be monitored during the project period through field surveys. Management of diseases to be coordinated closely with the health cluster and district officer and explore the use of biological controls and education on the health risks of standing water.
Risk of security concerns/conflicts between host communities and refugees in relation to project delivery Moderate risk	The project has been designed to improve the ecosystem that hosts (and provides for) both the host communities and refugees and increase available resources, and by doing so reduce the level of tension / conflict between these groups. Where possible, dialogues and coordinated implementation between refugees and host community groups for activities like beekeeping and mushroom growing, will be pursued. These serve as important opportunities for interaction that can help fill the vacuum left after the closure of the common markets. When consulted in September 2020, 'peaceful coexistence meetings' were highlighted as an important peacekeeping and conflict-resolution tool by both refugees and host communities. This was attributed to the improved dialogue between the two groups that was facilitated by these meetings (elaborated on in Annex 2.1, Section 9.1). Management arrangements for forest reserves and afforested woodlots for fuel will consider the needs of both the host and refugee communities, with a view to develop benefit sharing schemes co-designed with the refugee and host communities. Management mechanisms promote dialogue and shared use. This has the potential to improve host-refugee relations, as damage to agricultural fields caused by fuelwood collection was highlighted as a challenge by host communities during consultations in September 2020. The majority of the interventions are located in host communities. This will help to rebalance the perception that refugees have unfairly benefited from support (in comparison to the host communities). The project will be delivered under the guidance of the Regional government, District Councils, UNHCR and other agencies / NGOs experienced in working in the region and managing any disputes. The conflict sensitivity assessment planned under output 1 will contribute to further assess the risk of potential conflicts and tensions linked to project delivery between host community and refugees, as well as between direct beneficiaries and indirect beneficiaries outside the project target area, to identify suitable mitigation measures in the specific context of the project target areas.
Risk of displacement, dispossession or limitations on access to resources Low to moderate	The land use planning, including the allocation of grazing land / corridors for pastoralists such as the Sukuma, and conflict resolution mechanisms such as peaceful coexistence meetings in the project are expected to lower the risk. There are plans to set aside land for jointly managing forests for sustainable harvest of fuelwood that will be accessible to both groups. Consultations among both host communities and refugees have found interest in both groups. There are concerns to having clear and jointly agreed upon arrangements for responsibility and benefit sharing, which will be addressed by project through Activity 2.2. The feasibility study recognizes that agricultural development in riparian areas is an issue within the project area (in terms of its negative impact on water availability). The project does not seek to actively restore riparian areas, although restoration may be an outcome of providing additional benefits in terms of agriculture, livelihoods and water supply. However,

	there may still be a perception that the project is seeking to displace economic activity from riparian areas. It is proposed that the land use planning process will provide the mechanism by which local communities will be able to understand how agricultural and livelihood activities can be at least as productive away from riparian areas compared to within them. The process will also help communities to recognize the benefits to the wider landscape of restoring the riparian areas. All decisions regarding land use planning will be taken by local communities and there will be no involuntary restrictions on land use, livelihood resources or any loss of land (see Component 1.1). Sensitization on local bylaws relating to providing a 60m buffer between land under cultivation and the riverbanks can be promoted by the local authorities in this regard.
Potential increase of incidence of Sexual and Gender Based Violence and Sexual Exploitation, Abuse, and Harassment Moderate risk	The project is not introducing activities that are expected to result in increased risk for SEAH. SEAH protocols by partners on the ground are in place. The project may potentially increase the risk of SGBV, particularly intimate partner violence, in carrying out activities targeted to women's livelihoods. The beneficiary selection process will consider this risk and involve male partners at the outset. Continuous monitoring of women's risks at the beneficiary level will be implemented in carrying out activities. The project will likely reduce SGBV risks for women from fuelwood collection, by providing fuelwood sources within the 4km buffer zone and reducing distances travelled. The incidence of SGBV as a result of firewood collection outside camps was highlighted as a particular issue by refugees, during consultations in September 2020.

Approach to environmental and social management during project implementation

The risks considered in the ESERN risk screening checklist have been further elaborated in the Environmental and Social Management Plan (ESMP) in Annex 6. This shows how these risks have been considered in the fundamental design of the project and any additional mitigation measures that may be required.

In addition, given the structure of the project, whereby the exact siting and detailed design of the project interventions are being determined as part of the project itself (through the participatory land-use planning processes), the specific nature of any residual environmental and social risks that remain will be assessed as part of project implementation.

Risks will be managed through:
- Involving an experienced environmental and social safeguards specialist through the specific planning, design and implementation processes.
- Ensuring the project officers (forestry, livelihoods, agriculture and water) recruited onto the project are sufficiently experienced in safeguards relevant to their specialist technical areas.
- Establishing methodologies for the environmental and social assessment of each of the activities under the project.
- Ensuring the monitoring and evaluation (M&E) process includes adequate review of environmental and social risks (this will be done by an M&E specialist with specific experience in reviewing environmental and social considerations).
- Assigning formal roles and responsibilities for environmental and social management across the project team.
- Establishing a reporting procedure for environmental and social concerns related to the interventions.
- Incorporating environmental and social safeguard considerations in the training, capacity building and technical assistance delivered through the project.

The environmental and social management processes will draw upon the following documents:
- Green Climate Fund Environmental and Social Management System;
- UNEP safeguards systems;
- UNHCR safeguards systems;
- summaries of stakeholder consultations;
- ESMP and stakeholder management plan;
- Policy and regulatory environment; and
- Gender Assessment and Action Plan.

G.2. Gender assessment and action plan (max. 500 words, approximately 1 page)

During project preparation, a gender assessment was conducted to understand the differential vulnerability to, and impact of, climate change on women and men in Kigoma. From the findings of this assessment, and an analysis of the proposed project solution, a gender action plan (GAP) was developed. The process to produce these components included a desktop review of available literature and an analysis of relevant legislation, policy and programmes in the country, as well as interviews and consultations with project and beneficiary stakeholders.

The gender assessment and problem analysis identified causes and effects related to land tenure, participation in agricultural activities, access to knowledge and financial services, and broad participation in income-generating activities. A critical factor affecting women's adaptive capacity to climate change is related to insecure land tenure and exclusion from access to and control over land. This ultimately results in lower productivity and income generated from natural resources. Women's vulnerability to climate change is also exacerbated by their central role in collecting firewood and water and engaging in rain-fed agricultural activities. Gender-blind policies, and discriminatory social and cultural norms related to land use, agriculture, water and forestry activities, results in women having limited participation in income-generating activities and involvement in farmers' groups, as well as being underrepresented in forest management activities.

Women from refugee and host communities are also susceptible to protection issues like sexual and gender-based violence (SGBV) when they travel outside of their settlements, particularly to collect firewood. A 2015 field report from Refugees International found that 25% of 224 SGBV incidents reported by Burundian refugees in Tanzania over a five-month period occurred during firewood collection trips, and 90% of these incidents were rape or sexual assault[233]. Although culturally a gendered role, women refugees are also forced into this activity as they pose a lesser risk to host communities, whereas male refugees collecting firewood are perceived to be 'stealing' and may be violently attacked. While host community-refugee meetings had been reportedly effective in reducing incidences of SGBV, respondents during September 2020 consultations noted that SGBV during fuelwood collection remained an ongoing challenge. Other dynamics which have also been known to increase instances of SGBV are increasingly being experienced by refugees, including family conflict — particularly regarding returning to country of origin —, firewood collection beyond the 4km camp buffer zone and restrictions of freedom of movement and livelihood activities.

To mitigate against biasing against women as project beneficiaries during implementation and subsequent to project completion, the gender action plan outlines several interventions to ensure the project does not unintentionally contribute to perpetuating or worsening gender equality and gender-based climate vulnerability in Kigoma. The GAP seeks to promote gender equality by including activities which will increase women's representation and awareness, as well as ensuring that training programmes are gender-sensitive and -inclusive. It includes clear targets, sex-disaggregated data collection, gender-sensitive design features and quantifiable performance indicators to promote women's participation and benefits from the project.

The actions of the GAP include:
- promoting women's inclusion in all project aspects, including training and employment opportunities;
- pursuing representative participation in all consultations and workshop events;
- advancing gender diversity and challenging negative stereotyping in public awareness activities; and
- designing and implementing gender-sensitive training that considers the different learning methods and training accessibility of men and women.

The full Gender Assessment and project-level Gender Action Plan are included as Annex 8.

G.3. Financial management and procurement (max. 500 words, approximately 1 page)

The financial management and procurement within the Programme will be guided by applicable UN financial regulations, rules and practises. The financial rules of UNEP, which follow International Public Sector Accounting Standards (IPSAS), are promulgated pursuant to the Financial Regulations and Rules of the UN. Within this context, funding allocation mechanisms are managed in accordance with applicable UN rules and procedures, including eligibility criteria, proposal evaluation processes, quality assurance and control, project monitoring and supervision. UNEP is audited annually by the UN Board of Auditors and has established dedicated trust funds for Green Climate Fund resources.

In line with UNEP reporting procedures outlined in section E.7, reports to summarise the disbursement and projected demands for Programme funding will be prepared and submitted to a UNEP Programme Officer who will conduct

[233] Vigaud-Walsh, *Women and Girls Failed: The Burundian Refugee Response in Tanzania*, 2015. Available at: http://static1.squarespace.com/static/506c8ea1e4b01d9450dd53f5/t/5678aee07086d7cddecf1bab/1450749707001/20151222+Tanzania.pdf.

Programme supervision, in line with reporting standards and methodologies applied in past projects, such as those implemented using GEF modalities.

UN financial regulations and rules require the segregation of duties, and safeguards to ensure compliance with UN financial rules and regulations. All procurement will be undertaken in line with applicable UN procurement regulations, rules and policies. In line with UN regulations, rules and policies, funds will be transferred in tranches to the Executing Entities (EEs) once the EEs has satisfied the conditions that are defined under the Subsidiary Agreements to be signed between UNEP and the Government of Tanzania Vice-President's Office (Project Cooperation Agreement) and between UNEP and UNHCR (UN-to-UN Agreement) The Agreements will include specific obligations for the EE on financial management and reporting and will require periodic reporting from the EE. The Government of Tanzania Vice-President's Office (VPO), as the lead EE, will be responsible for the submission of consolidated financial reports to UNEP, including funding for activities executed by VPO and UNHCR. The coordination mechanisms to deliver consolidated reports will be specified in the Memorandum of Understanding signed between VPO and UNHCR.

G.4. Disclosure of funding proposal

☒ No confidential information: The accredited entity confirms that the funding proposal, including its annexes, may be disclosed in full by the GCF, as no information is being provided in confidence.

☐ With confidential information: The accredited entity declares that the funding proposal, including its annexes, may not be disclosed in full by the GCF, as certain information is being provided in confidence. Accordingly, the accredited entity is providing to the Secretariat the following two copies of the funding proposal, including all annexes:

- full copy for internal use of the GCF in which the confidential portions are marked accordingly, together with an explanatory note regarding the said portions and the corresponding reason for confidentiality under the accredited entity's disclosure policy, and
- redacted copy for disclosure on the GCF website.

The funding proposal can only be processed upon receipt of the two copies above, if containing confidential information.

H. ANNEXES

H.1. Mandatory annexes

☒	Annex 1	NDA no-objection letter(s) **(template provided)**
☒	Annex 2	Feasibility study - and a market study, if applicable
☒	Annex 3	Economic and/or financial analyses in spreadsheet format
☒	Annex 4	Detailed budget plan **(template provided)**
☒	Annex 5	Implementation timetable including key project/programme milestones **(template provided)**
☒	Annex 6	E&S document corresponding to the E&S category (A, B or C; or I1, I2 or I3): **(ESS disclosure form provided)** ☐ Environmental and Social Impact Assessment (ESIA) or ☐ Environmental and Social Management Plan (ESMP) or ☐ Environmental and Social Management System (ESMS) ☐ Others (please specify – e.g. Resettlement Action Plan, Resettlement Policy Framework, Indigenous People's Plan, Land Acquisition Plan, etc.)
☒	Annex 7	Summary of consultations and stakeholder engagement plan
☒	Annex 8	Gender assessment and project/programme-level action plan **(template provided)**
☒	Annex 9	Legal due diligence (regulation, taxation and insurance)
☒	Annex 10	Procurement plan **(template provided)**
☒	Annex 11	Monitoring and evaluation plan **(template provided)**
☒	Annex 12	AE fee request **(template provided)**
☒	Annex 13	Co-financing commitment letter, if applicable **(template provided)**
☒	Annex 14	Term sheet including a detailed disbursement schedule and, if applicable, repayment schedule

H.2. Other annexes as applicable

☒	Annex 15	Evidence of internal approval **(template provided)**
☒	Annex 16	Map(s) indicating the location of proposed interventions
☐	Annex 17	Multi-country project/programme information **(template provided)**
☐	Annex 18	Appraisal, due diligence or evaluation report for proposals based on up-scaling or replicating a pilot project
☐	Annex 19	Procedures for controlling procurement by third parties or executing entities undertaking projects financed by the entity
☐	Annex 20	First level AML/CFT (KYC) assessment
☒	Annex 21	Operations manual (Operations and maintenance)
☐	Annex 22	Assessment of GHG emission reductions and their monitoring and reporting (for mitigation and cross cutting-projects)[234]
☒	Annex X	Other references Annex 22.1 Tanzania country refugee response plan. Annex 22.2 UNHCR Financial Management Manual.

Annex 23. Letter of support Danish Embassy of Tanzania.

Annex 24. Government of Tanzania Vice President's Office capacity assessment certificate

** Please note that a funding proposal will be considered complete only upon receipt of all the applicable supporting documents.*

[234] Annex 22 is mandatory for mitigation and cross-cutting projects.

No-objection letter issued by the national designated authority(ies) or focal point(s)

UNITED REPUBLIC OF TANZANIA
VICE PRESIDENT'S OFFICE

Telegram: "MAKAMU", HQ
Telephone: +255 026 2329006
Fax. No. +255 026 2329007
E-mail: ps@vpo.go.tz

Government City,
Mtumba Area,
Vice President's Office Building,
P. O. Box 2502,
40406 DODOMA.

In reply please quote:

Ref. No. CBA. 78/90/03/39

18th November, 2021

The Executive Director,
Green Climate Fund,
Sondo Business District,
175 Art Centre-daero,
Yeonsu-gu, Incheon 22004
REPUBLIC OF KOREA

Re: Funding proposal for the GCF by the United Nations Environment Programme (UNEP) regarding Building Climate Resilience in the Landscape of Kigoma Region, Tanzania.

Dear Madam, Sir,

We refer to the project titled **Building Climate Resilience in the Landscape of Kigoma Region** in Tanzania as included in the funding proposal submitted by **the United Nations Environment Programme (UNEP)** to us on 29 October 2021.

The undersigned is the duly authorized representative of United Republic of Tanzania Vice President's Office, the National Designated Authority of Tanzania.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the project as included in the funding proposal.

By communicating our no-objection, it is implied that:

(a) The government of United Republic of Tanzania has no-objection to the project as included in the funding proposal;
(b) The project as included in the funding proposal is in conformity with the national priorities, strategies and plans of Tanzania;
(c) In accordance with the GCF's environmental and social safeguards, the project as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the project as included in the funding proposal has been duly followed.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

Mary N. Maganga
PERMANENT SECRETARY

Environmental and social safeguards report form pursuant to para. 17 of the IDP

Basic project or programme information	
Project or programme title	Building climate resilience in the landscapes of Kigoma region, Tanzania
Existence of subproject(s) to be identified after GCF Board approval	No
Sector (public or private)	Public
Accredited entity	United Nations Environment Programme (UNEP)
Environmental and social safeguards (ESS) category	Category B
Location – specific location(s) of project or target country or location(s) of programme	United Republic of Tanzania
Environmental and Social Impact Assessment (ESIA) (if applicable)	
Date of disclosure on accredited entity's website	Wednesday, September 6, 2023
Language(s) of disclosure	English and Kiswahili (or Swahili)
Explanation on language	English and Swahili are the official languages of Tanzania. They serve as working languages in the country, with Swahili being the official national language. Within the camps, Swahili and Kirundi are widely spoken as well as French. At the request of the Government, Swahili has been chosen as the local language for disclosure because it is the official language for the Government authorities, and it is widely spoken in the camps.
Link to disclosure	English: https://open.unep.org/docs/gcf/Annex_6_Kigoma_GCF_ESMS_English_rev23_08_2023_clean.pdf Kiswahili (or Swahili): https://open.unep.org/docs/gcf/Annex_6_Kigoma_GCF_ESMS_Kiswahili_rev23_08_2023_clean.pdf
Other link(s)	https://open.unep.org/project/GCF-GCF-Tanzania-FP
Remarks	An ESIA consistent with the requirements for a Category B project is contained in the "Environmental and Social Management System (ESMS)".
Environmental and Social Management Plan (ESMP) (if applicable)	
Date of disclosure on accredited entity's website	Wednesday, September 6, 2023
Language(s) of disclosure	English and Kiswahili (or Swahili)
Explanation on language	English and Swahili are the official languages of Tanzania. They serve as working languages in the country, with Swahili being the official national language. Within the camps, Swahili and Kirundi are widely spoken as well as French. At the request of the Government,

	Swahili has been chosen as the local language for disclosure because it is the official language for the Government authorities, and it is widely spoken in the camps.
Link to disclosure	English: https://open.unep.org/docs/gcf/Annex_6_Kigoma_GCF_ESMS_English_rev23_08_2023_clean.pdf Kiswahili (or Swahili): https://open.unep.org/docs/gcf/Annex_6_Kigoma_GCF_ESMS_Kiswahili_rev23_08_2023_clean.pdf
Other link(s)	https://open.unep.org/project/GCF-GCF-Tanzania-FP
Remarks	An ESMP consistent with the requirements for a Category B project is contained in the "Environmental and Social Management System (ESMS)".
Environmental and Social Management (ESMS) (if applicable)	
Date of disclosure on accredited entity's website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A
Link to disclosure	N/A
Other link(s)	N/A
Remarks	N/A
Any other relevant ESS reports, e.g. Resettlement Action Plan (RAP), Resettlement Policy Framework (RPF), Indigenous Peoples Plan (IPP), Indigenous Peoples Planning Framework (IPPF) (if applicable)	
Description of report/disclosure on accredited entity's website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A
Link to disclosure	N/A
Other link(s)	N/A
Remarks	The UNEP Environmental, Social and Economic Review Note (ESERN) is appended as addendum 2 of the "Environmental and Social Management System".
Disclosure in locations convenient to affected peoples (stakeholders)	
Date	Tuesday, September 5, 2023
Place	Swahili Summaries of the Environmental and Social Management Systems were made available to stakeholders during the stakeholder consultations. The full translated copies in Swahili and English were first posted by February 15, 2021 in District Offices of Kibondo, Kakonko, and Kasulu and the Kigoma Regional Administration Secretariat as well as in Nduta, Mtendeli, and Nyarugusu refugee camps. Following the update of the ESMS in November 2021, revised copies of the ESMS in Swahili and English were posted on January 5, 2022. Following the revision of the ESMS in August 2023, the ESMS Swahili and English version were posted on

	September 6, 2023 in the District Offices of Kibondo, Kakonko, and Kasulu, the Kigoma Regional Administration Secretariat as well as in several locations in the Nduta and Nyarugusu refugee camps. (Mtendeli camp was closed in January 2022). The following are the contact details of the District Offices and Regional Administration Secretariat: Kibondo District Office P.O.BOX 43, Kibondo, Kibondo District Council, Kibondo Township Ihulilo Street Telephone: +255 028 282 0084 Mobile: +255 028 282 0084 Fax: +255 028 282 0084 Email: ded@kibondodc.go.tz Kakonko District Office P.O.BOX 3 Halmashauri ya Wilaya ya Kakonko Telephone: +255 028 282 0137 Mobile: +255 075 747 0800 Fax: +255 028 282 0137 Email: ded@kakonkodc.go.tz Kasulu District Office S.L.P 97 Halmashauri Ya Wilaya Ya Kasulu Telephone: +255 028 281 0339 Mobile: +255 028 2810339 Fax: +255 028 2810339 Email: ded@kasuludc.go.tz Kigoma Regional Administration Secretariat Office Mkoa Wa Kigoma Postal Address: S.L.P 125 Telephone: +255 028 280 2330 Mobile: +255 738 192 977 Email: ras@kigoma.go.tz / ras.kigoma@tamisemi.go.tz / info@kigoma.go.tz
Date of Board meeting in which the FP is intended to be considered	
Date of accredited entity's Board meeting	N/A
Date of GCF's Board meeting	Monday, October 23, 2023

Note: This form was prepared by the accredited entity stated above.

Secretariat's assessment of FP218

Proposal name:	Building climate resilience in the landscapes of the Kigoma Region, Tanzania
Accredited entity:	United Nations Environment Programme (UNEP)
Country/(ies):	United Republic of Tanzania
Project/programme size:	Small

I. Overall assessment of the Secretariat

1. The funding proposal is presented to the Board for consideration with the following remarks:

Strengths	Points of caution
The project seeks to address the climate vulnerability of host communities in degraded landscapes occupied by displaced populations facing unique challenges and layered vulnerabilities in a shared landscape. The project intends to bridge the gap of the development–humanitarian–climate nexus.	The overall project risk rating is high, driven by the political risks of the project. The project necessitates close monitoring by the executing entities, accredited entity and GCF from the start of project implementation following early-stage active adaptive management.
The project is firmly anchored in government structures and ownership. The project will establish a team working through Government: Vice President's Office, Division of Environment, the Ministry of Home Affairs will play a key role in overseeing the implementation of activities in the refugee camps; The Regional Administration, which operates under the President's Office, Regional Administration and Local Government, oversee the Local Government Authorities' activities providing supervision and administrative instructions; The Tanzania Forest Service under the Ministry of Natural Resources and Tourism; and The National Environmental Management Council will ensure compliance with relevant environmental laws and regulations as well as providing technical advisory and compliance guidance on all environmental aspects of the project.	

2. The Board may wish to consider approving this funding proposal with the terms and conditions listed in the respective term sheet and addendum XXI, titled "List of proposed conditions and recommendations".

II. Summary of the Secretariat's assessment

2.1 Project background

3. The project will focus on the surrounding host communities and refugee camps of Nduta and Nyarugusu, and the former Mtendeli camp,[1] in the Kigoma Region of the United Republic of Tanzania. These communities are nature-dependent, living in an increasingly degraded landscape, with complex and varied needs. Using an integrated landscape approach, the project intends to complement the already existing development and humanitarian efforts in the area by focusing on climate change-related impacts.

4. The Kigoma Region covers about 37,000 km^2 in the north-west of the country and has a population of about 2.3 million people. Situated on a plateau, the region hosts about 280,000 refugees from neighbouring countries. The Kigoma Region is composed of a mosaic landscape, surrounded by areas of global biodiversity importance, with a predominance of Miombo forests and patches of agricultural areas covering varied landscape continuum features. In the north of the region, agropastoralist Sukuma people pass through in search of grazing land.

5. Historical climate trends show more intense and erratic precipitation events leading to an increasing frequency of flood events. Indeed, the project makes a relevant distinction between the pluvial flood risks (from extreme precipitation events) and the fluvial flood risk (from rivers and streams). The Nyarugusu camp is mainly affected by the pluvial flood risks, while the former Mtendeli camp and the Nduta camp are exposed to both pluvial and fluvial flooding. Pluvial flood risk will increase because the intensity of hourly rainfall is projected to rise from about 1 mm/hour (Representative Concentration Pathway 4.5) to about 1.5 mm/hour (Representative Concentration Pathway 8.5) at the end of this century.

6. Fluvial flood risk is expected to increase by substantially more, because climate change has altered, and will continue to alter, vegetation. The roughness index of land surfaces will decrease, resulting in higher erosion and flood risks. This is compounded by the fact that hydrological processes are highly non-linear. For example, an increase in rainfall of 10 per cent can result in an additional flood risk of 50 per cent. It is reasonably justified to assume that climate change is a major factor in increased flood risk, leading to secondary impacts such as gully erosion. The changes in the rainfall profile of the Kigoma Region show more prolonged dry spells. There is also evidence regarding mean annual temperature increase, leading to increased evapotranspiration and evaporation of surface water resources. These drying trends are affecting local livelihoods that rely on rainfall and ecosystem goods and services in the Kigoma Region. This climate context, compounded with the landscape degradation threats and impacts (described below), justifies the proposed interventions.

7. The project consists of two components with four outputs that will be implemented by two executing entities (EE): the Tanzania Vice-President Office (VPO) as lead EE and the United Nations High Commission for Refugees (UNHCR) as co-EE. According to the environmental and social safeguards (ESS) categorization provided by the United Nations Environment Programme (UNEP), which is the accredited entity (AE), this is a Category B project.

8. Component 1: Increased resilience of ecosystems and people to climate change threats:

(a) Output 1: Participatory Climate-resilient Land Use Plans developed (C-LUP) to support ecosystem-based adaptation;

[1] At the time of the Secretariat review, the Nduta and Nyarugusu camps are in existence, and the Mtendeli camp was closed in January 2022. Refugees in Mtendeli camp, except for those opting for voluntary repatriation, were relocated to Nduta camp after the Mtendeli camp has been decommissioned.

(b) Output 2: Degraded ecosystems restored to support climate change adaptation in host and refugee communities; and

(c) Output 3. Climate-resilient livelihoods practised to increase the capacity of host communities and refugees to better adapt in changing climatic conditions.

9. Component 2: Strengthened policies for climate-responsive planning and development:

(a) Output 4: Information on climate change adaptation disseminated and mainstreamed into policies, plans and strategies in the United Republic of Tanzania and in humanitarian programmes.

10. All activities will be executed, overseen and managed by the Division of Environment of VPO. UNHCR will execute activity 2.3, and subactivities 3.2.1 and 3.2.2. Most activities in outputs 2 and 3 will be co-implemented by the district councils and relevant technical and local partners in coordination with VPO.

11. The project is seeking only grant funding from the GCF, with a mix of co-financing in the form of an in-kind contribution from UNHCR, with a co-financing ratio of 1:0.24. Considering the least developed country (LDC) status of the country and the extreme political, social, economic and climate vulnerability of the target beneficiaries, this can be considered satisfactory.

2.2 Component-by-component analysis

Component 1: Increased resilience of ecosystems and people to climate change threats (total cost: USD 21,196,079; GCF USD 16,728,786, or 79 per cent)

12. The starting point for all project interventions will be the development of C-LUPs to support climate-resilient and ecosystem-based adaptation measures. This is proposed as a fundamental step in determining how the interventions described later in outputs 2–4 will be planned and delivered. Key steps will include the identification of the specific adaptation measures for each village and the intervention designs to account for climate change: activities 2.3 (flood and erosion control) and 3.2 (rainwater harvesting and irrigation) are clear illustrations of this. As each C-LUP identifies village forest areas, it is also an entry point for the preparation of community-based forest management (CBFM) plans and for agroforestry interventions (output 2).

13. **Output 1: Participatory climate-resilient land-use plans developed to support ecosystem-based adaptation (total cost: USD 1,602,805; GCF USD 1,602,805, or 100 per cent)**: Land-use planning activities will support overcoming challenges related to the relationships between the agropastoral communities (such as the Sukuma people), host communities, and refugees. While the host communities in the project areas are composed of farmers and livestock keepers (about 70 per cent) and are more tied to the land, the Sukuma people are generally not part of community planning and awareness-raising processes. This is due to their transient lifestyle and non-inclusion as part of the local social fabric. Currently, the lack of land-use planning and management of customary land can lead to conflict between farmers and pastoralists. For example, the lack of land assigned for grazing can also lead to encroachment onto village and national forest reserves outside camps and villages. The project will engage with the Sukuma people and other land-use groups in the land-use planning process, and in activities relating to environmental education and strengthening environmental management committees with overall processes reflected in the stakeholder engagement plan. Overall land-use planning processes at village level serve as the mechanism for designating appropriate land use with the participation of users and community members.

14. Another important group are the refugees, who have unique needs. Refugees living in the camps located within the Kigoma Region have limited social and economic freedoms (i.e. restricted movement beyond the camp). Although they practise agricultural activities to some

extent, they are also often reliant on food aid. However, many refugees are now in protracted situations where the temporary camps are becoming increasingly more permanent places of settlement. Therefore, incorporating climate-resilient approaches and measures (i.e. measures to prevent and control flood and erosion, and to increase access to alternative livelihoods) in camp planning is key for sustainable camp management.

15. Under output 1, the planning processes for host villages and camps will be separate, because these are discrete and well-defined areas, but will be coordinated to maximize landscape-level benefits (e.g. stream protection along streams in camp and host community land). The participatory land-use planning process under output 1 offers an opportunity to allocate grazing areas, in addition to land for crops, woodlots, natural forests, settlements and different land uses within host communities. Land-use planning processes serve as the mechanism to adjudicate boundaries for land uses, identify management regimes, achieve consensus, and negotiate across different and potentially competing interests. The stakeholder consultation and lessons learned from implementing these participatory planning measures under past initiatives suggest that increased clarity with land-use planning can: (i) successfully contribute to resolving community conflicts around land resource use; and (ii) enhance community capacity to share resources and resolve differences.

16. The long-term legal outcome of the village land-use planning process is the issuance of a certificate of customary right of occupancy. This document specifies rights conferred to a land occupier and user following tribal customs and traditions. The project will support this relatively small incremental cost to promote project ownership, engender support, and lead to sustainability of project outputs and outcomes. The certificate will provide long-term security for the villagers, and may be used as collateral for borrowing money to conduct other future village-related climate-resilient investments.

17. A conflict sensitivity assessment has been introduced under output 1 and will be conducted in the initial stages of project implementation (second quarter, first year). This will enable the project partners to gain a more in-depth understanding of potential conflict risks linked to the project delivery and to embed suitable mitigation measures in the project intervention design. The assessment will cover: (i) conflict related to land-use planning in host communities; (ii) tensions between indirect beneficiaries outside the project target area and direct beneficiaries; and (iii) risks related to the relocation of refugees from the camps. These risks have been assessed by UNHCR during project formulation (see annex 6 of the funding proposal).

18. **Output 2: Degraded ecosystems restored to support climate change adaptation in host and refugee communities (total cost: USD 13,038,743; GCF USD 9,978,657, or 77 per cent**): Activity 2.1 aims at protecting natural forest systems and enhancing the value that local communities place on the ecosystems and their services through CBFM. It aims to reduce climate vulnerability and negative pressure on biodiversity in nearby areas by: (i) improving forest protection and patrolling; (ii) allocating areas for CBFM; (iii) establishing dedicated woodlots for fuelwood use; and, more indirectly, (iv) enhancing ecosystem functions and services. Based on the feasibility assessment and economic analysis, the impacts of project activities are expected to be largely positive.

19. Activity 2.2 (forest monitoring) is expected to complement the CBFM interventions. Afforestation and agroforestry are expected to take place in the host communities. The afforestation labour force for the village woodlots will come from the village community members and will be mobilized by village leaders. Agroforestry systems are expected to mimic forest-like conditions in the long term. Thus, they will recover the ecosystem services that were initially lost when forests were converted to agricultural lands. The afforestation and agroforestry interventions are also expected to reduce the pressure on the native Miombo forests and allow them to regenerate so that they can resiliently experience climate change-related impacts such as ecological range shifts and alteration of the phenological cycles. Both

native and non-native forest species are expected to be used for afforestation. Native species will be used for recovery of degraded areas, while non-native species are preferred by locals as fodder trees and due to their fast-growth and multipurpose characteristics. While the integration of trees and shrubs in crop systems reduces economic vulnerability by diversifying production, it also acts as security against total crop failure.

20. Activity 2.3 (flood-reduction infrastructure) is integral to protecting the people living in and around the refugee camps. The feasibility study includes a rigorous and detailed analysis of flood risks based on the information available. It is reasonably justified to assume that climate change is a major factor in increased flood risk. The proposed impact infrastructure measures encompass: (i) micro-scale, stone-pitched drainage channels; and (ii) sustainable drainage systems. Flood and erosion control measures will also be implemented in the land area of the camps, as an integral component of the camp closure and environmental restoration plan, in combination of other restoration interventions implemented by UNHCR. The post-consolidation restoration activities are expected to result in increasing the resilience of host communities in terms of (i) reducing flood and erosion risk for downstream host communities; and (ii) enabling district government plans to use the consolidated camp infrastructure to improve host communities' access to basic services.

21. The beneficiaries of this activity are primarily the refugees of the former Mtendeli camp, and of the Nduta and Nyarugusu camps, given the local impact of this intervention. This complements the larger-scale agroforestry and village land afforestation interventions (activity 2.2) that will have a wider-ranging impact. It is expected that local communities and refugees will be involved in planning and constructing the flood-reduction infrastructure. Building on previous solutions in the camps, the effectiveness of stone-pitched drainage channels will be enhanced by 16 stone-filled gabions. Further fine-tuning is essential, and this is foreseen during the implementation of the project using local institutions and knowledge. The proposed sustainable drainage systems involve construction using human-made materials and/or natural systems that: (i) tend to preserve existing open spaces; (ii) protect natural systems (including groundwater and surface water) for improved drainage and filtration; and (iii) make use of existing land-use plans and maintenance to manage stormwater flows.

22. Fostering ownership will be key for ensuring continued maintenance of the proposed interventions and to function effectively. The operation and maintenance (O&M) costs basically relate to camp-based small-scale flood mitigation infrastructure that will be turned over to the Ministry of Home Affairs (MoHA); the rest will be turned over to respective district and regional authorities. The water and sanitation humanitarian cluster development agencies will support the MoHA in this role. Most of the maintenance tasks required are inspection, cleaning, desilting and weeding (for instance, through community labour). Materials for replacement are expected to be procured from the local markets. Moreover, an O&M plan has been developed and incorporated into the project design.

23. **Output 3: Climate-resilient livelihoods practised to increase the capacity of host communities and refugees to better adapt in changing climatic conditions (total cost: USD 7,094,532; GCF USD 5,687,324, or 80 per cent)**: Climate-resilient agricultural practices under activity 3.1 include: (i) use of cultivars resilient to pests, drought and heat stress; (ii) integrated soil management; (iii) integrated soil fertility management; (iv) intercropping; and (v) water harvesting and management training. These practices are to be implemented as appropriate according to the participatory needs and capacity evaluations undertaken with the beneficiaries. These activities will be implemented in alignment with the climate-smart agriculture approach and support the farmer field schools on climate-resilient agricultural practices. This also includes the use of traditional knowledge through training sessions co-designed between farmers, district officers and extension schools. Activity 3.1 will also support capacity-building of the savings association management offices (SAMOs) to manage savings

associations (SAs). SAMOs will support the distribution of agricultural and forestry inputs (e.g. seeds, cuttings and tools) to the beneficiaries using existing SAs in host communities.

24. Activity 3.2 aims at providing water for agriculture. The activity consists of four subactivities: (i) storing and retaining water; (ii) bringing water to the fields; (iii) training and awareness-raising; and (iv) strengthening institutions (water users associations). This is supported by an analysis of rainwater harvesting options for the project areas. A detailed assessment of the options will be done during implementation, because this depends on the micro-scale local conditions (soil, slopes, vegetation, community structure and capacity, among others). Treadle pumps are optimal investments given the local conditions.

25. Aside from rainfall, temperature is also considered. Higher temperatures will induce higher evaporation from natural lands, leaving less water available for crops. Moreover, crop water requirements will rise with increasing temperatures. Selecting the infrastructure (i.e. run-off harvesting micro-dams, unlined water pans, micro-ponds, micro-solar, and treadle pumps) is straightforward because the technologies are simple and easily maintained.

26. Ownership is essential in terms of maintenance and, thus, sustainability. It is recommended that training (component 2) concentrates on useful, practical, technical and hands-on lessons and practices so that sustainability and maintenance of the infrastructure will be secured.

Component 2: Strengthened policies for climate-responsive planning and development (total cost: USD 469,554; GCF USD 469,554, or 100.0 per cent)

27. **Output 4 (total cost: USD 469,554; GCF USD 469,554, or 100 per cent):** Climate change adaptation knowledge and experience will be disseminated and mainstreamed into policies, plans and strategies in the United Republic of Tanzania and in UNHCR humanitarian programmes. The results of the planning process and physical interventions implemented under outputs 1–3 will be used to demonstrate the economic, environmental and social benefits and value, thus contributing to the sustainability and scalability of the results and impacts. Activities under output 4 include: (i) gathering evidence of the economic benefits of ecosystem-based adaptation to host communities and refugee populations for use by policymakers and planners; (ii) developing communication products for policy and planning mainstreaming; and (iii) strengthening the development–humanitarian–climate nexus through the ecosystem-based adaptation model.

28. **Monitoring and evaluation (total cost: USD 1,421,283; GCF USD 1,421,283, or 100 per cent):** The monitoring and evaluation cost is about 6 per cent of the total project costs and is compliant with the Evaluation Policy for the GCF.

29. **Project management (total cost: USD 531,730; GCF USD 387,730, or 73 per cent):** The GCF portion of the project management cost is less than 5 per cent of the total requested GCF funding and is compliant with the GCF policy on fees.

III. Assessment of performance against investment criteria

3.1 Impact potential

30. The project aims to reach 1.9 per cent of the total national population and approximately half of the population in the Kigoma Region. The project's direct beneficiaries are estimated at 570,341 people who will benefit from the adoption of diversified, climate-resilient livelihoods based on the sustainable management, conservation and restoration of ecosystems, as well as on climate-smart agriculture practices and CBFM in the Kigoma landscape. These interventions will be anchored in land-use planning processes that aim to improve incentives for climate resilience for their effective implementation. An additional 711,516 people in the

region are also expected to benefit from enhanced knowledge and learning in climate-resilient activities as well as from the replication potential of local financial structures such as the SAs. The total expected beneficiaries amount to 1,281,857 people. Specific criteria for activities are to be used to realize project benefits. For example, the specific villages and land area to be included in the CBFM system (as well as for the afforestation and agroforestry under activity 2.1) will be chosen through the land-use planning process under output 1 following important environmental- and social-related indicators. For activity 3.3 (mushroom growing and beekeeping activities), vulnerable populations from the host communities will be targeted (e.g. female-headed households, people with disabilities). The criteria are deemed to be appropriate and are in line with the GCF investment criteria indicators.

31. On the basis of the information provided in the feasibility study, it has been shown that the Kigoma Region is experiencing rising temperatures as well as increased rainfall intensity and variability, which will contribute to an increase in the frequency and severity of floods and prolonged droughts, as well as heat and water stress. Already, observed changes in weather patterns have produced climate impacts that have compounded environmental degradation, as well as negatively affecting agricultural outputs, the incidence of pests and diseases, forest resources and deforestation, surface- and groundwater availability, and soil erosion and fertility. Projected climate changes will further exacerbate these problems, together with non-climatic development drivers of degradation. These changes are threatening local rain-fed and ecosystem-reliant livelihoods in the region. In support of these arguments, annex 2 to the proposal includes climate information based on the observational data that was used to validate the CORDEX regional climate models used for assessing projected climate changes. The validation[2] assessed seasonal signal, magnitude and discrepancies of the 10 models compared with the station data.

32. The climate analysis described in annex 2 notes that the current observational trends are projected to increase into the future. The most notable trends are the lowering of early season rainfall, and the focusing of rainfall into the mid and, particularly, late season. There is also a general increase in extreme precipitation days, leading to an increase in potential flooding events. The meteorological drought index (the Standard Precipitation Evapotranspiration Index) indicates that the variability in the rainfall will likely lead to more frequent and severe drought events. The average maximum and daily extreme temperatures are increasing in all seasons. The projections anticipate that the current climate exposure in the camps and host communities will continue and will further expose the area to enhanced climate impacts.[3]

33. The severe gully erosion problem in former Mtendeli camp has significantly increased flood risk for downstream communities. The camps are at the top of the watershed. Hydraulic analysis shows that flooding in Kizuvu, Kasanda and Kakonko Districts occurs primarily a result of precipitation falling within the camp area itself and flowing down. Both host communities and refugee camps reported the occurrence of flooding events and subsequent riverbank and other land erosion. Current land use state under the RCP4.5 2050 precipitation future shows an average of ~4% additional inundation area. Under the no adaptation scenario, this increase will be 4.9% or 380 ha of additional flooding (Annex 2, section 2.2.2).

34. The expected adaptation benefits for host population surrounding Mtendeli include:

(a) Flood risk reduction in villages located downstream of the degraded Mtendeli camp area;

[2] The validation process determined that the Institut Pierre Simon Laplace IPSL-CM5A-MR was the best-performing overall model. This model was therefore used to assess future climate change.

[3] Further information related to climate analyses can be found in the document describing the AE's discussion with independent Technical Advisory Panel.

(b) Protection of downstream agricultural lands and soils from further flooding, erosion and siltation, thereby improving food security;

(c) Reduced siltation in surface water resources in the catchment thus improving water quality;

(d) Protection of roads and services infrastructure in the decommissioned camp area and in downstream villages; and

(e) Opportunities for expansion of climate resilient productive activities in the restored former camp area.

35. Through this project, it is expected that the landscape and ecosystems of the Kigoma Region can continue to support their residents, including host communities, refugees and agropastoralist groups that pass through the region as this territory faces climate change impacts and environmental degradation. Overall, USD 14,191,680, which account of 83 per cent of GCF funding, will be spent on targeted interventions in the host communities within a 10-15 km radius surrounding the camps.

36. Project benefits for the host communities and the agropastoralist groups include: (i) improved tenure arrangements and clear land-use demarcation; (ii) increased provision of ecosystem services and land productivity; (iii) increased water availability for irrigation during dry periods; and (iv) improved food security through improved crop production.

~~37.~~ Project benefits for the population within the refugee camps include: (i) increased flood protection of crops, settlements and assets; (ii) improved health and safety from flood protection; (iii) improved nutrition from alternative livelihoods such as mushroom and honey production; and (iv) increased income and savings through climate-resilient agriculture.

38. This project aims to enhance the protection of 42,000 hectares of natural ecosystems, and to strengthen their response to climate change and environmental degradation. This includes 30,000 hectares of forests (dominated by Miombo forests) under CBFM, 2,000 hectares under afforestation, and 10,000 hectares under agroforestry (mainly current agriculture lands). The project activities (activities 2.3 and 3.2) also expect to have indirect positive impacts in terms of increases in ecosystem resilience, given the expected benefits from the flood management and rainwater harvesting interventions. These benefits relate to improved ecosystem health for the provision of ecosystem services, including: (i) enhanced infiltration; (ii) soil nutrient retention; (iii) freshwater ecosystem function; (iv) reduced surface run-off; and (v) reduced soil erosion and siltation. Valuation of these benefits and quantification of benefit streams is expected to be conducted under output 4 during project implementation. The sustainable management and restoration of these ecosystems is deemed critical for enhancing the connectivity of the landscape, given that the region hosts high conservation value ecosystems that are currently under fragmentation stress. As clearly shown in the baseline and supporting evidence, climate change has contributed to the range shift of the Miombo forests. These forests are critical for provision of non-timber forest products and ecosystem services for the local populations.

39. The project estimates mitigation co-benefits because of the emission reductions generated through the avoided deforestation, improved agricultural practices, afforestation and agroforestry practices. This estimation has been made using the Ex-Ante Carbon-balance Tool (EX-ACT). Sequestration of 762,073 t CO_2 eq and emission reductions of 3.2 million t CO_2 eq are expected throughout the project's lifetime. These are not included in the logical framework (logframe) because the emission reductions are not expected to be reported during project implementation. Given the complex setting for project implementation (i.e. refugee camps' priorities and vulnerability conditions), the required investments in resources to track carbon sequestration and emission avoidance for reporting purposes may detract resources and focus from the core project activities; namely, delivering adaptation interventions to generate

sustainable and resilient landscapes. During project implementation, and as part of the forest and agroecological monitoring activities, it is recommended that areas be identified where carbon accountability could potentially be incorporated or reported through proxies.

3.2 Paradigm shift potential

40. The project presents a robust climate rationale for adaptation. The climate baseline information includes well-referenced scientific information and sources. It also demonstrates how the project activities would help reduce vulnerabilities to climate change impacts, and how the climate hazards will affect the project's geographical area (the Kigoma Region, and three refugee camps) in terms of value of physical assets and livelihoods, and their nuanced impacts to specific beneficiary groups, given the shared landscape. Appropriate geographical and temporal scales have been used for planning the project activities; namely, observed historical climate trends and expected future climate change combined with the local data available and a timescale that matches the project's lifespan. The project also distinguishes between the drivers directly related to climate change (e.g. changes in precipitation and temperature) and non-climatic drivers (e.g. land-use change).

41. In relation to the interventions, the paradigm shift is demonstrated through different project features inherent to the climate, development, and humanitarian contexts, especially in community benefit-sharing. Fostering local integration, the project is expected to augment the recognition of refugee contributions to sustainable and resilient landscape management in the Kigoma Region. Mindful of post-project sustainability, the project intends to incentivize shifts in the forestry–water–agriculture sectors within the Kigoma landscape. In the long term, the sustainability of the project's activities is supported by high financial profitability. It is expected that farmers will use the Savings Associations support to purchase seedlings from the existing tree nurseries once they realize the benefits of including trees in their farming systems. Furthermore, to promote livelihood activities that preserve forestry resources the project will support beekeeping groups in improving product quality, packaging and marketing to facilitate access to the existing honey value chains in the Kigoma region. Such strong financial viability contributes to scalability of the activities and increases the potential for future replication by private sector sources of finance. Although the exit strategy is backed by the economic and financial models and linked to the income-generating activities, a more compelling financial mechanism can be developed to further incentivize communities and refugees to continue the resilient livelihood activities after the project ends.

42. A key delivery mechanism for the project's interventions is the use of SAs. These have the potential to improve access to sustainable funding mechanisms and to use grant financing in a more catalytic way, while also incentivizing smallholder farmers to continue the resilient land-use practices after the project in the host communities. The SAs operate the accounts or hold the money that is not lent or granted to members through physical cash boxes, mobile money, and, rarely, in in-bank accounts. SAMOs support the SAs and will be managed by implementing partners (e.g. Good Neighbours, Care International, or the United Nations Capital Development Fund). It is expected that the project procurement officer will also support the capacity-building and monitoring of SAs. Strengthening SAs through the project will build on the experience of the United Nations Capital Development Fund in developing digital platforms for data in SAs. The lack of penetration of private financial institutions in the project area can be addressed by working with the SAs, which typically serve as cooperatives in the region. The SA expansion model will ensure that communities have improved access to farm inputs, collectively negotiate prices improve marketing of products and have access to microfinance for the continuous implementation of climate-smart agriculture. Using SAs will also facilitate tax collection. It is recommended that the potential role of SAs and SAMOs for forestry-related activities be explored, because these are already locally legitimized models that allow for sectoral replication.

43. In terms of land-use planning, the proposal has an innovative approach, as the long-term, legal outcome of the VLUP is the issuance of a CCRO. This document specifies rights conferred to a land occupier and user following tribal customs and traditions on land. The CCRO will provide long-term security for the villagers and may be used as collateral for borrowing money to conduct other future village-related, climate-resilient investments. Moreover, by adopting a landscape approach with coordinated land-use planning across the project area, the project offers innovation and potential to generate new lessons learned in the practice of VLUPs in the country.

44. The project will also facilitate the generation of evidence that will then favour scaling up and replication based on concrete results and impacts from ecosystem-based adaptation interventions in host communities and refugee camps. The valuation of ecosystem services and of business models related to non-timber forest products, under different climate conditions as a result of implementation of the project activities, is seen as a strategic tool to further catalyse private and public funding; and also to inform policies and in particular those related to displaced populations and the opportunities for their formal inclusion in the local economy. It is seen as a positive that, for project implementation, the AE will work with VPO and UNHCR as EEs. The EEs will work through local government agencies and institutions such as the district councils as implementing partners. UNHCR remains committed to strengthen approaches and interventions that cut across the climate-humanitarian-development nexus. In addition, linkages and coordination with the regional administration, local governments, the National Land Use Planning Commission, the Tanzanian Forest Service (TFS) and other relevant agencies will be involved in all project activities, including the various policy activities under output 4.

45. In summary, the project also has potential for scaling up and replication within the target area, in other regions of the country, and in other countries facing similar challenges in relation to displaced populations. GCF grants are to be invested in knowledge management and sharing, and capacity-building. It is expected that, based on such knowledge management and capacity-building, the project will create a positive enabling environment for regional (local and subnational) and national stakeholders to replicate and scale up some of the market-based solutions and models for income-generating activities. It is also important to better understand the role and potential involvement of the private sector in agricultural and forestry value chains (including non-timber forest products), expanding opportunities for smallholder farmers and refugees to add value to their products and create sustainable financing opportunities, which may gradually reduce the need for full grant support in the future. Therefore, this project is not seen as a temporary humanitarian relief project but as a long-term strategy to ensure that the landscape and ecosystem of the Kigoma Region can continue to support its residents, including refugees and host communities.

3.3 Sustainable development potential

46. The project will create several economic, environment, social and gender benefits. In terms of environmental benefits, the project will promote the conservation, restoration and sustainable management of 42,000 hectares of ecosystems, encompassing 30,000 hectares of forests under CBFM approaches, 10,000 hectares of agroforestry systems implemented, and 2,000 hectares of lands afforested. These interventions are expected to generate mitigation co-benefits through the sequestration of 762,073 t CO_2 eq and emission reductions of 3.2 million t CO_2 eq over 20 years. Moreover, the project will deliver significant benefits related to the enhancement and maintenance of key hydrological services.

47. The project aims to reach 35 per cent of the agricultural/crop land in the project area at a density of 500/ha for a total of 10,000ha. Spatial land use analysis indicates that there are around 28,000 ha of agricultural/crop land. The selection of species to be planed will be made in discussion with farmers, in consideration of farmers’ preference of certain non-native species

Leucaena (*Leucaena diversifolia, Leucaena trichandra*) and Calliandra (*Calliandra calothyrsus*) as fodder trees. A species suitability assessment will be carried out further to inform the final species selection.

48. The preservation of ecosystem services has wide benefits over the multiple dimensions of sustainable development. The project will deliver positive impacts on food security and current levels of malnutrition. Greater access to inputs and enhanced water and soil management, as supported by the project, will increase agricultural production and reduce farmers' vulnerabilities to climate shocks and stresses. By improving agricultural production, increasing the availability of food locally, and by improving households' income from agriculture, access to food will be improved by enabling these beneficiaries to purchase more diverse food items. Moreover, through the development of mushroom and beekeeping activities, both host communities and refugees will increase their local diet diversity. Project activities such as strengthening farmer field schools, establishing market linkages, providing agricultural inputs and on-training, and strengthening SAMOs and agroprocessing activities through secondary transformation, as well as access to microfinance, will contribute to increased food security.

49. Social benefits for host communities include improved tenure arrangements and clear land-use demarcation. For refugees, they include improved health and safety through flood protection, as well as reduced risk of water- and vector-borne diseases during flood events. Overall social benefits relate to improved social cohesion among host communities, refugees and agropastoralists.

3.4 Needs of the recipient

50. The project targets an LDC with high vulnerability to climate change, as evidenced in the funding proposal and feasibility study. The project will be implemented in the Kigoma Region, which is one of the poorest regions in the United Republic of Tanzania, with more than 32 per cent of its population living below the poverty line and with more than 34 per cent of the region's population underemployed. Moreover, 90 per cent of the region's population is engaged in agriculture, and most smallholder farmers are dependent on rain-fed agriculture, which is highly vulnerable to climate change impacts such as damage to crops, soil nutrient loss, water availability for irrigation, occurrence of pests and diseases, and crops being pushed beyond their thermal limits.

51. The protracted refugee situation began more than 20 years ago, and is unlikely to change. Identified recipient needs include the need to address refugees' unique vulnerabilities, as they are almost universally excluded from national or local adaptation plans, education and development interventions. Therefore, the project intends to develop a model for meeting the climate change adaptation needs of refugees, host communities and agropastoralists sharing the same landscape. It is worth noting that repatriation is encouraged but voluntary and remains regulated by the Tripartite Commission including the Government of Tanzania, Government of Burundi and UNHCR. It is understood that the closure of Mtendeli camp was motivated by resource efficiency for service provision in the camps considering the current reduction in refugee population in both Mtendeli and Nduta camps.

52. There are several factors limiting the capacity of the United Republic of Tanzania to adapt to climate change. These include the need for improved economic opportunities, the need to address environmental degradation in rural areas such as the Kigoma Region, and the need for financial services and institutional strengthening. The project will improve access to financial services by strengthening SAs to deliver climate-resilient interventions and by scaling up financial inclusion. The project will address the overarching institutional capacity deficit in climate-resilient and land-use planning through capacity-building and transfer of best practices

in order to support institutionalization and capitalization of lessons learned and results from local to national levels.

3.5 Country ownership

53. The project is firmly anchored in government structures and ownership. The Government of the United Republic of Tanzania has actively participated in the project design, ensuring alignment with its relevant policies and plans. Exchanges and discussions between the Secretariat and the national designated authority confirm prioritization by the Government. The project will be aligned with the following policies and plans: the intended nationally determined contribution of the United Republic of Tanzania; the National Development Plan (2016–2021); the National Adaptation Programme of Action (2007); the National Climate Change Strategy (2012); and the Guidelines for Integrating Climate Change Adaptation into National Sectoral Policies, Plans and Programmes (2012). In particular, the project will contribute with the adaptation contributions of the intended nationally determined contribution – aiming to reduce climate change impacts related to the spatial and temporal variability of declining rainfall, and the frequent droughts and floods, which have long-term implications for all productive sectors and ecosystems, particularly the agriculture sector.

54. Although the project does not include separate co-financing from the Government, the national designated authority (VPO) has expressed its continued support for and prioritization of the project. VPO is the lead EE for this project, working with UNHCR. As mentioned in the previous section, both VPO and UNHCR will work through local government agencies and institutions such as the district offices to implement this project. The Ministry of Home Affairs (MoHA) will play a key role in overseeing the implementation of activities in the refugee camps. The Government of the United Republic of Tanzania will be represented at ministerial and district levels in the project steering committee, providing oversight and guidance. Furthermore, The National Environmental Management Council (NEMC) will ensure compliance with relevant environmental laws and regulations as well as providing technical advisory and compliance guidance on all environmental aspects of the project.

55. Recently, the United Republic of Tanzania has developed its own Climate-Smart Agriculture Guidelines, which identifies several areas where capacity-building interventions are required. These areas are well aligned with the interventions proposed by the project. In turn, the Guideline contributes to the country's commitment to create a resilient, climate-smart agriculture sector by 2030.

56. The policies of the United Republic of Tanzania towards refugees have been more restrictive in recent years. However, globally, it is important to note that the country has, for decades, accepted a comparatively high number of refugees compared with the rest of the world, particularly given its status as an LDC. While refugees do not currently form part of the formal economy, the Kigoma Region will probably continue to host refugees in the long term. Given its international legal obligations, the Government is committed to protecting refugees and asylum seekers.

57. Risk mitigation measures are in place, particularly to ensure the Government's capacity to continuously adhere to best practices on international refugee. The country's Refugee Response Plan (2019–2020) acknowledges the need for wider efforts within the environment and energy sectors, and for strengthened links with climate change adaptation and mitigation interventions. For this to materialize, it is crucial to collect evidence on project benefits in terms of climate change adaptation and sustainable landscapes in the region.

58. The Government of Tanzania foresees ensuring sustainability thanks to the enhanced capacity of local stakeholders built by the project as well as provisions to the District budgets

towards financial sustainability of interventions requiring government funding, both during and beyond the project duration.

3.6 Efficiency and effectiveness

59. The proposed level of concessionality (based on a GCF grant plus a minimum level of in-kind co-financing from UNHCR) is based on three underlying characteristics of the proposed investment: (i) high poverty level of the recipient country; (ii) the targeted group, which is extremely resources-constrained; and (iii) the "public good" nature of the target group. For this project, it is not appropriate to make use of reimbursable grants as a financial instrument. A limited amount of the GCF funding requested is to be spent on equipment with the potential to directly generate revenue. Even then, the benefits will be primarily for subsistence or to supplement the incomes of very poor, vulnerable populations, particularly in districts that host and share resources with refugees. Any increases in income and revenue will be re-channelled back to the operation of SAs in the host communities as well as to support maintenance costs of rainwater harvesting and irrigation infrastructure. In the face of climate shocks and their impacts, such as low crop yields and damage to property, savings offer an important safety net for very poor households. The level of concessionality is properly justified in the economic and financial analysis.

60. The project is primarily considered as one to create public goods. It is to be co-financed by UNHCR through an in-kind contribution of USD 4.61 million. Non-cash additional funding will be generated. It is to be noted that the GCF Co-financing Policy does not set any requirement from the Government of the host country to provide co-financing to the project. The project will generate substantial positive externalities, including carbon dioxide sequestration and reduction in carbon dioxide emissions. However, these are not included in the economic analysis, given the scope and approach of the project. The returns generated by the project are expected to be high enough to justify the investment from both a financial and an economic point of view. The proposal demonstrates the economic viability of the project, with an increased net present value (NPV) of USD 6,586,348 million and an economic internal rate of return (IRR) of 13 per cent when the value of greenhouse gas mitigation benefits is excluded and 20 per cent when all the benefits are included (using the higher discount rate of 6.2 per cent in both cases). The results of the economic analysis are robust to sensitivity analysis on costs and benefits. The economic analysis shows that the proposed agroforestry activities will be economically viable over the 20-year project lifetime, with a high IRR of 50 per cent. The proposal also demonstrates cost-effectiveness using a financial model. The project is expected to generate an NPV of USD 17,890,067, with an IRR ranging from 16 to 787 per cent. The high returns across all the activities reassure the Secretariat of the financial viability in the long-run of the activities as well the scalability potential. A sensitivity analysis has shown that the results of the financial analysis are robust.

61. While the project will be approximately 20 per cent co-financed, parallel financing is expected to be about double the project size. Letters of support from the Royal Danish Embassy in Dar es Salaam have been received and presented as annexes to the funding proposal. This support demonstrates the overall coherence and complementarity of the project with other investments in the Kigoma Region.

62. In terms of cost-effectiveness, benchmarks are provided for water pumped using the proposed technology (treadle pumps), which are less expensive than solar-powered pumps (the alternative considered). Consequently, the project will install three treadle pumps for every solar-powered pump. A comparison benchmark is not reported for other activities. This is understandable given the inherent difficulties in providing benchmarks for the proposed activities. The project exhibits a cost per beneficiary of USD 33, which is at the low end of projects in the GCF portfolio.

63. In-kind and cash contributions from the Government of Tanzania have not been quantified in the proposal as co-finance commitment however, government are expected to be provided to enable project implementation and future sustainability of interventions.

IV. Assessment of consistency with GCF safeguards and policies

4.1 Environmental and social safeguards

64. **Environmental and social risk category**. The project primarily aims to improve the resilience of refugee camps in the Kigoma Region and their host communities against the threats of climate change through an integrated landscape ecosystem approach. The project will have two components. Component 1 will consist of three outputs/objectives: (output 1) scaling up and developing new C-LUPs for villages to support ecosystem-based adaptation; (output 2) restoring ecosystems to support climate change adaptation by implementing CBFM, agroforestry and afforestation in village lands, and flood and erosion control system in densely populated areas; and (output 3) provision of and diversification of livelihoods for refugees and host communities through agricultural extension, provision of production inputs, water management systems, infrastructure support such as water harvesting and micro-irrigation systems. Component 2 will focus on strengthening policies for climate-responsive planning and development. In addition to mitigation benefits through CO_2 sequestration, other potential environmental and social co-benefits include enhancing landscape biodiversity and the ecosystem services that it provides, and the possibility of addressing potential conflict relating to competing use of resources between the host communities and the refugees, such as those between farmers and pastoralists.

65. The potential environmental and social (E&S) risks and impacts of the project will come mainly from activities under Component 1. The activities involve only potential limited adverse environmental or social risks and/or impacts that are few in number, generally site-specific, largely reversible, and readily addressed through mitigation measures. The AE has assigned the project Category B or moderate risk based on its Environmental, Social, and Economic Review Note tool. The Secretariat confirms the E&S risk category (Category B) of the project and that the level of risk of the activities is within the level to which the AE is accredited.

66. **Safeguards instruments.** The AE has submitted an environmental and social management system (ESMS) document that contains an environmental management framework and a social assessment and management plan which identifies the risks and potential impacts of the project activities, and the corresponding mitigation/management measures to be adopted. The ESMS document also contains a detailed description of the stakeholder engagement plan and the grievance redress mechanism for the project.

67. **Compliance with GCF's environmental and social safeguards standards.** The paragraphs below identify the key risks/impacts regarding the GCF ESS standards and how they are addressed in the proposal.

68. **ESS 1: Assessment and management of environmental and social risks and impacts.** The AE's E&S risk screening tool, augmented by items relevant to the GCF ESS standards, was used for undertaking the E&S risk screening of the project proposal, which underwent adequate assessment and mitigation planning. The ESMS document includes substantive descriptions of the environmental and social profile of the project area, assessment of the E&S risks and impacts, and the corresponding mitigation/management plans. For the site-specific project activities, the ESMS outlines the process for screening and assessing the project's individual site-specific ground activities. During project implementation, each site-specific project activity under outputs 2 and 3 would be subject to: (i) E&S risk screening using the AE's Environmental, Social and Economic Screening process augmented with requirements

for GCF projects; (ii) review and continuous validation (every six months) of the screening results by the Safeguards and Gender Officer; (iii) securing the necessary permits and certifications from the Government of Tanzania; (iv) determining the need for and preparation of, specific activity-level assessments and management planning; and (v) implementing mitigation measures and reporting.

69. **ESS 2: Labour and working conditions.** The GCF standard on labour and working conditions applies to activities under output 2 such as implementing CBFM schemes, agroforestry, afforestation and construction of small-scale water management structures; and under output 3, which may involve construction of small structures for beekeeping and mushroom production. The risks including prevalent labour issues in the project area, and occupational health and safety concerns, have been adequately assessed and corresponding mitigation measures have been identified. "Labor Management Guidelines" have been incorporated into the ESMS which include the assessment of contractors' processes and capacities for managing health and safety, and for protecting labour and other rights.

70. **ESS 3: Resource efficiency and pollution prevention.** The project would involve use of water resources for irrigation of farmland and nurseries. There is thus a risk of reduced water resources in the project area due to water pumping, surface run-off and rainwater harvesting. However, this risk is assessed to be low because the small water infrastructure to be installed would generally be aimed at efficiently managing water resources by increasing water availability, reducing flooding and increasing infiltration through restoration of degraded lands, improving soil stability and providing rainwater harvesting infrastructure to reduce the pressure on surface and groundwater resources. In terms of pollution, the small-scale construction activities could result in temporary generation of dusts, noise and sediments that may pollute the waterways during construction, while nursery operations and agroforestry may require the use of pesticides and/or agrochemicals. These risks have been adequately assessed and mitigation measures are identified in the ESMS. Although the project will not promote the use of pesticides/herbicides and none will be provided by the project to community members, a "Pesticide Management Guidelines" has also been prepared and incorporated into the ESMS.

71. **ESS 4: Community health, safety and security.** The project's risks and potential impacts on community health, safety and security include: (i) potential exposure of project communities to water-based and/or vector-borne from water capture/impounding structures and the forests; (ii) exposure of communities to communicable diseases from workers carrying out the small-scale construction works; (iii) potential increase of incidence of gender-based violence (GBV) and sexual exploitation, abuse and harassment; (iv) possible conflicts between host villagers and refugees in relation to project delivery, resulting in security concerns; (v) exposure of residents, especially children, to safety hazards relating to project water management structures and accident hazards during construction; and (vi) possible exposure of communities to pesticides. All of these risks/impacts have been adequately addressed in the ESMS. A pre-approval conflict sensitivity analysis was conducted that will be further detailed at the start of project implementation during the land-use planning process and to assess and mitigate conflicts related to tensions between the indirect beneficiaries outside the project target area and direct beneficiaries within the target area. The analysis will also be evaluated at project mid-term to assess the effectiveness of the conflict mitigation measures.

72. **ESS 5: Land acquisition and involuntary resettlement.** Although the project will not pursue activities that would intentionally cause physical displacement (i.e. loss of homes) or economic displacement (i.e. loss of a substantial portion of livelihood), the following risks may remain:

(a) *Possible loss of access to land due to formalization of land tenure*. The project will support formalization of traditional tenure (i.e. issuance of Certificates of Customary Rights of Occupancy). The formalization of ownership rights is expected to improve tenure security for current occupants/users (including refugees) who are engaged in informal land-use arrangements with recognized landowners. However, the risk of displacement remains for current informal occupants/claimants who might be displaced in favour of the deemed rightful claimant. This risk is related to the risk of setting off new or stirring up old land boundary conflicts. The village-based process of formalization within the highly participatory C-LUP process, plus the village-based land adjudication system, will greatly manage and reduce this risk;

(b) *Possible economic/income loss due to land use restrictions under C-LUP*. The purpose of all land-use plans is inevitably to restrict or regulate land use for a particular site or zone. Hence, some existing land uses, such as pasture areas and sloping land plots, may be restricted or changed. Baseline information and the stakeholder consultation also indicate that there are current land uses along the riverbanks that may be changed or restricted. Participatory planning that involves all stakeholders (i.e. farmers, pastoralists and other land users) reduces this risk. However, "voluntary restriction" is generally difficult to ascertain even in community-based planning because individual land users could be pressured by their peers or group leaders. The planning process has to adopt some objective criteria for ensuring that restriction is voluntary. The project has to ensure that any losses should be compensated or otherwise waived by the individual. The ESMS provides guidelines and options as to how individuals or groups who would lose access can be compensated, should this become necessary; and

(c) *Possible loss of land and assets attached to land due to small infrastructure*. The project will involve construction of small public/communal infrastructure and facilities (e.g. water harvesting, water reservoirs, irrigation canals, drainage structures, flood control structures, pumps and nurseries). While these activities may not cause involuntary resettlement impacts (i.e. loss of homes and/or loss of livelihoods), they would likely require acquisition of rights-of-way, easements and piece of lands for the facilities, resulting in losses of land and assets. However, a compensation framework is included in the ESMS to address this risk.

73. **ESS 6: Biodiversity conservation and sustainable management of living natural resources.** The project is aimed at restoring ecosystems to improve resilience of the landscape to both climate change-driven hazards and non-climate drivers and hence is expected to provide a positive impact on biodiversity and habitats. However, the following potential risks have been identified: (i) risk of invasive species or adverse impacts of introduction of non-native species in afforestation activities and fuelwood plantations; (ii) risk of indirect impacts of project interventions on the Moyowosi Game Reserve and Malagarasi-Moyowozi Wetlands, which is a Ramsar site, and the Kigosi Game Reserve (IUCN CAT IV); (iii) possible fragmentation of fish habitat due to in-stream structures; and (iv) risk of impacts on ecosystems from possible intensification of agricultural production. Nevertheless, the ESMS has adequately addressed these risks as follows:

(a) *Risk associated with introduction of non-native species.* A species suitability assessment will be conducted within the first year of the project. Fast-growing and highly productive species will be prioritized – which will help to meet the fuelwood demand and, by doing so, reduce the pressure on the native Miombo forest. Options for species, from an SNV Netherlands Development Organization project, include *Calliandra calothyrsus*, *Sesbania sesban*, *Leucaena leucocephala*, *Maesopsis eminii* and *Senna siamea*. Other species to be considered include *Acacia auriculiformis*, *A. crassicarpa*, *A. leptocarpa, A. mangium, Albizia lebbeck, Glyricidia sepium* and *Leucaena pallida*;

(b) *Risk of indirect impacts on the protected areas*. Restoration and conservation in forest reserves within a 10 km radius to the Moyowosi Game Reserve would be undertaken. There would be no afforestation with fast-growing species for fuelwood in this area. The project team will monitor any potential impacts of the rainwater harvesting structures on streamflow into the wetlands by monitoring extraction rates and streamflow over time relative to a baseline that will be determined at the start of the project;

(c) *Risk of impact on passage of fish from in-stream structures.* The in-stream structures would be designed to incorporate fish ramps to allow for passage of fish. This will, however, need to be monitored closely to ensure the effectiveness of the measures and provide for adaptive management should that be necessary to achieve the expected outcomes; and

(d) *Risk of impact on ecosystems from more intensive agriculture.* The implementation of activities under activity 3.1 (climate-resilient agriculture) will be consistent with the Climate Smart Agriculture Guidelines of Tanzania that promote sustainable practices and will be done in coordination with the Agricultural District Officer. Sustainable practices will be promoted in the Farmer Field School approach.

74. **GCF Indigenous Peoples Policy and ESS 7: Indigenous Peoples.** The ESMS provides an analysis of the different groups that might be considered under the scope of the GCF Indigenous Peoples Policy. It concludes that it is unlikely that these groups would be based in the Kigoma Region where project activities are to be undertaken, thus a separate Indigenous Peoples plan is not essential and the ESMS provides a sufficient framework to consider relevant social issues. The ESMS further highlights the presence of agropastoralists, mainly from a majority Sukuma community that may pass through Kigoma Region. The project will engage with agropastoralists and other land-users in the land-use planning process, and in activities relating to environmental education and strengthening environmental management committees and overall in the project, as will be reflected in the stakeholder engagement plan. The land-use planning process provides opportunity to allocate grazing areas and, by engaging in this process, the stakeholders, including agropastoralists, will have an opportunity to participate in mechanisms to adjudicate boundaries for land uses, identify management regimes, achieve consensus and negotiate across different interests. Crucial to the opportunity will be ensuring that the different community stakeholders are identified, engaged and consulted throughout the process, including in conflict sensitivity analyses. In line with its roles and functions, the GCF Indigenous Peoples Advisory Group is available to provide advice to the AE.

75. **ESS 8: Cultural heritage.** The ESMS provides for strict avoidance of activities that affect cultural heritage sites or structures. Moreover, only small physical structures such as flood control infrastructure and small-scale irrigation infrastructure will be constructed by the project. Nevertheless, since there are small-scale excavation activities that are expected, a section on Cultural Heritage and Chance Finds Procedures has been incorporated in the ESMS to guide the project should it affect cultural heritage sites.

76. **Sexual exploitation, abuse and harassment.** The revised GCF Environmental and Social Policy adopted by decision B.BM-2021/18 requires safeguarding from sexual exploitation, abuse and harassment (SEAH) in GCF-financed activities. The AE provided SEAH safeguarding in its submission for this funding proposal. The ESMS of the project identifies a number of SEAH issues including sexual and GBV that may occur in the project and put in measures to prevent and address/mitigate the same. Among others, the project will prevent SEAH and/or GBV in alignment with the United Nations policies and practices for the prevention of sexual and GBV and will embark on various awareness and capacity-building initiatives on the subject matter. These policies are complementary to the GCF SEAH provisions. The project will further develop project-level guidelines and contractual requirements related to SEAH and its investigation and disciplinary procedures. The AE recognizes that incidences of sexual GBV in the refugee operations in Kigoma is high, especially rape cases. Likewise, the inter-agency standard operational procedures and protocols will be applied in managing the potential risk of SEAH and to guide SEAH complaints handling and grievance mechanisms. In addition to this, the AE will work closely with other stakeholders including the Ministry of Community Development, Gender and Women as per the National Plan of Action to end

Violence against Women and Children in Tanzania; protection committees at village, district and national levels; and One Stop Centers for case referrals among other service providers and collaborators. Overall, the project is in alignment with GCF SEAH provisions.

77. **Implementation arrangements**. In addition to the project steering committee, which provides the high-level project direction, a project management unit will serve as the secretariat at the operational level and will be staffed by technical experts including on safeguards. A Safeguards and Gender Specialist (including a part-time international expert consultant) will be engaged and will be responsible for identifying emerging risks, developing adaptive management and mitigation measures, including monitoring and reporting on the social and environment safeguards plan. The management plans will also be built into the contracts and agreements with contractors and implementing partners of the project.

78. **Stakeholder engagement and grievance redress mechanism.** The AE has submitted a stakeholder engagement plan (annex 7 of the funding proposal "Consultations Summary and Stakeholder Engagement Plan"). The document describes in detail the results of the series of consultations conducted on the project and provides a stakeholder mapping/analysis as well as a detailed plan for continued engagement with stakeholders during the project implementation. For individual project activities during implementation, the ESMS provides for the timely disclosure of all project-level and activity-level safeguard instruments and other relevant information locally by the project's safeguards and gender specialist. In line with the GCF Indigenous Peoples Policy, the GCF Indigenous Peoples focal point will be available for assistance at any stage, including before a claim has been made.

79. The ESMS provides for the establishment of a grievance redress mechanism that will build on an existing UNHCR feedback and complaints mechanism. The AE also has its Stakeholder Response Mechanism, which allows stakeholders to provide feedback or report concerns, complaints or grievance issues about its projects.[4] The ESMS also refers to the GCF Independent Redress Mechanism where stakeholders can file requests and complaints through a dedicated website. In line with the GCF Indigenous Peoples Policy, the GCF Indigenous Peoples focal point will be available for assistance at any stage, including before a claim has been made.

4.2 Gender policy

80. The AE has provided a gender assessment and gender action plan and, therefore, complies with the requirements of the Gender Policy of the GCF.

81. The gender assessment demonstrates the existence of an enabling environment conducive to promoting gender equality and women's empowerment in the country. The Ministry of Community Development, Gender and Children was established, and a number of policies and legislative acts have been passed, such as the Land Act and the Village Land Act, as well as two key poverty reduction strategy papers, MKUKUTA I and II. Such development plans and strategies recognize the importance of gender equality and women's empowerment. In addition, the National Plan of Action to end Violence against Women and Children in Tanzania aims to include all people, including refugees, in its effort to curb and address violence against women in the country and the refugee camps.

82. The gender assessment was based on a desk review and stakeholder consultation among both the refugees and the Tanzanian communities surrounding the refugee camps. The assessment covered the differentiated issues for women in the refugee camps and host communities. It highlighted the difference in the challenges for women and men within the refugee camps as well as the differences in challenges in the host community between women

[4] Feedback and stakeholder responses can be sent through the following website https://www.unep.org/about-un-environment/why-does-un-environment-matter/un-environment-project-concern.

and men. While in both cases women are disproportionately affected, refugee women seem to have many more challenges in coping with the climate change-induced challenges to their nearby environment. Similarly, despite a supportive policy environment, Tanzanian women continue to struggle in accessing land and land rights. Cultural and religious institutions tend to favour men and perpetuate views that men are the sole owners of the land, and have and control power over the income generated. This could be a source of conflict within the home if women attempt to control land. Such views and practices leave women with little or no incentive to invest and improve the land, while on the other hand women do not possess the resources and inputs required to ensure resilience and resilient agricultural practices. Further, in the refugee camps, economic activities, such as selling food aid rations, non-food items, tailoring/embroidering, small-trade barbering and hairstyling, cooking, soap-making, and repairs, are mostly done by women while men are engaged mostly in carpentry or masonry in addition to labour-intensive agricultural activities.

83. There are GBV risks within and outside of the refugee camps. Refugees in the camps are exposed to violence, as well as when they leave the camps to search for work, to trade, or to gather firewood; and their children inside the camps are at risk of GBV, sexual exploitation and abuse, and early or forced marriage. Refugees are restricted and endure congested living conditions and face restrictions on cash-based interventions and livelihood opportunities. This forces some to engage in risky coping mechanisms or behaviours (transactional sex, illegally travelling outside of camp boundaries, or engaging in prohibited livelihoods) to support their households, maintain food and energy security, and obtain necessary resources. For both the host and refugee communities, firewood collection is the responsibility of women and, for refugees, with travel limited to within a 4-km buffer around the camps, which falls short of fulfilling their needs as a result of depletion, women are forced to travel further beyond this boundary.

84. The assessment indicates that gender issues in forestry, agriculture and land use are critical challenges and need to be addressed. Regarding land, issues of insecure land tenure and exclusion from access and control over land exist, compounded with gender-blind policies, and discriminatory social and cultural norms. Further, challenges to women exist in relation to access to land use and agricultural practices while access to water and forestry is limited. Women also have fewer skills and less access to information and knowledge, which limits their ability to diversify their livelihoods in a manner that is climate resilient. In addition to productive work, women bear the burden of unpaid care and domestic work, which puts greater demands on time available for collecting fuelwood and water. Women tend to have lower productivity and incomes, rendering them economically weaker and dependent. This in turn forces some women to engage in behaviours and practices that they do not want to engage in. The assessment is thorough and clearly outlines challenges and opportunities; while the project puts in place mechanisms to address causes of gender inequalities within the refugee and host communities as they become resilient to climate change-related challenges.

85. The AE has provided a gender action plan and therefore complies with the requirement of the GCF Gender Policy. The gender action plan includes activities addressing the challenges to women in the refugee camps and in host communities. The plan includes baseline, indicators, targets (for host and refugees), timelines and budgets. Activities will ensure women engage in decision-making processes and project activities. Activities aim to enhance capacities and skills while creating opportunities for access to assets transferred to refugee and host community women. In addition, activities ensure women can benefit from accessing climate-smart agriculture training, and inputs and irrigation investments; improved environmental management; and conservation of water sources. The land-use planning and forestry-related interventions will support women to better manage natural resources, and reduce their work burden and the time needed for collecting fuelwood; while providing crèche facilities and moving meetings to more convenient times for women to participate have been factored in to the project. Moreover, for refugee women, exposure to the threat of GBV and other forms of

violence is expected to be reduced when conflicts over fuelwood and deforestation have been mitigated, and when increased engagement in alternative livelihoods activities contributes to increased economic empowerment. Further collaboration with existing mechanisms and institutions addressing GBV will be maintained, while awareness and sensitization on GBV will be mainstreamed in various training programmes. The project will have a safeguards and gender officer who will support the implementation of the gender action plan and will engage in the grievance redress mechanism and its implementation, including on GBV. The AE is expected to ensure the officer is always available to support the implementation of the gender action plan and the mainstreaming of gender throughout project implementation.

4.3 Risks

4.3.1. Overall proposal assessment (high risk)

86. This project aims to address the need for climate change adaptation for both host communities and refugees in the Kigoma Region of the United Republic of Tanzania. The project targets four outputs: (i) participatory, climate-resilient land-use planning in refugee camps and surrounding villages; (ii) restoration of degraded ecosystems; (iii) climate-resilient agriculture and livelihood diversification; and (iv) information on climate change adaptation disseminated and mainstreamed into policies, plans and strategies in the country and in humanitarian programmes. The total project cost is USD 23.62 million, of which it is proposed that GCF provide USD 19.01 million in grants. Co-financing will be provided by UNHCR (the EE) in the form of an in-kind contribution of USD 4.61 million. The project is exposed to political risks related to government policies towards refugees. The AE has identified a high probability of such risks: these risks can weaken the project's climate impact and will therefore need active adaptive management from an early stage.

4.3.2. Accredited entity/executing entity capability to execute the current project (high risk)

87. The AE (UNEP) has an extensive track record of implementing climate change projects through, for example, the Least Developed Countries Fund, the Special Climate Change Fund, the Adaptation Fund, and bilateral funds such as those of the European Union. UNEP has a regional presence and operations. At the AE level, the task manager, with the support of the portfolio manager, will take the lead role in monitoring risks in cooperation with the EEs. Accordingly, the Secretariat requires the AE to confirm that any political risks arising out of the project will be borne by the AE.

88. VPO will act as the lead EE of the project. VPO has experience with projects of a similar size and scope with bilateral and multilateral donors, including the Asian Development Bank and the International Fund for Agricultural Development, and the AE. VPO will execute project activities through various mechanisms, including direct execution by the project management unit (PMU), and subsidiary agreements with district governments and TFS, as well as contracts with service providers and consultants. UNHCR, one of the EEs, will execute the project by overseeing and managing the implementing partners involved in the implementation of project interventions. UNHCR has three field offices in the project area (in Kigoma city, Kasulu and Kibondo) that work with the MoHA and local government authorities. In addition to its humanitarian activities, UNHCR has a track record of supporting local communities with climate change projects, including in forestry, water conservation and sustainable energy. As reported by the AE, UNHCR Tanzania has provision for a senior-level, in situ risk adviser, whose work covers all facets of risks that the operations may face, including political risks.

4.3.3. Project-specific execution risks (high risk)

89. **Political risk:** The AE and UNHCR have identified a number of risks related to government policy. These risks relate to implementation by the United Republic of Tanzania of its asylum policy in line with international standards; access for asylum seekers fleeing conflict from neighbouring countries; risks related to voluntary repatriation; refugee data; and an overall restrictive protection environment. Through its own risk analysis UNHCR notes that government policies towards refugees will continue to be restrictive. Materialization of political risk or civil unrest could create a reputational risk for the AE, UNHCR and GCF. Options for halting disbursements in the event of adverse circumstances could be limited because of the humanitarian consequences of stopping the flow of funds. Some comfort can be derived from the assessment by the AE that the project will, upon completion, contribute to the recognition that refugees can contribute to host communities, and from the UNHCR's active engagement with local authorities. However, these points would not substantially reduce the political risk. The project implementation period is five years; any adverse changes in government policies pertaining to refugees before the project implementation has been completed could erode the project's climate impact. The lead EE of the project is a government entity, which could reduce political risks but could also add complexity in case of political risks materializing. The Secretariat has undertaken an interdivisional analysis and acknowledges that the project needs an active adaptive management approach from the outset.

90. **O&M:** post-implementation O&M costs are not included in project costs and are to be borne by the asset owners. Constructed assets such as small-scale flood mitigation infrastructure in the camps will be turned over to MoHA. Water-harvesting infrastructure in the host communities will be turned over to district and regional authorities, with regular maintenance activities to be performed by water users' associations, which will rely on SAs or devise schemes for small user fees to support O&M. Payment structures and viability of these schemes will be determined during project implementation.

91. **Land use:** Formalization of land tenure could lead to the disturbance of informal arrangements between refugees and host communities. However, the EE does not foresee this because host communities will have better knowledge of where they can enter into agreements with refugees on diverse land-use practices in respect of village by-laws. In addition, there may be risks of displacing economic activities and livelihoods that are incompatible with climate adaptation and environmental objectives, such as farming along floodplains and within national and village forest reserves. Suitable alternative areas will be identified, such as communal village land, and affected populations will have the option to be engaged in project activities under output 3 and employment on the small-scale water infrastructure that is to be built. The benefits received by host communities in the project could create a source of tension with other communities outside the project area. Close monitoring is advised on progress on land-use agreements.

4.3.4. Compliance (medium risk)

92. The United Republic of Tanzania is not the subject of any current United Nations Security Council (UNSC) resolutions. UNEP, as the AE, has confirmed that no individual or entity that is listed on any UNSC sanctions list, including the United Nations Consolidated Sanctions list, will be involved in any manner with the project or its activities, either as a counterparty or implementer, or as a beneficiary.

93. UNEP has stated that the project will be operated in accordance with United Nations regulations, rules and policies, including the Anti-Fraud and Anti-Corruption Framework of the United Nations Secretariat and UNEP's Misconduct and Anti-Fraud Policies. The financial management and procurement for the project will be guided by United Nations financial regulations, rules and practices, as well as UNEP's Operations Manual. UNEP will also closely monitor the financial management of the project using the established monitoring and evaluation

procedures and financial reporting mechanism, including an annual audit. UNEP will also establish internal controls and project fund management.

94. According to UNEP, the risk of GCF proceeds being utilized for money laundering or terrorist financing has been assessed as low, and will be mitigated through appropriate legal instruments which will include warranties and caveats by the EEs (i.e. VPO and UNHCR).

95. Project cooperation agreements (PCAs) between UNEP and VPO, and between UNEP and UNHCR, will include warranties and caveats by the EEs on required anti-fraud and anti-corruption safeguards, guided by the Anti-Fraud and Anti-Corruption Framework of the United Nations Secretariat, as well as the GCF Policy on Prohibited Practices. In turn, VPO and UNHCR will require its implementing entities to ensure compliance with the same policies.

96. UNEP has also explained that, in line with the standards of its internal practices, it will provide training on prohibited practices, anti-money laundering and counter-financing of terrorism to the EEs as part of the project inception.

97. VPO will act as the lead EE for this project, ensuring alignment with national policies and plans as well as complementarity with other existing adaptation projects and Tanzania's national adaptation plan (NAP) process. As the lead EE, VPO will oversee overall project implementation and will be accountable to UNEP for the effective use of resources during project implementation.

98. The PCA signed between VPO and UNEP will serve as a legal instrument (the Subsidiary Agreement) to regulate the obligations of the parties. VPO will be responsible for the efficient disbursement of financial resources for the implementation of the designated project activities and responsible for the coordination with relevant partners, and compilation of the required information from all implementing partners, including UNHCR as secondary EE, to provide consolidated project progress reports and expenditure reports to UNEP.

99. Execution by UNHCR is necessary to leverage its mandate on refugee and host community relations and its convening power among the different actors on the ground. As secondary EE, UNHCR will execute GCF-funded project activities in collaboration with MoHA and VPO. UNHCR will be responsible for overseeing and managing the partners involved in implementing the designated project interventions, prioritizing implementation by local government authorities where possible.

100. UNHCR has in place a Strategic Framework for the Prevention of Fraud and Corruption and a policy on anti-money laundering, which also encompasses counter-terrorism financing measures. Although fraud and corruption remain as risks, mitigation measures are in place and constantly strengthened. UNHCR Tanzania is one of the UNHCR operations with senior-level risk management capacity in situ as a means to ensure the operation strengthens anti-fraud and corruption measures.

101. UNEP has provided its capacity assessment of VPO in its position as lead EE on the project. Although Tanzania has adopted dedicated legislation against money laundering and terrorist financing (ML/TF), UNEP's assessment of "moderate risk" for anti-money laundering and counter-financing of terrorism (AML/CFT) due diligence is of some concern.

102. VPO appears to have sufficient capacity at the organizational level and in the capital city, Dodoma, to support project implementation. However, most activities will be conducted in Kigoma, 690 km away from Dodoma, which is why the funding proposal states that the PMU will be based in Kigoma and supported by VPO staff in Dodoma. Furthermore, experience with previous UNEP projects with VPO indicates challenges related to recruitment, staff replacement and backstopping.

103. According to the Financial Action Task Force (FATF) Mutual Evaluation Report (MER) on Tanzania[5] published in June 2021, and based on the National Risk Assessment (NRA) Action Plan[6] conducted by Tanzania in 2015–2016, national "AML/CFT policies to address identified ML/TF risks and its activities do not adequately address the identified ML/TF risks." The NRA Action Plan is divided into three categories: high priority; medium priority; and quick wins. Within the high priority category are the following planned activities:

(a) Create AML awareness;

(b) Amend the AML Act;

(c) Amend the Anti-Money Laundering and Proceeds of Crime Act (for Zanzibar) (AMLPOCA);

(d) Amend the AML and AMLPOCA Regulations;

(e) Review the AML/CFT Guidelines;

(f) National Identification Authority to fast-track issuance of national identification documents;

(g) Employ adequate AML/CFT supervision staff;

(h) Collect the required AML/CFT data and statistics for mutual evaluation; and

(i) Formulate policy and conduct a national ML/TF risk assessment.

(j) URT has conducted AML/CFT awareness for reporting entities;

(k) LEAs, judiciary, and Members of Parliament.

104. Overall, the FATF MER has identified significant gaps in the legal framework, and the NRA Action Plan has not put much focus on areas of high ML/TF vulnerabilities and risks identified. Notwithstanding this, competent authorities have continued to take action to address some of the identified risks. In an effort to address the abovementioned capacity gaps, UNEP has confirmed that it will ensure that the PCA requires VPO to:

(a) Take steps to ensure compliance with the Anti-Fraud and Anti-Corruption Framework of the United Nations Secretariat, as well as the GCF Policy on Prohibited Practices; and

(b) Seek clearance from UNEP prior to final selection or contracting of personnel (including subcontractors).

105. UNEP confirms that there will be no planned disbursements or distribution of cash, vouchers, commodities, or other items of value directly or indirectly to beneficiaries as part of any activities in this project. Materials for the project implementation include agricultural hand tools, seeds, construction of small-scale flood infrastructure, environmental monitoring devices, and the like. These have low likelihood of being misused. Drone equipment will be provided to TFS, a government agency. Use of equipment and prevention of misuse will be embedded in a memorandum of understanding with TFS.

106. The PCA obliges VPO and UNHCR to ensure that any materials or technology procured are utilized in line with the approved GCF project. In turn, VPO and UNHCR agreements with implementing entities will include the same requirement. UNEP, VPO and UNHCR's reporting and review mechanisms are expected to identify any unauthorized use of funds.

107. UNEP states that it adheres to United Nations regulations, rules and policies related to whistleblower protection, as documented in "Administrative issuance on addressing discrimination, harassment, including sexual harassment and abuse of authority"

[5] Available at https://www.fatf-gafi.org/media/fatf/documents/reports/mer-fsrb/ESAAMLG-Mutual-Evaluation-Report-Tanzania-June-2021.pdf.
[6] Available at https://www.fiu.go.tz/TanzaniaNRA(Main)ReportDec2016.pdf.

(ST/SGB/2019/8).[7] UNHCR's policies are also compliant with United Nations rules, regulations and policies. Specifically, UNHCR handles allegations of misconduct through its independent Inspector General's Office.

108. The UNEP Environmental and Social Sustainability Framework and Stakeholder Response Mechanism provide an avenue for stakeholders to provide feedback or report concerns, complaints or grievance issues on proposed or ongoing UNEP projects.

109. UNEP has confirmed that it is committed to avoiding or minimizing unintended harm to stakeholders that may directly or indirectly result from its work. Stakeholders are strongly advised to make an effort to raise any concerns, complaints or grievances to the relevant UNEP Project Manager, its local project partners, consultants or the related UNEP Regional Office. Stakeholders may also submit their grievance through the UNEP website.

110. **Recommended risk rating**: The Secretariat has conducted a review of the project in accordance with relevant GCF Board approved policies and does not find any material issue or deviation with respect to compliance issues. Based on available information for this funding proposal, the Secretariat has determined a compliance risk rating of "medium" and has no objection to this proposal proceeding.

111. Although the FATF MER and NRA Action Plan suggest potential concern regarding Tanzania's (and hence the VPO's) capacity with respect to AML/CFT risks, UNEP has identified such risks through its capacity assessment of VPO and provided sufficient assurance that compliance-related risks can be appropriately mitigated via PCAs, with warranties and caveats supporting transparency and oversight.

4.3.5. GCF portfolio concentration risk (low risk)

112. In the event of approval, the impact of this proposal on the GCF portfolio concentration in terms of results areas and single proposal would not be material.

4.3.6. Recommendation

113. It is recommended that the Board consider the above factors in its decision.

Summary risk assessment		Risk assessment
Overall project/programme	High	Driven by the political and reputational risks of the project, the project necessitates close monitoring by the EE, AE and GCF from the start of project implementation.
AE/EE capability to implement the project/programme	Medium	
Project-specific execution	High	
GCF portfolio concentration	Low	
Compliance	Medium	

4.4 Fiduciary

114. UNEP will be the AE, and the United Republic of Tanzania acting through its VPO and UNHCR will function as the EEs for this project.

[7] Available at https://digitallibrary.un.org/record/3827518.

115. As the AE, UNEP will carry out fiduciary and safeguards oversight and provide the necessary scientific expertise and technical support to the project formulation, start-up, implementation, evaluations and closure.

116. VPO will act as the lead EE of the project, ensuring alignment with the country's national policies and plans as well as complementarity with other existing adaptation projects and Tanzania's NAP process. As the lead EE, VPO will oversee overall project implementation and will be accountable to the AE for the effective use of resources during project implementation. A PCA signed between VPO and UNEP will serve as the Subsidiary Agreement to regulate the obligations of the parties. Compliance with applicable United Nations rules, regulations and policies will be reflected in the Subsidiary Agreement. VPO will be responsible for the efficient disbursement of financial resources for the implementation of the designated project activities, and for the compilation of the required information from all implementing partners, including UNHCR as second EE, to provide consolidated project progress reports and expenditure reports to the AE. VPO will execute project activities through different mechanisms including direct execution by PMU, cooperation agreements with district government (known as district councils) and TFS as well as contracts with service providers and consultants. The district councils, under the supervision of VPO, will execute delegated activities as the district-level directly and may be authorized to contract NGOs, SAMOs and service providers.

117. As the EE, UNHCR will execute the project by overseeing and managing the implementing partners involved in the implementation of project interventions, prioritizing implementation by local government authorities where possible. UNHCR will be accountable to the AE for the effective use of resources during project implementation. All operating policies and procedures will follow United Nations rules, regulations and policies, which include provisions for financial management and procurement. UNHCR will be responsible for the efficient disbursement of financial resources for the implementation of all project activities. The transfer of funds between the AE and EE will be governed by a "United Nations to United Nations" agreement. A memorandum of understanding between VPO and UNHCR will define the principles of collaboration, team coordination and information-sharing mechanisms between the two EEs.

118. The flow of funds between UNHCR and implementing partners such as NGOs and other service providers is governed by a Project Partnership Agreement or, in the case of firms or individuals, other applicable contracting arrangements. The flow of funds between VPO and NGOs will be governed by cooperation agreements.

119. Coordination mechanisms will be put in place to ensure lessons learned are leveraged in the project. Following the engagement of implementing partners, UNHCR will implement risk-based approaches to implementation, including a strong monitoring framework for financial management and procurement by implementing partners, options for UNHCR to procure on behalf of implementing partners, and adequate controls on ensuring resources are used for their intended purposes and right-of-use agreements.

120. The financial management and procurement within the programme will be guided by United Nations financial regulations, rules and practices. The financial rules of UNEP, which follow International Public Sector Accounting Standards, are promulgated pursuant to the financial regulations and rules of the United Nations. Within this context, funding allocation mechanisms are managed in accordance with United Nations rules and procedures, including eligibility criteria, proposal evaluation processes, quality assurance and control, and project monitoring and supervision.

121. UNEP is audited annually by the United Nations Board of Auditors and has established dedicated trust funds for GCF resources. United Nations financial regulations and rules require the segregation of duties, and safeguards to ensure compliance with United Nations financial rules and regulations.

122. All procurement will be undertaken in line with applicable United Nations procurement regulations, rules and policies. In line with these regulations, rules and policies, funds will be transferred in tranches to an EE once the EE has satisfied the conditions that are defined under the Subsidiary Agreement (i.e. the PCA) to be signed between UNEP and VPO and between UNEP and UNHCR. The agreements will include specific obligations for each EE on financial management and reporting, and will require periodic reporting from the EE. VPO, as the lead EE, will be responsible for the submission of consolidated financial reports to UNEP, including funding for activities executed by VPO and UNHCR. The coordination mechanisms to deliver consolidated reports will be specified in the memorandum of understanding signed between VPO and UNHCR.

4.5 Results monitoring and reporting

123. The funding proposal is an adaptation project focusing on three results areas of the Integrated Results Management Framework: (i) most vulnerable people and communities, (ii) Health and well-being, and food and water security and (iii) ecosystems and ecosystem services. Through application of integrated landscape approach in Kigoma region of Tanzania, the project aims to benefit a total of 1.28 million beneficiaries, of which 0.57 million are direct and 0.71 million are indirect beneficiaries.

124. The theory of change (TOC) clearly defines the ultimate project goal and explicitly states the causal logic that informs the project's design, interventions and stated outcomes. Hence, the TOC itself has been crafted in a manner that is clearly supportive of meeting the ultimate project goals. The "if, then, because" logic is clearly shown in the diagram provided in the funding proposal. The ToC diagram identifies co-benefits, which can be achieved along the way towards climate results and captures the causal relationship between outcomes and co-benefits. The assumptions capture the most important factors underlying the achievement of results, including how the project's benefits will diffuse to indirect beneficiaries. The barriers and risks that may hinder the achievement of these changes are identified and linked to pertinent activities.

125. The logical framework is structured in alignment with the GCF integrated results management framework. It shows how the project can contribute to three GCF adaptation results areas with project outcomes and outputs, including paradigm shift potential and enabling environment indicators as well as core outcome indicators. Relevant GCF core and supplementary indicators have been chosen for monitoring and reporting the key performance of the project. The project-specific indicators also seem to be adequate to measure the progress of project implementation along with project deliverables. A continuous feedback mechanism has enabled refinement of the elements of the logical framework in terms of identification of results areas, indicators, means of verification, targets, assumptions, etc. They are found to be adequate and are expected to allow smooth monitoring and reporting of project results.

126. The implementation timetable has been provided in the required structure and format. The milestones and deliverables that are envisaged at different stages of project implementation are adequately planned and they are consistent with the logframe. Adherence to the proposed implementation timetable will facilitate smooth implementation of the project as well as timely delivery of the results at planned intervals.

127. The arrangements for monitoring, evaluation, and reporting (annex 11) have been strengthened with the introduction of a monitoring and evaluation (M&E) specialist who will work with the M&E officer to create and implement the M&E Plan. The overall monitoring and evaluation budget has been identified in the budget plan, and it is in the range of 2–5 per cent of the total programme budget as requested by the GCF Evaluation Policy. It is also confirmed that the costs of interim and final evaluations will be covered by the AE fee.

4.6 Legal assessment

128. The original AMA was signed with the Accredited Entity on 15 December 2016, and became effective on 20 February 2017. The Accredited Entity's first term of accreditation expired on 20 February 2022. The Accredited Entity was re-accredited by the Board pursuant to decision B.33/10. The first amendment and restatement agreement amending and restating the AMA was signed on 13 July 2023 and it became effective on 2 August 2023.

129. The Accredited Entity has provided a certificate confirming that it has obtained all internal approvals and that it has the capacity and authority to implement the project.

130. The proposed project will be implemented in the United Republic of Tanzania, a country in which GCF is not provided with privileges and immunities. This means that, among other things, GCF is not protected against litigation or expropriation in this country, the risks of which need to be further assessed. The GCF Secretariat provided a draft agreement on privileges and immunities and a background note to the Government of the United Republic of Tanzania on 13 November 2015. The latest communication was sent by the GCF Secretariat to the Government of the United Republic of Tanzania on 24 August 2016.

131. The Heads of the Independent Redress Mechanism and Independent Integrity Unit have both expressed the view that it would not be legally feasible to undertake their redress activities and/or investigations, as appropriate, in countries where GCF is not provided with relevant privileges and immunities. Therefore, it is recommended that disbursements by GCF be made only after GCF has obtained satisfactory protection against litigation and expropriation in the United Republic of Tanzania, or has been provided with appropriate privileges and immunities.

4.7 List of proposed conditions (including legal)

132. To address the matters raised in this section, it is recommended that any approval by the Board be made subject to the following conditions:

(a) Signature of the funded activity agreement in a form and substance satisfactory to the GCF Secretariat within 180 days from the date of Board approval; and

(b) Completion of the legal due diligence to the satisfaction of the GCF Secretariat.

Independent Technical Advisory Panel's assessment of FP218

Proposal name:	Building Climate Resilience in the Landscapes of the Kigoma Region, United Republic of Tanzania
Accredited entity:	United Nations Environment Programme (UNEP)
Country/(ies):	United Republic of Tanzania
Project/programme size:	Small

I. Assessment of the independent Technical Advisory Panel

1.1 Impact potential *Scale: N/A*

1. Note: This project proposal was evaluated by the independent Technical Advisory Panel (independent TAP) for the twenty-ninth meeting of the Board (B.29). However, the Government of the United Republic of Tanzania decided to withdraw the project just before the publication deadline for the Board meeting. Since then, the project changed some relevant aspects, including: (i) the inclusion of the Vice President's Office (VPO) of the United Republic of Tanzania (the GCF national designated authority) as lead executing entity (EE), which will hire and supervise the project management unit (PMU); (ii) the change of the role of the United Nations High Commissioner for Refugees (UNHCR), which will only execute some components of the project related to refugee camps; (iii) the definition of the roles of the Ministry of Home Affairs (MoHA) and the National Environmental Management Council (NEMC) in the project; (iv) the closure of one of the three refugee camps, transferring the refugees from the Mtendeli camp to the Nduta camp; (v) the removal of both the savings association (SA) opportunity and the funding of livelihood alternatives (mushroom and bee keeping) with GCF resources from the refugees; (vi) the removal of the inclusion of land-use planning activities inside the refugee camps; (vii) the inclusion of a technical partner to enhance project quality; and (viii) the change in budget and overall indicators in different components. The effects of these changes are addressed in the different sections of this assessment.

2. Kigoma is a region with a population of about 2.3 million people, out of a total of 56.3 million in the United Republic of Tanzania. The region spans about 37,000 square kilometres (km^2) in the north-west of the country. Refugees have been present in the region since the early 1960s. About 86 per cent of the refugee population in the country are accommodated in the three camps supported by the UNHCR in the Kigoma Region. The region hosts about 280,000 refugees from neighbouring countries, the majority of whom are living in the refugee camps of Mtendeli, Nduta and Nyarugusu. These settlements have added to the pressure of the already densely populated region on the surrounding degraded agroecological landscapes.

3. **The Government of the United Republic of Tanzania has decided to close the Mtendeli camp and relocate the refugees to Nduta camp, with the exception of those opting for voluntary repatriation**. Therefore, the final beneficiaries of the project will be defined as the refugees in the Nduta and Nyarugusu camps and host communities.

4. About 90 per cent of the host community[1] population in the areas adjacent to the refugee camps and a substantial number of refugees are engaged in small-scale agriculture. As a result, they rely heavily on the goods and services provided by the region's ecosystems. Climate and non-climate drivers have degraded and are further degrading these ecosystems. Easing tensions and misunderstandings between the refugee/migrant population and the host population has been part of the focus of the humanitarian–development response from United Nations agencies.

5. The project strategy is to ensure that the landscape and ecosystem of the Kigoma Region can continue to support its residents, including refugees and host communities. The project proposes responding to the climate change impacts affecting an already vulnerable and degraded landscape, while considering the complex needs of host communities and refugees in an environment where climate, humanitarian and development approaches to problems and solutions intersect.

6. The project consists of four components. Component 1 aims to increase the resilience of ecosystems and people to climate change threats. It proposes the development of participatory, climate-resilient land-use plans and adapting the existing village land-use plans (VLUPs) by acknowledging and addressing the role of the changing climate in shaping local ecosystems and landscapes. The planning processes for host villages and camps will be separate, as these are discrete and well-defined areas, but coordinated to maximize landscape-level benefits. However, the objective of developing them in a participatory manner is to ensure the most effective use of land resources through negotiation between different interest groups. The project proposes to scale up the VLUPs to other villages and to result in the issuance of a certificate of customary right of occupancy (CCRO) that specifies rights conferred to a land occupier and user following tribal customs and traditions on land. It is also aiming to build the capacity of local and regional government officials to support ecosystem-based adaptation (EbA) and landscape restoration, improve public awareness of climate change and support district, ward and village-level environmental management committees (EMCs).

7. In the proposal prepared for B.29, there was a component to support climate-resilient refugee camp land-use planning, but MoHA objected to the inclusion of land-use planning activities inside the camps. In the absence of land-use plans in camps, the location and type of flood and erosion control measures inside the camps and surrounding areas (activity 2.3) will be informed by consultations with relevant stakeholders and the refugee population as well as the results of a technical assessment.

8. Component 2, which focuses on restoring degraded ecosystems to support climate change adaptation in host and refugee communities, aims to protect natural forest systems and enhance the value that local communities place on the ecosystem services that supports them. It will establish and manage 30,000 hectares (ha) of village land forest reserves (VLFRs), including dedicated lots for fuelwood in forest areas such as the Miombo woodlands near beneficiary host communities. Pressure upon natural forest systems is predominantly driven by the local demand for fuelwood, which is the main source of energy for cooking in both the refugee and host communities. In Kigoma Region, 98 per cent of households rely on fuelwood for cooking, with 81 per cent using firewood and the remainder using charcoal.

9. This component will involve providing equipment and training for 10 forestry and natural resources officers for forest monitoring in national forest reserves. It also plans to implement 12,000 ha of agroforestry and village land afforestation to increase the resilience of

[1] The local Tanzanian communities living within 10–15 km of the camps. To clarify, the host community includes various groups, such as farmers, agropastoralists and other groups engaged in various livelihood activities. In other words, the term host community refers to all groups, including agropastoralists, that are not formally settled in villages.

land use, including the establishment of nurseries in target villages. district councils, in collaboration with the Tanzania Forest Service Agency (TFS), will lead the implementation of this activity with support from the technical partner and local non-governmental organizations (NGOs)/community-based organizations for community facilitation. Moreover, it will involve undertaking afforestation on 2,000 ha of village land to (i) restore critical ecosystem services to meet climate adaptation needs and improve ecosystem services such as flood control; and (ii) establish a sustainable supply of fuelwood that will take some of the pressure off the native Miombo forest and allow it to regenerate so it can provide its ecological and livelihood support functions. To mitigate the increasing risk of wildfire and human-induced fires in the afforested area, TFS will conduct an assessment, establish the necessary firebreak areas through casual labour and provide fire management training for patrol teams and communities.

10. The second part of this component is to implement flood and erosion control in densely populated areas of the refugee camps to reduce the exposed and flood-prone ferrosol-type soils, which have little or no vegetation cover. As the camps were established rapidly, within a context of humanitarian pressures, land-use planning was suboptimal, involving the removal of vegetation to provide space for the camps. The increased pressure of development combined with increased intensity of climate change-induced rainfall events resulted in gully erosion, soil loss and degradation in and around the camps. The proponents aim to construct 10 km of micro-scale, stone-pitched drainage channels and sustainable drainage system interventions – including 16 stone-filled gabions and micro-infiltration trenches/ponds/soakaways – in the two refugee camps and in the lands of the consolidated camp.

11. In the Mtendeli decommissioned camp area, the flood and erosion control measures will be conducted after the camp closure in combination with other restoration interventions implemented by UNHCR. According to the proponents, the post-closure restoration will result in adaptation benefits for the host community in terms of (i) reduced flood and erosion risk for downstream host communities; and (ii) potential for expansion of agricultural and agroforestry activities in the camp area. The land administration will be handed back to the Kakonko district government, which has the intention to use various camp facilities for the provision of services such as education and healthcare as well as to host district offices. Other areas of the decommissioned camp may be used by the surrounding villages as communal land for agriculture, grazing or forestry activities. To the independent TAP, it is still unclear why GCF should spend resources in erosion and flood control measures for an empty camp, especially if UNHCR is tasked with developing the camp rehabilitation plan for its handover to the Government of the United Republic of Tanzania, including soil conservation and reforestation. The independent TAP believes that the original motivation for accepting this investment was to support the adaptive capacity and the quality of life of the refugees in the camps, but in the absence of refugees, this investment with GCF resources is not sufficiently justified under climate change adaptation rationale. Moreover, there is a risk that the Government of the United Republic of Tanzania could decide to close the other two camps, in which case the investments from GCF will not have a solid climate change and humanitarian rationale.

12. Component 3 on climate-resilient livelihoods, which is to be implemented by the district councils in coordination with the VPO, aims to increase the capacity of host communities and refugees to better adapt to changing climatic conditions. The goal is to promote modern technologies and management practices to strengthen the capacity of farmers, district officials and agricultural extension workers in climate-resilient agriculture through farmer field schools (FFS) and the provision, through SA management offices, of resilient seeds, tools and necessary equipment for implementing climate-resilient agricultural practices.

13. Component 3 also aims to increase water availability and use efficiency through water harvesting and efficient irrigation interventions. The project expects to construct micro-rainwater harvesting systems, including 94 check dams, 20 runoff harvesting microdams (charco dams, 120 unlined water pans and 120 lined microponds). The proponents expect to

establish water pumping and irrigation through 50 microsolar and 172 treadle pumps that will draw water from the rainwater harvesting locations and pump it to where it is needed.

14. This component will conduct training and awareness-raising for 400 members of water user associations (WUAs) and farmers on water management and efficient water use through FFS. It will also support the establishment of new and strengthen existing WUAs through organizing and conducting 20 meetings/trainings.

15. Furthermore, component 3 aims to promote climate-resilient livelihood diversification to strengthen food security and nutrition as well as increase livelihood options. The district councils, in coordination with the VPO, will implement the livelihood interventions in the host communities with resources from GCF, but decided to remove the support to refugees using GCF resources. The project states that UNHCR, through the project co-funding, will implement livelihood interventions in the camps, opening them up to host communities if and where possible, with the final livelihood activities to be coordinated and agreed with MoHA. The livelihood alternatives include the promotion of (i) beekeeping to incentivize forest conservation in the forests surrounding the camps and villages by providing 52 beehives, equipment and training for 850 people; and (ii) mushroom cultivation for marginalized groups with training and mushroom-growing facilities by providing equipment and training to 26 mushroom groups. However, the proposal is not explicit with regard to the number of livelihood interventions and the resources needed to support the refugee communities, all subject to the approval of MoHA.

16. Component 4 focuses on information and on mainstreaming climate change adaptation in policies, plans and strategies in the United Republic of Tanzania. It will use the results of the planning processes and physical interventions implemented under output 1.3 to demonstrate the economic, environmental, and social value of this model to stakeholders. This approach will contribute to the sustainability and scalability of the outputs and GCF fund-level outcomes at the national level. The benefits of scaling the model up to global level are less clear in terms of humanitarian programming through UNHCR, as the project has substantially lowered support to the refugees using resources from GCF.

17. The project beneficiaries (drawn from both the host communities and refugee camps) are estimated at 1,281,800 people, consisting of approx. 570,300 direct (292,800 women and 277,500 men) and approx. 711,500 indirect beneficiaries (366,635 women and 344,865 men).[2] Of the direct beneficiaries, 36 per cent are refugees, with a population in November 2021 of 205,596 persons. The estimate is that 15,000 additional households (approx. 80,000 people, mostly from host communities) adopt, at the end of the project, diversified, climate-resilient livelihoods in agriculture, beekeeping and mushroom-growing.

18. The project's approach to adaptation is, on the one hand, building small-scale infrastructure to support flood control, water harvesting and efficient irrigation interventions, and, on the other hand, adopting ecosystem-based approaches that support ecosystem functions and services. In terms of flood control interventions, as mentioned above, the development of the camps was done within the context of humanitarian pressures, with suboptimal land-use planning. Therefore, these proposed interventions respond both to a failure in initial planning and to an increase in precipitation intensity due to climate change.

19. In terms of ecosystem-based adaptation at the landscape level, it is well noted that climate change and variability in the United Republic of Tanzania – in combination with land use drivers, such as deforestation and forest encroachment – have resulted in the alteration of

[2] Considering the 51 per cent women population in the three camps (UNHCR Tanzania Refugee Situation Statistical Report, September 2021) and 51.5 per cent women population in the three target districts (United Republic of Tanzania population and housing census, 2012). Calculations can be consulted in Annex 4– Detailed assumptions or sources.

ecosystem functions and agroecological systems, consequently affecting livelihoods across the country. In the case of Kigoma Region, host communities expressed the view during consultations that the establishment of refugee camps had contributed to this accelerated deforestation as a result of increased fuelwood harvesting and illegal cultivation. The host communities are also expanding the agricultural areas, driving deforestation, putting pressure on the ecosystems and exacerbating drought and flood impacts, thus resulting in lower agricultural yields.

20. The project proponents developed an accurate analysis to provide a sound climate change rationale. The observational station that serves Kigoma Region was used to validate the CORDEX (Coordinated Regional Climate Downscaling Experiment) regional climate models used in the assessment of projected climate changes. After a validation assessment of seasonal signal, magnitude, and discrepancies of 10 models to the station data, the proponents selected the IPSL-IPSL-CM5A-MR model as the best-performing model overall. Therefore, this model was used to assess future climate changes.

21. The past and current observational trends were used to project the climate trends into the future. These notable trends are the decline in early season rainfall and the concentration of rainfall in the mid-season and late season. There is a projected general increase in extreme precipitation days (ninety-fifth percentile), leading to an increase in potential flooding events. The meteorological drought index of the Standardized Precipitation Evapotranspiration Index indicates that the variability in the rainfall will probably lead to more frequent and severe drought events. The average maximum and daily extreme temperatures (ninety-fifth percentile) are projected to increase in all seasons.

22. The overall analysis indicates that the current climate exposure noted in Kigoma Region is a result of current climate changes. The projected future anticipates these will continue and further expose the area to enhanced climate impacts. Therefore, all project interventions respond to present and projected climate impacts.

23. Finally, the adaptation measures in the areas where the host communities live are well designed to ensure practical adaptive capacity under a landscape approach. The integrated set of interventions will build local resilience to climate change by avoiding land degradation and soil erosion and by improving hydrology, landscape restoration and water quality, resulting in more-resilient livelihoods for the host communities. However, the erosion and flood control interventions in the Mtendeli decommissioned camp area are not fully justified in terms of climate change. Moreover, as the refugees from this camp will be transferred to the Nduta camp, there is a risk of having less space per family to ensure adaptative livelihood conditions. The revised funding proposal after consultation with the Government of Tanzania has diminished the livelihood alternatives for the refugees with resources from GCF and has a narrower nexus between climate change and humanitarian support to ensure the adaptive capacity of refugee communities. Overall, the climate change impact is medium to high.

1.2 Paradigm shift potential *Scale: N/A*

1.2.1. Innovation

24. The project acknowledges the mosaic of land uses in the project site as well as the complex and varying relationships between the human settlements (refugees and host communities) and the ecosystems available, taking into consideration the increased challenges associated with climate change.

25. The landscape-level approach to project design and the challenges of both host communities and refugees to support their livelihoods while adapting to the changing climate are well analysed in the project. However, as explained above, the current funding proposal has

reduced support to refugee communities and therefore the majority of the interventions in land use, agriculture, water and livelihoods will end up mostly benefiting the host communities. Moreover, there is a risk that the refugee camps will end up closing as was the case with the Mtendeli camp. Therefore, the paradigm shift of having the first GCF project with a significant humanitarian–climate nexus is not as promising or transformational as in the original FP assessed by iTAP back in February 2021.

26. The sustainability of the project interventions needs further refinement and thought. The project is relying on external technical support to deliver some of the interventions and is staffing the project management unit (PMU), including district officers, with resources from GCF. Moreover, the project lacks viable marketing alternatives to support the communities with agroforestry interventions.

27. Due to the encampment policy in the United Republic of Tanzania, refugees are legally required to remain in the camps, and they are not allowed to be engaged in formal employment either within or outside the camps, thus having to rely on humanitarian assistance or the informal economy.

28. The United Republic of Tanzania has been generous in accepting refugees, mainly from Burundi and the Democratic Republic of the Congo since the 1960s. However, the country has the right to open or close refugee camps based on its own policies towards refugees. It has also experienced an increased influx of refugees when neighbouring countries have faced political instability. On some occasions, it has closed camps due to the repatriation of refugees. For example, in 2012, some camps in Kigoma Region were closed, and then reopened in 2015. Between February 2021 and now, the Government of the United Republic of Tanzania decided to close one camp. The permanence of these camps in the future will depend on the regional refugee situation and on the willingness of the government to maintain them.

29. Although the refugee camps are relatively well situated topographically, Tanzanian authorities and camp planners were probably unable to account for long-term development pressures and climate change impacts when designating the area for the settlements. Increasingly frequent floods are damaging the shelters, roads and communal facilities. These floods cause latrine pits to overflow, resulting in standing water, with negative impacts on health and sanitation. Floods also lead to deep gully erosion, which places community safety at risk of unstable ground. Consequences are being felt at the landscape level with increasing soil erosion. To respond to this challenge, the project provided detailed plans and designs to implement flood and erosion control infrastructure on public lands in and around the camps.

30. **This opens a question to the Board in terms of investing GCF resources in public goods for temporary humanitarian relief projects.** In response to the question of the independent TAP on the strategy of the project in the event that refugee camps are either closed or reallocated, the project proponents answered that livelihood activities would be absorbed by host communities, and the flood and erosion control measures would help to protect future occupants against climate-driven flood events. However, in the case of the Mtendeli camp, the future occupants are yet to be defined, and even if they end up in the hands of the district authorities for health or educational purposes, there is a disconnection with the original purpose of the investments, which is to support the adaptive livelihood of refugee communities.

31. The productive activities proposed for vulnerable host and refugee communities, such as beekeeping and mushroom-growing, are well thought out in that they support healthier diets on reduced land area and ensure some income generation. Participants will receive training as well as inputs and shared facilities managed by the beekeeping and mushroom-growing groups. According to the proponents, in the stakeholder consultations, refugees showed a lot of interest in these activities, seeing that the knowledge to be gained will continue to help them if/when they return to their country of origin, thus positively contributing to their self-reliance. However, the current funding proposal has erased the GCF funds allocated to these activities in

refugee camps, which now rely on co-financing at a more limited scale. Therefore, the beneficiaries of the GCF-funded livelihood interventions will mainly be host communities.

32. In terms of land-use planning, the proposal has an innovative approach, as the long-term, legal outcome of the VLUP is the issuance of a CCRO. This document specifies rights conferred to a land occupier and user following tribal customs and traditions on land. The CCRO will provide long-term security for the villagers and may be used as collateral for borrowing money to conduct other future village-related, climate-resilient investments. Moreover, by adopting a landscape approach with coordinated land-use planning across the project area, the project offers innovation and potential to generate new lessons learned in the practice of VLUPs in the country.

33. The agroforestry component is based on the need to ensure basic subsistence crops, such as casava and beans, and less focused on achieving regenerative agriculture that takes climate change into account. The project proponents argue that these crops were selected based on climate resilience, potential to increase incomes and contribution to food security. The farmers will have access to seeds and FFS, aiming to transfer better productive technologies as well as sustainable and conservation practices. This will be complemented with agroforestry, afforestation and forest conservation activities that contribute to erosion control and limit the loss of topsoil rich in organic material.

34. Benefits to refugee and host communities from the landscape interventions would come in the form of common ecosystem services, such as restored critical forest ecosystem services to manage floods and a sustainable supply of fuelwood to relieve pressure on the native Miombo forest. These restored ecosystem services would facilitate ecological and livelihood support functions.

35. The project will implement community-based forest management (CBFM) measures for resilient ecosystems. Under CBFM, villagers take full ownership and management responsibility for an area of forest as a VLFR. Following this legal transfer of rights and responsibilities from central to village government, villagers may harvest timber and forest products, collect, and retain forest royalties and undertake patrols (including arresting and fining offenders). Ensuring that villagers take ownership and responsibility for their natural resources, instead of there being a centralized "command and control" type of arrangement, is one of the key paradigm shift elements in the proposal.

36. The establishment of VLFRs is a provision of the Forest Act of 2002. However, most of the VLFRs in Kakonko, Kasulu and Kibondo Districts are still in the stage of declaration by the village council. The project aims to facilitate the establishment and management arrangements of 20 village forest reserves over 30,000 ha.

37. In Kigoma Region, 98 per cent of households rely on fuelwood for cooking, with 81 per cent using firewood and the remainder using charcoal.[3] The project proposes establishing 2,000 ha of forest lots producing a sustainable supply of fuelwood. This will take some of the pressure off the native Miombo forest and allow it to regenerate so that it can provide its ecological and livelihood support functions, benefiting both refugee and host communities.

38. The project presents a good mix of alternatives (micro-rainwater harvesting systems, run-off harvesting microdams, unlined water pans and lined microponds) to increase water availability and use water efficiently through water harvesting and efficient irrigation interventions, thus supporting the nursery, agroforestry and afforestation interventions. These water systems are common and do not constitute an innovation per se. However, if they are well

[3] Kigoma Region Basic Demographic and Socio-Economic Profile 2016.

placed and managed, they could be one of the most important activities for ensuring landscape restoration under climate change conditions.

39. The project is also proposing the use of SAs as a key delivery mechanism for the project. They allow individuals to pool financing together to save and borrow money based on the consensus of members. This project will seek to strengthen and scale up existing structures by channelling funds through SAs for the purchase of climate-resilient equipment. The independent TAP believes that strengthening these organizations with knowledge on climate change and allowing them to reach the communities will be a key component in the sustainability of this project. However, in the revised proposal, this alternative is not made available to the refugee communities, which will have no financing options.

40. Finally, benefit-sharing schemes would be based on agreements among four parties: the host community; the Government of the United Republic of Tanzania represented by MoHA; UNHCR; and the refugee communities, with the involvement of TFS. Representation of the host community and refugees will be based on existing governance structures, such as village leadership in the host community. The development of appropriate benefit-sharing agreements will be facilitated by representatives from all three levels of EMCs established under subactivity 1.1.4. The independent TAP believes that a key parameter of success of the project is related to the benefit-sharing arrangements, ensuring peaceful coexistence between the host and refugee communities.

41. The project's theory of change does not include any assumptions about the fact that the refugee camps could be closed, thus making the project a common EbA project for Kigoma Region. Therefore, the Government of the United Republic of Tanzania, together with the United Nations Environment Programme (UNEP) and UNHCR, should consider the real expected long-term changes of the project, including the option of a region with no refugee camps.

1.2.2. Potential for knowledge and learning

42. The project will support the transfer and adoption of low-carbon agroforestry and forestry systems to change the current deforestation and landscape degradation patterns.

43. Project interventions will include extensive training and capacity-building, which will allow knowledge of the benefits of these activities, such as resilient agriculture and alternative livelihoods, to be sustained and expanded to wider communities.

44. The project will use FFS as a methodology to share knowledge on EbA. The FFS will involve the village-level EMCs, with clear mechanisms for capacity-building. Moreover, the project has clear targets in terms of the number of people trained in FFS and the number of people (host and refugee communities) trained in beekeeping and mushroom-growing facilities.

1.2.3. Contribution to the creation of an enabling environment and regulatory framework

45. The project will conduct training and awareness-raising on climate change and good environmental management practices while involving district governments. The training will include the understanding of projected climate changes, impacts such as wildfire risks, adaptation strategies and the importance of local ecosystems and the services they provide.

46. Public awareness-raising will also target Sukuma agropastoralist communities that contribute to deforestation and ecosystem degradation in the project area but are not currently included in wider community conservation or land-use planning processes. The training that would be conducted by project staff and district governments, in coordination with land-use planners and consultants, would contribute to having a better inclusion strategy to allow these groups to be part of the landscape restoration goals and solutions.

47. The project will support district, ward and village-level EMCs and improve coordination between the different levels. This will be achieved through the development of coordination protocols and structures, regular reporting and monitoring systems, and the provision of upskilling to EMC members at all levels. Training on conflict resolution and benefit-sharing will also be provided to EMCs to ensure that the project contributes to reduced potential for conflict between natural resources users.

48. In terms of water management, the Lake Tanganyika Basin Water Board, which foresees and plans the use of water resources (surface water, rivers and lakes), will support the establishment and capacity development of the WUAs through a participatory process.

49. The EMCs/village land-use management committees and the WUAs are currently not performing as they should. This is largely due to a lack of capacity and support linked to underfunding at the district level to provide the foreseen support functions. The project will provide capacity-building on planning and activities to safeguard shared natural resources, including ensuring community adherence to the village environmental bylaws, during the project duration and beyond. Targeted training will be provided to support conflict resolution capacities, roles in supporting benefit-sharing arrangements, and coordination protocols between WUAs, EMCs and relevant institutions at the local level.

1.2.4. Scalability and replicability

50. The project's EbA approach in the host communities could be scaled to other regions in the United Republic of Tanzania. However, the sustainability and scalability of the project interventions will require financial investments by the Government of Tanzania and the regional district offices. Moreover, since a technical partner and NGOs will provide training and technical advisory services to the local government and forestry officials in the technical design, implementation and monitoring of agroforestry systems, there is a need to ensure that TFS and the Kigoma officials ultimately have the skills and capacity to scale the model to other regions in the country as well as maintain and scale the interventions in the Kigoma region after project completion.

51. In its original version, the project had the potential to address adaptation needs of host communities and refugees in an integrated way. However, the current proposal has a limited climate–humanitarian nexus, with few components within the refugee camps that could be documented to replicate the experience and approach by UNHCR in other countries.

52. Overall, the project has a limited paradigm shift in terms of the climate–humanitarian nexus, but it has an interesting EbA approach in the selected Kigoma landscape that needs more thought in terms of sustainability and scalability.

1.3 Sustainable development potential *Scale: N/A*

1.3.1. Environmental co-benefits

53. The benefits associated with the EbA approach will include the functionality of the overall ecosystem services. Some of these ecological services include the retention of water that will support the evapotranspiration regime, the limiting of the effects of floods, and the reduction of soil erosion. Moreover, if assisted restoration is done with a variety of species, it could increase biodiversity. It could also support natural habitats to allow forest species to regrow, enable animal species to increase, and ensure ecosystem services as important as pollination, pest control and the provision of food and non-timber products.

54. The project will establish forestry and agroforestry systems that will supply wood and food to selected communities, thereby preventing the expansion of the agricultural frontier and

allowing the regeneration of ecosystems at the landscape level. It will also reduce the pressure on forests and the consequent fragmentation of the fauna habitats of these ecosystems, thus contributing to the conservation of biodiversity.

55. The agroforestry and forestry systems promoted could significantly increase soil fertility. However, the selection of trees is very important. As scientists from the Royal Botanic Gardens, Kew, point out, tree planting is a brilliant solution to tackle climate change and protect biodiversity, but the wrong tree in the wrong place can do more harm than good.[4] In a biodiversity-rich country such as the United Republic of Tanzania, the selection of tree species matters. Therefore, there is a need to ensure the planting of native species with the potential to restore degraded soils and maintain essential ecological services.

56. Controlling soil erosion with flood control interventions and reforestation will improve water retention, which will in turn serve to ensure ecosystems restoration. At the same time, working on flood control will lower the risk of waterborne diseases, including diarrhoea, malaria, bilharzia and urinary tract infections, during the rainy season.

57. The geographic information system (GIS)-based forest monitoring data management system will support TFS in monitoring forest cover and fire risk management. The system will assist in controlling illegal deforestation activities in the national forest reserves (approx. 219,642 ha) in the target districts and assessing the impact of VLUPs.

58. The independent TAP asked about the selection of beans and cassava as the main crops to be used in implementing agroforestry systems. The proponents replied that the selection was based on an assessment of the main options for climate-adaptive agricultural technologies and practices, and the market potential of the two crops. Cassava performs well in poor soils and dry conditions but is also able to tolerate increased rainfall. Studies examining the impacts of predicted climate change on cassava production in Africa indicate that it is likely to be positively impacted under most climate change scenarios. Beans have high nutritional value and contribute to soil fertility. While most common varieties of the crop are susceptible to both drought and heat stress, improved varieties with resistance to these climate stressors are available and will be provided to local communities as part of the proposed project's interventions. However, it will be advisable to involve a wider variety of choices to restore soils and ensure crop diversity and nutrition.

1.3.2. Economic co-benefits

59. The project expects to create livelihood alternatives with agroforestry systems and small endeavours such as mushroom-growing facilities and beekeeping, thus creating decent direct and indirect green jobs, most of which are expected to persist even after the end of the project. However, these alternatives will mostly benefit the host communities. The refugees will have the possibility to have some livelihood alternatives with the support of the UNHCR co-financing, but these will be limited and based on agreements with MoHA. Improved and diversified agricultural production systems will ensure food security and provide opportunities to connect to markets, thus generating revenues to achieve a better quality of life for the communities. By diversifying crop production, the communities would be able to have more stable incomes throughout the year.

60. The project will result in ecosystem services that would be supporting (e.g. nutrient cycling and soil formation); provisioning (e.g. timber and non-timber forest products); regulating (e.g. water regulation and flood control); and cultural (e.g. recreation and education).

[4] BBC. 2021. *Scientists address myths over large-scale tree planting*. Available at <https://www.bbc.com/news/science-environment-55795816>.

All of these services will serve to support income generation and job creation and ensure long-term rural livelihood sustainability.

61. The economic value of ensuring water availability through harvesting systems, which provide members of the host communities with access to a more reliable source of water, is potentially very high. This will allow communities to extend the growing season and ensure their crop production and revenues. The infrastructure put in place for these purposes will also have the added benefit of reducing the impacts of floods through water retention and soil conservation.

62. The project will also support the SAs in allowing individuals from the host communities to pool finances together to save and borrow money and maintain their agroforestry systems. This will ensure the long-term sustainability of the initial investments by the project, allowing the communities to continue the activities beyond the end of the project.

1.3.3. Social co-benefits

63. The project expects to ensure social cohesion between host and refugee communities, enabling a more peaceful and safer environment while creating a sense of "belonging" and respect for nature. However, the options for refugee communities are limited as the project provides few options to support their quality of life and they are obliged to stay within the camp limits.

64. The host households could ensure increased food security with the implementation of agroforestry systems, and they will have a better standard of living from increased income. Having village plans in place will allow the communities to work together and ensure long-term landscape arrangements that will benefit the population at large.

65. Communities will have greater knowledge of climate change and be able to decide on their productive systems and the conservation of natural resources. When forest and agroforestry systems are restored, the effect on microclimates will benefit the community livelihoods around the selected areas.

66. The project will improve the quality of life of host and refugee communities by helping to avoid the diseases related to water contamination and storage, thus reducing risks of water- and vector-borne illnesses during flood incidents. Moreover, the project will help reduce exposure to tsetse flies and the risk of contracting sleeping sickness.

1.3.4. Gender-sensitive development

67. In the United Republic of Tanzania, gender inequality is a salient issue. Although the overall labour-force participation rate (including the informal sector) of women in the country is 88.7 per cent, only 29 per cent of women participate in the formal economy.

68. Women also participate in low-productivity sectors. Specifically, women account for most of the agricultural workforce, with about 90 per cent of women living in rural areas participating in the sector.

69. A five-year National Plan of Action to End Violence Against Women and Children (2017/18–2021/22) has been developed by consolidating eight different action plans addressing violence against women and children to create a single, comprehensive, national plan of action to eradicate violence against women and children in the country.

70. The Village Land Act introduced in 1999 has broken new ground in women's rights, with some sections rendering invalid any customary practices that discriminate against women. The act upholds the right of every woman to acquire, hold, use, and deal with land; voids the role of customary law in determining land tenure; and recognizes women's rights to land in the event

of the death of their spouse or upon divorce. There are also requirements for female representation in key decision-making bodies. The project will make sure that the law is effectively implemented through the support of village plans.

71. Despite efforts by UNHCR and its partners, refugee women and children continue to be exposed to a range of risks. These include sexual and gender-based violence, child abuse and early and forced marriages. Some of these risks relate to existing negative cultural norms and practices related to power imbalances that bring about gender inequality.

72. The gender action plan has an overall impact statement to "improve land management and increased resilience of host communities and refugees, including poor and female-headed households, to climate change through land-use planning, land use and forestry interventions, resilient agriculture and livelihood diversification and improved water management and drainage."

73. The project involves the following outcomes: (i) women are represented, and participate meaningfully, in land-use planning at the village and camp levels; (ii) CBFM and agroforestry practices engage women in resilience-building and decrease their vulnerability; (iii) the adaptive capacity of women farmers is increased and women's resilience to climate change is increased both economically and through increased food security; and (iv) women and their needs, interests and knowledge are meaningfully captured in mainstreaming climate change adaptation in policies, plans, strategies and programmes. The plan is well developed and has specific activities, indicators and a budget to ensure its development.

74. Overall, the plan is more oriented to support women of the host communities. The project is operating in the context of an increased population of the Nduta refugee camp, with the negative impacts of densely populated camps on the security and livelihoods of women. Moreover, refugee women and men will not have access to the SAs and will have just a few livelihood alternatives facilitated through resources from UNHCR (if agreed with the Government of the United Republic of Tanzania). The plan includes refugee women in the agroforestry and afforestation trainings, but it is not clear if they will be able to benefit from these trainings to support their livelihoods.

1.4 Needs of the recipient *Scale: N/A*

1.4.1. Vulnerability of the country and vulnerable groups

75. The United Republic of Tanzania is a country in East Africa with an area of about 945,000 km^2. Following two decades of sustained growth, the country reached an important milestone in July 2020 when it formally graduated from low-income country to lower-middle-income country status.[5] In 2019, it was one of the top three growth performers in East Africa, enjoying a stable economy. However, pandemic-induced shocks slowed its gross domestic product (GDP) growth rate from 5.8 per cent in 2019 to an estimated 2.0 per cent in 2020, as shocks to export-oriented sectors such as tourism, manufacturing and related services diminished business revenue and labour income, which adversely affected domestically oriented firms of all sizes across all sectors.[6]

76. President Samia Suluhu Hassan was sworn in on 19 March 2021 to serve as the United Republic of Tanzania's sixth, and first woman, president, following the death of President John

[5] World Bank Group. 2021. Economic Outlook of Tanzania. Available at <https://www.worldbank.org/en/country/tanzania/overview#1>.
[6] Ibid.

Maguful on 17 March 2021.[7] The government is prioritizing the implementation of a new strategy to contain the COVID-19 pandemic and has a renewed approach to regional trade and cooperation and a commitment to domestic policy reforms designed to improve the business environment.

77. In the period 2011–2018, the agriculture sector's contribution to GDP grew much more slowly than those of the rest of the economy, averaging 4.4 per cent a year, or 1.4 per cent per capita. With about 25 per cent of the population of about 56.3 million living below the poverty line, of which 75.5 per cent of the poor are dependent on agriculture for their livelihoods, agriculture sector growth is crucial for poverty reduction. Despite having fertile land and water resources, the agriculture sector is hampered by low productivity and persistent poverty.

78. Rural areas such as Kigoma Region face unique developmental challenges, including changing climate, socioeconomic inequality and a fluctuating population of refugees from neighbouring countries, in particular from Burundi and the Democratic Republic of the Congo. Moreover, infrastructure constraints, low agricultural productivity, a limited fiscal base, insufficient human capital and environmental factors contribute to inhibiting the region's development.

79. The country has been one of the most generous to refugees, not only in East Africa, but also globally. The country has received refugees mainly from Burundi and the Democratic Republic of the Congo. The United Nations estimates that by 2003, 500,000 refugees were living in camps. However, in 2012 some of the camps in Kigoma Region closed and the remaining 37,000 refugees living there returned to Burundi. In 2015, political instability again resulted in civil war in Burundi, with a mass influx of refugees from there into the United Republic of Tanzania. To meet this increased demand, refugees were transferred to the Mtendeli and Nduta camps, which were reopened in 2016 and 2015, respectively.

80. About 25 per cent of the current refugees come from the Democratic Republic of the Congo.[8] The Nyarugusu camp hosts the largest proportion of refugees from the Democratic Republic of the Congo: about 82,000 in 2018.[9] Although several drivers of conflict in Burundi and the Democratic Republic of the Congo remain present, there is no certainty as to the lifetime of the refugee camps. The project proponents believe that it is unlikely that all the refugees currently accommodated in the three camps will return to their own countries in the long run. However, the United Republic of Tanzania is actively encouraging the repatriation of refugees, and is signalling, with the closure of the Mtendeli camp and the reduced support in this project to the refugee communities, that the refugees should not receive special support.

81. Furthermore, the three camps do not follow the exact parameters for the classification of refugee accommodation as camps or settlements. Refugees living there are restricted in terms of movement beyond the borders of the camps. They must rely on food aid to a large extent, and there are restrictions placed on their socioeconomic freedoms. In terms of policy response, the locations are officially considered camps. However, for many refugees, the camps are permanent places of residence rather than merely temporary places of safety or transit centres.

82. The regional instability, with persistent ethnic tensions and an enduring culture of violence within communities, can be exacerbated by chronic poverty, malnutrition, low levels of development and the countries' young populations (especially in Burundi). In addition to these

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[7] Ibid.
[8] Ibid.
[9] UNHCR. 2018. *Congolese Situation: Responding to the Needs of Displaced Congolese and Refugees. Annex - The United Republic Of The Tanzania. Supplementary Appeal, January - December 2018*. Available at <http://reporting.unhcr.org/sites/default/files/2018%20Congolese%20Situation%20SB%20-%20Tanzania.pdf>.

challenges, several environmental drivers of conflict remain in the region due to the scarcity of land and the overexploitation of resources, which are mainly used for subsistence agriculture.

83. In terms of livelihoods and economic activity, refugees are compelled by ever-decreasing funding to engage in activities to support themselves financially. Although there have been efforts by the Government of the United Republic of Tanzania to reduce the economic activity within the camps, the Government has been lenient in enforcing restrictions on the refugees about working outside the camps. The feasibility study explains that many people – Burundians in particular – work on farms in the surrounding communities, and that laws against employment are not always enforceable in practice.

84. The project places a priority on supporting the host communities neighbouring the camps, namely those directly impacted by the establishment of the camps. Despite the encampment policy of the United Republic of Tanzania, interactions between refugees and host communities occur. There are informal employment arrangements and agricultural land-lease/land-use agreements. Interactions take place through some trade activities despite the closure of common markets. Refugees leave the camps frequently to collect fuelwood from the land of the host communities.

85. The host community population in the project consists of farmers and livestock keepers. About 70 per cent of the population in the region are engaged in small-scale farming, with others involved in livestock keeping. Community members access non-timber forest products, such as food, medicines, fuelwood and other items from village forests, especially in Miombo woodlands. The current lack of land-use planning and management of customary land can lead to conflict between farmers and pastoralists, especially pastoralist groups that pass through Kigoma, such as the Sukuma people, in search of fertile grazing land towards the south.

86. Climate and non-climate drivers have degraded, and are further degrading, the ecosystems in Kigoma Region. According to the feasibility analysis, Kigoma is experiencing rising temperatures as well as increased rainfall intensity and variability. These will contribute to an increase in the frequency and severity of floods and prolonged droughts, as well as heat and water stress. Observed changes in weather patterns have already produced climate impacts that have compounded environmental degradation and negatively affected agricultural outputs, the incidence of pests and diseases, forest resources and deforestation, surface- and groundwater availability, and soil erosion and fertility. Projected climate changes will further exacerbate these already noted climate problems, together with non-climatic development drivers of degradation. These changes are threatening local rain-fed and ecosystem-reliant livelihoods in the region.

87. In conclusion, the needs of the recipients are high, as both vulnerable host and refugee communities need to find sustainable livelihood alternatives in the same landscape, sharing scarce resources under climate change.

1.5 Country ownership *Scale: N/A*

1.5.1. Alignment with the national climate strategy

88. The Government of the United Republic of Tanzania has put in place sectoral policies and legislation that focuses on enhancing agricultural growth, natural resources management and climate change interventions. These include the National Agriculture Policy (2013), which aims to develop the agriculture industry through practices that sustain the environment.

89. The country has several policies in place to support the project and its climate goals. The Agriculture Climate Resilience Plan (2014–2019) proposes a range of adaptation options, including, but not limited to: improving agricultural land and water management; accelerating

the uptake of climate-smart agriculture; reducing impacts of climate-related shocks through risk management; and strengthening knowledge and systems for targeted climate action. Moreover, the goal of the Tanzania Climate Smart Agriculture Programme 2015–2025 is to have an "agricultural sector that sustainably increases productivity and enhances climate resilience and food security for national economic development". The programme focuses on six core objectives, including increasing the productivity of the agricultural sector through (appropriate) climate-smart agriculture practices that consider gender and enhance climate resilience of agricultural and food systems.[10]

90. The country's intended nationally determined contribution identifies adaptation as a national priority. The severity of climate-related disasters and the impacts associated with spatial and temporal variability of rainfall and droughts and higher incidence of floods are all recognized. The interventions outlined in the project are aligned with the country's National Adaptation Programme of Action (2007), the National Climate Change Strategy (2012), the Guidelines for Integrating Climate Change Adaptation into National Sectoral Policies, Plans and Programmes of Tanzania (2012), the country's intended nationally determined contribution (2015) and the Tanzania Development Vision 2025.

91. The National Five-Year Development Plan 2016/17–2020/21 defines specific sectoral interventions in natural resources management and agricultural development. In particular, the document refers to a number of key interventions, including having in place participatory climate change adaptation measures at catchment/WUA level and introducing and adopting crop and livestock varieties suited to adverse conditions brought about by climate change.

92. The Tanzania Country Refugee Response Plan (January 2019–December 2020)[11] for refugees from Burundi and the Democratic Republic of the Congo includes: "wider efforts within the environment and energy sector and associated links with climate change adaptation and mitigation. This will be undertaken in the context of both the host community and the refugee community. Implementation priorities in this regard will include climate-smart agriculture, forest landscape restoration, sustainable forest management, community-based natural resource management, land-use planning, water conservation though the promotion of water harvesting technologies, Integrated Water Resource Management, river catchment conservation and the adoption of renewable energy technologies."

93. In the project area, the Regional Commissioner and the Regional Administrative Secretary of Kigoma have identified the region's highest priorities for development. These priorities include improving extension services to farmers to increase productivity and community participation in forest management and reforestation, both of which are addressed through this project.

94. However, the independent TAP has noted that the project does not include any type of co-funding by the Government of the United Republic of Tanzania. This appears unusual as many entities will be engaged to support activities. In response to the question of the independent TAP on this matter, the proponents have explained that the project is very cognizant of investments made by district and regional authorities, TFS and other government entities.

95. In the United Republic of Tanzania, all public funds are centrally controlled. District budgets encompassing the various sectors are submitted for consideration to the central government through the district executive director, and are then confirmed and allocated annually. The district, regional and TFS budget requirements for environment- and forestry-

[10] Tanzania Climate Smart Agriculture programme. Available at https://marlo.cgiar.org/data/ccafs/projects/56/caseStudy/TANZANIA-CSA-PROGRAM-Final-version-3-August-2015.pdf.

[11] Available at <https://data2.unhcr.org/en/documents/details/68448>.

related activities are reported to be quite modest in the three districts concerned. However, the actual amounts allocated are believed to be even smaller due to the government's competing priorities and limited financial resources. Due to the annual public budgeting cycles, the various government entities contributing to environmental issues, and the related administrative complications of having these contributions reflected as co-financing in the project budget, it is suggested that the government contribution be considered as parallel financing.

96. However, this parallel financing from the government is not identified and valued in the proposal, so it cannot be considered as official co-financing. The project sustainability will greatly depend on the ability of the VPO to allocate human and financial resources increasingly over the time of the project to allow the district offices, TFS and other relevant institutions to gradually manage and scale up the project interventions without the financial support of GCF and technical support from NGOs and technical partners. This includes the management and operation of flood and water harvest interventions, the GIS system and the forestry and agroforestry landscapes.

1.5.2. Capacity of accredited entities and executing entities to deliver

97. UNEP will be the accredited entity (AE). The VPO of the Government of the United Republic of Tanzania will act as the lead EE of the project, ensuring alignment with the national policies and plans as well as complementarity with other existing adaptation projects and the country's NAP process. As the lead EE, VPO will oversee overall project implementation and be accountable to the AE for the effective use of resources during project implementation.

98. VPO will execute project activities through different mechanisms including direct execution by PMU, cooperation agreements with district governments and TFS, as well as contracts with a technical partner, service providers and consultants and, if and when required, the engagement of technical experts and national/international NGOs. The district governments, under the supervision of VPO, will execute delegated activities at the district level directly and may be authorized to contract NGOs, Savings Association Management Offices (SAMOs) and service providers. The selection of the technical experts and national and international NGOs will be undertaken with UNEP guidance to ensure that United Nations requirements are met and with the aim of accelerating project implementation.

99. The new project proposal includes the engagement of a technical partner with expertise in landscape approach, agroforestry, afforestation and GIS to provide technical advisory and capacity-building services to district officials, TFS officials and participant communities in the implementation of Output 1 and Output 2. The independent TAP believes that there is a need to have a careful technical and knowledge transfer plan in place to ensure that the national entities end up having the capacity to manage and scale up the project activities after project completion.

100. UNHCR will be the EE for the project activity 2.3 (Implement flood and erosion control within and in the vicinity of the of the Nduta and Nyarugusu refugee camps and the Mtendeli decommissioned camp) and subactivities 3.2.1 and 3.2.2 (Construct micro-rainwater harvesting systems and establish water pumping and irrigation through approx. 50 micro-solar and 172 treadle pumps). Through committed co-financing, UNHCR will implement activities in the camps targeting the refugee population (contributing to activities 2.2 (forestry) and 3.3 (livelihoods)). However, the livelihood support needs to be further explained in the current proposal, which has eliminated investments from GCF in these activities for the refugee camps.

101. MoHA will play a key role in overseeing the implementation of activities in the refugee camps by coordinating the planning and monitoring of camp activities with UNHCR. This will include consultation on the location and design of infrastructure in the camps. MoHA will also

provide technical guidance and oversight on the performance of NGOs and service providers executing activities in the camps.

102. The VPO and UNHCR will be accountable to the AE for the effective use of resources during project implementation. All operating policies and procedures will follow applicable United Nations rules, regulations and policies, which include provisions for financial management and procurement.

103. As the AE, UNEP will oversee the efficient and effective delivery of the project's objectives. UNEP has a broad global portfolio of climate change projects funded through different funds including the Adaptation Fund, the Least Developed Countries Fund, the Special Climate Change Fund, and bilateral funds such as those involving the European Commission. The goal of the climate change-related activities of UNEP in Africa includes supporting countries to reduce vulnerability and build resilience to the impacts of climate change through EbA. The UNEP knowledge base is derived from completed and ongoing projects that support: (i) methods and tools for decision-making; (ii) prioritizing, designing and implementing adaptation interventions; (iii) enhancing climate resilience by restoring vulnerable ecosystems that underpin community livelihoods; and (iv) monitoring the socioeconomic and environmental benefits of adaptation interventions.

104. UNHCR has a long history in managing refugee camps. In the United Republic of Tanzania, it has managed refugee and host community relations with good convening power among the different actors on the ground.

105. UNHCR has been working to address climate change issues in refugee contexts, especially through promoting the inclusion of refugees in local climate change adaptation programming. UNHCR has three field offices in the project area (in Kigoma town, Kasulu and Kibondo), which work with MoHA and local government authorities. In addition to its humanitarian activities, UNHCR also supports local communities with a large portfolio of relevant projects that include forestry, water conservation and sustainable energy.

106. In line with the principles of the Global Compact on Refugees, UNHCR United Republic of Tanzania has forged partnerships with a range of development partners, United Nations agencies and the private sector to address energy and environment challenges in the camps and those faced by the host communities. UNHCR will build on these experiences and partnerships while implementing the project. Project compliance with the UNHCR Policy on Age, Gender and Diversity[12] is mandatory. UNHCR applies an age, gender and diversity approach to all aspects of its work, applying participatory methodologies to promote the roles of women, men, girls and boys of all ages and backgrounds as agents of change in their families and communities.

107. The new institutional arrangement allows for better country ownership as the VPO will work with the district councils that have the mandated authority in land-use planning and development at the district level and will therefore ensure that the project is anchored in local governance processes and that results are sustainable beyond the project duration. VPO will also sign one cooperation agreement with TFS, the authority responsible for forestry interventions and forest monitoring, for the implementation of activities 2.1 and 2.2 in coordination with the respective district councils.

108. NEMC, particularly the NEMC office in Kigoma, will be responsible for ensuring compliance with relevant environmental laws and regulations as well as providing technical advisory and compliance guidance on all environmental aspects of the project to the two EEs and the implementing partners.

[12] <https://www.unhcr.org/protection/women/5aa13c0c7/policy-age-gender-diversity-accountability-2018.html>.

109. The Kigoma Regional Authority will host the PMU, provide support to the delivery of activities in the region and oversee the implementation of activities delegated to the district councils. The Kigoma Regional Authority where the PMU will be located, will convene all technical officers, including land-use planning officers, agriculture officers (representatives of the Ministry of Agriculture and food security at district level), engineering officers, etc., at regular intervals to inform the project's annual workplan and discuss project implementation progress and adaptive management approaches.

110. The project steering committee will be formed during the inception of the project and be responsible for high-level project direction. It will be composed of nominated representatives from the VPO as the national designated authority and lead EE, MoHA, the President's Office – Regional Administration and Local Government, the Ministry of Water and Irrigation, the Ministry of Agriculture and Food Security, the Ministry of Community Development, Gender and Children, the Ministry of Livestock and Fisheries, the Ministry of Lands, Housing and Human Settlements, the Ministry of Tourism and Natural Resources, NEMC, TFS, district government representatives and refugee leaders, UNHCR and UNEP.

111. Overall, the new proposal has a better institutional arrangement, with more country ownership and a mix of national and international specialized entities working together with the national and regional governments to ultimately deliver the project at the community level.

1.5.3. Engagement with civil society organizations and other relevant stakeholders

112. The project design has been based on prior consultations, including with national and district government officials, communities and partners. The first round took place in March 2017 and involved key stakeholders in the project area, including the government, NGOs, implementing partners and potential beneficiaries. The first visit to the three refugee camps also involved focus groups with potential beneficiaries as well as meetings with each camp commander and delivery partners currently responsible for implementation interventions in each camp.

113. The second round of consultations was carried out during the feasibility study visit in September 2018. The purpose of the visit was to assess the technical feasibility of interventions and gather information about necessary management and arrangements structures for the implementation of the project.

114. The third round of consultations took place in November 2018. It involved a series of strategic meetings with stakeholders, including the Permanent Secretary and senior officials within the VPO. The VPO outlined the review process and issuance of a no-objection letter.

115. A fourth round of community consultations with project beneficiaries was undertaken between 7 and 11 September 2020 to validate the project design. Consultations sought to: (i) introduce beneficiary communities to the project and validate the relevance and applicability of project interventions (socioeconomic consultation); and (ii) ensure the validity of the technical and environmental assumptions (technical consultation).

116. The fifth round of consultations started on 9–10 September 2021, when representatives from the VPO, UNHCR, MoHA and UNEP met in person in Dar es Salaam to discuss the project execution arrangements and the necessary project proposal revisions. The parties agreed on the changes in the proposal, as explained in this assessment. Consultations with TFS officers took place on 25 October 2021 to gain further understanding of the forest monitoring systems in place and the existing gaps in terms of equipment and information management systems, as well as to identify the appropriate investments in fire risk management. Between 15 September and 25 October, several meetings took place between VPO, UNHCR and UNEP to review the funding proposal and annexes based on the agreements reached in Dar es Salam and subsequent meetings.

117. However, during 2021, no consultations were undertaken with the communities (both host and refugee communities), and the changes in the project were decided using a more top-down approach.

118. The project has presented socioeconomic questionnaires used during consultation, including the answers by relevant stakeholders. However, it has not presented numbers on the men and women consulted and involved in each focus group.

119. The objective of the stakeholder engagement and social inclusion plan is to engage all relevant stakeholders early in the land-use planning processes. It lists objectives and has a stakeholder engagement table. However, based on the consultation that has already taken place, the plan is very basic and it does not specify the roles of the communities and refugees.

120. The project will need to design a more robust consultation process at the beginning of project implementation and continue the consultations on an ongoing basis, while including women's groups, refugee groups, farmers' groups, implementing partners, supporting organizations (e.g. the Kibondo Beekeeping Cooperative) and district officials.

1.6 Efficiency and effectiveness *Scale: N/A*

1.6.1. Cost-effectiveness and efficiency

121. The total project cost is estimated at USD 23,618,646, of which GCF is expected to provide USD 17,198,340 as grant resources. UNHCR will be the only co-financier, providing USD 4,467,293 of grant resources.

122. Component 1 on VLUPs involves an investment of USD 1,062,805 from GCF, representing a reduction in the costs of developing climate-resilient refugee camp land-use planning that was included in the February 2021 proposal. Component 2, to restore ecosystems and deliver flood reduction infrastructure, will require USD 13,038,743 (USD 9,978,657 from GCF). Component 3, on intercropping systems and water harvest interventions, is estimated to cost USD 7,094,532 (USD 5,687,324 from GCF). Finally, Component 4, on mainstreaming climate change in plans and policies, is estimated to cost USD 469,554 in GCF funding.

123. The new proposal is investing more resources in (i) equipment and training for forest monitoring; (ii) a technical partner to support 10,000 ha of agroforestry and 2,000 ha of afforestation systems; (iii) fire risk management and FFS; and (iv) efforts to increase the coverage of farmers targeted from 16 per cent to 25 per cent, resulting in the increase of agricultural input beneficiaries from 10,000 to 15,000 households and the provision of more improved seeds (revised upwards from 1 kilogram to 20 kilograms).

124. It is noted that the erosion and flood infrastructure for the three camps account for USD 3,084,315, or 13 per cent of the requested amount from GCF. Even though this is not a big proportion of the project, it is important to realize that if the refugee camps are closed, the use of these resources might not be justified on climate change grounds.

125. In general, the budget is well balanced and provides good value for money. The independent TAP enquired about the low costs of the forestry and agroforestry systems. The proponents responded that the costing was based on actual local costs from the field or benchmarked against existing projects.

126. The cost of protecting village forest reserves is estimated at USD 420,000 for 20 villages (estimated at 30,000 ha), or USD 21,000 per village and USD 14/ha. It is noted that these are facilitation costs for establishing reserves and their management arrangements, and that no input and labour costs are included. Therefore, there is a need to understand whether the government will pay for labour costs, and how they are to be accounted for in a parallel budget.

127. A financial analysis of key project activities and subactivities was carried out, along with an economic analysis of the full project. The project proposal presents the internal rate of return for subactivities, with all of them having a positive internal rate of return as shown in the following table.

Table. 1

Project subactivity	Internal rate of return	Increase in net present value (NPV) for scenario with project relative to scenario without project (USD)	
		NPV	NPV per hectare
2.2.1. Establish and maintain (a) agroforestry systems; and (b) nurseries for production of tree seedlings	98%	9,828,099	343
2.2.2. Undertake afforestation on village land to (a) restore critical ecosystem services; and (b) establish a sustainable supply of fuelwood	50%	2,933,973	1,467
3.1.2. Provide inputs for climate-resilient agriculture	541%	5,877,457	19,592
3.2. Increase water availability and use efficiency through water harvesting and efficient irrigation interventions	197%	3,509,447	8,158
3.3.1. Promote beekeeping to incentivize forest conservation in the forests surrounding the camps and villages	Undefined	996,463	N/A
3.3.2. Promote mushroom cultivation for marginalized groups with training and mushroom growing facilities.	787%	553,763	N/A

128. The overall project's economic internal rate of return is estimated to be 13 per cent when the value of greenhouse gas mitigation benefits are excluded and 18 per cent when they are factored in.

129. The project involves an operation and maintenance plan that is well developed, explaining the arrangements to maintain interventions during and after project completion.

130. In terms of project sustainability, the project's most important strategies include:

(a) SAs, which are critical to the financial sustainability and reinvestment in productive interventions in the project. The SAs are proposed as a key delivery mechanism for the project. They allow individuals to pool finances together to save and borrow money based on the consensus of members. This project seeks to strengthen and scale up these existing structures by channelling funds through SAs for the purchase of climate-resilient equipment. However, the SAs will not be offered to the refugee communities, losing out on an opportunity to give them some economic relief.

(b) Long-term land-use plans and landscape interventions; and

(c) Support for the mainstreaming of climate change adaptation in policies, plans and strategies in the United Republic of Tanzania.

131. However, the absence of co-financing from the Government of the United Republic of Tanzania might hinder the sustainability of the interventions after project completion. Moreover, the project will need to design a plan to gradually ensure that the district councils, TFS and other relevant institutions have the technical and financial capacity to manage and scale up the project in order to avoid the dependency on the technical partner and NGOs hired with GCF resources to support the project implementation.

132. Finally, and as explained above, the investments in flood and erosion control in areas that will no longer be refugee camps poses a question relating to the climate and humanitarian rationale of the interventions. Moreover, the reduction of options for the refugee communities, including the lost opportunity for access to SAs and the lack of livelihood alternatives funded by GCF resources, diminished the effectiveness of the interventions to support the adaptive capacity of refugees.

133. In conclusion, the independent TAP raises a point of precaution in relation to the infrastructure investments with GCF resources for temporary refugee camps, and the lack of GCF resources to support livelihood alternatives of refugee communities. Moreover, although the project provides good value for money, the project proponents need to further analyse the sustainability of the project interventions during and after project completion.

II. Overall remarks from the independent Technical Advisory Panel

134. The independent TAP would like to stress the fact that the interventions in the refugee camps will be subject to political decisions around the closure of the camps, as it is happening with the Mtendeli camp. Moreover, the restrictive policies of the Republic of Tanzania towards refugees, might hinder the overall spirit of the project to respond to the climate change impacts of hosts and refugee in an environment where climate, humanitarian and development approaches to problems and solutions intersect. Therefore, with the aim of safeguarding GCF investment in activities to be implemented within the camps' boundaries, the independent TAP points out to the Board that the investments made on flood and erosion control measures might not be validated under a climate-humanitarian nexus rationale, in the event of the camp's closure.

135. The independent TAP recommends that the board approves the project, subject to the following conditions:

136. Prior to the second disbursement of funds by GCF under the funded activity agreement, the AE shall deliver, in a form and substance satisfactory to the GCF Secretariat:

(a) An updated stakeholder engagement plan including:

 (i) Detailed direct beneficiary mapping in the host communities and in the refugee camps situated in the project area, and evidence of the effective consultations and participation of project beneficiaries -drawn from both the host communities and refugee camps- in the project activities;

137. Prior to the third disbursement of funds by GCF under the funded activity agreement, the AE shall deliver, in a form and substance satisfactory to the GCF Secretariat:

(a) A sustainability plan, including:

 (i) the human resources and financial arrangements with governmental bodies at national and sub-national levels, to ensure the effective implementation of the project activities during and after project completion.

 (ii) the benefit-sharing arrangements in host communities and refugee camps that evidence how the benefits derived from the project are accrued and shared among the host community beneficiaries and among refugees within the camps in the project Area.

Response from the accredited entity to the independent Technical Advisory Panel's assessment (FP218)

Proposal name:	Building Climate Resilience in the Landscapes of the Kigoma Region, United Republic of Tanzania
Accredited entity:	United Nations Environment Programme (UNEP)
Country/(ies):	Tanzania
Project/programme size:	Small

Impact potential

We thank the independent Technical Advisory Panel (iTAP) for the review. UNEP is in general agreement with the points raised, with some clarifications, as follows:

1. UNHCR executes activity 2.3, which includes flood and erosion control measures within the refugee camps and surrounding host community villages as well as activities 3.2.1 (micro-rainwater harvesting systems) and 3.2.2 (irrigation) which are implemented in host community villages.

2. As described in Point 7, land use planning processes will not take place in the camps. Participatory plans will be drawn up for the flood and erosion control outlining the locations and mix of grey and green interventions in the camps and immediate surrounding areas.

3. Ministry of Home Affairs (MoHA) and UNHCR rehabilitation plans after camp decommissioning do not have the sufficient scale to address the severe gully erosion problem in Mtendeli camp, which has significantly increased flood risk for downstream communities. This requires an investment to prevent further gully erosion, given climate change. The FP describes how the land degradation associated with the presence of refugee camps has significantly increased exposure of surrounding host communities to climate change risks, particularly flooding, jeopardizing climate resilience of host communities downstream. Therefore, the impact of the camps on the surrounding host population (364,745) remains and needs to be addressed as part of the integrated landscape approach to enhance resilience.

4. The project before and after restructuring was designed to support communities living in the Kigoma region, whether outside or inside the camps, considering the ecosystem connectivity between communities and overall vulnerability to climate change. Overall, USD 14,191,680 (83% of GCF project intervention grant) will be spent on targeted interventions in the host communities within a 10-15 km radius surrounding the camps. The project will directly benefit 364,745 people in the host community (64% of the target population) and 205,596 people in refugee camps (36% of the target population).

5. The type and coverage of livelihoods is contained in the logical framework in the FP (e.g 15,000 people practicing climate-resilient farming). UNHCR co-finance for livelihood support to refugees under Output 3 is estimated at 1,047,183 USD in the Detailed budget (Annex 4).

6. The amount of project funding for the refugee community remains the same. The smaller GCF grant to refugees is compensated by a higher investment co-financing by UNHCR for the refugee community. UNHCR continues to hold a high interest in disseminating learning and

benefits from this integrated landscape approach to inform its humanitarian programming globally. *7.* The FP sets out the climate rationale for the camp areas and surrounding communities. The camps are at the top of the watershed. Hydraulic analysis shows that flooding in Kizuvu, Kasanda and Kakonko Districts occurs primarily a result of precipitation falling within the camp area itself and flowing down. Both host communities and refugee camps reported the occurrence of flooding events and subsequent riverbank and other land erosion. Current land use state under the RCP4.5 2050 precipitation future shows an average of ~4% additional inundation area. Under the no adaptation scenario, this increase will be 4.9% or 380 ha of additional flooding (Annex 2, section 2.2.2). The expected adaptation benefits for host population surrounding Mtendeli include: • Flood risk reduction in villages located downstream of the degraded Mtendeli camp area. • Protection of downstream agricultural lands and soils from further flooding, erosion and siltation, thereby improving food security. • Reduced siltation in surface water resources in the catchment thus improving water quality. • Protection of roads and services infrastructure in the decommissioned camp area and in downstream villages. • Opportunities for expansion of climate resilient productive activities in the restored former camp area. *8.* Nduta has a capacity for 130,000 refugees and with the consolidation of Mtendeli refugees the camp population is estimated at 80,000 refugees, therefore still under capacity. UNHCR co-finance will support same small-scale livelihood activities that promote refugee households' food security in Nduta camp, including refugees reallocated from Mtendeli. Only the support to the saving associations in the camps with GCF grant funding has been removed in the revision.
Paradigm shift potential
1. There are currently 205,000 refugees in the camps covered by the project. In the unlikely event of refugee numbers dramatically decreasing, closure of the camp(s) and re-allocation of the land to the host communities or change to other land use, project activities are designed to anchored to the landscape and benefit communities living in that landscape. *2.* It is expected that farmers will use the Savings Associations loans to purchase seedlings from the existing tree nurseries once they realize the benefits of including trees in their farming systems. Furthermore, to promote livelihood activities that preserve forestry resources the project will support beekeeping groups in improving product quality, packaging and marketing to facilitate access to the existing honey value chains in the Kigoma region. *3.* The original purpose of the investment is to enhance resilience of communities at the landscape level and in so doing to address the climate adaptation needs of both refugees and host communities living in that landscape. Flood and erosion control interventions in the decommissioned Mtendeli camp area address the most critical adaptation need (reduced exposure to flood risk) of host communities located downstream of the camp. *4.* The long-term assumption is that Tanzania will continue to be refugee hosting country. As a stable and peaceful country in a volatile region, Tanzania's long history of hosting refugees is expected to remain intact for the future. With regards to the current population, while ongoing repatriation to Burundi supported by both governments is ongoing, there is a reasonable

assumption and shared understanding that a large refugee population will remain in the foreseeable future. *5.* Skills and capacity will be strengthened with the inclusion of a technical partner, a chief technical advisor, a water resources expert and a climate resilient land use planning expert that will provide continuous technical advisory and skills transfer. *6.* The integrating factor between host and refugee communities is ecosystem connectivity and services to mitigate floods and droughts. Livelihoods, forestry and flood and erosion control components within the refugee camps and surrounding areas have been maintained in the revised proposal. UNHCR remains committed to strengthen approaches and interventions that cut across the climate-humanitarian-development nexus. Experiences and lessons learned from this project would feed into wider organizational efforts to expand understanding and impact in this domain.
Sustainable development potential
With respect to the comment in paragraph 73 *(i) women are represented, and participate meaningfully, in land-use planning at the village and camp levels*: Kindly note that land use planning will not take place at camp level.
Needs of the recipient
Repatriation is encouraged but voluntary and remains regulated by the Tripartite Commission including the Government of Tanzania, Government of Burundi and UNHCR. The closure of Mtendeli camp was motivated by resource efficiency for service provision in the camps considering the current reduction in refugee population in both Mtendeli and Nduta camps.
Country ownership
The project is firmly anchored in government structures and ownership. The project will establish a team working through Government: • Vice President's Office, Division of Environment (DoE) which has the mandate to implement climate change actions in the country and is the designated lead agency for environment and climate change-related activities in the country. • Ministry of Home Affairs (MoHA) will play a key role in overseeing the implementation of activities in the refugee camps • The Regional Administration, which operates under the President's Office, Regional Administration and Local Government (PO-RALG), oversee the Local Government Authorities' (LGAs) activities providing supervision and administrative instructions. • Kibondo, Kakonko and Kasulu districts and all technical district officers including land use planning officers, agriculture officers, engineering officers, etc. at regular intervals, to inform the project annual workplan and to discuss project implementation progress and adaptive management approaches. • The Tanzania Forest Service (TFS), a semi-autonomous organisation under the Ministry of Natural Resources and Tourism. • The National Environmental Management Council (NEMC) will ensure compliance with relevant environmental laws and regulations as well as

providing technical advisory and compliance guidance on all environmental aspects of the project.

The Government of Tanzania foresees ensuring sustainability thanks to the enhanced capacity of local stakeholders built by the project as well as provisions to the District budgets towards financial sustainability of interventions requiring government funding, both during and beyond the project duration. This will be detailed in the sustainability plan (condition B).

Efficiency and effectiveness

1. Kindly note the total grant requested to GCF is $ 19,007,353 USD and UNHCR co-finance amount is $4,611,293

2. In the unlikely event that the refugee camps do close, the flood and erosion control infrastructures will be absorbed by the host communities. Please refer to the responses 4 and 7 under 'Impact Potential'.

3. Regarding the development of technical and financial capacity to manage and scale up the project, please refer to response under 'Country Ownership'. The shift towards greater involvement of the government through direct execution is believed to be a strong factor in ensuring future sustainability of the project. In-kind and cash contributions from the Government of Tanzania have not been quantified in the proposal as co-finance commitment at the time of the proposal, however, government contributions (ex. human resources time and District budget allocations) will be provided to enable project implementation and future sustainability of interventions. Please refer to response under 'Country Ownership'.

4. The proposal has a strong climate and humanitarian rationale. There are clear adaptation benefits to both the refugee and host communities. Please refer to responses 6 and 8 under 'Impact Potential' and responses 4 and 6 under 'Paradigm Shift'.

5. The recommendation to further analyse the sustainability of the project interventions after project completion is well received and will be addressed in the sustainability plan (in response to the proposed ITAP condition B).

Overall remarks from the independent Technical Advisory Panel:

We note the iTAP's concern related to the closure of the Mtendeli camp. Kindly note that the Mtendeli camp closure responds to a change in the size of the refugee population and the need to optimize limited humanitarian resources. Furthermore, we note that the refugee population is subject to increase or decrease as a result of changes in the regional context. Tanzania is expected to continue to play an important role as a refugee hosting country, and the full closure of all camps in Kigoma is not foreseeable.

We thank the ITAP for its careful review of the funding proposal and we accept the following conditions.

(a) Prior to the second disbursement of funds by GCF under the funded activity agreement, the AE shall deliver, in a form and substance satisfactory to the GCF Secretariat:

(i) An updated stakeholder engagement plan including:

1) Detailed direct beneficiary mapping in the host communities and in the refugee camps situated in the project area, and evidence of the effective consultations and participation of project beneficiaries -

drawn from both the host communities and refugee camps- in the project activities;

(b) Prior to the third disbursement of funds by GCF under the funded activity agreement, the AE shall deliver, in a form and substance satisfactory to the GCF Secretariat:

(i) A sustainability plan, including:

1) the human resources and financial arrangements with governmental bodies at national and sub-national levels, to ensure the effective implementation of the project activities during and after project completion.

2) the benefit-sharing arrangements in host communities and refugee camps that evidence how the benefits derived from the project are accrued and shared among the host community beneficiaries and among refugees within the camps in the project Area.

Gender Documentation for FP218

Building climate resilience in the landscapes of Kigoma Region, Tanzania

Annex 8

Gender Assessment and Gender Action Plan

FP0009_UNEP_Tanzania_2023 06 30

Contents

1. Introduction

1.1. The context: gender issues in Tanzania

Gender inequality is a salient issue in Tanzania. Over the last decades, the government has enacted and promoted a series of legislation and policies aimed at achieving gender equality. These policies have focused on greater representation of women in government and business, the importance of gender equality to the country's development and land rights for women. However, implementation has been challenging, due in large part to cultural norms and the application of customary law.

There is significant discrepancy in the economic roles of men and women. Although the overall labour force participation rate (including the informal sector) of women in Tanzania is 88.7%, only 29% of women participate in the formal economy. Gender segregation is ubiquitous, with women accounting for the majority of the workforce in the agriculture (52%) and the trading sector (55%) whilst being in the minority in the manufacturing, construction, transport and finance sectors. Important gender differences persist in the remuneration of workers, whereby only 4% of employed women in Tanzania are in paid jobs in either the formal or informal sector and women on average earn 35% less than their male counterparts.

Women are also participating in low productivity sectors, specifically women account for the majority of the agricultural workforce, with approximately 90% of women living in rural areas participating in the sector. However, returns in the agriculture sector have declined over time, with the real growth rate of the sector declining from 7.5% in 2008 to 3.4% in 2013. Low productivity in the sector is largely a consequence of inadequate investment and lack of access to farm inputs, extension services, credit, modern technology application, trade and marketing support. Heavy dependency on rain-fed agriculture and unsustainable use of natural resources also play a role in reducing productivity[1].

Presently in Tanzania, women are responsible for the majority of domestic work. Men have traditional expectations of women's role in household chores. Women who have jobs outside of the home are still expected to take responsibility of domestic work. This gendered division of labour begins early, with young girls, even those in school, being expected to shoulder their share of household chores. The same is not expected of their brothers.[2] Women in Tanzania spend 28.8% of their time on unpaid care work whereas men spend 9% of their time on such work.[3]

[1] Hali ya Uchumi wa Taifa katika Mwaka 2014; Bank of Tanzania Annual Report 2013/14
[2] Fienstein et. al, "Gender Inequality in the Division of Household Labour in Tanzania," *African Sociological Review*, Vol. 14, No. 2, 2010. Available at: https://www.ajol.info/index.php/asr/article/viewFile/70239/58428
[3] UN Joint Programme for Kigoma

25% of Tanzanian households are headed by women, approximately half of whom have been widowed. In rural areas, 24% of all households are female-headed versus 27% in urban areas.[4] Female headed households (FHHs) have a larger number of dependents than male headed households (MHHs) with an average of 1.6 dependents in rural areas as compared to 1.3 for MHHs.[5] The wellbeing of a household is linked with the education level of its head; higher levels of education correlate with better wellbeing conditions. This is critical for rural women, 49% of whom have no educational attainment at all, as compared to 40% of rural men.[6] Because FHHs in Tanzania typically have lower levels of education than MHHs, FHHs have poorer wellbeing conditions. This pattern continues for members of FHHs, who have, on average, fewer years of education than members of MHHs.[7]

Women in Tanzania face additional prejudice and, in some cases, violence. According to UN Women, discrimination, abuse and Violence Against Women and Girls (VAWG) is rife in Tanzania, owing to "patriarchal and traditional norms." Two out of 15 women between the ages of 15 to 49 have experienced physical violence and one out of ten women of the same age range has experienced rape.

In terms of needs, women farmers require increased access to capital and educational resources to increase their adaptive capacity. Land tenure is also a critical limitation on women's ability to become climate resilient.

1.1.1. Kigoma region

Kigoma is among the poorest regions in Tanzania, with a poverty rate of 34,5%.[8] 76% of the population are farmers.[9] Many of the trends seen at the national level are evident in Kigoma. The percentage of FHHs in Kigoma is higher in both rural and urban areas at 34.8% and 38.5% respectively.[10] The average size of FHHs is 7.8 as compared to 4.0 for MHHs. The 2011/2012 Tanzania Household Budget found that larger household sizes correlate with increased poverty rates, suggesting that FHHs in Kigoma are likely to be poorer than MHHs.[11] The literacy rate is higher amongst men (71%) than women (67%).[12]

Nearly 25% of people in Kigoma have never attended school. There is a significant gender gap with regards to school attendance. According to the 2012 census, 28.1% of females in Kigoma have never attended school versus 21.5% of males. 28.1% of girls were attending school as compared to 32.6% of males.[13] National level trends suggest that educational attainment is

[4] Osorio et al., "Gender Inequalities in Rural Employment in Tanzania Mainland," 2014. Available at: http://www.fao.org/3/a-i4083e.pdf
[5] Ibid.
[6] Ibid.
[7] Ibid.
[8] Tanzania National Bureau of Statistics, Household Budget Survey 2017/18
[9] Ibid.
[10] Kigoma Regional Profile
[11] Ibid.
[12] Ibid.
[13] Ibid.

positively linked to wellbeing at the household level.[14] Thus it is likely that FHHs will have lower levels of wellbeing than MHHs.

Adherence to traditional gender roles results in women and girls assuming responsibility for domestic work, whereas, men are recognized as breadwinners expected to provide for their families. Women and girls are responsible for the collection of water and firewood as well as farming activities. Consequently, their risk of exposure to sexual and gender-based violence is high. The risk increases as these resources become scarcer, requiring women and girls to travel ever greater distances.[15] Additionally, dependency on firewood for cooking fuel negatively impacts women. Inefficient cooking practices result in poor indoor air quality and affect the health of all household members, but especially women, who are responsible for food preparation[16] and the related challenge of obtaining cooking fuel, often in the form of fuelwood.

1.2. Legal and administrative framework for women and gender equality

At the policy level, Tanzania has demonstrated a significant commitment to achieving gender equality. In 1990, the Ministry of Community Development, Gender and Children (MCDGC), formerly known as the Ministry of Community Development, Women Affairs and Children (MCDWAC) was established.

Over the following years, a number of policies and legislative acts have been passed to this effect, emphasizing non-discrimination and the use of affirmative action. These include:

- **The Land Act and Village Land Act of 1999** repealed customary and traditional practices with the aim of increasing gender equality in land tenure, use and management.
- A year later, in 2000, **legislation was passed to increase the number of women in government positions** to 30% by 2005. At present, women hold 36% of parliamentary seats and 20% of ministerial positions.[17] 33% of local government seats are also reserved for women.[18]
- The **Small and Medium Enterprise (SME) Development Policy of 2003** highlights the significance of women's role in SMEs and includes provisions to support women entrepreneurs.[19]
- Two key poverty reduction strategy papers, **MKUKUTA I and II** (2005 and 2010, respectively), recognized the significance of gender equality and women's

[14] Osorio et al., "Gender Inequalities in Rural Employment in Tanzania Mainland," 2014. Available at: http://www.fao.org/3/a-i4083e.pdf
[15] UN Joint Programme for Kigoma
[16] Ibid.
[17] World Bank, Tanzania, *Gender Data Portal*, http://datatopics.worldbank.org/gender/country/tanzania
[18] Ministry of Health, Community Development, Gender, Elderly and Children, Women Political Empowerment, http://www.mcdgc.go.tz/index.php/issues/women_political_empowerment_and_decision_making
[19] UNIDO, *Tanzania SME Development Policy 2003*, 2012. Available at: https://open.unido.org/api/documents/5403996/download/TANZANIA%20SME%20DEVELOPMENT%20POLICY%202003%20-%20%E2%80%9Cten%20years%20after%E2%80%9D%20-%20Implementation%20Review

empowerment and identified both as key national development concerns. Additionally, the plans include gender-specific objectives to reduce poverty and achieve equality.

- The **Five-Year National Development Plan** emphasizes women's economic empowerment as a driver of economic equality between genders.
- **Tanzania Vision 2025** highlights equality between genders, in reference to Tanzania's Constitutions. Additionally, equality between genders is recognized as a key strategy for achieving the path outlined in the document.

The Tanzanian government has undertaken a number of additional activities to promote gender equality. Gender responsive budgeting was introduced with the aim of ensuring that public funds advance gender equality. In 2010, Tanzania became a signatory to the Southern African Development Community Protocol on Gender and Development, which calls for 50/50 representation in all decision-making bodies.[20] A five-year National Plan of Action to End Violence Against Women and Children (NPA-VAWC 2017/18 – 2021/22), has been developed by consolidating eight different action plans addressing violence against women and children to create a single comprehensive, National Plan of Action to eradicate violence against women and children in the country. The NPA-VAWC addresses the problem of violence against women and children by focusing on building preventive systems and responding to the needs of survivors. The implementation of this Plan is the direct responsibility of the Ministry of Health, Community Development, Gender, Elderly and Children (MoHCDGEC).

The National Plan of Action to end Violence against Women and Children in Tanzania envisages to include all persons in Tanzania including refugees. Under this initiative, Gender Desks manned by the Tanzania Police Force have been established in all 3 refugee camps and Social Welfare Officers have been deployed from the District to support case management in both the camps and surrounding host community villages.

1.3. Gender issues in land, agriculture and forestry

1.3.1. Gender and land access

Access to land and productive resources are essential to empower women economically. However, in Tanzania, women continue to face discrimination in land rights, in spite of positive reforms. Though laws may not formally discriminate against women, societal norms interfere with their implementation. Gender discrimination in land access is evidenced by low levels of land ownership amongst women: only 20% of Tanzanian women owned land in their own names.[21] Women who do own land often have smaller plots and less livestock. They also have less access to critical financial, educational and technological resources.

The Village Land Act (VLA) was introduced in 1999 with the aim of negating customary laws that discriminated against women's land tenure. The VLA breaks new ground in women's rights with Section 3(2) and Sections 3, 18, 22, and 20(2) rendering as invalid any customary practices that discriminate against women. It also states (Section 3(2)): "The right of every

[20] United Nations Tanzania, "Gender Issues". http://tz.one.un.org/who-we-are/7-un-programmes/87-gender?showall=&start=2

[21] Ibid.

woman to acquire, hold, use and deal with, the land shall be to the same extent and subject to the same restrictions treated as a right of any man." The Act also voids the role of customary law in determining land tenure and it recognized women's rights to land in the event of the death of her husband or upon divorce. There are also requirements for female representation in key decision-making bodies. In the Land Tribunal Act (No. 2/2002) and its regulation (of 2004), it is clearly stated under Section 5 that at least three of the seven members of a Village Land Council should be women. The Land Use Planning Act states that land adjudication committees should be composed of at least four female members out of nine, and there should be at least 25% female representation on VCs (as in the Local Government Act).

Despite a supportive policy environment, Tanzanian women continue to struggle to access land rights. Cultural and religious institutions tend to favour men and may perpetuate the view that men are the sole owners of the land and that they should control the income generated from it. Should a woman attempt to claim her right to land, household and community conflicts may arise, discouraging women from seeking such rights.[22] Social norms perpetuate the patriarchal view that men are the primary decision-makers and as such, should have disproportionate influence in land related decisions and practices related to ownership and control of the land.

Insecure land rights offer women little incentive to improve the resilience of their agriculture.[23] Without any assurances, women are hesitant to make investments that would enhance their land's productivity.

1.3.2. Gender and agricultural activities

The agriculture sector is a mainstay of Tanzania's economy, employing nearly 67% of the labour force.[24] 90% of rural women are employed in agriculture and they are the main producers of food and cash crops.[25] [26] In all regions in Tanzania, rural women have lower earnings than rural men despite being more active within the sector.[27] Nearly 50% of women work as unpaid family labour in agriculture.[28] Agriculture accounts for nearly a quarter of Tanzania's GDP. The sector is underperforming in part due to prevailing gender inequality in land rights and in access to and control of resources (assets, capital, extension services, etc.). Farming is practised by men and women of all ages, but due to the physical nature of work, elderly people, and people with health problems do not typically participate.

Women in Tanzania are more vulnerable to climate change than men, especially within the agricultural sector. This is due to the gender roles and norms but also because of a dependency on rain-fed agriculture. Women's vulnerability to climate change is also exacerbated by their central role in water collection. Increased water scarcity will increase the

[22] UNA Tanzania, "Land rights in Tanzania- A Gender Issue," http://una.or.tz/land-rights-tanzania-gender-issue/
[23] Ibid.
[24] CIA, "Tanzania," *The World Factbook*, https://www.cia.gov/library/publications/the-world-factbook/geos/tz.html.
25 UNA Tanzania, "Land rights in Tanzania- A Gender Issue," http://una.or.tz/land-rights-tanzania-gender-issue/
26 https://www.climatelearningplatform.org/sites/default/files/resources/tanzania_country_climate_risk_assessment_report_-final_version.pdf
27 UNA Tanzania, "Land rights in Tanzania- A Gender Issue," http://una.or.tz/land-rights-tanzania-gender-issue/
28 UN Joint Programme for Kigoma

time women spend traveling to collect water and result in opportunity costs, including less time to attend school, vocational training, or engage in income generating activities.[29]

The economy and livelihoods of the population in Kigoma region rely on natural resources, in particular agricultural production. About 90% of the region's population, including refugees, are engaged in rainfed, subsistence agriculture. Farmers groups in Kigoma region, especially in the targeted districts, are generally in need of further capacity development and extension support. The group leader is usually male, whereas supporting leadership (treasurer and in some cases the secretary) are women. Some groups supported by NGOs have more female membership - especially those cultivating cassava.

As noted, refugees also engage in agriculture activities. Within the camps, land plots allocated per household are small. Families engage in small gardening projects around their plots to supplement monthly food rations received. Home gardening in the camp has been restricted to kitchen gardening whereby maize crops are banned. Women play a key role in tending to the kitchen gardens both by way of taking charge of gardening operations including weeding, harvesting, post-harvest handling (including drying) and to a small extent sale/exchange of the excess food that is not consumed by the household. Irrigation and watering of the gardens is also largely carried out by women in the household. They are supported to practice innovative ways of producing vegetables through the use of sacks/ basins and kitchen gardens to improve dietary diversity.[30]

Refugees also engage in agriculture activities outside the camps. As Tanzania follows a strict encampment policy, these are informal land use/land lease arrangements which mainly involve renting a farm or farming for Tanzanians. Refugee agricultural labourers include both men and women and Congolese and Burundians.

1.3.3. Gender and forestry

Women tend to be underrepresented in forest management activities.[31] Gender relations shape patterns that in turn reinforce women's vulnerability to deforestation and forest degradation. In order to develop appropriate interventions such as Community Based Forestry (CBF) that address the location-specific vulnerabilities and are context specific, there is need to understand how men and women are affected by deforestation and degradation[32]. Despite the accepted role that gender plays in determining a person's vulnerability to deforestation and forest degradation in Tanzania there has been limited emphasis specifically on the linkages between forestry and gender. The hardship brought by deforestation and forest

29 Ibid.

30 A full-time agronomist is employed by UNHCR's implementing partner Danish Refugee Council to provide technical support, conduct demonstration gardening sessions and worked closely with project animators (whose main role are to educate and sensitize the refugee community on the importance and management of kitchen gardens, including disease management).

31 Sellers, "Gender and Climate Change: A Closer Look at Existing Evidence," November 2016. Available at: https://wedo.org/wp-content/uploads/2016/11/GGCA-RP-FINAL.pdf

32 FAO (2017). Training Manual on Community Forest Management and Diversification

degradation affects men and women differently and exacerbates the already existing inequalities between the sexes[33].

In the host community villages where land use planning has been undertaken, women are represented in village environmental management committees. With regard to ongoing tree planting and forest related activities in the refugee camps, these are undertaken in close collaboration with women. Through annual distribution of seedlings, women are involved in the distribution plans as well as in the tree seedling preparation and planting phases.

As noted, refugee women bear the primary responsibility of collecting fuelwood. In host communities it is also the women are mostly involved in the collection of fuelwood[34] and other non-forest timber products (NFTPs) such as mushrooms. In some cases these activities serve as an adaptation strategy or as an effort to secure an additional income when normal income generating activities cannot be undertaken. Men, however, are becoming increasingly involved in NFTP collection, as their traditional livelihood activities are under increasing climate-related stress.[35]

Women also face cultural barriers when engaging in forest-related livelihood activities, such as beekeeping. As part of Enabel's work in the Kibondo area, women have been supported to become beekeepers. Some of them raised issues related to cultural prejudices towards women engaging with hives, as traditionally they are believed to damage bees by touching the hives.[36]

During community consultations conducted in September 2020, many of the women, from both populations, expressed interest in the forest-related activities aimed at supporting sustainable alternative livelihoods and were enthusiastic about the idea of mushroom farming and beekeeping. In a number of the group discussions, women expressed their particular enthusiasm for mushroom farming. When probed they indicated that beekeeping is considered an activity more likely to be undertaken by men.

1.3.4. Access to finance, information and decision-making

Rural women in Tanzania have limited decision-making power and access to credit, and thus face challenges investing in resilience measures.[37] Only 13.2% of FHHs have access to credit, as compared to 86.7% of MHHs.[38] Women dominate the informal economy, resulting in their limited control over land. The resulting inability to provide collateral affects their ability to secure finance. At the village level, women are involved in women-only village savings associations.

33 Sangeda et al. (2016). Understanding the Driver-Commodities – Gender Nexus in Tanzania. Project Report. Sokoine University of Agriculture, Tanzania and IIED, UK.
34 National Forest Resources Monitoring and Assessment of Tanzania Mainland (2008)
35 Ibid.
36 Source: Enabel's video https://www.youtube.com/watch?v=lInw5pZN71k
37 Irish Aid, Resilience and Economic Inclusion Team, Policy Unit, *Tanzania Country Climate Change Risk Assessment Report*, February 2018. Available at: https://www.climatelearningplatform.org/sites/default/files/resources/tanzania_country_climate_risk_assessment_report_-final_version.pdf
38 Ibid.

Rural women have limited access to educational and technological resources. Literacy rates are lower in rural areas of Tanzania, with 39% of women being illiterate and 23% of men.[39] 49% of rural women have no educational attainment at all, as compared to 40% of rural men.[40] Though there is evidence to suggest that there are equal levels of participation in climate change awareness raising activities, discrepancies in literacy levels and ability to access information between genders suggest that "it is unlikely that there is equality in awareness."[41]

During community focus group discussions notable variations were evident in the level of participation in and the financial contributions and gains obtained from the savings associations. In the host communities, many respondents indicated to be members of one of the numerous savings associations. Particularly women appeared to be well represented in these savings groups. Amounts obtained through the savings groups in some cases are quite substantial with women indicating to have received amounts of up to TZs 800,000. The finances obtained through the savings associations are frequently spent on expenses such as school fees, household expenditures, small business enterprises and agriculture inputs.

The picture obtained in the camps differs slightly. For instance, in Nduta camp the men interviewed were notably more actively participating in the savings associations than women. There may be some cultural aspects at play whereby one hypothesis is that saving associations in Tanzania were traditionally initiated and used by women only. Tanzanian men may give money to or allow their wives to invest their money in savings associations, whereas refugee men may prefer to be participating directly themselves. In the camps it was noted that in some cases, mostly by the men, substantial investments were made, though not at the scale as noted in the host communities. Investments made included agriculture activities (e.g. land leasing) and livestock trading. Women in Mtendeli remarked that the importance and strength of the savings associations has decreased substantially since the closure of the common markets in early 2019. The women explained that food rations are sometimes monetized to be able to participate in the savings associations. In some cases, women pool portions of their rations and take turns in selling the food thereby obtaining some cash. Recent cuts in WFP food rations are expected to further erode the functioning of these savings groups, without additional interventions to support livelihoods.[42]

The community leadership structures in the camps are yet to achieve gender parity in decision making positions even if the current refugee leader in Nyarugusu is a Congolese woman. At the household level, decision making power also tends to be skewed in favour of men. Community consultations revealed that intimate partner conflicts occur with some regularity between husbands and wives on how household money is being spent. Refugee women in Mtendeli camp remarked on the impact of life skills trainings that the refugee community, both men and women, had received, stating that this improved women's position in the household and decision-making power. Women in both host community villages and within

[39] Osorio et al., 2014
[40] Ibid.
[41] Hepworth, *Climate Change Vulnerability and Adaptation Preparedness in Tanzania*, 2010. Available at; http://www.tzdpg.or.tz/fileadmin/_migrated/content_uploads/TZ_CC_Adaptation_Preparedness_-HBS_2010_02.pdf
[42] As of November 2020, WFP food rations in the camps stand at 72%

the camps indicated that should their incomes increase, or even surpass that of their husbands, they do not fear strained relationships or increased violence within the household.

1.3.5. Risks of Sexual and Gender-based Violence

Kigoma is home to three refugee camps: Nduta, Mtendeli and Nyarugusu. As of September 2021, the camps host 205,596 refugees, 78 percent of whom are women or children and just over half (55 percent) are children. According to UNHCR 54.3 percent of refugee households in Tanzania are female headed households.[43]

The major economic activity within the camps is the selling of food rations and non-food items received from aid agencies. There is some skills-based economic activity in the camps which includes tailoring/embroidering, barbering and hairstyling, cooking, soap-making, repairs, carpentry or masonry (done primarily by men) and small trade (women).[44] Refugees also undertake labour-intensive activities, which are largely centred around agriculture. Both women and men engage in farming.[45]

Despite efforts by UNHCR and its partners, refugee women and children continue to be exposed to a range of risks including SGBV, child abuse, early and forced marriages. Some of these risks relate to existing negative cultural norms and practices related to power imbalance that bring about gender inequality. Gender discrimination often sustains systems that perpetuate GBV against women and girls in the refugee camps. Typical culturally driven harmful traditional practices include physical abuse (wife beating) as a form of disciplining, child marriages, and forced marriages among others. UNHCR GBV reports between January and September 2020 indicate over 1,700 women and girls suffered from different forms for GBV including physical abuse (intimate partner violence), rape, emotional/psychological abuse, denial of resources and opportunities and forced/child marriages. The risk of GBV are faced both within and outside of the refugee camps. UNHCR and partners have established GBV multisectoral referral systems to support all GBV survivors based on their needs.

Refugees living in the camps are particularly exposed to violence, including GBV when they leave the camps to search for work, to trade, or to gather firewood in the villages near the camps. In the zones around the camps both host communities and refugees fear and experience occasional violence, leading to conflict and mistrust, although this is more acute for the Burundians than for the more-established Congolese. Refugee children inside the camps are further at risk of GBV, sexual exploitation and abuse, and early or forced marriage due to the congested living conditions especially in collective shelters, and the limited opportunities for post-primary education. The existing restrictions on freedom of movement, restrictions on cash-based interventions and the limited livelihood opportunities have further complicated the search for self-reliance among refugees, consequently increasing their vulnerability. Women and girls may become compelled to engage in risky coping mechanisms or behaviours (such as sex work/transactional sex, illegally travelling outside of camp boundaries, or engaging in prohibited livelihoods) to support their households, maintain food and energy security, and obtain necessary resources. Firewood collection is considered a

[43] UNHCR Tanzania Refugee Situation Statistical Report as of 30 September 2021
[44] Socioeconomic Assessment of North West Tanzania
[45] Ibid.

particular risky activity and exposes women to protection issues, especially SGBV. Firewood collection is a gendered activity and is mostly undertaken by women. Although refugees are technically only permitted to travel within a 4km buffer around the camps to collect wood fuel, these areas cannot presently meet the supply needs for the camp populations, forcing women to travel further beyond this boundary and exposing them to risk of GBV. Women engaged in collection risk violence, torture, rape and murder perpetrated by locals.[46] A 2015 field report from Refugees International found that 25% of 224 GBV incidents reported by Burundian refugees in Tanzania over a five month period occurred during firewood collection trips, and 90% of these incidents were rape or sexual assault.[47] Although culturally a gendered role, women refugees are also forced into this activity as they pose a lesser risk to host communities, whereas male refugees collecting firewood are perceived to be 'stealing' and may be violently attacked. Men who sometimes accompany the women during firewood collection for security purposes, rarely report any assaults as culturally this would make them seem weak and incapable of defending themselves as well as their loved ones. As such, statistics in this regard are not readily available.

Within the camp setting, a number of other factors can be identified that exacerbate the risk of GBV amongst refugees, which include:

- Alcohol use and abuse
- Family conflict and intimate partner violence, particularly where there is dissent about voluntary return to country of origin
- Households practicing polygamy
- Poor lighting in camps leading to insecurity
- Restrictions of freedom of movement and livelihood activities, which contribute to negative coping mechanisms such as sex work and transactional sex
- Empowerment projects where female household members begin to earn money, or save through VSLA, which can trigger inter-partner violence from men

Peaceful co-existence meetings are held on a quarterly basis between representatives from the refugee community and the host communities. These meetings systematically address GBV issues and incidences, often related to fuelwood collection, and have contributed to sensitizing the host and refugee communities and mitigating any conflict that could jeopardize the relationships between the two communities. Issues relating to the apprehension of perpetrators also feature in the peaceful coexistence meetings as the local law enforcement officers with jurisdiction outside the camps are also invited to these forums. Camp law enforcement officers are also present during these meetings to ensure cooperation with local authorities.

Preventive measures in place include promotion of lockable doors in shelters use of partitions to improve privacy, proper design of sanitary facilities, and engaging men and community leaders in creating safe spaces for women's participation in economic programs. Increase in food rations are reported to reduce engagement in risky coping strategies. While limited at

[46] Ibid.

[47] Vigaud-Walsh, *Women and Girls Failed: The Burundian Refugee Response in Tanzania*, 2015. Available at: http://static1.squarespace.com/static/506c8ea1e4b01d9450dd53f5/t/5678aee07086d7cddecf1bab/1450749707001/20151222+Tanzania.pdf.

the moment, long-term investments in livelihood activities for income generation and alternative fuel sources are expected to reduce SGBV risks.

Inter-agency standard operating procedures and protocols (SOPs) exist for managing risks of sexual and gender-based violence (SGBV)[48] and sexual exploitation, abuse, and harassment (SEAH)[49]. The SEAH protocol covers roles and responsibilities, task force membership, procedures for receiving complaints and reports, conducting investigations, confidentiality, reporting to national authorities, interagency code of conduct and other processes.

A community-based feedback mechanism has been set up in all the camps where help desks are staffed by different humanitarian partners. Refugees can report incidents as well as obtain information on services and assistance. Complaints can be made in person, in writing, or by email depending on the preference of the complainant. Complaint boxes have been established in all three camps to receive feedback from persons of concern and are managed by UNHCR and Partners. Referral pathways to address incidents including feedback from complaint boxes have also been established and are operational.

Data is collected in an information management system. Incidents of sexual exploitation and abuse have been reported by persons of concern where the perpetrators are security personnel. Incidents that involve humanitarian workers and refugee incentive workers are also treated as sexual exploitation and abuse. UNHCR is also starting to develop disability inclusion in SGBV programming, such as developing tools to support non-verbal communication with a view to make SGBV referral pathways more accessible to people with disability.

Moreover, UNHCR activities prioritized for 2021 include mobilizing the refugee community in awareness-raising campaigns on SGBV prevention and response. In addition, capacity building will be provided to police, judicial and partner staff to strengthen Survivor’s access to legal recourse including mobile courts. To address psycho-social needs SGBV survivors will be provided with counselling. These are parallel activities that will be done by UNHCR, and that will be complemented by the gender action plan in the project by reinforcing messages on SGBV in training activities and increasing awareness on incident reporting mechanisms.

Outside the camps in host community settings, the Kigoma Joint Programme has undertaken and planned under the theme “Violence Against Women and Children (VAWC)”, under the leadership of the International Organization for Migration (IOM), UN Population Fund (UNFPA), UNICEF, and UN Women.

The VAWC theme obtained funding of $1,129,617 in 2019-2020. Under this funding, the main achievements include:

[48] The inter-agency standard operating procedures for prevention and response to gender-based violence in Nyarugusu, Nduta and Mtendeli refugee camps, Tanzania, was updated in April 2018 and is periodically revised.

[49] The inter-agency protocol for protection from sexual exploitation and abuse was endorsed on October 24, 2018. Parties to this protocol include UNHCR, WFP, UNICEF, IOM, Plan International, Medecins Sans Frotieres, International Rescue Committee, Caritas Kigoma Diocese, ICRC, Norwegian Refugee Council, Danish Refugee Council, OXFAM International, Good Neighbors, World Vision, Water Mission, Tanzanian Red Cross Society, HelpAge International, African Initiatives for Relief and Development, Tanzanian Ministry of Home Affairs, Women’s Legal Aid Center, REDESO, Baba Watoto Centre, and Tanganyika Christian Relief Services.

- around 12,410 community members were trained on the effects of child marriage and GBV;
- 60,971 people were reached with VAWC messages and advocacy materials;
- Active Protection Committees in 4 districts were established;
- 105 victims of trafficking and survivors of domestic violence were accommodated in safe house and provided with basic needs;
- 201 women and children received legal aid services.

Priorities include strengthening of coordination structures, developing online training, improving response services, continued prevention advocacy activities with focus on men as actors to prevent violence. In support of these activities, the project's gender action plan will include activities to include SGBV modules for women and men participants of training activities, including how to access support and incident reporting mechanisms. Moreover, the lessons learned from the community feedback mechanism and complaints boxes within the camps will be extended to the host community setting with the support of the Protection Committees established in the districts.

In the local communities of Kigoma Region, the most commonly reported types of SGBV are physical violence, emotional violence, economic violence, human trafficking and child labour. Several factors contribute to the drivers of SEAH in the Kigoma region: traditional gender roles and cultural norms perpetuate practices like early marriage and gender-based violence; high poverty levels and limited livelihood opportunities increase vulnerability; insufficient infrastructure, limited access to services, and low awareness of support services hinder survivors' ability to seek help; overcrowding, resource scarcity, and tensions between communities contribute to the risk.

The government of Tanzania has taken measures to prevent and address SGBV and SEAH at all levels. This includes adopting a National Plan of Action and establishing Women and Children Protection Committees at various levels, staffed by professionals from different backgrounds. These committees serve as primary points for collecting cases of violence against women and children. One Stop Centers, including four in the Kigoma region, act as referral points and provide services to survivors. However, more centers are needed throughout the country.

To raise awareness, sensitization programs are conducted by various entities, including Women and Children Protection Committees, the Ministry of Social Affairs, Gender and Children Desks, and civil society organizations. Reporting mechanisms are in place, allowing victims to report to the committees, social welfare officers, police, and health officers. There is also a Toll-Free number available for reporting SGBV cases, ensuring confidentiality and anonymity.

For case management and referral, victims are referred to receive clinical and psychosocial support through One Stop Centers, Police Gender and Children Desks, Women and Children Protection Committees, and social welfare officers. Psychological counselling is provided by professionals in healthcare facilities or One Stop Centers.

1.4. Gender considerations in the project design

1.4.1. Approach and problem analysis

Without a thorough understanding of the specific roles, responsibilities and opportunities that are ascribed to women and men, the project's activities may add to women's 'time poverty' (the opportunity cost of the non-remunerated hours women spend performing domestic/care/household work, which could have otherwise been spent on economically remunerative activities). From the overview of gender issues above, it becomes clear that women, both from the refugee as well as the host communities, are bearing the responsibility for a substantial number of household tasks. This entails productive and reproductive tasks and includes agriculture production and post-harvest activities, collecting fuelwood and water, food preparation and child care. Through the implementation of the project activities, women are expected to adopt climate-resilient agricultural techniques, sustainable forest management practices and alternative livelihoods – altering and in some cases increasing their time-use.

The assessment of gender issues prevalent among host communities and refugees in Kigoma region, and the considerations on gender-related issues for forestry, agriculture and land use have been used to identify the main problems related to gender. The table below provides a problem analysis with regards to gender issues, while the following sections describe how these are mainstreamed into the project design.

Root causes	**Insecure land tenure and exclusion from access and control over land**	**Gender-blind policies, discriminatory social and cultural norms related to land use, agriculture, water and forestry**	**Gender gaps in skills and access to information and knowledge**	**Women and girls' burden of unpaid care and domestic work**
Causes	Limited and inadequate adaptation to climate change and insufficient resilience measures	Women's low participation in decision-making processes related to land, agriculture and water. Limited participation in farmers groups	Lack of knowledge and skills to diversify livelihoods in a climate-resilient way	Greater demand on their time to collect firewood and fetch water for domestic use and irrigation
Immediate effects	Land and forest degradation, compromised agriculture yield	Limited access to markets and to income-generating opportunities	Lower adaptive capacity	Women have limited time to engage in income generating activities
Effects	Lower productivity and income generated from natural resources	Lower productivity and income generated from	Individual and household resilience to climate change is compromised, food	Women have less opportunity to earn an income and are economically

		agricultural activities	insecurity is exacerbated	disadvantaged and more dependent

1.4.2. Gender mainstreaming in the project design

Gender mainstreaming means deliberately giving visibility and support to both women's and men's participation, contribution and benefit, and its ultimate goal is to achieve gender equality. The rationale for considering gender issues in the project is that men and women not only play different roles in the communities, with distinct levels of control over resources, but they often have different needs too. It is, therefore, important to treat gender issues in an integrated development strategy intended to reverse natural resource depletion in general and address ecosystem services too. Thus, to address gender inequalities, it is of crucial importance to take into account the particular needs of women in the framework of ecosystem-based adaptation activities.

It is also necessary to remain sensitive to not exposing women to greater risk as a result of their empowerment through project activities. UNHCR identified that, if managed poorly, empowerment projects where refugee women begin to earn income/engage in SAs, can expose them to inter-partner violence, as male partners struggle to accept these changes. The project interventions under the action plan should remain cognisant and sensitive to these dynamics, to ensure women are sufficiently capacitated to exercise their agency and empowerment, and that male community members are not ‘left behind’ but are engaged participants in women’s empowerment.

The gender assessment and problem analysis identified gaps in access and control over resources and opportunities. Mainstreaming gender in the project design is achieved through the use of sex-disaggregated data, together with developing strategies and actions to close those gaps, devoting resources and expertise for implementing equality strategies and monitoring results using gender sensitive indicators.

The VLUP processes supported by the project in host communities under output 1 are participatory in nature and the national guidelines provide suggestions to ensure that women are represented in the decision-making process. These will be followed, and additional efforts are detailed in the gender action plan. The VLUP seeks to include agreements that change traditional land management practices, thus improving the position of women in controlling land and natural resources.

Mainstreaming gender considerations in forestry-related activities can be done at different levels[50] [51] and the project’s gender mainstreaming activities under output 2 will build on good practices that have proven effective. The project will ensure that staff who provide training (i.e. extension staff) also receive gender training, in order to avoid that they follow socio-cultural norms that inhibit women’s participation in the project forestry activities. Also, project aims to set aside land for jointly managing forests for sustainable harvest of fuelwood

[50] FAO (n.d.) How to mainstream gender in forestry. A practical field guide.

[51] Gender and sustainable forest management: entry points for design and implementation. https://www.climateinvestmentfunds.org/sites/cif_enc/files/knowledge-documents/gender_and_sustainable_forest_management.pdf

that will be accessible to refugees and host communities. Consultations among both groups have found interest by women in this activity among both refugees and host communities.

Gender mainstreaming will consider the capacity building activities provided under Output 3 (e.g. Farmer Field Schools), and are geared towards improving women's access to inputs and resources delivered under the project (e.g. access to agriculture inputs and provision of beekeeping equipment). The project will include awareness raising and skills to extension officers in promoting gender equity throughout the training and agricultural support activities. The project will ensure that both men and women extension officers are equipped with participatory tools and skills to manage diverse cultural contexts and power dynamics. The project agriculture support will prioritise two crops – cassava and beans – and will focus on increasing their productivity. In the analysis of crops' supply chains, cassava and beans scored the highest in terms of market potential, potential for promoting gender equity and women's economic empowerment, contributing to food security, and income potential.[52] Agriculture support activities under the project only target host community members. The project will learn from successful practices in Tanzania where extension services and FFS have effectively performed gender-sensitive training and access to information (such as Farmers' Groups Network (MVIWATA).[53]

The project acknowledges the intersectional systems of privilege and access that operate within the target sites, that may exclude women — particularly refugees or poor farmers — from consultative and decision-making processes and spaces. Additionally, women's needs, interests and knowledge are often marginalised when planning for, and implementing adaptation interventions. To address this inequality, the project will work to ensure women are included in the activities of Output 4, their needs, knowledge and interests are captured, and communication products are gender-sensitive in their content, design and distribution.

1.4.3. Project expected benefits promoting gender equality

The project's gender sensitive approach seeks to achieve practical and tangible benefits to women as well as bring about strategic changes in gender relations. For women to meaningfully benefit from the project intervention it is crucial they are included in the various decision-making processes. The gender action plan outlines concrete steps and activities that will ensure inclusion of women in decision making processes and project activities. Through enhanced capacities, skills, access to assets transferred to refugee and host community women by the project, they will be capacitated to improve their decision-making power both at the household and the community level. This effect is expected to extend beyond the duration of the project.

Women rely heavily on natural resources to be able to fulfil their various productive and reproductive tasks. Women, like men, engage in agriculture activities and by ensuring they

[52] Enabel Programme's document: Sustainable Agriculture Kigoma Regional Project (SAKIRP) - comparative analysis of the main crops

[53] Mbo'o-Tchouawou, M. and Colverson, K. 2014. Increasing access to agricultural extension and advisory services: How effective are new approaches in reaching women farmers in rural areas? Nairobi, Kenya: International Livestock Research Institute (ILRI).

are able to benefit from accessing agriculture trainings and inputs and irrigation investments, women in the host communities will increase their resilience to current and projected future climate change impacts. Furthermore, for women in the host communities, improved environmental management including conservation of water sources will improve their year-round access to water for domestic and productive uses and thereby reducing the burden on women who have to fetch water.

The land use planning in host communities and forestry related interventions are expected to lead to better managed natural resources which will substantially reduce women's work burden as less time will be spend collecting fuelwood. Moreover, and particularly for refugee women, the exposure to threat and GBV or other forms of violence will be reduced as they will no longer be compelled to move far from the camp boundaries in search of fuelwood. Finally, host community and refugee relations will improve as existing and expected future conflicts over fuelwood and deforestation will be mitigated.

Pro-actively targeting women in the project's alternative livelihoods activities and enabling their full participation, is expected to contribute to their economic empowerment. The strengthening of Savings Association is expected to bring positive externalities as, particularly in the host communities, women are generally well represented in these groups. Money obtained by women in host communities through these savings associations is not only used to invest in productive assets and activities, but also to meet household needs and pay for school fees. Increased incomes may enable some women to purchase fuelwood or other energy sources to meet (part of) their cooking energy requirements.

2. Gender Action Plan

Impact Statement: Improved land management and increased resilience of host communities and refugees, including poor and female-headed households, to climate change through land use planning, land use and forestry interventions, resilient agriculture and livelihood diversification and improved water management and drainage. **Outcome Statement:** The project will catalyse a paradigm shift by mainstreaming climate change adaptation concepts into the planning functions of multiple bodies whose mandate includes agriculture, water use, flood prevention and forestry and through the introduction of much-needed integrated land use planning tools.					
Project Outcome 1			Women are represented in and participate meaningfully in land use planning at the village level		
Project Output 1			Participatory Climate-resilient Land Use Planning (C-LUP) to support ecosystem-based adaptation		
Project Activity		**Gender Activity**	**Indicator**	**Target**	**Cost**
1.1	Support climate-resilient village land-use planning	1. Ensure women are represented at all levels and during all stages of the land use planning processes. This includes village leadership roles relating to land use planning (Village Land Councils (VLCs); newly formed District-level Participatory Land-Use Management (PLUM) teams; Village Land Use Management (VLUM) committees and District, Ward and Village level Environmental Management	1.a % of Village Councils adhering to gender participation requirements (quotas) outlined in national legislation[54].	1.a 100% by end of year 1 1.b 50% (for each) by end of year 1	$ 45,600 / year annual cost of Safeguards and Gender Officer) $19,800 /year part-time cost of Social & Environmental safeguards expert (IC) $ 25,000 / year (implementation of ESMS and Gender action plan)

[54] The project will promote this target to the extent possible, but recognizing that the Village Councils are independently organized and the project may not be able to influence this process.

		Committees (EMCs). (Year 1 Q3 – Year 2 Q2). Sub-activities will include: • Assess current levels of women's participation in these structures • Identify the specific barriers to women's participation • Propose measures to ensure full and equal participation of women • Develop and implement a strategy to assess whether LUPs fully capture women's needs and priorities (e.g. through focus group discussions, key information interviews). 2. Include women's needs and interests in finalising the number and distribution of interventions under Outputs 2 and 3 as part of the participatory land use planning process. (Year 1 Q3 – Year 2 Q2). Sub-activities may include: • Hold separate meetings to capture women's views during the LUP process and have representatives of women's groups publicly represent their voices during decision-making processes. • Organize meetings at a time of day and location conducive to their schedules and sensitive to their domestic responsibilities. • Ensure the LUP includes agreements that change traditional land management practices, thus improving the position of women in controlling land and natural resources. • Facilitate dialogue between men and women during the LUP process to address cultural barriers to women's involvement,	1.b % of VLUM committees and EMCs representatives that are women. 2 % of VLUPs that reflect women's needs and interests 3. Gender capacity assessment completed and number of trainings conducted	2. 100% by end of year 1 3. 1 gender capacity assessment completed and x[55] number of trainings conducted by the end of year 1	Other costs are embedded in activity costs, particularly in workshops, meetings, and training amounting to $ 156,000 for Output 1.1

[55] Final number to be determined through assessment.

		and ensure women are not perversely affected by these actions • Promote awareness among women and men on incidence of and avoiding gender-based violence as well as access to community feedback mechanisms, complaints boxes, and referral pathways to obtain support • Ensure awareness raising activities are sensitive to the different learning and language dynamics and preferences of men and women • Provide appropriate information to women regarding their rights to access to resources under the project (e.g. distribution of agricultural inputs). 3. Safeguards and gender officer to assess capacity of district staff to address gender issues in the project and conduct trainings accordingly to ensure that gender expertise is present at the district level, and that the District PLUM team has sufficient awareness of gender equality related to land access, participation in decision-making processes, control over resources. (Years 1, 2 and 3) 4. Conflict sensitivities around Village Land Use Planning that could heighten SEAH risk will be mitigated through a participatory, conflict sensitive and gender responsive approach to Village Land Use Planning. that includes different different different stakeholder groups such as pastoralists, agropastoralists, farmers and other land users. 5.Reduce the risk of intimate partner violence by involving male partners in beneficiary selection processes from the outset to promote understanding of the positive impact of empowering women and the benefits to the family. Continuous monitoring of			

		women's risks at the beneficiary level will be implemented in carrying out activities. 6.The project will mitigate SGBV/SEAH incidents by applying SEAH due diligence in selecting personnel, contractors, and partners. Contracts will include provisions against SGBV/SEAH, and training will be provided to ensure awareness and accountability. 7. The project will ensure target groups are aware of and have access to the SGBV and SEAH protocols in place inside the camps and in the host villages/districts, which include prevention, protection, management and referral measures.			
Project Outcome 2			Community based forest management (CBFM) and agroforestry practices engage women in resilience building and decrease their vulnerability		
Project Output 2			Degraded ecosystems restored to support climate change adaptation in host and refugee communities		
Activity		**Gender Component**	**Indicator**	**Target**	**Cost**
2.1	Implement Community Based Forest Management (CBFM) for resilient ecosystems	1.Ensure women are included in and meaningfully participate in CBFM activities. The Safeguards and Gender expert will support the local government authorities in the development of inclusive work plans to ensure women's participation in planning and execution of community-based forestry management (CBFM). (Year 2 and Year 3) This may include: • Organise women-only training sessions and where possible invite female extension	1. % of villagers (host community) involved in CBFM are women	1.50% at project completion	$ 45,600 / year (annual cost of Safeguards and Gender Officer) $19,800/year part-time cost of Social & Environmental safeguards expert (IC) $ 25,000 / year (implementation of ESMS and gender action plan)

		workers to allow participants to feel comfortable to express their ideas and needs. Include training modules for women and men training participants on GBV for women and men and access to community feedback mechanisms, complaints boxes, and referral pathways to obtain support for survivors of GBV. • Implement capacity development activities by ensuring that all participants have the same knowledge to work together more effectively. Activities can also be targeted to the needs of the women participants, in particular considering women's use of NFTPs. • Engage women stakeholders by planning activities and meetings at a time of day and in a location conducive to their schedules and sensitive to their domestic responsibilities. • Hold trainings in locations with childcare services, or in spaces where children may be accommodated during training. 2. Reduce the risk of intimate partner violence by involving male partners in beneficiary selection processes from the outset to promote understanding of the positive impact of empowering women and the benefits to the family. Continuous monitoring of women's risks at the beneficiary level will be implemented in carrying out activities. 3.The project will mitigate SGBV/SEAH incidents by applying SEAH due diligence in selecting personnel, contractors, and partners. Contracts will include provisions against SGBV/SEAH, and training will be provided to ensure awareness and accountability.			Other costs are embedded in activity costs, particularly in workshops, meetings, and training amounting to $ 18,750 for Output 2.1

		4.The project will ensure target groups are aware of and have access to the SGBV and SEAH protocols in place inside the camps and in the host villages/districts, which include prevention, protection, management and referral measures.			
2.2	Implement 12,000 hectares of agroforestry and village land afforestation to increase resilience of land use	1. Provide agroforestry and village land afforestation training to women (Year 2) 2. Support women farmers to increase their resilience through agroforestry and village land afforestation. The Safeguards and Gender expert will support the local government authorities in the development of inclusive work plans (Years 2, 3, 4 and 5) to ensure women's participation in agroforestry activities following the following principles: • Organise women-only training sessions and where possible invite female extension workers to allow participants to feel comfortable to express their ideas and needs. Include training modules for women and men training participants on GBV and access to community feedback mechanisms, complaints boxes, and referral pathways to obtain support for survivors of GBV • Implement capacity development activities by ensuring that all participants have the same knowledge to work together more effectively. Activities can also be targeted to the needs of the women participants, in particular considering women's responsibility in collecting fuelwood • Engage women stakeholders by planning activities and meetings at a time of day and in a location conducive to their	1. % of women receiving training 2 % of women benefitting from agroforestry activities 3 % of men and women beneficiaries with reduced time devoted to fuelwood collection	1. 50% of women (including host community and refugee population) receiving training at project completion 2. 50% of women (including host community and refugee population) benefitting from agroforestry activities at project completion 3. 20% at project completion (including host community and refugee population)	$ 45,6000/ year (annual cost of Safeguards and Gender Officer) $19,800/year part-time cost of Social & Environmental safeguards expert (IC) $ 25,000 / year (implementation of ESMS and gender action plan)

		schedules and sensitive to their domestic responsibilities. • Hold trainings in locations with childcare services, or in spaces where children may be accommodated during training. 3.Reduce the risk of intimate partner violence by involving male partners in beneficiary selection processes from the outset to promote understanding of the positive impact of empowering women and the benefits to the family. Continuous monitoring of women's risks at the beneficiary level will be implemented in carrying out activities. 4.The project will mitigate SGBV/SEAH incidents by applying SEAH due diligence in selecting personnel, contractors, and partners. Contracts will include provisions against SGBV/SEAH, and training will be provided to ensure awareness and accountability. 5.The project will ensure target groups are aware of and have access to the SGBV and SEAH protocols in place inside the camps and in the host villages/districts, which include prevention, protection, management and referral measures.			
2.3	Implement flood and erosion control in densely populated areas to reduce the exposed and flood-prone ferrosol-type soils which have little or no vegetation cover	1.Ensure participation of women in flood and erosion control investment planning during the stakeholder consultations that will be held in Year 2, Q3. Separate meetings with women will be held to ensure women are consulted on the investments are flood and erosion control measure are prioritized based on their contribution to reducing women's vulnerability to climate change. The Safeguards and Gender Officer will develop and support the implementation of an assessment (e.g. through focus group discussions or interviews with key informants) to determine the impact of the flood and erosion investments on women's vulnerability to climate change	1.% of people consulted in flood and erosion investment planning	1.50% of refugee and host population consulted are women by year 3	$ 45,600 / year (annual cost of Safeguards and Gender Officer) $19,800/year part-time cost of Social & Environmental safeguards expert (IC) $ 25,000/ year (implementation of ESMS and gender action plan)

		2.The project will mitigate SGBV/SEAH incidents by applying SEAH due diligence in selecting personnel, contractors, and partners. Contracts will include provisions against SGBV/SEAH, and training will be provided to ensure awareness and accountability. 3.The project will ensure target groups are aware of and have access to the SGBV and SEAH protocols in place inside the camps and in the host villages/districts, which include prevention, protection, management and referral measures.			

Project Outcome 3			Adaptive capacity of women farmers increased and women's resilience to climate change increased economically and through increased food security		
Project Output 3			**Climate-resilient livelihoods practiced to increase the capacity of host communities and refugees to better adapt in changing climatic conditions**		
Activity		**Gender Component**	**Indicator**	**Target**	**Cost**
3.1	Promote modern technologies and management practices to strengthen the capacity of farmers, district officials and agricultural extension workers in climate-resilient agriculture	1. Build gender capacity of extension staff. The Safeguards and Gender Officer will be tasked to assess the capacity staff who provide training (i.e. extension staff) to address gender issues and will provide gender training as required, to equip them with the knowledge and skills to target and support men and women farmers equitably (Year 2 Q1) This will include: • Assess the capacity the SAMOs to address gender issues and will provide gender training as required, to equip them with the knowledge and skills to target and support men and women farmers equitably. • Support to the development of gender-responsive communication and outreach materials for the community trainings to ensure communities are aware of the benefits of participating and expected results, etc. • Identify preferred and trusted communication channels and methods for women. 2. Build women's capacity in climate resilient agriculture and increase their productivity through targeted training in Years 2- 5, including through the following:	1. Number of extension staff that are participating in gender training 2. % of FFS training that are women 3.a. % of direct beneficiaries receiving agriculture inputs that are women 3.b # and % of women practicing climate smart agriculture.	1. X extension staff are given gender training by Year 2 2. 50% of beneficiaries receiving FFS training are women at year 5 (host community) 3.a 50% at project completion (host community) 3.b 750 (50%) by mid-term; 7,500 (50%) by and of project (host community)	$ 45,600 / year (annual cost of Safeguards and Gender Officer) $19,800/year part-time cost of Social & Environmental safeguards expert (IC) $ 25,000/ year (implementation of ESMS and gender action plan) Other costs are embedded in activity costs, particularly in workshops, meetings, and training amounting to $ 207,550 for Output 3.1

		• Take a flexible approach in forming the groups for coordination and capacity building, considering both mixed-sex groups and single – sex groups. • If deemed beneficial, organise women-only training sessions and where possible invite female extension workers to allow participants to feel comfortable to express their ideas and needs. • Schedule training at times when women can attend. This may include organizing training outside of lunch hours to avoid conflicts with women's household responsibilities and limiting the length of training so that women could attend and still see to other activities. Include training modules for women and men training participants on GBV and access to community feedback mechanisms, complaints boxes, and referral pathways to obtain support for survivors of GBV. • Provide transportation if travel distances would hamper their participation. • Organize childcare services, or in spaces where children may be accommodated during training. • Ensure the location of the demo plots allows women to easily access these without spending too much time travelling also considering any protection/GBV risks that may arise from travelling to and from the demo plots. 3. Ensure women are receiving agricultural inputs and are participating in climate smart agriculture activities (Years 2 -5), including through the following actions: • Ensuring women are fully able to participate in the saving associations			

		• Ensuring women are equally benefitting from the distribution of agriculture inputs, including vulnerable women who may not be part of a savings association. 4.Reduce the risk of intimate partner violence by involving male partners in beneficiary selection processes from the outset to promote understanding of the positive impact of empowering women and the benefits to the family. Continuous monitoring of women's risks at the beneficiary level will be implemented in carrying out activities. 5.The project will mitigate SGBV/SEAH incidents by applying SEAH due diligence in selecting personnel, contractors, and partners. Contracts will include provisions against SGBV/SEAH, and training will be provided to ensure awareness and accountability. 6.The project will ensure target groups are aware of and have access to the SGBV and SEAH protocols in place inside the camps and in the host villages/districts, which include prevention, protection, management and referral measures.			
3.2	Increase water availability and use-efficiency through water harvesting and efficient irrigation interventions	1. Ensure women are benefitting from water harvesting and irrigation schemes. The Safeguards and Gender Officer will support inputs to designing the exercise to determine the beneficiary profile and select the technologies to ensure these are appropriate to address specific needs of women in year 2, Q3 This may include having separate meetings with women to determine their needs and discuss various technology options and how they would benefit them. The Safeguards and Gender Officer will develop and support the implementation of an assessment (e.g. through	1. % of women that were targeted by the water harvesting and/or irrigation schemes that have experienced improved water availability 2. % of WUA members are women.	1. 100% at project completion (host community) 2. 50% at project completion (host community)	$ 45,600 / year (annual cost of Safeguards and Gender Officer) $19,800/year part-time cost of Social & Environmental safeguards expert (IC) $ 25,000/ year (implementation of ESMS and gender action plan) Other costs are embedded in activity costs, particularly in

		focus group discussions or interviews with key informants) to determine the impact of water harvesting and irrigation schemes on water availability and women's access to water in year 5. 2. Ensure women are able to participate in Water User Associations. Efforts will include: • As required, implement capacity development activities specifically targeting women to ensure that all participants have the same knowledge to work together more effectively. Include training modules for women and men training participants on GBV and access to community feedback mechanisms, complaints boxes, and referral pathways to obtain support for survivors of GBV. • Schedule training/meetings at times when women can attend. This may include organizing training outside of lunch hours to avoid conflicts with women's household responsibilities and limiting the length of training so that women could attend and still see to other activities. • Provide transportation if travel distances would hamper their participation. • Organize childcare services, or in spaces where children may be accommodated during training. 3.The project will mitigate SGBV/SEAH incidents by applying SEAH due diligence in selecting personnel, contractors, and partners. Contracts will include provisions against SGBV/SEAH, and training will be provided to ensure awareness and accountability,			workshops, meetings, and training amounting to $ 76,400 for Output 3.2

		4.The project will ensure target groups are aware of and have access to the SGBV and SEAH protocols in place inside the camps and in the host villages/districts, which include prevention, protection, management and referral measures.			
3.3	Promote climate-resilient livelihood diversification to strengthen food security and nutrition, provide alternative income as a safety net, and to sustain the implementation of climate-resilient agricultural and forest management practices	1. Target women participants in beekeeping. The Safeguards and Gender Specialist will work with implementing partners, including the Tanzania Forest Service, to develop selection criteria are targeted to vulnerable and marginalised groups, including women. Female headed households will be specifically targeted, particularly as the alternative livelihood activities are considered to have a relative low workload. (Year 3 Q1) 2. Target women participants for mushroom growing. The Safeguards and Gender Specialist will work with implementing partners, including the Tanzania Forest Service, to develop selection criteria are targeted to vulnerable and marginalised groups, including women. Female headed households will be specifically targeted, particularly as the alternative livelihood activities are considered to have a relative low workload. (Year 3 Q1) 3. Provide training to women (Years 2-5), taking the following into consideration: • Take a flexible approach in forming the groups for coordination and capacity building, considering both mixed-sex groups and single – sex groups.	1. # and % of participants in beekeeping that are women 2. # and % of participants in mushroom growing activities that are women 3. % of beneficiaries trained in diversified livelihood activities are women	1. 40 (50%) by mid-term; 425 (50%) by end of project[56] (host community and refugee population) 2. 25 (80%) by mid-term; 250 (80%) by end of project (host community and refugee population) 3. 50 % of beneficiaries trained in diversified livelihood activities are women by end of project (host community and refugee population)	$ 45,600/ year (annual cost of Safeguards and Gender Officer) $19,800/year part-time cost of Social & Environmental safeguards expert (IC) $ 25,000 / year (implementation of ESMS and gender action plan) Other costs are embedded in activity costs, particularly in workshops, meetings, and training amounting to $ 19,590 for Output 4.1

[56] Beekeeping may be an activity traditionally associated with men in the target communities. Targets will be revised at the baseline / inception phase of the project.

		• If deemed beneficial, organise women-only training sessions and where possible invite female extension workers to allow participants to feel comfortable to express their ideas and needs. Include training modules for women and men training participants on GBV and access to community feedback mechanisms, complaints boxes, and referral pathways to obtain support for survivors of GBV. • Schedule training at times when women can attend. This may include organizing training outside of lunch hours to avoid conflicts with women's household responsibilities and limiting the length of training so that women could attend and still see to other activities. • Provide transportation if travel distances would hamper their participation. • Organize childcare services, or in spaces where children may be accommodated during training. 4. Reduce the risk of intimate partner violence by involving male partners in beneficiary selection processes from the outset to promote understanding of the positive impact of empowering women and the benefits to the family. Continuous monitoring of women's risks at the beneficiary level will be implemented in carrying out activities. 5.The project will mitigate SGBV/SEAH incidents by applying SEAH due diligence in selecting personnel, contractors, and partners. Contracts will include provisions against SGBV/SEAH, and training will be provided to ensure awareness and accountability.			

		6.The project will ensure target groups are aware of and have access to the SGBV and SEAH protocols in place inside the camps and in the host villages/districts, which include prevention, protection, management and referral measures.			
Project Outcome 4			Women and their needs, interests and knowledge are meaningfully captured in mainstreaming climate change adaptation into policies, plans, strategies and programmes.		
Project Output 4			Information on climate change adaptation disseminated and mainstreamed into policies, plans and strategies in Tanzania and in humanitarian programmes		
Activity		**Gender Component**	**Indicator**	**Target**	**Cost**
4.1	Generate evidence of the economic benefits of ecosystem-based adaptation to host and refugee populations, for use by policymakers and planners	1. Include gender consideration, emphasising women's empowerment and gender equality in reports and recommendations in Years 4 and 5 through: • Ensuring women are represented in working sessions for report distribution, by encouraging participation and scheduling and locating meetings that are sensitive to women's duties and child minding responsibilities. • Ensuring women are represented in providing inputs to the modelling methodology to value ecosystems, by encouraging participation and scheduling and locating stakeholder engagements that are sensitive to women's duties. • Ensuring women are represented in working sessions for report distribution, by encouraging participation and	1. Full and summary reports include sections on 'Gender Considerations' with gender-sensitive reporting, and recommendations for enhancing women's empowerment and gender equality in the application of findings. 1. b IASC Gender with Age marker score for all project reports [57]	1. a 100% by end of project 1.b All project reports have a IASC GAM score of 3 or higher.	$ 45,600 / year (annual cost of Safeguards and Gender Officer) $19,800/year part-time cost of Social & Environmental safeguards expert (IC) $ 25,000/ year (implementation of ESMS and gender action plan) Other costs are embedded in activity costs, particularly in workshops, meetings, and training amounting to $ 33,900 for Output 4.1

[57] https://www.iascgenderwithagemarker.com/en/home/

		scheduling and locating meetings that are sensitive to women's duties and child minding responsibilities. • The Safeguards and Gender Officer will screen all full and summary project reports for the IASC GAM score (using the IASC online tool) and will provide inputs and advise to ensure they reach the agreed rating. 2.The project will mitigate SGBV/SEAH incidents by applying SEAH due diligence in selecting personnel, contractors, and partners. Contracts will include provisions against SGBV/SEAH, and training will be provided to ensure awareness and accountability. 3.The project will ensure target groups are aware of and have access to the SGBV and SEAH protocols in place inside the camps and in the host villages/districts, which include prevention, protection, management and referral measures.			
4.2	Develop communication products to disseminate project results	1.Ensure that communication products (policy briefs and guidelines) prepared in Years 4 and 5 are gender-responsive in their content, design and distribution, including through the ensuring that: • Women, and their interests, needs and knowledge, are represented in consultations to compile lessons-learned, project outcomes, and guidelines. • Women, women's groups and institutions working in women's interests are represented in targeted stakeholders for receiving communication products. • 2.The project will mitigate SGBV/SEAH incidents by applying SEAH due diligence in selecting personnel, contractors, and partners. Contracts will include	1.Sections on 'Gender Considerations' included in every communication product developed and distributed, with recommendations for including women in scaling project interventions.	1.100% by end of project	$ 45,600 / year (annual cost of Safeguards and Gender Officer) $19,800/year part-time cost of Social & Environmental safeguards expert (IC) $ 25,000 / year (implementation of ESMS and gender action plan) Other costs are embedded in activity costs, particularly in workshops, meetings, and training amounting to $ 22,000 for Output 4.2

		provisions against SGBV/SEAH, and training will be provided to ensure awareness and accountability.			
4.3	Draft revisions to key plans and policies and support their integration into national and district government planning processes, to promote up-scaling of the EbA model	1.Draft guidelines and other inputs prepared in Years 4 and 5 to be gender-responsive in their content, design and distribution, by taking the following actions: • The Safeguards and Gender Officer will support the design of the review methodology for reviewing key plans, policies and processes to ensure women's adaptation needs and interests are fully considered. • Women and women's groups are represented in stakeholder consultations for conducting the review of key plans, policies and processes, including women-specific focus groups and/or workshops. • The Safeguards and Gender Officer will screen all project guidelines or other inputs to be produced for the IASC GAM score (using the IASC online tool) and will provide inputs and advise to ensure they reach the agreed rating. • The lead consultant for Activity 4.3, in collaboration with the Safeguards and Gender Officer will lead the development of evidence based advocacy messages that showcase the effects of the project's gender responsiveness and women's participation and how it is contributing to the protection and realization of women's rights. • 2. The project will mitigate SGBV/SEAH incidents by applying SEAH due diligence in selecting personnel, contractors, and	1. a 'Gender Considerations' sections are included in the outputs of the reviews of key plans, policies and processes, as well as in guidelines and other inputs developed. 1.b. IASC Gender with Age marker score for all project guidelines or other inputs	1. a 100% by end of project, with integrated evidence-based advocacy messages 1.b All project reports have a IASC GAM score of 3 or higher at project completion.	$ 45,600 / year (annual cost of Safeguards and Gender Officer) $19,800/year part-time cost of Social & Environmental safeguards expert (IC) $ 25,000 / year (implementation of ESMS and gender action plan) Other costs are embedded in activity costs, particularly in workshops, meetings, and training amounting to $ 39,000 for Output 4.3

		partners. Contracts will include provisions against SGBV/SEAH, and training will be provided to ensure awareness and accountability			
	Sub-total of dedicated costs for Safeguards and Gender Officer, Safeguards expert consultant and operating costs of ESMS and Gender Action Plan	The Safeguards and Gender Officer will oversee the implementation of the Gender Action Plan and the implementation of the Social and Environmental Management System (ESMS) with the technical advisory support of a Social and Environmental Safeguards international expert engaged part-time in support of the PMU. The Safeguards and Gender Officer will ensure that the project-level grievance redress mechanism (GRM) (See Annex 6: ESMS) will be accessible to women and men, including to women who are subject to SGBV, SEAH and sensitive to their special confidentiality and protection needs and refer cases to existing protection mechanisms accordingly.			$ 452,000
	Sub-total of project activity costs including workshops, training and meetings				$ 573,190
	Total amounts related to implementation of gender action plan				$ 1,025,190

Progress on the implementation of the Gender Action Plan will be reported in the Annual Performance Reports (APR), including:

- Description of gender activities implemented.
- Gender and population type disaggregation of activity beneficiaries and progress towards the gender targets.

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